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(54) **MAGEB2 BINDING CONSTRUCTS**

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ABSTRACT

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The present invention relates to binding constructs comprising a domain which binds to MAGEB2. Moreover, the invention provides polynucleotides encoding the binding constructs, a vector comprising said polynucleotides and a host cell transformed or transfected with said polynucleotides or vectors. Furthermore, the invention provides processes for producing the binding constructs, methods of treatment using the binding constructs, diagnostic uses of the binding constructs, and kits comprising the binding constructs.

Specification includes a Sequence Listing.

Gating Hierarchy Used in the Enrichment of MAGEB2 N-Term Helix, N-Term Peptide, and Middle Loop Region Peptide Binders

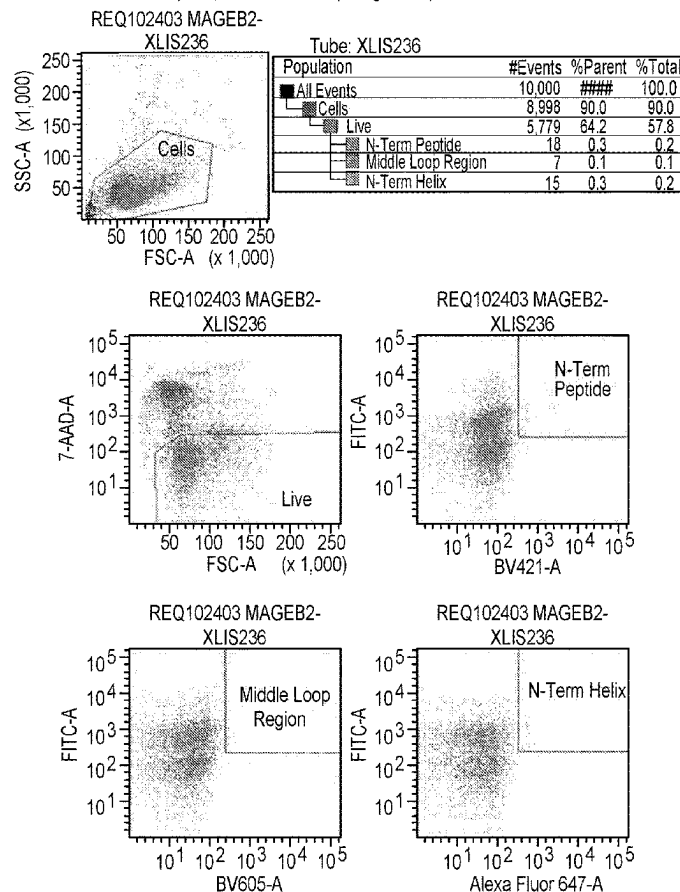


FIG. 1

Gating Hierarchy Used in the Enrichment of MAGEB2 N-Term Helix, N-Term Peptide, and Middle Loop Region Peptide Binders

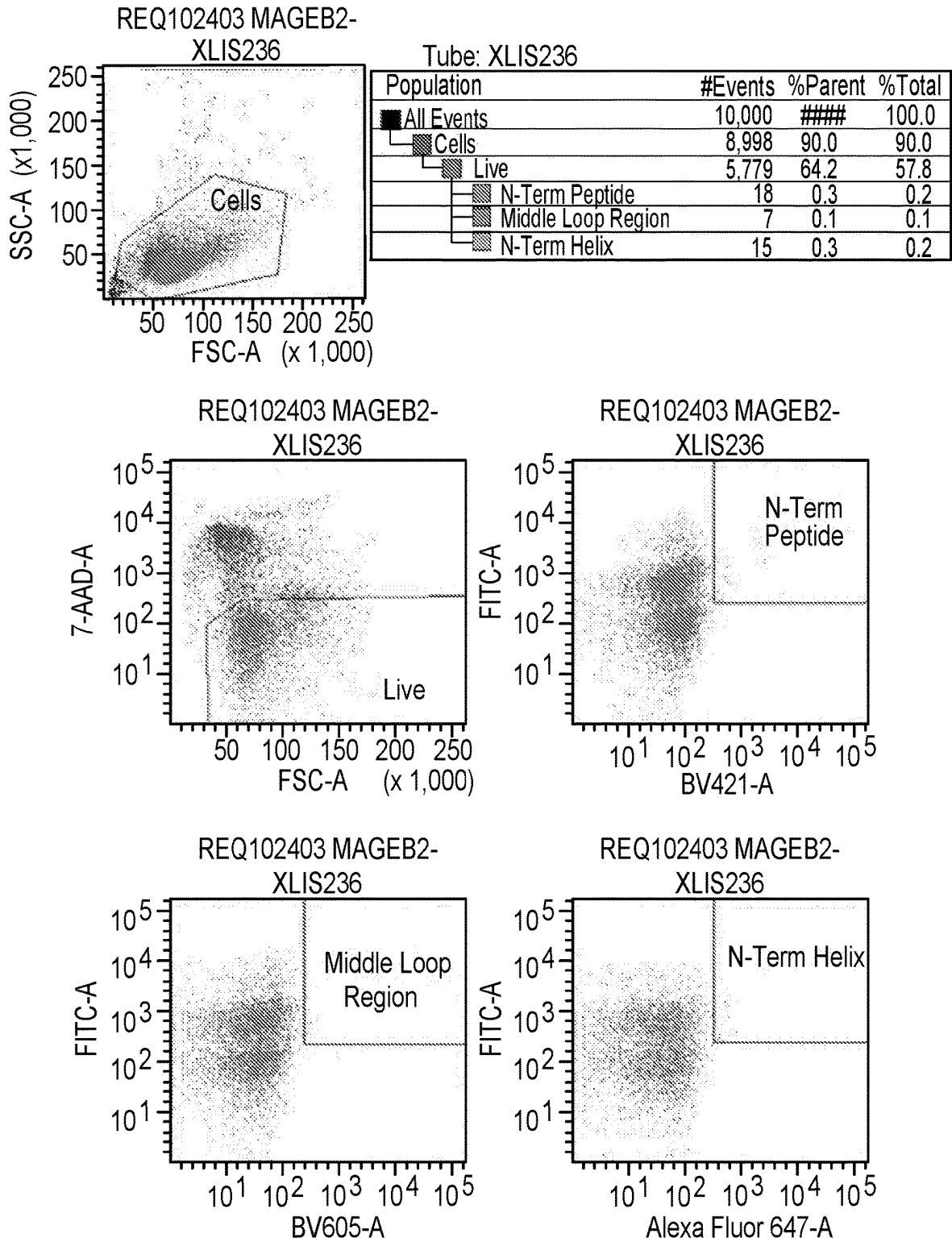


FIG. 2

ANTIBODIES THAT BIND TO FULL LENGTH MAGEB2 PROTEIN

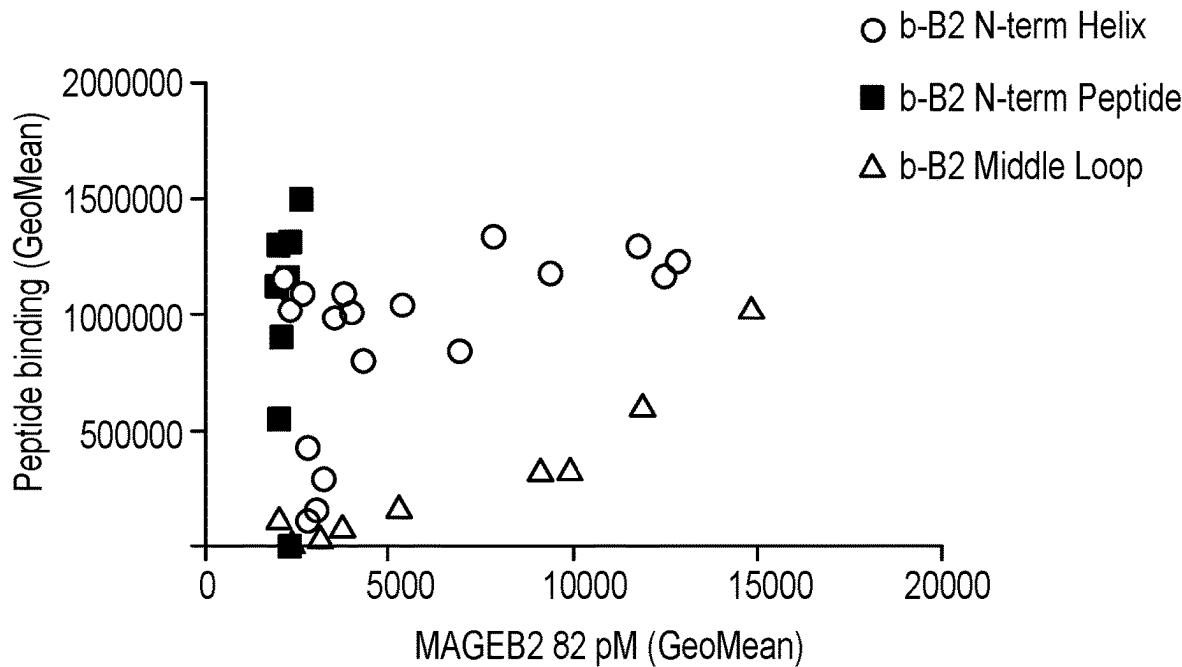
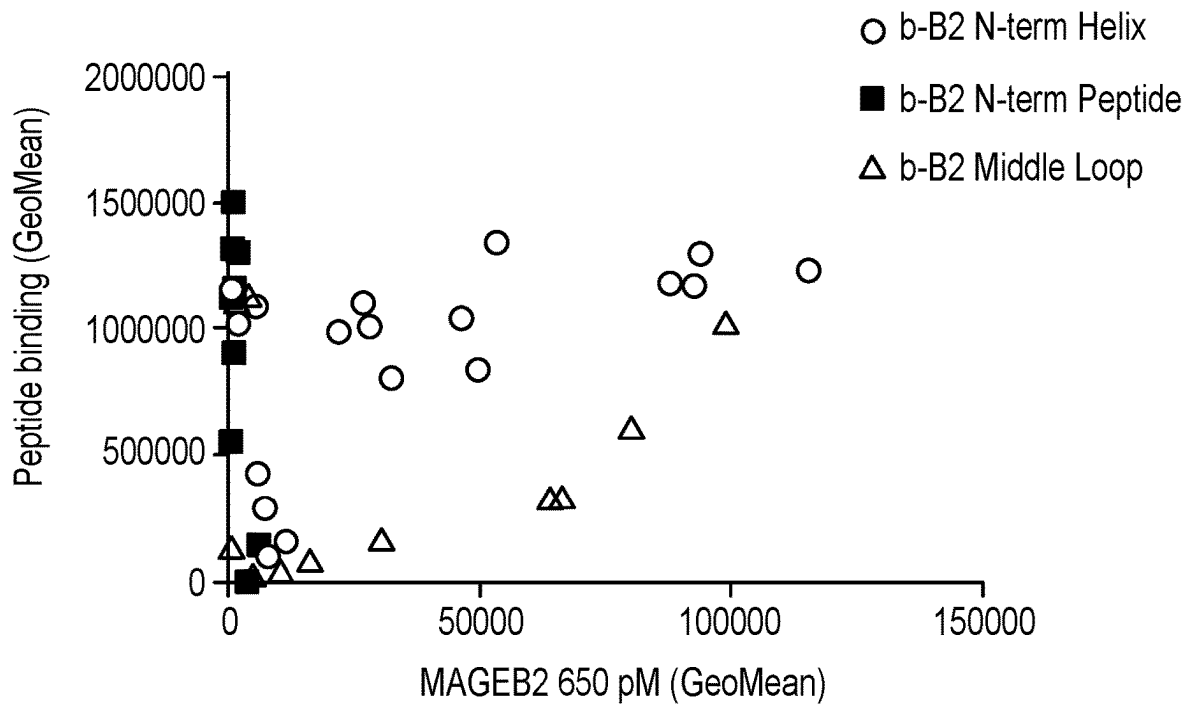


FIG. 3

IHC Assay in MAGEB2 Transfected Cells

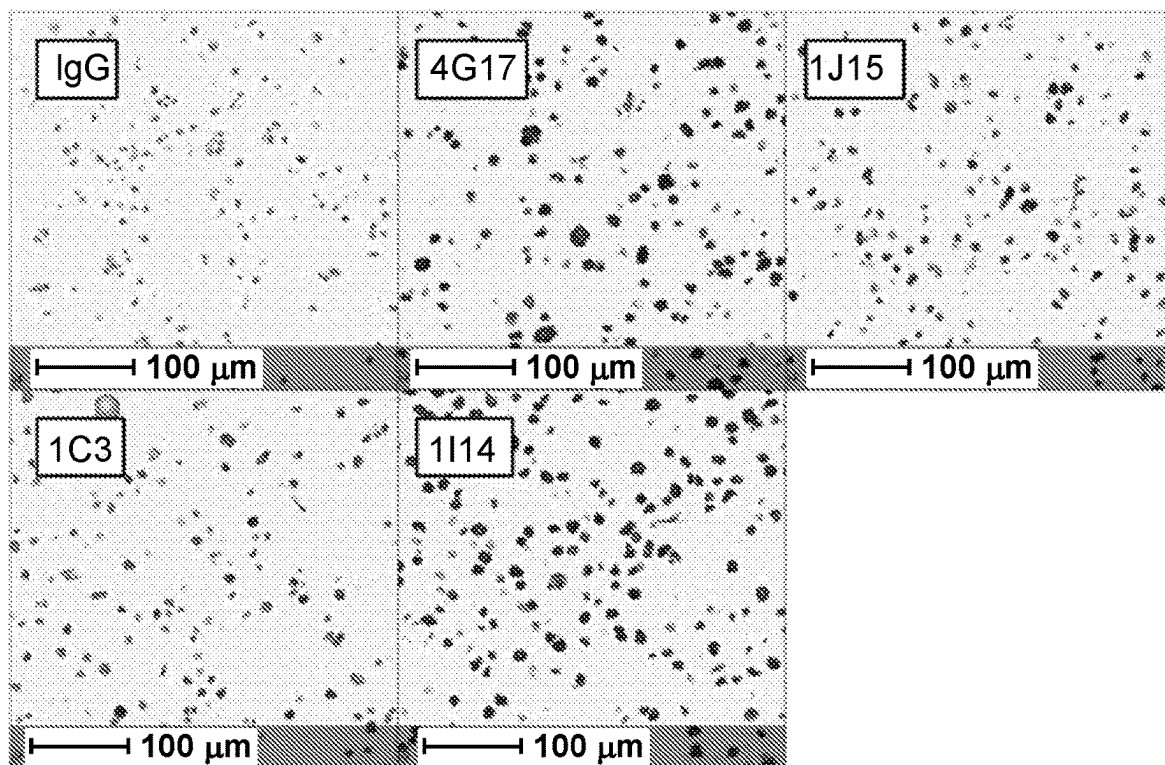


FIG. 4

IHC Assay in MAGEB5 Transfected Cells

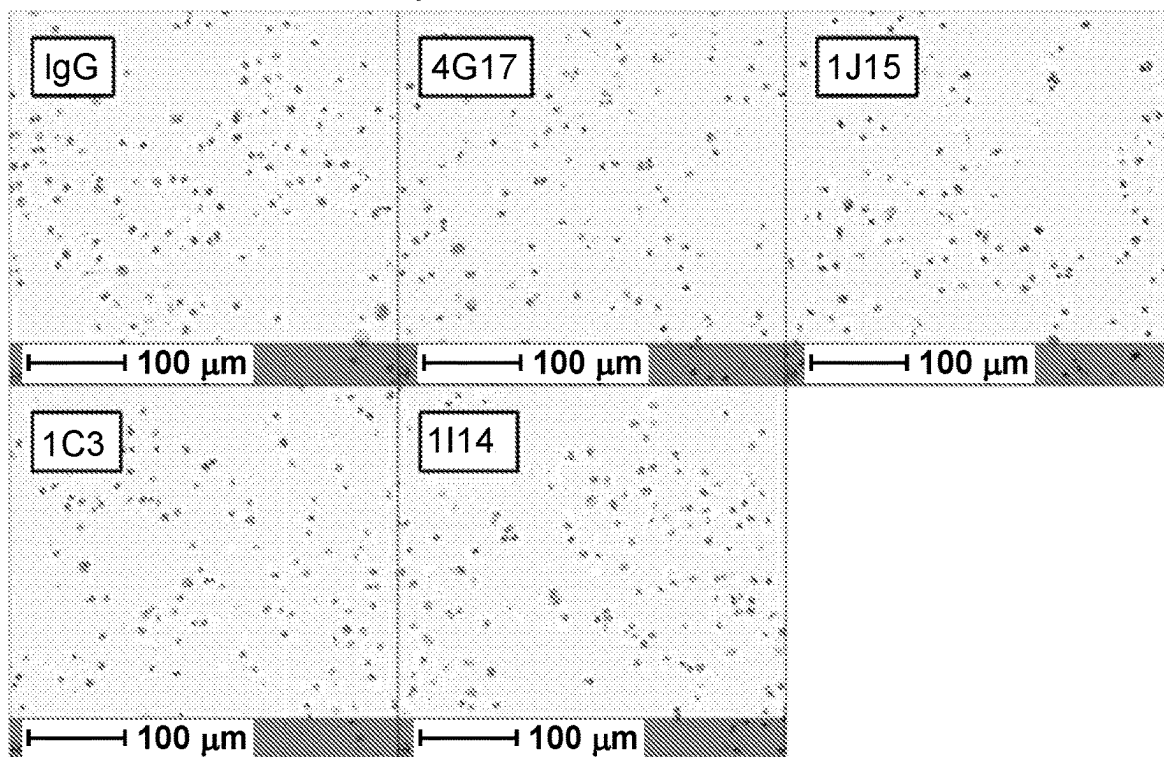


FIG. 5

IHC Assay in Control Testis Tissue

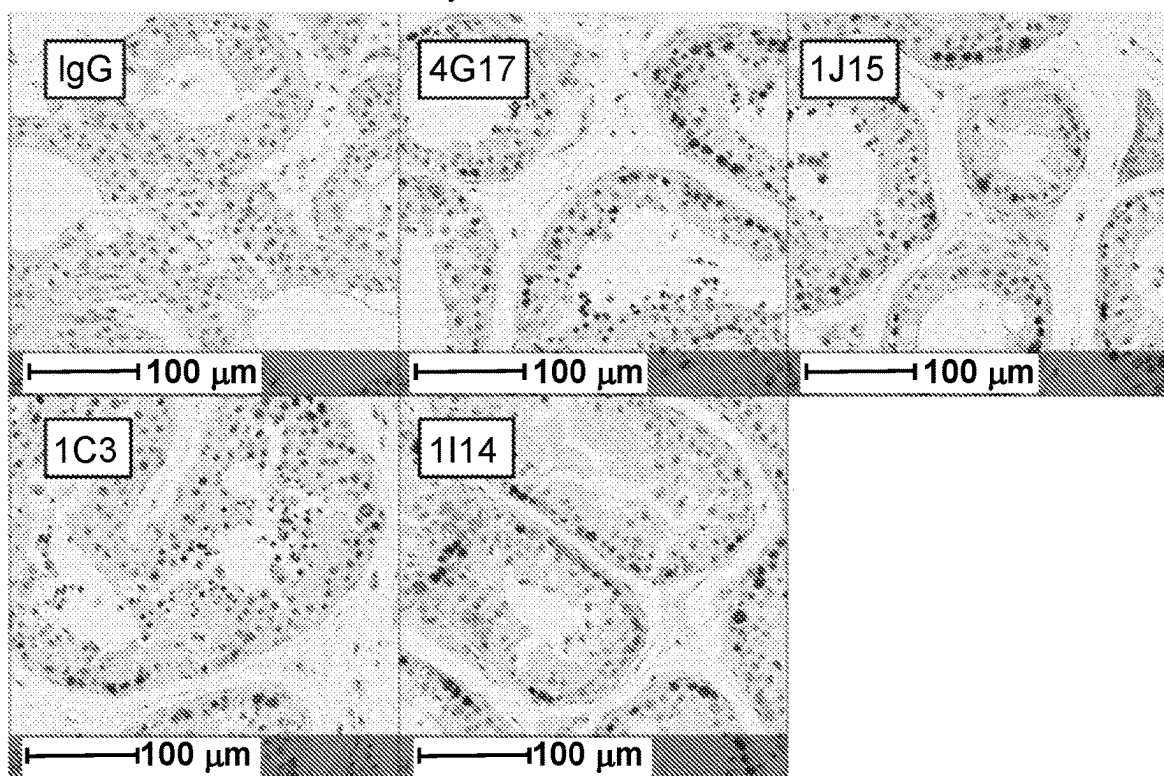


FIG. 6

IHC Assay in Normal Human Tissue

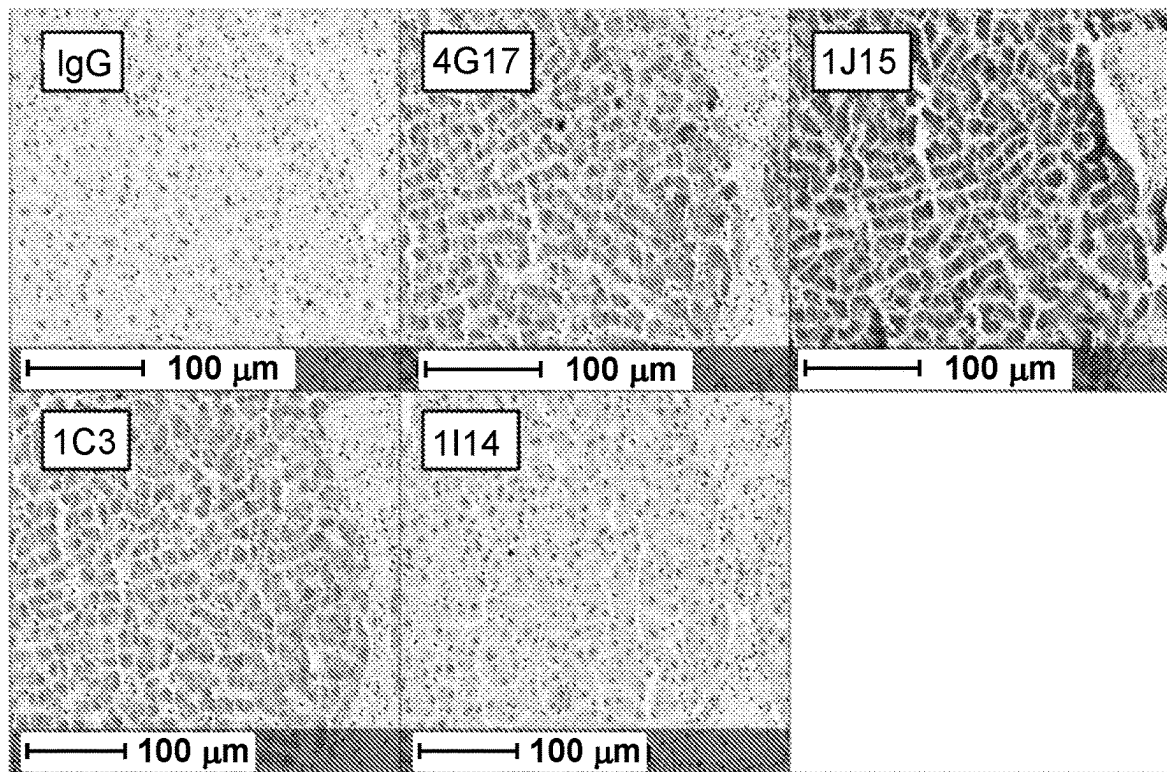
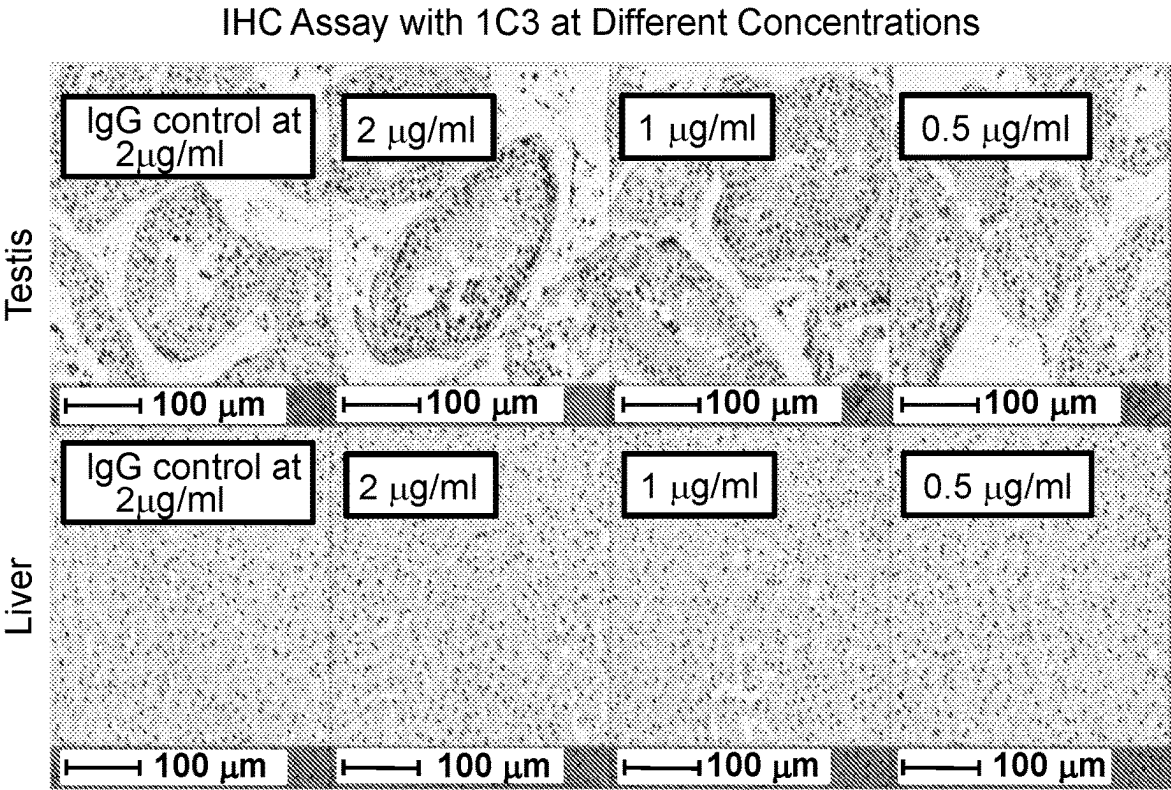


FIG. 7



MAGEB2 BINDING CONSTRUCTS**SEQUENCE LISTING**

[0001] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on May 26, 2022, is named A-2784-WO01-SEC_SL.txt and is 298,117 bytes in size.

FIELD OF THE INVENTION

[0002] The field of this invention relates to compositions and methods related to cancer treatments and diagnostics, including binding constructs that bind to MAGEB2.

BACKGROUND OF THE INVENTION

[0003] The MAGE (melanoma antigen genes) family contains about 60 genes that are categorized into several subfamilies. The MAGE-A, -B, and -C subfamilies are expressed mainly in the testis and are aberrantly expressed in various cancer types. The MAGE-D, -E, -F, -G, -H, -L, and -N subfamilies are expressed in a wide variety of tissues. See, e.g., Lee and Potts, J. Mol. Biol., 2018. One of the MAGE family members, MAGEB2, is typically only expressed in normal testis. MAGEB2, which may function to enhance ubiquitin ligase activity of RING-type zinc finger containing E3 ubiquitin protein ligases, has been found to be aberrantly expressed in a variety of human tumors such as lung carcinoma, breast carcinoma, melanoma, and others. Given this aberrant expression, MAGEB2 is a potential target for new therapeutic agents.

[0004] Early detection and classification of cancer is a crucial factor in successful treatment of the disease. A sensitive and precise diagnostic assay that allows detection and quantification of tumor antigens, e.g., MAGEB2, would aid in earlier detection and classification of cancer in patients and could also predict clinical response and outcome for appropriate cancer therapeutics.

[0005] The ability to reliably detect these tumor antigens may provide early indication of the disease and/or the disease progression. Immunological diagnostic assays are an important tool for detecting a variety of disease conditions, including cancer. However, such assays may not always be sensitive and/or specific enough to reliably detect particular tumor antigens located on tumor cells, particularly if they are expressed in low levels and/or are not expressed on the surface of tumor cells.

[0006] In some instances, a molecular diagnostic assay may be desired and may provide the required specificity and sensitivity, and therefore be the best option to detect a particular tumor antigen. However, in other instances it may also be desired to confirm proper expression of the tumor antigen within the tumor tissue sample and thus an immunohistochemical assay may be better suited.

[0007] Accordingly, there remains a need for sensitive and precise diagnostic assays to detect cancer antigens that are useful for detecting malignant cells and/or to help predict efficacy and to improve safety of the relevant therapeutic.

SUMMARY OF THE INVENTION

[0008] In one embodiment, the invention provides an isolated antigen binding construct that binds MAGEB2,

wherein the binding construct binds to an epitope comprising a sequence selected from SEQ ID NOs: 2, 3, 4, 562, or 388-554.

[0009] In another embodiment, the invention provides an isolated antigen binding construct, wherein the antigen binding construct comprises a CDRL1, a CDRL2, a CDRL3, a CDRH1, a CDRH2, and a CDRH3, wherein the CDRL1 comprises a sequence set forth in SEQ ID NO: 85; the CDRL2 comprises a sequence set forth in SEQ ID NO: 86; the CDRL3 comprises a sequence set forth in SEQ ID NO: 87; the CDRH1 comprises a sequence set forth in SEQ ID NO: 229; the CDRH2 comprises a sequence set forth in SEQ ID NO: 230; and the CDRH3 comprises a sequence set forth in SEQ ID NO: 231.

[0010] In another embodiment, the invention provides an antigen binding construct, wherein the antigen binding construct comprises a CDRL1, a CDRL2, a CDRL3, a CDRH1, a CDRH2, and a CDRH3, wherein the CDRL1 comprises a sequence set forth in SEQ ID NO: 73; the CDRL2 comprises a sequence set forth in SEQ ID NO: 74; the CDRL3 comprises a sequence set forth in SEQ ID NO: 75; the CDRH1 comprises a sequence set forth in SEQ ID NO: 217; the CDRH2 comprises a sequence set forth in SEQ ID NO: 218; and the CDRH3 comprises a sequence set forth in SEQ ID NO: 219.

[0011] In another embodiment, the invention provides an isolated antigen binding construct, wherein the antigen binding construct comprises a CDRL1, a CDRL2, a CDRL3, a CDRH1, a CDRH2, and a CDRH3, wherein the CDRL1 comprises a sequence set forth in SEQ ID NO: 91; the CDRL2 comprises a sequence set forth in SEQ ID NO: 92; the CDRL3 comprises a sequence set forth in SEQ ID NO: 93; the CDRH1 comprises a sequence set forth in SEQ ID NO: 235; the CDRH2 comprises a sequence set forth in SEQ ID NO: 236; and the CDRH3 comprises a sequence set forth in SEQ ID NO: 237.

[0012] In a further embodiment, the invention provides an isolated antigen binding construct, wherein the antigen binding construct comprises a light chain variable region comprising a sequence set forth in SEQ ID NO: 346 and a heavy chain variable region comprising a sequence set forth in SEQ ID NO: 347. In yet a further embodiment, the invention provides an isolated antigen binding construct, wherein the antigen binding construct comprises a light chain variable region comprising a sequence set forth in SEQ ID NO: 338 and a heavy chain variable region comprising a sequence set forth in SEQ ID NO: 339.

[0013] In another embodiment, the invention provides an antigen binding construct, wherein the antigen binding construct comprises a light chain variable region comprising a sequence set forth in SEQ ID NO: 350 and a heavy chain variable region comprising a sequence set forth in SEQ ID NO: 351.

[0014] In another embodiment, the invention provides a method of making an antibody that binds to MAGEB2 comprising immunizing an animal with a peptide comprising a sequence selected from SEQ ID NO: 2, 3, 4, or 562, and isolating from said animal antibodies that bind to MAGEB2.

[0015] In another embodiment, the invention provides a method for treating a tumor in a subject, said method comprising: determining the subject as responsive to treatment with an anti-MAGEB2 therapeutic by obtaining a sample from the subject, wherein the sample comprises a

cell from the tumor, measuring the level of MAGEB2 in the sample using an antigen binding construct provided herein, and determining the subject as responsive to treatment with an anti-MAGEB2 therapeutic, and administering to the subject an effective amount of the anti-MAGEB2 therapeutic.

[0016] In another embodiment, the invention provides a method of identifying a subject as needing an anti-MAGEB2 therapeutic comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct provided herein; and b) identifying the subject as needing the anti-MAGEB2 therapeutic when the level of MAGEB2 is increased relative to a control.

[0017] In another embodiment, the invention provides a method of determining treatment for a subject with a MAGEB2 positive tumor comprising: determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct provided herein; and determining the treatment as comprising an anti-MAGEB2 therapeutic when the level of MAGEB2 is increased, relative to a control. In another embodiment, the invention provides a method of determining efficacy of treatment with an anti-MAGEB2 therapeutic in a subject comprising: determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct provided herein before treatment with an anti-MAGEB2 therapeutic and after treatment with an anti-MAGEB2 therapeutic; and determining the treatment as effective when the level of MAGEB2 positive tumor cells is decreased after treatment with the anti-MAGEB2 therapeutic.

[0018] In another embodiment, the invention provides a method of diagnosing a subject with a tumor, comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct provided herein; and b) diagnosing the subject as having a MAGEB2 positive tumor when the level of MAGEB2 is increased relative to a control.

[0019] In another embodiment, the invention provides a method of identifying a subject having a MAGEB2 positive tumor comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct provided herein; and b) identifying the subject as having a MAGEB2 positive tumor when the level of MAGEB2 is increased relative to a control.

[0020] In another embodiment, the invention provides a method of identifying a subject as needing an anti-MAGEB2 therapeutic comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct provided herein; and b) identifying the subject as needing the anti-MAGEB2 therapeutic when the level of MAGEB2 is increased relative to a control.

[0021] In another embodiment, the invention provides a method of determining treatment for a subject with a MAGEB2 positive tumor comprising: determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct provided herein; and determining the treatment as comprising an anti-MAGEB2 therapeutic when the level of MAGEB2 is increased, relative to a control.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1. This figure shows the gating hierarchy used in the enrichment of MAGEB2 N-Term Helix, N-Term Peptide, and Middle Loop Region peptide binders as described in Example 2 herein.

[0023] FIG. 2. This figure shows two plots that show binding by various anti-MAGEB2 antibodies binding to immunogen peptide and full length MAGEB2 protein. The MAGEB2 panel was run in a limited antigen style assay (only highest concentration shown) to identify highest affinity binders to full length MAGEB2 protein.

[0024] FIG. 3. This figure shows immunohistochemical results for a MAGEB2 IHC assay that measured immunoreactivity in MAGEB2 transfected cells. Intense MAGEB2 IHC staining was observed in CHO-MAGE-B2+ cells with the four anti-MAGEB2 antibodies 4G17, 1J15, 1C3, 1I14, but not with IgG control.

[0025] FIG. 4. This shows immunohistochemical results for a MAGEB2 IHC assay that measured immunoreactivity in MAGEB5 transfected cells. No MAGEB2 IHC staining was observed in CHO-MAGEB5+ cells with the four anti-MAGEB2 antibodies 4G17, 1J15, 1C3, 1I14, and with IgG control.

[0026] FIG. 5. This figure shows immunohistochemical results for a MAGEB2 IHC assay that measured immunoreactivity in control testis tissues. Intense MAGEB2 IHC staining in spermatogonia cells was observed in testis with the four anti-MAGEB2 antibodies 4G17, 1J15, 1C3, 1I14, but not IgG control at 2 µg/ml. Some nuclear stain was seen with antibodies 4G17, 1J15, 1C3; Leydig cell stain was seen with antibody 1J15.

[0027] FIG. 6. This figure shows immunohistochemical results for a MAGEB2 IHC Assay that measured immunoreactivity in normal human tissue. Non-specific particulate staining in hepatocytes was seen in liver tissue core (in normal human tissue with anti-MAGEB2 antibodies 4G17, 1J15, 1C3, but not with anti-MAGEB2 antibody 1I14, and IgG control at 5 µg/ml).

[0028] FIG. 7. This figure shows immunohistochemical results for a MAGEB2 IHC assay that measured immunoreactivity in testis and liver tissue, with titration down of the 1C3 antibody. At 2 µg/ml Ab concentration, anti-MAGEB2 antibody 1C3 clone has weak to mild intensity specific staining in appropriate proportions of spermatogonia cells as previously observed. There is rare nuclear staining and the intensity is slightly less than that observed at higher concentrations. However, at this concentration there is no non-specific background staining in liver.

DETAILED DESCRIPTION

[0029] MAGEB2 is aberrantly expressed in a variety of human cancer types. MAGEB2 is not expressed on the surface of cells. MAGEB2 peptides, however, are displayed on the surface of tumor cells by the MHC class I molecule. In particular, the MAGEB2 peptide GVDGEEHSV (SEQ ID NO: 561) is displayed on the surface of tumor cells by the MHC class I molecule as a peptide-MHC complex. See, for example, U.S. Patent Appl. Publ. No. US2016/0250307A1 (U.S. patent application Ser. No. 14/975,952) and US2017/0080070A1 (U.S. patent application Ser. No. 15/357,757). Although the MAGEB2 peptide-MHC complex is displayed on the surface of cells and would potentially be a target for a diagnostic agent, the number of copies on the cell surface

of this peptide-MHC complex is far too low to be detected. Accordingly, binding constructs (e.g., antibodies) that are able to bind and detect MAGEB2 that is intracellularly expressed are described herein.

Binding Constructs

[0030] The present invention provides binding constructs comprising domains which bind to a MAGEB2 protein. These binding constructs are alternatively referred to as antigen binding constructs.

[0031] The term binding construct refers to a construct that is capable of binding to its specific target or antigen and comprises the variable heavy chain (VH) and/or variable light chain (VL) domains of an antibody or fragment thereof. Typically, a binding domain according to the present invention comprises the minimum structural requirements of an antibody which allow for the target binding. This minimum requirement may e.g. be defined by the presence of at least the three light chain CDRs (i.e. CDR1, CDR2 and CDR3 of the VL region) and/or the three heavy chain CDRs (i.e. CDR1, CDR2 and CDR3 of the VH region), preferably of all six CDRs. An alternative approach to define the minimal structure requirements of an antibody is the definition of the epitope the antibody binds within the structure of a specific target, respectively, the protein domain of the target protein composing the epitope region (epitope cluster) or by reference to a specific antibody competing with the epitope of the defined antibody. Alternatively, the minimal structure requirements may be defined by the paratope sequences within the binding domain of the antibody.

[0032] Binding constructs of the present invention comprise at least one binding domain. The term “binding domain” characterizes in connection with the present invention a domain which specifically binds to, interacts with, or recognizes a given target epitope or a given target region on the target molecule, e.g., MAGEB2 or a specific region within MAGEB2. The structure and function of the binding domain is based on the structure and/or function of an antibody, e.g. of a full-length or whole immunoglobulin molecule, and is from the variable heavy chain (VH) and/or variable light chain (VL) domains of an antibody or fragment thereof. In certain embodiments, the binding domain comprises the presence of three light chain CDRs (i.e. CDR1, CDR2 and CDR3 of the VL region) and/or three heavy chain CDRs (i.e. CDR1, CDR2 and CDR3 of the VH region). In certain embodiments, the binding domain is produced by or obtainable by phage-display or library screening methods rather than by grafting CDR sequences from a pre-existing (monoclonal) antibody into a scaffold.

[0033] The binding domain of a binding construct according to the invention may e.g. comprise the above referred groups of CDRs. Preferably, those CDRs are comprised in the framework of an antibody light chain variable region (VL) and an antibody heavy chain variable region (VH); however, it does not have to comprise both. Fd fragments, for example, have two VH regions and often retain some antigen-binding function of the intact antigen-binding domain. Additional examples for the format of antibody fragments, antibody variants or binding domains include (1) a Fab fragment, a monovalent fragment having the VL, VH, CL and CH1 domains; (2) a F(ab')₂ fragment, a bivalent fragment having two Fab fragments linked by a disulfide bridge at the hinge region; (3) an Fd fragment having the two VH and CH1 domains; (4) an Fv fragment having the VL

and VH domains of a single arm of an antibody, (5) a dAb fragment (Ward et al., (1989) Nature 341:544-546), which has a VH domain; (6) an isolated complementarity determining region (CDR), and (7) a single chain Fv (scFv), the latter being preferred (for example, derived from an scFv-library). Examples for embodiments of binding constructs according to the invention are e.g. described in WO 00/006605, WO 2005/040220, WO 2008/119567, WO 2010/037838, WO 2013/026837, WO 2013/026833, US 2014/0308285, US 2014/0302037, WO 2014/144722, WO 2014/151910, and WO 2015/048272.

[0034] In a specific embodiment of the invention, the binding construct is a full-length antibody, as described herein below.

[0035] Also within the definition of “binding domain” or “domain which binds” are fragments of full-length antibodies, such as VH, VHH, VL, (s)dAb, Fv, light chain (VL-CL), Fd (VH-CH1), heavy chain, Fab, Fab', F(ab')₂ or “r IgG” (“half antibody” consisting of a heavy chain and a light chain). Binding constructs according to the invention may also comprise modified fragments of antibodies, also called antibody variants or antibody derivatives. Examples include, but are not limited to, scFv, di-scFv or bi(s)-scFv, scFv-Fc, scFv-zipper, scFab, Fab₂, Fab₃, diabodies, single chain diabodies, tandem diabodies (Tandab's), tandem di-scFv, tandem tri-scFv, “minibodies” exemplified by a structure which is as follows: (VH-VL-CH3)₂, (scFv-CH3)₂, ((scFv)₂-CH3+CH3), ((scFv)₂-CH3) or (scFv-CH3-scFv)₂, multibodies such as triabodies or tetrabodies, and single domain antibodies such as nanobodies or single variable domain antibodies comprising merely one variable region, which might be VHH, VH or VL, that specifically binds to an antigen or target independently of other variable regions or domains. Further possible formats of the binding constructs according to the invention are cross bodies, maxi bodies, hetero Fc constructs, mono Fc constructs and scFc constructs. Examples for those formats will be described herein below.

[0036] According to the present invention, binding domains are in the form of one or more polypeptides. Such polypeptides may include proteinaceous parts and non-proteinaceous parts (e.g. chemical linkers or chemical cross-linking agents such as glutaraldehyde). Proteins (including fragments thereof, preferably biologically active fragments, and peptides, usually having less than 30 amino acids) comprise two or more amino acids coupled to each other via a covalent peptide bond (resulting in a chain of amino acids).

[0037] The term “polypeptide” as used herein describes a group of molecules, which usually consist of more than 30 amino acids. Polypeptides may further form multimers such as dimers, trimers and higher oligomers, i.e., consisting of more than one polypeptide molecule. Polypeptide molecules forming such dimers, trimers etc. may be identical or non-identical. The corresponding higher order structures of such multimers are, consequently, termed homo- or heterodimers, homo- or heterotrimers etc. An example for a heteromultimer is an antibody molecule, which, in its naturally occurring form, consists of two identical light polypeptide chains and two identical heavy polypeptide chains. The terms “peptide”, “polypeptide” and “protein” also refer to naturally modified peptides/polypeptides/proteins wherein the modification is effected e.g. by post-translational modifications like glycosylation, acetylation, phosphorylation and the like. A “peptide”, “polypeptide” or

“protein” when referred to herein may also be chemically modified such as pegylated. Such modifications are well known in the art and described herein below.

[0038] The definition of “antibody” according to the invention comprises full-length antibodies, also including camelid antibodies and other immunoglobulins generated by biotechnological or protein engineering methods or processes. These full-length antibodies may be for example monoclonal, recombinant, chimeric, deimmunized, humanized and human antibodies, as well as antibodies from other species such as mouse, hamster, rabbit, rat, goat, or non-human primates.

[0039] For the antibodies provided herein, the variable regions of immunoglobulin chains generally exhibit the same overall structure, comprising relatively conserved framework regions (FR) joined by three hypervariable regions, more often called “complementarity determining regions” or CDRs. The CDRs from the two chains of each heavy chain/light chain pair mentioned above typically are aligned by the framework regions to form a structure that binds specifically to the target epitope. From N-terminal to C-terminal, naturally-occurring light and heavy chain variable regions both typically conform with the following order of these elements: FR1, CDR1, FR2, CDR2, FR3, CDR3 and FR4. A numbering system has been devised for assigning numbers to amino acids that occupy positions in each of these domains.

[0040] This numbering system is defined in Kabat Sequences of Proteins of Immunological Interest (1987 and 1991, NIH, Bethesda, Md.), or Chothia & Lesk, 1987, J. Mol. Biol. 196:901-917; Chothia et al., 1989, Nature 342: 878-883.

[0041] The term “variable” refers to the portions of the antibody or immunoglobulin domains that exhibit variability in their sequence and that are involved in determining the specificity and binding affinity of a particular antibody (i.e., the “variable domain(s)”). The pairing of a variable heavy chain (VH) and a variable light chain (VL) together forms an antigen-binding domain.

[0042] Variability is not evenly distributed throughout the variable domains of antibodies; it is concentrated in sub-domains of each of the heavy and light chain variable regions. These sub-domains are called “hypervariable regions” or “complementarity determining regions” (CDRs). The more conserved (i.e., non-hypervariable) portions of the variable domains are called the “framework” regions (FRM or FR) and provide a scaffold for the six CDRs in three-dimensional space to form an antigen-binding surface. The variable domains of naturally occurring heavy and light chains each comprise four framework (FRM) regions (FR1, FR2, FR3, and FR4), largely adopting a β -sheet configuration, connected by three hypervariable regions, which form loops connecting, and in some cases forming part of, the β -sheet structure. The hypervariable regions in each chain are held together in close proximity by the framework regions and, with the hypervariable regions from the other chain, contribute to the formation of the antigen-binding site.

[0043] The terms “CDR”, and its plural “CDRs”, refer to the complementarity determining region of which three make up the binding character of a light chain variable region (CDR-L1, CDR-L2 and CDR-L3) and three make up the binding character of a heavy chain variable region (CDR-H1, CDR-H2 and CDR-H3). CDRs contain most of

the residues responsible for specific interactions of the antibody with the antigen and hence contribute to the functional activity of an antibody molecule, i.e., they are the main determinants of binding specificity to a particular target.

[0044] The exact definitional CDR boundaries and lengths are subject to different classification and numbering systems. CDRs may therefore be referred to by Kabat, Chothia, contact or any other boundary definitions, including the numbering system described herein. Despite differing boundaries, each of these systems has some degree of overlap in what constitutes the so called “hypervariable regions” within the variable sequences. CDR definitions according to these systems may therefore differ in length and boundary areas with respect to the adjacent framework region. See for example Kabat (an approach based on cross-species sequence variability), Chothia (an approach based on crystallographic studies of antigen-antibody complexes), and/or MacCallum (Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md., 1991; Chothia et al., J. Mol. Biol. 1987, 196: 901-917; and MacCallum et al., J. Mol. Biol. 1996, 262: 732). Still another standard for characterizing the antigen binding side is the AbM definition used by Oxford Molecular’s AbM antibody modeling software. See, e.g., Protein Sequence and Structure Analysis of Antibody Variable Domains, Antibody Engineering Lab Manual (Ed.: Duebel, S. and Kontermann, R., Springer-Verlag, Heidelberg). To the extent that two residue identification techniques define regions of overlapping, but not identical regions, they can be combined to define a hybrid CDR. However, the numbering in accordance with the Kabat system is preferred.

[0045] Typically, CDRs form a loop structure that can be classified as a canonical structure. The term “canonical structure” refers to the main chain conformation that is adopted by the antigen binding (CDR) loops. Each canonical structure can be characterized by the torsion angles of the polypeptide backbone. Correspondent loops between antibodies may, therefore, have very similar three dimensional structures, despite high amino acid sequence variability in most parts of the loops (Chothia and Lesk, J. Mol. Biol., 1987, 196: 901; Chothia et al., Nature, 1989, 342: 877; Martin and Thornton, J. Mol. Biol. 1996, 263: 800). Furthermore, there is a relationship between the adopted loop structure and the amino acid sequences surrounding it. The conformation of a particular canonical class is determined by the length of the loop and the amino acid residues residing at key positions within the loop, as well as within the conserved framework (i.e., outside of the loop). Assignment to a particular canonical class can therefore be made based on the presence of these key amino acid residues.

[0046] The term “canonical structure” may also include considerations as to the linear sequence of the antibody, for example, as catalogued by Kabat (Kabat et al.). The Kabat numbering scheme (system) is a widely adopted standard for numbering the amino acid residues of an antibody variable domain in a consistent manner and is the preferred scheme applied in the present invention as also mentioned elsewhere herein. Additional structural considerations can also be used to determine the canonical structure of an antibody. For example, those differences not fully reflected by Kabat numbering can be described by the numbering system of Chothia et al. and/or revealed by other techniques, for

example, crystallography and two- or three-dimensional computational modeling. Accordingly, a given antibody sequence may be placed into a canonical class which allows for, among other things, identifying appropriate chassis sequences (e.g., based on a desire to include a variety of canonical structures in a library). Kabat numbering of antibody amino acid sequences and structural considerations as described by Chothia et al., loc. cit. and their implications for construing canonical aspects of antibody structure, are described in the literature. The subunit structures and three-dimensional configurations of different classes of immunoglobulins are well known in the art. For a review of the antibody structure, see *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory, eds. Harlow et al., 1988.

[0047] As used herein, the terms “single-chain Fv,” “single-chain antibodies” or “scFv” refer to single polypeptide chain antibody fragments that comprise the variable regions from both the heavy and light chains, but lack the constant regions. Generally, a single-chain antibody further comprises a polypeptide linker between the VH and VL domains which enables it to form the desired structure which would allow for antigen binding. Single chain antibodies are discussed in detail by Pluckthun in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds. Springer-Verlag, New York, pp. 269-315 (1994). Various methods of generating single chain antibodies are known, including those described in U.S. Pat. Nos. 4,694, 778 and 5,260,203; International Patent Application Publication No. WO 88/01649; Bird (1988) *Science* 242:423-442; Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883; Ward et al. (1989) *Nature* 334:544-545; Skerra et al. (1988) *Science* 242:1038-1041. In specific embodiments, single-chain antibodies can also be bispecific, multispecific, human, and/or humanized and/or synthetic.

[0048] In some embodiments, the binding constructs of the present invention are “in vitro generated binding constructs”. This term refers to a binding construct according to the above definition where all or part of the variable region (e.g., at least one CDR) is generated in a non-immune cell selection, e.g., an in vitro phage display, protein chip or any other method in which candidate sequences can be tested for their ability to bind to an antigen. In other embodiments, the binding construct sequences are generated by genomic rearrangement in an immune cell in an animal. This term thus preferably excludes sequences generated solely by genomic rearrangement in an immune cell in an animal. It is envisaged that the first and/or second domain of the binding construct is produced by or obtainable by phage display or library screening methods rather than by grafting CDR sequences from a pre-existing (monoclonal) antibody into a scaffold.

[0049] A “recombinant antibody” is an antibody made through the use of recombinant DNA technology or genetic engineering.

[0050] The term “monoclonal antibody” (mAb) or monoclonal antibody construct as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible naturally occurring mutations and/or post-translation modifications (e.g., isomerizations, amidations) that may be present in minor amounts. Monoclonal antibodies are highly specific, being directed against a single antigenic site or epitope on the antigen, in contrast to conventional (polyclonal) anti-

body preparations which typically include different antibodies directed against different antigenic sites or epitopes. In addition to their specificity, the monoclonal antibodies are advantageous in that they are synthesized by clonal cell culture and are uncontaminated by other immunoglobulins. The modifier “monoclonal” indicates the character of the antibody as being obtained from a substantially homogeneous population of antibodies, and is not to be construed as requiring production of the antibody by any particular method.

[0051] For the preparation of monoclonal antibodies, any technique providing antibodies produced by continuous cell line cultures can be used. For example, monoclonal antibodies to be used may be made by the hybridoma method first described by Koehler et al., *Nature*, 256: 495 (1975), or may be made by recombinant DNA methods (see, e.g., U.S. Pat. No. 4,816,567). Examples for further techniques to produce human monoclonal antibodies include the trioma technique, the human B-cell hybridoma technique (Kozbor, *Immunology Today* 4 (1983), 72) and the EBV-hybridoma technique (Cole et al., *Monoclonal Antibodies and Cancer Therapy*, Alan R. Liss, Inc. (1985), 77-96).

[0052] Hybridomas can then be screened using standard methods, such as enzyme-linked immunosorbent assay (ELISA) and surface plasmon resonance analysis, e.g. Biacore™ to identify one or more hybridomas that produce an antibody that specifically binds with a specified antigen. Any form of the relevant antigen may be used as the immunogen, e.g., recombinant antigen, naturally occurring forms, any variants or fragments thereof, as well as an antigenic peptide thereof. Surface plasmon resonance as employed in the Biacore system can be used to increase the efficiency of phage antibodies which bind to an epitope of a target cell surface antigen (Schier, *Human Antibodies Hybridomas* 7 (1996), 97-105; Malmberg, *J. Immunol. Methods* 183 (1995), 7-13).

[0053] Another exemplary method of making monoclonal antibodies includes screening protein expression libraries, e.g., phage display or ribosome display libraries. Phage display is described, for example, in Ladner et al., U.S. Pat. No. 5,223,409; Smith (1985) *Science* 228:1315-1317; Clackson et al., *Nature*, 352: 624-628 (1991) and Marks et al., *J. Mol. Biol.*, 222: 581-597 (1991).

[0054] In addition to the use of display libraries, the relevant antigen can be used to immunize a non-human animal, e.g., a rodent (such as a mouse, hamster, rabbit or rat). In one embodiment, the non-human animal includes at least a part of a human immunoglobulin gene. For example, it is possible to engineer mouse strains deficient in mouse antibody production with large fragments of the human Ig (immunoglobulin) loci. Using the hybridoma technology, antigen-specific monoclonal antibodies derived from the genes with the desired specificity may be produced and selected. See, e.g., XENOMOUSE™, Green et al. (1994) *Nature Genetics* 7:13-21, US 2003-0070185, WO 96/34096, and WO 96/33735.

[0055] A monoclonal antibody can also be obtained from a non-human animal, and then modified, e.g., humanized, deimmunized, rendered chimeric etc., using recombinant DNA techniques known in the art. Examples of modified binding constructs include humanized variants of non-human antibodies, “affinity matured” antibodies (see, e.g. Hawkins et al. *J. Mol. Biol.* 254, 889-896 (1992) and Lowman et al., *Biochemistry* 30, 10832-10837 (1991)) and

antibody mutants with altered effector function(s) (see, e.g., U.S. Pat. No. 5,648,260, Kontermann and Dubel (2010), loc. cit. and Little (2009), loc. cit.).

[0056] In immunology, affinity maturation is the process by which B cells produce antibodies with increased affinity for antigen during the course of an immune response. With repeated exposures to the same antigen, a host will produce antibodies of successively greater affinities. Like the natural prototype, the *in vitro* affinity maturation is based on the principles of mutation and selection. The *in vitro* affinity maturation has successfully been used to optimize binding constructs, e.g., antibodies or antibody fragments. Random mutations inside the CDRs are introduced using radiation, chemical mutagens or error-prone PCR. In addition, the genetic diversity can be increased by chain shuffling. Two or three rounds of mutation and selection using display methods like phage display usually results in antibody fragments with affinities in the low nanomolar range.

[0057] A preferred type of an amino acid substitutional variation of the binding constructs involves substituting one or more hypervariable region residues of a parent antibody (e. g. a humanized or human antibody). Generally, the resulting variant(s) selected for further development will have improved biological properties relative to the parent antibody from which they are generated. A convenient way for generating such substitutional variants involves affinity maturation using phage display. Briefly, several hypervariable region sides (e. g. 6-7 sides) are mutated to generate all possible amino acid substitutions at each side. The antibody variants thus generated are displayed in a monovalent fashion from filamentous phage particles as fusions to the gene III product of M13 packaged within each particle. The phage-displayed variants are then screened for their biological activity (e. g. binding affinity) as herein disclosed. In order to identify candidate hypervariable region sides for modification, alanine scanning mutagenesis can be performed to identify hypervariable region residues contributing significantly to antigen binding. Alternatively, or additionally, it may be beneficial to analyze a crystal structure of the antigen-antibody complex to identify contact points between the binding domain and, e.g., the MAGEB2. Such contact residues and neighboring residues are candidates for substitution according to the techniques elaborated herein. Once such variants are generated, the panel of variants is subjected to screening as described herein and antibodies with superior properties in one or more relevant assays may be selected for further development.

[0058] The antibodies of the present invention specifically include "chimeric" antibodies (immunoglobulins) in which a portion of the heavy and/or light chain is identical with or homologous to corresponding sequences in antibodies derived from a particular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is/are identical with or homologous to corresponding sequences in antibodies derived from another species or belonging to another antibody class or subclass, as well as fragments of such antibodies, so long as they exhibit the desired biological activity (U.S. Pat. No. 4,816,567; Morrison et al., Proc. Natl. Acad. Sci. USA, 81: 6851-6855 (1984)). Chimeric antibodies of interest herein include "primitized" antibodies comprising variable domain antigen-binding sequences derived from a non-human primate (e.g., Old World Monkey, Ape etc.) and human constant region sequences. A variety of approaches for making chi-

meric antibodies have been described. See e.g., Morrison et al., Proc. Natl. Acad. Sci. U.S.A. 81:6851, 1985; Takeda et al., Nature 314:452, 1985, Cabilly et al., U.S. Pat. No. 4,816,567; Boss et al., U.S. Pat. No. 4,816,397; Tanaguchi et al., EP 0171496; EP 0173494; and GB 2177096.

[0059] "Humanized" antibodies, variants or fragments thereof (such as Fv, Fab, Fab', F(ab')₂ or other antigen-binding subsequences of antibodies) are antibodies or immunoglobulins of mostly human sequences, which contain (a) minimal sequence(s) derived from non-human immunoglobulin. For the most part, humanized antibodies are human immunoglobulins (recipient antibody) in which residues from a hypervariable region (also CDR) of the recipient are replaced by residues from a hypervariable region of a non-human (e.g., rodent) species (donor antibody) such as mouse, rat, hamster or rabbit having the desired specificity, affinity, and capacity. In some instances, Fv framework region (FR) residues of the human immunoglobulin are replaced by corresponding non-human residues. Furthermore, "humanized antibodies" as used herein may also comprise residues which are found neither in the recipient antibody nor the donor antibody. These modifications are made to further refine and optimize antibody performance. The humanized antibody may also comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. For further details, see Jones et al., Nature, 321: 522-525 (1986); Reichmann et al., Nature, 332: 323-329 (1988); and Presta, Curr. Op. Struct. Biol., 2: 593-596 (1992).

[0060] Humanized antibodies or fragments thereof can be generated by replacing sequences of the Fv variable domain that are not directly involved in antigen binding with equivalent sequences from human Fv variable domains. Exemplary methods for generating humanized antibodies or fragments thereof are provided by Morrison (1985) Science 229:1202-1207; by Oi et al. (1986) BioTechniques 4:214; and by U.S. Pat. Nos. 5,585,089; 5,693,761; 5,693,762; 5,859,205; and 6,407,213. Those methods include isolating, manipulating, and expressing the nucleic acid sequences that encode all or part of immunoglobulin Fv variable domains from at least one of a heavy or light chain. Such nucleic acids may be obtained from a hybridoma producing an antibody against a predetermined target, as described above, as well as from other sources. The recombinant DNA encoding the humanized antibody molecule can then be cloned into an appropriate expression vector.

[0061] Humanized antibodies may also be produced using transgenic animals such as mice that express human heavy and light chain genes, but are incapable of expressing the endogenous mouse immunoglobulin heavy and light chain genes. Winter describes an exemplary CDR grafting method that may be used to prepare the humanized antibodies described herein (U.S. Pat. No. 5,225,539). All of the CDRs of a particular human antibody may be replaced with at least a portion of a non-human CDR, or only some of the CDRs may be replaced with non-human CDRs. It is only necessary to replace the number of CDRs required for binding of the humanized antibody to a predetermined antigen.

[0062] A humanized antibody can be optimized by the introduction of conservative substitutions, consensus sequence substitutions, germline substitutions and/or back mutations. Such altered immunoglobulin molecules can be made by any of several techniques known in the art, (e.g., Teng et al., Proc. Natl. Acad. Sci. U.S.A., 80: 7308-7312,

1983; Kozbor et al., *Immunology Today*, 4: 7279, 1983; Olsson et al., *Meth. Enzymol.*, 92: 3-16, 1982, and EP 239 400).

[0063] The term “human antibody”, “human antibody construct” and “human binding domain” includes antibodies, antibody fragments, and binding domains having antibody regions such as variable and constant regions or domains which correspond substantially to human germline immunoglobulin sequences known in the art, including, for example, those described by Kabat et al. (1991) (*loc. cit.*). The human antibodies, antibody fragments or binding domains of the invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis *in vitro* or by somatic mutation *in vivo*), for example in the CDRs, and in particular, in CDR3. The human antibodies, antibody fragments or binding domains can have at least one, two, three, four, five, or more positions replaced with an amino acid residue that is not encoded by the human germline immunoglobulin sequence. The definition of human antibodies, antibody fragments and binding domains as used herein also contemplates fully human antibodies, which include only non-artificially and/or genetically altered human sequences of antibodies as those can be derived by using technologies or systems such as the Xenomouse. Preferably, a “fully human antibody” does not include amino acid residues not encoded by human germline immunoglobulin sequences.

[0064] The ability to clone and reconstruct megabase-sized human loci in yeast artificial chromosomes YACs and to introduce them into the mouse germline provides a powerful approach to elucidating the functional components of very large or crudely mapped loci as well as generating useful models of human disease. Furthermore, the use of such technology for substitution of mouse loci with their human equivalents could provide unique insights into the expression and regulation of human gene products during development, their communication with other systems, and their involvement in disease induction and progression.

[0065] An important practical application of such a strategy is the “humanization” of the mouse humoral immune system. Introduction of human immunoglobulin (Ig) loci into mice in which the endogenous Ig genes have been inactivated offers the opportunity to study the mechanisms underlying programmed expression and assembly of antibodies as well as their role in B-cell development. Furthermore, such a strategy could provide an ideal source for production of fully human monoclonal antibodies (mAbs)—an important milestone towards fulfilling the promise of antibody therapy in human disease. Fully human antibodies are expected to minimize the immunogenic and allergic responses intrinsic to mouse or mouse-derivatized mAbs and thus to increase the efficacy and safety of the administered antibodies. The use of fully human antibodies can be expected to provide a substantial advantage in the treatment of chronic and recurring human diseases, such as inflammation, autoimmunity, and cancer, which require repeated compound administrations.

[0066] One approach towards this goal was to engineer mouse strains deficient in mouse antibody production with large fragments of the human Ig loci in anticipation that such mice would produce a large repertoire of human antibodies in the absence of mouse antibodies. Large human Ig fragments would preserve the large variable gene diversity as

well as the proper regulation of antibody production and expression. By exploiting the mouse machinery for antibody diversification and selection and the lack of immunological tolerance to human proteins, the reproduced human antibody repertoire in these mouse strains should yield high affinity antibodies against any antigen of interest, including human antigens. Using the hybridoma technology, antigen-specific human mAbs with the desired specificity could be readily produced and selected. This general strategy was demonstrated in connection with the generation of the first XenoMouse mouse strains (see Green et al. *Nature Genetics* 7:13-21 (1994)). The XenoMouse strains were engineered with YACs containing 245 kb and 190 kb-sized germline configuration fragments of the human heavy chain locus and kappa light chain locus, respectively, which contained core variable and constant region sequences. The human Ig containing YACs proved to be compatible with the mouse system for both rearrangement and expression of antibodies and were capable of substituting for the inactivated mouse Ig genes. This was demonstrated by their ability to induce B cell development, to produce an adult-like human repertoire of fully human antibodies, and to generate antigen-specific human mAbs. These results also suggested that introduction of larger portions of the human Ig loci containing greater numbers of V genes, additional regulatory elements, and human Ig constant regions may recapitulate substantially the full repertoire that is characteristic of the human humoral response to infection and immunization. The work of Green et al. was recently extended to the introduction of greater than approximately 80% of the human antibody repertoire through introduction of megabase sized, germline configuration YAC fragments of the human heavy chain loci and kappa light chain loci, respectively. See Mendez et al. *Nature Genetics* 15:146-156 (1997) and U.S. patent application Ser. No. 08/759,620. The production of the XenoMouse® animals is further discussed and delineated in U.S. patent application Ser. No. 07/466,008, Ser. No. 07/610,515, Ser. No. 07/919,297, Ser. No. 07/922,649, Ser. No. 08/031,801, Ser. No. 08/112,848, Ser. No. 08/234,145, Ser. No. 08/376,279, Ser. No. 08/430,938, Ser. No. 08/464,584, Ser. No. 08/464,582, Ser. No. 08/463,191, Ser. No. 08/462,837, Ser. No. 08/486,853, Ser. No. 08/486,857, Ser. No. 08/486,859, Ser. No. 08/462,513, Ser. No. 08/724,752, and Ser. No. 08/759,620; and U.S. Pat. Nos. 6,162,963; 6,150,584; 6,114,598; 6,075,181, and 5,939,598 and Japanese Patent Nos. 3 068 180 B2, 3 068 506 B2, and 3 068 507 B2. See also Mendez et al. *Nature Genetics* 15:146-156 (1997) and Green and Jakobovits *J. Exp. Med.* 188:483-495 (1998), EP 0 463 151 B1, WO 94/02602, WO 96/34096, WO 98/24893, WO 00/76310, and WO 03/47336.

[0067] In an alternative approach, others, including GenPharm International, Inc., have utilized a “minilocus” approach. In the minilocus approach, an exogenous Ig locus is mimicked through the inclusion of pieces (individual genes) from the Ig locus. Thus, one or more VH genes, one or more DH genes, one or more JH genes, a mu constant region, and a second constant region (preferably a gamma constant region) are formed into a construct for insertion into an animal. This approach is described in U.S. Pat. No. 5,545,807 to Surani et al. and U.S. Pat. Nos. 5,545,806; 5,625,825; 5,625,126; 5,633,425; 5,661,016; 5,770,429; 5,789,650; 5,814,318; 5,877,397; 5,874,299; and 6,255,458 each to Lonberg and Kay, U.S. Pat. Nos. 5,591,669 and 6,023,010 to Krimpenfort and Berns, U.S. Pat. Nos. 5,612,

205; 5,721,367; and U.S. Pat. No. 5,789,215 to Berns et al., and U.S. Pat. No. 5,643,763 to Choi and Dunn, and Gen-Pharm International U.S. patent application Ser. No. 07/574,748, Ser. No. 07/575,962, Ser. No. 07/810,279, Ser. No. 07/853,408, Ser. No. 07/904,068, Ser. No. 07/990,860, Ser. No. 08/053,131, Ser. No. 08/096,762, Ser. No. 08/155,301, Ser. No. 08/161,739, Ser. No. 08/165,699, Ser. No. 08/209,741. See also EP 0 546 073 B1, WO 92/03918, WO 92/22645, WO 92/22647, WO 92/22670, WO 93/12227, WO 94/00569, WO 94/25585, WO 96/14436, WO 97/13852, and WO 98/24884 and U.S. Pat. No. 5,981,175. See further Taylor et al. (1992), Chen et al. (1993), Tuailon et al. (1993), Choi et al. (1993), Lonberg et al. (1994), Taylor et al. (1994), and Tuailon et al. (1995), Fishwild et al. (1996).

[0068] Kirin has also demonstrated the generation of human antibodies from mice in which, through microcell fusion, large pieces of chromosomes, or entire chromosomes, have been introduced. See European Patent Application Nos. 773 288 and 843 961. Xenerex Biosciences is developing a technology for the potential generation of human antibodies. In this technology, SCID mice are reconstituted with human lymphatic cells, e.g., B and/or T cells. Mice are then immunized with an antigen and can generate an immune response against the antigen. See U.S. Pat. Nos. 5,476,996; 5,698,767; and 5,958,765.

[0069] In some embodiments, the binding constructs of the invention are “isolated” or “substantially pure” binding constructs. “Isolated” or “substantially pure”, when used to describe the binding constructs disclosed herein, means a binding construct that has been identified, separated and/or recovered from a component of its production environment. Preferably, the binding construct is free or substantially free of association with all other components from its production environment. Contaminant components of its production environment, such as that resulting from recombinant transfected cells, are materials that would typically interfere with diagnostic or therapeutic uses for the polypeptide, and may include enzymes, hormones, and other proteinaceous or non-proteinaceous solutes. It is understood that the isolated protein may constitute a wide range of percent concentration, e.g., from 5% to 99.9% by weight of the total protein content, depending on the circumstances. The polypeptide may be made at a significantly higher concentration through the use of an inducible promoter or high expression promoter, such that it is made at increased concentration levels. The definition includes the production of a binding construct in a wide variety of organisms and/or host cells that are known in the art. In preferred embodiments, the binding construct will be purified (1) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning cup sequenator, or (2) to homogeneity by SDS-PAGE under non-reducing or reducing conditions using Coomassie blue or, preferably, silver stain. Ordinarily, however, an isolated binding construct will be prepared by at least one purification step.

[0070] Peptides are short chains of amino acid monomers linked by covalent peptide (amide) bonds. Hence, peptides fall under the broad chemical classes of biological oligomers and polymers. Amino acids that are part of a peptide or polypeptide chain are termed “residues” and can be consecutively numbered. All peptides except cyclic peptides have an N-terminal residue at one end and a C-terminal residue at the other end of the peptide. An oligopeptide

consists of only a few amino acids (usually between two and twenty). A polypeptide is a longer, continuous, and unbranched peptide chain. Peptides are distinguished from proteins on the basis of size, and as an arbitrary benchmark can be understood to contain approximately 50 or fewer amino acids. Proteins consist of one or more polypeptides, usually arranged in a biologically functional way. While aspects of the lab techniques applied to peptides versus polypeptides and proteins differ (e.g., the specifics of electrophoresis, chromatography, etc.), the size boundaries that distinguish peptides from polypeptides and proteins are not absolute. Therefore, in the context of the present invention, the terms “peptide”, “polypeptide” and “protein” may be used interchangeably, and the term “polypeptide” is often preferred.

[0071] Polypeptides may further form multimers such as dimers, trimers and higher oligomers, which consist of more than one polypeptide molecule. Polypeptide molecules forming such dimers, trimers etc. may be identical or non-identical. The corresponding structures of higher order of such multimers are, consequently, termed homo- or heterodimers, homo- or heterotrimers etc. An example for a heteromultimer is a full-length antibody or immunoglobulin molecule, which, in its naturally occurring form, consists of two identical light polypeptide chains and two identical heavy polypeptide chains. The terms “peptide”, “polypeptide” and “protein” also refer to naturally modified peptides/polypeptides/proteins wherein the modification is accomplished e.g. by post-translational modifications like glycosylation, acetylation, phosphorylation and the like. A “peptide”, “polypeptide” or “protein” when referred to herein may also be chemically modified such as pegylated. Such modifications are well known in the art and described herein below.

[0072] A binding construct is said to “specifically bind” or “immunospecifically bind” to its antigen when the binding construct binds its antigen with a dissociation constant (KD) is $\leq 10^{-7}$ M as measured via a surface plasma resonance technique (e.g., BIACore, GE-Healthcare Uppsala, Sweden) or Kinetic Exclusion Assay (KinExA, Sapidyn, Boise, Idaho). In accordance with this invention a binding construct specifically binds or immunospecifically binds to MAGEB2.

[0073] Because of the sequence similarity between homologous proteins in different species, a binding construct or a binding domain that specifically binds to its target (such as a human target) may, however, cross-react with homologous target molecules from different species (such as, from non-human primates). The term “specific/immunospecific binding” can hence include the binding of a binding construct or binding domain to epitopes or structurally related epitopes in more than one species.

[0074] The term “epitope” refers to a region on an antigen, or specific amino acid residues, to which a binding domain, such as an antibody or immunoglobulin, or a derivative, fragment or variant of an antibody or an immunoglobulin, specifically binds. An “epitope” is antigenic and thus the term epitope is sometimes also referred to herein as “antigenic structure” or “antigenic determinant”. Thus, the binding domain is an “antigen interaction site”. Said binding/interaction is also understood to define a “specific recognition”.

[0075] “Epitopes” can be formed both by contiguous amino acids or non-contiguous amino acids juxtaposed by tertiary folding of a protein. A “linear epitope” is an epitope

where a contiguous amino acid primary sequence comprises the recognized epitope. A linear epitope typically includes at least 3 or at least 4, and more typically, at least 5 or at least 6 or at least 7, and frequently, about 8 to about 10 amino acids in a unique sequence, or even longer than 10 amino acids.

[0076] A “conformational epitope”, in contrast to a linear epitope, is an epitope wherein the primary sequence of the amino acids comprising the epitope is not the sole defining component of the epitope recognized (e.g., an epitope wherein the primary sequence of amino acids is not necessarily recognized by the binding domain). Typically, a conformational epitope comprises an increased number of amino acids relative to a linear epitope, and comprises noncontiguous amino acid sequences. With regard to recognition of conformational epitopes, the binding domain paratope recognizes a three-dimensional structure of the antigen, preferably a peptide or protein or fragment thereof (in the context of the present invention, the antigenic structure for one of the binding domains is comprised within the target cell surface antigen protein). For example, when a protein molecule folds to form a three-dimensional structure, certain amino acids and/or the polypeptide backbone forming the conformational epitope become juxtaposed enabling the antibody to recognize the epitope. Methods of determining the conformation of epitopes include, but are not limited to, x-ray crystallography, two-dimensional nuclear magnetic resonance (2D-NMR) spectroscopy and site-directed spin labelling and electron paramagnetic resonance (EPR) spectroscopy.

[0077] Various methods for identifying proteins, regions of proteins, or peptides that are useful as immunogens or for screening assays are known in the art. In one embodiment, and as described herein in the Examples, a Shannon entropy analysis of the MAGEB2 protein amino acid sequence can identify particular peptide sequences that can be used as immunogens.

[0078] Exemplary MAGEB2 peptides that were identified to be used as immunogens included: MAGEB2 peptide a.a.43-76: SSVSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV (N-terminal region peptide) (SEQ ID NO: 2), MAGEB2 peptide a.a.95-125: SSSQASTSTKSPSEDPLTRKSGSLVQFLLYK (MHD N-terminal helix peptide) (SEQ ID NO: 3), MAGEB2 peptide a.a.185-200: DLTDEESLLSSWDFPR (MHD middle loop peptide) (SEQ ID NO: 562).

[0079] Accordingly, in one embodiment of the present invention, provided is a method of immunizing an animal comprising administration to the animal any of the peptides of SEQ ID NOs: 2, 3, 4, or 562.

[0080] In another embodiment, provided is a method of generating an isolated antibody comprising immunizing an animal with any of the peptides of SEQ ID NOs: 2, 3, 4, or 562 and isolating said antibody. In further embodiments, provided is a method of generating an isolated monoclonal antibody comprising immunizing an animal with any of the peptides of SEQ ID NOs: 2, 3, 4 or 562 and further generating a monoclonal antibody using art recognized steps described herein.

[0081] In yet another embodiment, provided is an isolated antibody generated by a process comprising immunizing an animal with any of the peptides of SEQ ID NOs: 2, 3, 4, or 562.

[0082] In yet other embodiments, fragments of the peptides of SEQ ID NOs: 2, 3, 4 or 562 can be used as the immunogen. Accordingly, in one embodiment of the present invention, the peptide immunogen comprises a fragment of the MAGEB2 peptide a.a.43-76: SSVSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV (N-terminal region peptide) (SEQ ID NO: 2) that is 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, or 33 amino acids long.

[0083] In another embodiment of the present invention, the peptide immunogen comprises a fragment of the MAGEB2 peptide a.a.95-125: SSSQASTSTKSPSEDPLTRKSGSLVQFLLYK (MHD N-terminal helix peptide) (SEQ ID NO: 3) that is 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30, amino acids long.

[0084] In another embodiment of the present invention, the peptide immunogen comprises a fragment of the MAGEB2 peptide a.a.185-200: DLTDEESLLSSWDFPR (MHD middle loop peptide) (SEQ ID NO: 562) that is 10, 11, 12, 13, 14, or 15 amino acids long.

[0085] In further embodiments of the present invention, provided are binding constructs that compete for binding with any of the binding constructs described in the present invention. Competition assays are well known in the art and exemplary assays are described further herein.

[0086] In other embodiments, alternative peptides that can be used as immunogens for screening are provided in Table 1 below. Accordingly, the invention provides

TABLE 1

MAGEB2 Peptide Amino Acid Sequence	SEQ ID NO:
SVSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV	388
VSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV	389
SGGAASSSPAAGIPQEPQRAPTTAAAAAAGV	390
SGGAASSSPAAGIPQEPQRAPTTAAAAAAGV	391
GGAASSSPAAGIPQEPQRAPTTAAAAAAGV	392
GAASSSPAAGIPQEPQRAPTTAAAAAAGV	393
AASSSPAAGIPQEPQRAPTTAAAAAAGV	394
ASSSPAAGIPQEPQRAPTTAAAAAAGV	395
SSSPAAGIPQEPQRAPTTAAAAAAGV	396
SSPAAGIPQEPQRAPTTAAAAAAGV	397
SPAAGIPQEPQRAPTTAAAAAAGV	398
PAAGIPQEPQRAPTTAAAAAAGV	399
AAGIPQEPQRAPTTAAAAAAGV	400
AGIPQEPQRAPTTAAAAAAGV	401
GIPQEPQRAPTTAAAAAAGV	402
IPQEPQRAPTTAAAAAAGV	403
PQEPQRAPTTAAAAAAGV	404
QEPQRAPTTAAAAAAGV	405
EPQRAPTTAAAAAAGV	406
PQRAPTTAAAAAAGV	407

TABLE 1-continued

MAGEB2 Peptide Amino Acid Sequence	SEQ ID NO:
QRAPTTAAAAAGV	408
RAPTTAAAAAGV	409
APTTAAAAAGV	410
PTTAAAAAGV	411
TTAAAAAGV	412
TAAAAAGV	413
AAAAAGV	414
AAAAAGV	415
AAAAGV	416
AAAGV	417
SSVSGGAASSSPAAGIPQEPQRAPTTAAAAAG	418
SSVSGGAASSSPAAGIPQEPQRAPTTAAAAA	419
SSVSGGAASSSPAAGIPQEPQRAPTTAAAAA	420
SSVSGGAASSSPAAGIPQEPQRAPTTAAAAA	421
SSVSGGAASSSPAAGIPQEPQRAPTTAAA	422
SSVSGGAASSSPAAGIPQEPQRAPTTAA	423
SSVSGGAASSSPAAGIPQEPQRAPTTA	424
SSVSGGAASSSPAAGIPQEPQRAPTT	425
SSVSGGAASSSPAAGIPQEPQRAPT	426
SSVSGGAASSSPAAGIPQEPQRAP	427
SSVSGGAASSSPAAGIPQEPQRA	428
SSVSGGAASSSPAAGIPQEPQR	429
SSVSGGAASSSPAAGIPQEPQ	430
SSVSGGAASSSPAAGIPQEP	431
SSVSGGAASSSPAAGIPQE	432
SSVSGGAASSSPAAGIPQ	433
SSVSGGAASSSPAAGIP	434
SSVSGGAASSSPAAGI	435
SSVSGGAASSSPAAG	436
SSVSGGAASSSPA	437
SSVSGGAASSSPA	438
SSVSGGAASSSP	439
SSVSGGAASSS	440
SSVSGGAASS	441
SSVSGGAAS	442
SSVSGGAA	443
SSVSGGA	444

TABLE 1-continued

MAGEB2 Peptide Amino Acid Sequence	SEQ ID NO:
SSVSGG	445
SSVSG	446
SVSGGAASSSPAAGIPQEPQRAPTTAAAAAG	447
VSGGAASSSPAAGIPQEPQRAPTTAAAAA	448
SGGAASSSPAAGIPQEPQRAPTTAAAAA	449
GGAASSSPAAGIPQEPQRAPTTAAAA	450
GAASSSPAAGIPQEPQRAPTTAAAA	451
AASSSPAAGIPQEPQRAPTTAA	452
ASSSPAAGIPQEPQRAPTTA	453
SSSPAAGIPQEPQRAPTT	454
SSPAAGIPQEPQRAPT	455
SPAAGIPQEPQRAP	456
PAAGIPQEPQRA	457
AAGIPQEPQR	458
AGIPQEPQ	459
GIPQEP	460
IPQE	461
SSQASTSTKSPSEDPLTRKSGSLVQFLLYK	462
SQASTSTKSPSEDPLTRKSGSLVQFLLYK	463
QASTSTKSPSEDPLTRKSGSLVQFLLYK	464
ASTSTKSPSEDPLTRKSGSLVQFLLYK	465
STSTKSPSEDPLTRKSGSLVQFLLYK	466
TSTKSPSEDPLTRKSGSLVQFLLYK	467
STKSPSEDPLTRKSGSLVQFLLYK	468
TKSPSEDPLTRKSGSLVQFLLYK	469
KSPSEDPLTRKSGSLVQFLLYK	470
SPSEDPLTRKSGSLVQFLLYK	471
PSEDPLTRKSGSLVQFLLYK	472
SEDPLTRKSGSLVQFLLYK	473
EDPLTRKSGSLVQFLLYK	474
DPLTRKSGSLVQFLLYK	475
PLTRKSGSLVQFLLYK	476
LTRKSGSLVQFLLYK	477
TRKSGSLVQFLLYK	478
RKSGSLVQFLLYK	479
KSGSLVQFLLYK	480
SGSLVQFLLYK	481
GSLVQFLLYK	482

TABLE 1-continued

MAGEB2 Peptide Amino Acid Sequence	SEQ ID NO:
SLVQFLLYK	483
LVQFLLYK	484
VQFLLYK	485
QFLLYK	486
FLLYK	487
SSSQASTSTKSPSEDPLTRKSGSLVQFLLY	488
SSSQASTSTKSPSEDPLTRKSGSLVQFLL	489
SSSQASTSTKSPSEDPLTRKSGSLVQFL	490
SSSQASTSTKSPSEDPLTRKSGSLVQF	491
SSSQASTSTKSPSEDPLTRKSGSLVQ	492
SSSQASTSTKSPSEDPLTRKSGSLV	493
SSSQASTSTKSPSEDPLTRKSGSL	494
SSSQASTSTKSPSEDPLTRKSGS	495
SSSQASTSTKSPSEDPLTRKSG	496
SSSQASTSTKSPSEDPLTRKS	497
SSSQASTSTKSPSEDPLTRK	498
SSSQASTSTKSPSEDPLTR	499
SSSQASTSTKSPSEDPLT	500
SSSQASTSTKSPSEDPL	501
SSSQASTSTKSPSEDP	502
SSSQASTSTKSPSED	503
SSSQASTSTKSPSE	504
SSSQASTSTKSPS	505
SSSQASTSTKSP	506
SSSQASTSTKS	507
SSSQASTSTK	508
SSSQASTST	509
SSSQASTS	510
SSSQAST	511
SSSQAS	512
SSSQA	513
SSQASTSTKSPSEDPLTRKSGSLVQFLLY	514
SQASTSTKSPSEDPLTRKSGSLVQFLL	515
QASTSTKSPSEDPLTRKSGSLVQFL	516
ASTSTKSPSEDPLTRKSGSLVQF	517
STSTKSPSEDPLTRKSGSLVQ	518
TSTKSPSEDPLTRKSGSLV	519

TABLE 1-continued

MAGEB2 Peptide Amino Acid Sequence	SEQ ID NO:
STKSPSEDPLTRKSGSL	520
TKSPSEDPLTRKSGS	521
KSPSEDPLTRKSG	522
SPSEDPLTRKS	523
PSEDPLTRK	524
SEDPLTR	525
EDPLT	526
LTDEESLLSSWDFPR	527
TDEESLLSSWDFPR	528
DEESLLSSWDFPR	529
EESLLSSWDFPR	530
ESLLSSWDFPR	531
SLSSWDFPR	532
LLSSWDFPR	533
LSSWDFPR	534
SSWDFPR	535
SWDFPR	536
WDFPR	537
DLTDEESLLSSWDFP	538
DLTDEESLLSSWDF	539
DLTDEESLLSSWD	540
DLTDEESLLSSW	541
DLTDEESLLSS	542
DLTDEESLLS	543
DLTDEESLL	544
DLTDEESL	545
DLTDEES	546
DLTDEE	547
DLTDE	548
LTDEESLLSSWDFP	549
TDEESLLSSWDF	550
DEESLLSSWD	551
EESLLSSW	552
ESLLSS	553
SLLS	554

Affinity

[0087] The interaction between the binding domain and the epitope or the region comprising the epitope implies that

a binding domain exhibits appreciable affinity for the epitope and/or the region comprising the epitope on a particular protein or antigen, and unless otherwise specified, does not exhibit significant binding or reactivity with proteins or antigens other than MAGEB2. This affinity can be measured by various techniques known to one skilled in the art, such as in a surface plasmon resonance assay, such as a Biacore assay, or in a cell based assay.

[0088] “Appreciable affinity” includes binding with an affinity of about 106 M (KD) or stronger. Preferably, binding is considered specific when the binding affinity is about 10^{-12} to 10^{-8} M, 10^{-12} to 10^{-9} M, 10^{-12} to 10^{-10} M, 10^{-11} to 10^{-8} M, preferably of about 10^{-11} to 10^{-9} M. Whether a binding domain specifically reacts with or binds to a target can be tested readily by, inter alia, comparing the reaction of said binding domain with a target protein or antigen with the reaction of said binding domain with proteins or antigens other than the MAGEB2. Preferably, a binding domain of the invention does not essentially or substantially bind to proteins or antigens other than MAGEB2.

Specificity

[0089] The term “does not significantly bind” means that a binding construct or binding domain of the present invention does not bind to a protein or antigen other than MAGEB2. For example, a binding construct or binding domain that exhibits binding to proteins with similar amino acid sequences, e.g., MAGEA4 or MAGEA8, would not be a desirable binding construct.

[0090] In the present invention, the MAGEB2 binding constructs possess surprising levels of specificity and selec-

tivity to their target, as evidenced by its lack of binding to target negative cells. See, e.g., Examples 2 and 3 herein.

[0091] This specificity and selectivity are highly desired, yet difficult to achieve, properties for a binding construct in a diagnostic assay as it limits, reduces, or eliminates off-target binding and any potential false readouts.

[0092] Specific binding is believed to be effected by specific motifs in the amino acid sequence of the binding domain and the antigen. Thus, binding is achieved as a result of their primary, secondary and/or tertiary structure as well as the result of secondary modifications of said structures. The specific interaction of the antigen-interaction-side with its specific antigen may result in a simple binding of said side to the antigen. Moreover, the specific interaction of the antigen-interaction-side with its specific antigen may alternatively or additionally result in the initiation of a signal, e.g. due to the induction of a change of the conformation of the antigen, an oligomerization of the antigen, etc.

MAGEB2 Binding Constructs

[0093] The invention provides binding constructs comprising a domain which binds to MAGEB2, and in specific embodiments, to the MAGEB2 peptides SSVSG-GAASSSPAAGIPQEPQRAPTTAAAAAAGV (N-terminal region peptide) (SEQ ID NO: 2), a.a.95-125: SSSQAST-STKSPSEDPLTRKSGSLVQFLLYK (MHD N-terminal helix peptide) (SEQ ID NO: 3), or a.a.185-200: DLTDEE-SLLSSWDFPR (SEQ ID NO: 562).

[0094] Table 2 below provides amino acid sequences of exemplary MAGEB2 binding constructs VH-CDRs and VL-CDRs. Table 3 below provides amino acid sequences of exemplary MAGEB2 binding construct VH and VL domains.

TABLE 2

Exemplary Light Chain CDR and Heavy Chain CDR Sequences						
Molecule	CDR-L1	CDR-L2	CDR-L3	CDR-H1	CDR-H2	CDR-H3
1I14	QSSQSVYDNNAL A (SEQ ID NO: 85)	GASTLAS (SEQ ID NO: 86)	QCTYYVSSYQND (SEQ ID NO: 87)	SYAMS (SEQ ID NO: 229)	SIGGGGSAVYASW AKG (SEQ ID NO: 230)	GFYSIDL (SEQ ID NO: 231)
1C3	QASQNISSYLA (SEQ ID NO: 73)	RASTLAS (SEQ ID NO: 74)	QSYDDSRSSNFFY (SEQ ID NO: 75)	NYVIC (SEQ ID NO: 217)	CIDNANGRTYYAS WAKG (SEQ ID NO: 218)	SLATPL (SEQ ID NO: 219)
1H17	QSSKSVYNKNWL S (SEQ ID NO: 91)	GASTLAS (SEQ ID NO: 92)	AGGYSSSDTFA (SEQ ID NO: 93)	SGQLMC (SEQ ID NO: 235)	CIGSGSNAISTFYAS WAQG (SEQ ID NO: 236)	VGSDDYGSDVFPDP (SEQ ID NO: 237)

TABLE 3

Exemplary Variable Light Chain and Variable Heavy Chain Sequences		
Molecule	VL	VH
1I14	DVVMQTTPASVEATVGGTVTIKQSSQSVYDNNALA WYQNGAQRPRLLIYGASTLASGVPSPRFSASGSGTEF TLTISDLECAADAATYYCQCTYYVSSYQNDFFGGGTEVVV K (SEQ ID NO: 346)	QSVEESGGRLVTPGTPLTLTCTISGFSLSYAMSWVRQAPGKGL EWIGSIGGGGSAVYASWAKGRFTISKSTTVDLRITSPTTEDTAM YFCGRGFYSIDLWGPGLTVTVSS (SEQ ID NO: 347)
1C3	DIVMTQTPSSVEAAVGGTVTIKQASQNISSYLAWYQ QKPGQPPKLLIYRASTLASGVPSPRFGSGSGTQFTLTIS DLECAADAATYYCQSYDDSRSSNFFYAFGGGTEVVVK (SEQ ID NO: 338)	QSLEESGGGLVQPEGLTLTCTAFGVTLNYYICWVRQAPGKGL EWVGCIDNANGRTYYASWAKGRFTISKSTTGTGLQMTSLTAA DTATYFCARSLATPLWGPGLTVTVSS (SEQ ID NO: 339)

TABLE 3-continued

Exemplary Variable Light Chain and Variable Heavy Chain Sequences		
Molecule	VL	VH
1H17	AAVLTQTSPSPVSAAVGGTVSASCQSSKSVYNKNWLS WFQKPGQPPKLLIYGASTLASGVPSRFKSGSGTQF TLTISDVQCDDAATYYCAGGYSSSDTFAFGGGTEVVV K (SEQ ID NO: 350)	QEQLVESGGGLVKPGASLTLTCKASGFSFSSGQLMCWVRQAPG KGLEWIAICIGSGSNAISTFYASWAQGRFTISKSSSTVTTLQLTSLT AADTATYFCARVGSDDYGDSDVFDPWGPGTLVTVSS (SEQ ID NO: 351)

[0095] In one embodiment, the invention provides an isolated binding domain that binds to MAGEB2, wherein the binding domain comprises:

[0096] a) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 229, CDR-H2 as depicted in SEQ ID NO: 230, and CDR-H3 as depicted in SEQ ID NO: 231, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 85, CDR-L2 as depicted in SEQ ID NO: 86 and CDR-L3 as depicted in SEQ ID NO: 87; or

[0097] b) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 217, CDR-H2 as depicted in SEQ ID NO: 218, and CDR-H3 as depicted in SEQ ID NO: 219, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 73, CDR-L2 as depicted in SEQ ID NO: 74 and CDR-L3 as depicted in SEQ ID NO: 75; or

[0098] c) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 235, CDR-H2 as depicted in SEQ ID NO: 236, and CDR-H3 as depicted in SEQ ID NO: 237, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 91, CDR-L2 as depicted in SEQ ID NO: 92 and CDR-L3 as depicted in SEQ ID NO: 93.

[0099] In one embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain comprises:

[0100] a) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 229, CDR-H2 as depicted in SEQ ID NO: 230, and CDR-H3 as depicted in SEQ ID NO: 231, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 85, CDR-L2 as depicted in SEQ ID NO: 86 and CDR-L3 as depicted in SEQ ID NO: 87; or

[0101] b) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 217, CDR-H2 as depicted in SEQ ID NO: 218, and CDR-H3 as depicted in SEQ ID NO: 219, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 73, CDR-L2 as depicted in SEQ ID NO: 74 and CDR-L3 as depicted in SEQ ID NO: 75; or

[0102] c) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 235, CDR-H2 as depicted in SEQ ID NO: 236, and CDR-H3 as depicted in SEQ ID NO: 237, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 91, CDR-L2 as depicted in SEQ ID NO: 92 and CDR-L3 as depicted in SEQ ID NO: 93.

[0103] In one embodiment, the invention provides an isolated binding domain that binds to MAGEB2, wherein the binding domain binds to the same epitope as an antibody or binding construct that comprises:

[0104] a) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 229, CDR-H2 as depicted in SEQ ID NO: 230, and CDR-H3 as depicted in SEQ ID NO: 231, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 85, CDR-L2 as depicted in SEQ ID NO: 86 and CDR-L3 as depicted in SEQ ID NO: 87; or

[0105] b) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 217, CDR-H2 as depicted in SEQ ID NO: 218, and CDR-H3 as depicted in SEQ ID NO: 219, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 73, CDR-L2 as depicted in SEQ ID NO: 74 and CDR-L3 as depicted in SEQ ID NO: 75; or

[0106] c) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 235, CDR-H2 as depicted in SEQ ID NO: 236, and CDR-H3 as depicted in SEQ ID NO: 237, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 91, CDR-L2 as depicted in SEQ ID NO: 92 and CDR-L3 as depicted in SEQ ID NO: 93.

[0107] In one embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain binds to the same epitope as an antibody or binding construct that comprises:

[0108] a) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 229, CDR-H2 as depicted in SEQ ID NO: 230, and CDR-H3 as depicted in SEQ ID NO: 231, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 85, CDR-L2 as depicted in SEQ ID NO: 86 and CDR-L3 as depicted in SEQ ID NO: 87; or

[0109] b) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 217, CDR-H2 as depicted in SEQ ID NO: 218, and CDR-H3 as depicted in SEQ ID NO: 219, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 73, CDR-L2 as depicted in SEQ ID NO: 74 and CDR-L3 as depicted in SEQ ID NO: 75; or

[0110] c) a VH region comprising CDR-H1 as depicted in SEQ ID NO: 235, CDR-H2 as depicted in SEQ ID NO: 236, and CDR-H3 as depicted in SEQ ID NO: 237, and a VL region comprising CDR-L1 as depicted in SEQ ID NO: 91, CDR-L2 as depicted in SEQ ID NO: 92 and CDR-L3 as depicted in SEQ ID NO: 93.

[0111] In one embodiment, the invention provides an isolated binding domain that binds to MAGEB2, wherein the binding domain comprises:

[0112] a) a VH region comprising SEQ ID NO: 347 and a VL region comprising SEQ ID NO: 346; or

[0113] b) a VH region comprising SEQ ID NO: 339 and a VL region comprising SEQ ID NO: 338; or

[0114] c) a VH region comprising SEQ ID NO: 351 and a VL region comprising SEQ ID NO: 350.

[0115] In one embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain comprises:

[0116] a) a VH region comprising SEQ ID NO: 347 and a VL region comprising SEQ ID NO: 346; or

[0117] b) a VH region comprising SEQ ID NO: 339 and a VL region comprising SEQ ID NO: 338; or

[0118] c) a VH region comprising SEQ ID NO: 351 and a VL region comprising SEQ ID NO: 350.

[0119] In one embodiment, the invention provides an isolated binding domain that binds to MAGEB2, wherein the binding domain binds to the same epitope as an antibody or binding construct that comprises:

- [0120] a) a VH region comprising SEQ ID NO: 347 and a VL region comprising SEQ ID NO: 346; or
- [0121] b) a VH region comprising SEQ ID NO: 339 and a VL region comprising SEQ ID NO: 338; or
- [0122] c) a VH region comprising SEQ ID NO: 351 and a VL region comprising SEQ ID NO: 350.

[0123] In one embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain binds to the same epitope as an antibody or binding construct that comprises:

- [0124] a) a VH region comprising SEQ ID NO: 347 and a VL region comprising SEQ ID NO: 346; or
- [0125] b) a VH region comprising SEQ ID NO: 339 and a VL region comprising SEQ ID NO: 338; or
- [0126] c) a VH region comprising SEQ ID NO: 351 and a VL region comprising SEQ ID NO: 350.

[0127] In another embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain comprises a VH region selected from the group consisting of any VH region and a corresponding VL region as depicted in any of the sequences in Tables 21-25 herein.

[0128] In another embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain comprises a VH region selected from the group consisting of any VH region and a corresponding VL region as depicted in any of the sequences in Table 23 herein.

[0129] In another embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain comprises a CDR-H1, CDR-H2, CDR-H3 from a VH region selected from the group consisting of any VH region as depicted in any of the sequences in Table 22 herein and further comprises a corresponding CDR-L1, CDR-L2, CDR-L3 from a VL region selected from the group consisting of any VL region as depicted in any of the sequences in Table 21 herein.

[0130] In another embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain binds to the same epitope as an antibody or binding construct that comprises a CDR-H1, CDR-H2, CDR-H3 from a VH region selected from the group consisting of any VH region as depicted in any of the sequences in Table 22 herein and further comprises a corresponding CDR-L1, CDR-L2, CDR-L3 from a VL region selected from the group consisting of any VL region as depicted in any of the sequences in Table 21 herein.

[0131] In another embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain comprises a VH and a VL region consensus sequence selected from any of the sequences in Table 25 herein.

[0132] In another embodiment, the invention provides an isolated binding construct comprising a binding domain that binds to MAGEB2, wherein the binding domain comprises a CDR-H1, CDR-H2, CDR-H3 from a VH region consensus

sequence selected from the group consisting of any VH region as depicted in any of the sequences in Table 25 herein and further comprises a corresponding CDR-L1, CDR-L2, CDR-L3 from a VL region consensus sequence selected from the group consisting of any VL region as depicted in any of the sequences in Table 25 herein.

[0133] Exemplary binding construct full sequences are presented below:

1C3_LC

(SEQ ID NO: 555)

MDMRVPAQLLGLLLWLRGARCIVMTQTPSSVEAAVGGTVTIKCQASQN
ISSYLAWYQQKPGQPPKLLIYRASTLASGVPSRFKSGSGTQFTLTISDL
ECADAATYYCQSYDDSRSSNFFYAFGGGTEVVVKGDVPVPTVLLFPSSD
EVATGTVTIVCVANKYFPDVTVTWEVDGTTQTGTIENSKTPQNSADCTYN
LSSTLTLTSTQYNHKEYTCKVTQGTTSVVQSFPSRKNC

1C3_HC

(SEQ ID NO: 556)

MDMRVPAQLLGLLLWLRGARCQSLSESGGLVQPEGSLTLCFAFGVTL
TNYIICWVRQAPGKGLEWVGCIDNANGRTYYASWAKGRFTISKTSSTGT
LQMTSLTAADTATYFCARSLATPLWGPGLTVTVSSGQPKAPSVFPLAPCC
GDTSPSTVTLGCLVKGYLPEPVTVTWNSGTLTNGVTRFPSPVRQSSGLYSL
SSVSVTSSSQPVTCNVAPATNTKVDKTVAPSTCSKPTCPPPELLGGPS
VFIPPKPKDLMISRTPEVTCVVVDVSQDDPEVQFTWYINNEQVTRARP
PLREQQFNSTIRVVSSTLPVIAHQDWLRGKEFKCKVHNKALPAPIEKTISK
RGQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISVEWEKNGKAED
NYKTPAVLSDSGSYFLYSKLSVPTSEWQRGDVFTCSVMHEALHNHYTQK
SISRSPGK

1I14_LC

(SEQ ID NO: 557)

MDMRVPAQLLGLLLWLRGARCIVMTQTPASVEATVGGTVTIKCQSSQS
VYDNNALAWYQQNAGQPRRLIYGASTLASGVPSRFASGSGTEFTLTIS
DLECAADAATYYCQCTYYVSSYQNDFFGGGTEVVVKGDVPVPTVLLFPSSD
EVATGTVTIVCVANKYFPDVTVTWEVDGTTQTGTIENSKTPQNSADCTYN
LSSTLTLTSTQYNHKEYTCKVTQGTTSVVQSFPSRKNC

1I14_HC

(SEQ ID NO: 558)

MDMRVPAQLLGLLLWLRGARCQSVESGGRLVTPGTLTCTISGFSL
SSYAMSWVRQAPGKGLEWIGSIGGGSAVYASWAKGRFTISKSTTVDLR
ITSPTEDTAMYFCGRGFYSIDLWGPGLTVTVSSGQPKAPSVFPLAPCCG
DTPSSTVTLGCLVKGYLPEPVTVTWNSGTLTNGVTRFPSPVRQSSGLYSL
SVSVTSSSQPVTCNVAPATNTKVDKTVAPSTCSKPTCPPPELLGGPSV
FIFPPKPKDLMISRTPEVTCVVVDVSQDDPEVQFTWYINNEQVTRARP
LREQQFNSTIRVVSSTLPVIAHQDWLRGKEFKCKVHNKALPAPIEKTISK
RGQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISVEWEKNGKAED
YKTPAVLSDSGSYFLYSKLSVPTSEWQRGDVFTCSVMHEALHNHYTQKS
ISRSPGK

1H17_LC

(SEQ ID NO: 559)

MDMRVPAQLLGLLLWLRGARCAAVLTQTPSPVSAVGGTVSASCQSSKS
VYNKNWLSWFQKPGQPPKLLIYGASTLASGVPSRFKSGSGTQFTLTIS
DVQCDAATYYCAGGYSSSDTFAGGGTEVVVKGDVPVPTVLLFPSSD
EVATGTVTIVCVANKYFPDVTVTWEVDGTTQTGTIENSKTPQNSADCTYN
LSSTLTLTSTQYNHKEYTCKVTQGTTSVVQSFPSRKNC

1H17_HC

(SEQ ID NO: 560)

MDMRVPAQLLGLLLWLRGARCQEQLVESGGGLVKGASLTLCCKASGFS
FSSGQLMCWVRQAPGKGLEWIAICIGSGSNAISTFYASWAKGRFTISKSS
TTVTLQLTSLTAADTATYFCARVSGDDYGDSDVDPPWGPGLTVTVSSGQ
KAPSVFPLAPCCGDTSPSTVTLGCLVKGYLPEPVTVTWNSGTLTNGVTRFP
PSVRQSSGLYSLSSVSVTSSSQPVTCNVAPATNTKVDKTVAPSTCSKPT
TCPPPELLGGPSVFIIPPKPKDLMISRTPEVTCVVVDVSQDDPEVQFTW
YINNEQVTRARPPLEQQFNSTIRVVSSTLPVIAHQDWLRGKEFKCKVHNK
LPAPIEKTISKARGQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDI
SVEWEKNGKAEDNYKTPAVLSDSGSYFLYSKLSVPTSEWQRGDVFTCSV
MHEALHNHYTQKSISRSPGK

Competitive Binding

[0134] Whether or not an antibody or binding construct competes for binding to an antigen (such as MAGEB2) with

another given antibody or binding construct can be measured in a competition assay such as a competitive ELISA. Avidin-coupled microparticles (beads) can also be used. Similar to an avidin-coated ELISA plate, when reacted with a biotinylated protein, each of these beads can be used as a substrate on which an assay can be performed. Antigen is coated onto a bead and then precoated with the first antibody. The second antibody is added, and any additional binding is determined. Read-out occurs via flow cytometry. Preferably a cell-based competition assay that allows binding to intracellularly expressed MAGEB2 is used, using either cells that naturally express MAGEB2, or cells that were stably or transiently transformed with MAGEB2. The term “competes for binding”, in the present context, means that competition occurs between the two tested antibodies of at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80% or at least 90%, as determined by any one of the assays disclosed above, preferably the cell-based assay.

[0135] Competitive antibody binding assays include assays determining the competitive binding of two antibodies/binding constructs to a cell surface bound antigen. Common methods aim to detect binding of two antibodies/binding constructs, A and B, to the same antigen expressed by a cell may include steps of:

[0136] blocking of the antigen by pre-incubation of cells with antibody/binding construct A followed by a sub-maximal addition of labeled antibody/binding construct B and detecting the binding of B compared with binding in the absence of A;

[0137] titration (i.e. adding different amounts) of antibody/binding construct A in the presence of sub-maximal amounts of labeled antibody/binding construct B and detecting the effect on binding of B; or co-titration of A and B, wherein both antibodies/binding constructs are incubated together at maximal concentration and detecting whether the total binding equals or exceeds that of either A or B alone, i.e. a method which cannot be affected by the order of addition or relative amounts of the antibodies/binding constructs.

[0138] When two antibodies/binding constructs A and B compete for an antigen, the antibodies will very often compete with each other in blocking assays independently from the order of the addition of the antibodies. In other words, competition is detected if the assay is carried out in either direction. However, this is not always the case, and under certain circumstances the order of the addition of the antibodies or the direction of the assay may have an effect on the signal generated. This may be due to differences in affinities or avidities of the potentially competing antibodies/binding constructs. If the order of the addition has a significant effect on the signal generated, it is concluded that the two antibodies/binding constructs do compete if competition is detected in at least one order.

Epitope Amino Acid Residues

[0139] Structural analysis of the interaction between the target polypeptide and binding domain of, for example, a construct contemplated herein can provide amino acid residues of the target epitope involved with the binding interaction. In certain embodiments, the crystal structure of the target-binder interaction can provide these amino acid residues. In other embodiments, various other analyses can be performed to ascertain amino acid residues involved in the

binding interaction. For example, Xscan, alanine scanning, arginine scanning, and other techniques known to those skilled in the art. In the present invention, the binding constructs were generated using specific peptide immunogens, and must therefore bind to these specific peptide sequences.

[0140] In embodiments where a peptide is used as the immunogen, such as those provided herein in SEQ ID NOs 2, 3, 4, or 562, antibodies that are isolated will be expected to bind an epitope that comprises all of, or a fragment of, the peptide immunogen. This epitope can be a linear epitope with that comprises all of, or a fragment of, the peptide immunogen amino acid sequences. Alternatively, and more likely in the case of longer peptide immunogens, this epitope can be a conformational epitope that comprises discontinuous amino acid sequences from within the peptide immunogen. Accordingly, in one embodiment of the present invention, the epitope that the binding constructs (e.g., antibodies) bind to comprises the MAGEB2 peptide a.a.43-76: SSVSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV (N-terminal region peptide) (SEQ ID NO: 2).

[0141] In another embodiment, the epitope that the binding constructs (e.g., antibodies) bind to comprises a fragment of the MAGEB2 peptide a.a.43-76: SSVSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV (N-terminal region peptide) (SEQ ID NO: 2) that is 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, or 33 amino acids long.

[0142] In another embodiment, the epitope that the binding constructs (e.g., antibodies) bind to comprises at least two, at least three, at least four, or at least five discontinuous amino acid sequence fragments of the MAGEB2 peptide a.a.43-76: SSVSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV (N-terminal region peptide) (SEQ ID NO: 2), wherein each discontinuous amino acid sequence fragment is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 amino acids long.

[0143] In another embodiment of the present invention, the epitope that the binding constructs of the invention bind to comprises the MAGEB2 peptide a.a.95-125: SSSQASTSTKSPSEDPLTRKSGSLVQFLLYK (MHD N-terminal helix peptide) (SEQ ID NO: 3).

[0144] In another embodiment, the epitope that the binding constructs (e.g., antibodies) bind to comprises a fragment of the MAGEB2 peptide a.a.95-125: SSSQASTSTKSPSEDPLTRKSGSLVQFLLYK (MHD N-terminal helix peptide) (SEQ ID NO: 3) that is 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30, amino acids long.

[0145] In another embodiment, the epitope that the binding constructs (e.g., antibodies) bind to comprises at least two, at least three, at least four, or at least five discontinuous amino acid sequence fragments of the MAGEB2 peptide a.a.95-125: SSSQASTSTKSPSEDPLTRKSGSLVQFLLYK (MHD N-terminal helix peptide) (SEQ ID NO: 3), wherein each discontinuous amino acid sequence fragment is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 amino acids long.

[0146] In another embodiment of the present invention, the epitope that the binding constructs of the invention bind to comprises the MAGEB2 peptide a.a.185-200: DLTDEE-SLLSSWDFPR (MHD middle loop peptide) (SEQ ID NO: 562).

[0147] In another embodiment, the epitope that the binding constructs (e.g., antibodies) bind to comprises a fragment of the MAGEB2 peptide a.a.185-200: DLTDEE-SLLSSWDFPR (MHD middle loop peptide) (SEQ ID NO: 562) that is 10, 11, 12, 13, 14, or 15 amino acids long.

[0148] In another embodiment, the epitope that the binding constructs (e.g., antibodies) bind to comprises at least two, at least three, at least four, or at least five discontinuous amino acid sequence fragments of the MAGEB2 peptide a.a.185-200: DLTDEESLLSSWDFPR (MHD middle loop peptide) (SEQ ID NO: 562), wherein each discontinuous amino acid sequence fragment is 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids long.

[0149] In one embodiment, the invention provides a binding construct that binds to an epitope comprising the peptides of any of SEQ ID NOs: 2, 3, 4, 562, or 388-554.

[0150] The region of the binding domain that binds to the epitope is called a “paratope.” Specific binding is believed to be accomplished by specific motifs in the amino acid sequence of the binding domain and the antigen. Thus, binding is achieved as a result of their primary, secondary and/or tertiary structure as well as the result of potential secondary modifications of said structures.

[0151] The term “Fc portion” or “Fc monomer” means in connection with this invention a polypeptide comprising at least one domain having the function of a CH2 domain and at least one domain having the function of a CH3 domain of an immunoglobulin molecule. As apparent from the term “Fc monomer”, the polypeptide comprising those CH domains is a “polypeptide monomer”. An Fc monomer can be a polypeptide comprising at least a fragment of the constant region of an immunoglobulin excluding the first constant region immunoglobulin domain of the heavy chain (CH1), but maintaining at least a functional part of one CH2 domain and a functional part of one CH3 domain, wherein the CH2 domain is amino terminal to the CH3 domain.

[0152] In one embodiment of this definition, an Fc monomer can be a polypeptide constant region comprising a portion of the Ig-Fc hinge region, a CH2 region and a CH3 region, wherein the hinge region is amino terminal to the CH2 domain. It is envisaged that the hinge region of the present invention promotes dimerization. Such Fc polypeptide molecules can be obtained by papain digestion of an immunoglobulin region (of course resulting in a dimer of two Fc polypeptide), for example and not limitation. In another aspect of this definition, an Fc monomer can be a polypeptide region comprising a portion of a CH2 region and a CH3 region. Such Fc polypeptide molecules can be obtained by pepsin digestion of an immunoglobulin molecule, for example and not limitation.

[0153] In one embodiment, the polypeptide sequence of an Fc monomer is substantially similar to an Fc polypeptide sequence of: an IgG1 Fc region, an IgG2 Fc region, an IgG3 Fc region, an IgG4 Fc region, an IgM Fc region, an IgA Fc region, an IgD Fc region and an IgE Fc region. (See, e.g., Padlan, Molecular Immunology, 31(3), 169-217 (1993)). Because there is some variation between immunoglobulins, and solely for clarity, Fc monomer refers to the last two heavy chain constant region immunoglobulin domains of IgA, IgD, and IgG, and the last three heavy chain constant region immunoglobulin domains of IgE and IgM. As mentioned, the Fc monomer can also include the flexible hinge N-terminal to these domains. For IgA and IgM, the Fc monomer may include the J chain. For IgG, the Fc portion

comprises immunoglobulin domains CH2 and CH3 and the hinge between the first two domains and CH2. Although the boundaries of the Fc portion may vary an example for a human IgG heavy chain Fc portion comprising a functional hinge, CH2 and CH3 domain can be defined e.g. to comprise residues D231 (of the hinge domain—corresponding to D234 in Table 4 below) to P476, respectively L476 (for IgG4) of the carboxyl-terminus of the CH3 domain, wherein the numbering is according to Kabat. The two Fc portion or Fc monomer, which are fused to each other via a peptide linker define the third domain of the binding construct of the invention, which may also be defined as scFc domain.

[0154] In one embodiment of the invention it is envisaged that a scFc domain as disclosed herein, respectively the Fc monomers fused to each other are comprised only in the third domain of the binding construct. In some embodiments an IgG hinge region can be identified by analogy using the Kabat numbering as set forth in Table 4. In line with the above, it is envisaged that for a hinge domain/region of the present invention the minimal requirement comprises the amino acid residues corresponding to the IgG1 sequence stretch of D231 D234 to P243 according to the Kabat numbering. It is likewise envisaged that a hinge domain/region of the present invention comprises or consists of the IgG1 hinge sequence DKTHTCPPCP (SEQ ID NO: 4) (corresponding to the stretch D234 to P243 as shown in Table 4 below—variations of said sequence are also envisaged provided that the hinge region still promotes dimerization). In a preferred embodiment of the invention the glycosylation site at Kabat position 314 of the CH2 domains in the third domain of the binding construct is removed by a N314X substitution, wherein X is any amino acid excluding Q. Said substitution is preferably a N314G substitution. In a more preferred embodiment, said CH2 domain additionally comprises the following substitutions (position according to Kabat) V321C and R309C (these substitutions introduce the intra domain cysteine disulfide bridge at Kabat positions 309 and 321).

TABLE 4

Kabat numbering of the amino acid residues of the hinge region		
IMGT numbering for the hinge	IgG ₁ amino acid translation	Kabat numbering
1	(E)	226
2	P	227
3	K	228
4	S	232
5	C	233
6	D	234
7	K	235
8	T	236
9	H	237
10	T	238
11	C	239
12	P	240
13	P	241
14	C	242
15	P	243

[0155] In further embodiments of the present invention, the hinge domain/region comprises or consists of the IgG2 subtype hinge sequence ERKCCVECP (SEQ ID NO: 563), the IgG3 subtype hinge sequence ELKTPLDTHTCPRCP (SEQ ID NO: 564) or ELKTPLDTHTCPRCP (SEQ ID NO: 565), and/or the IgG4 subtype hinge sequence

ESKYGPPCPSCP (SEQ ID NO: 567). The IgG1 subtype hinge sequence may be the following one EPKSCDKTH-TCPPCP (SEQ ID NO: 568). These core hinge regions are thus also envisaged in the context of the present invention. **[0156]** The location and sequence of the IgG CH2 and IgG CH3 domain can be identified by analogy using the Kabat numbering as set forth in Table 5:

TABLE 5

Kabat numbering of the amino acid residues of the IgG CH2 and CH3 region				
IgG subtype	CH2 aa translation	CH2 Kabat numbering	CH3 aa translation	CH3 Kabat numbering
IgG ₁	APE . . . KAK	244 . . . 360	GQP . . . PGK	361 . . . 478
IgG ₂	APP . . . KTK	244 . . . 360	GQP . . . PGK	361 . . . 478
IgG ₃	APE . . . KTK	244 . . . 360	GQP . . . PGK	361 . . . 478
IgG ₄	APE . . . KAK	244 . . . 360	GQP . . . LGK	361 . . . 478

[0157] In one embodiment of the invention, the emphasized bold amino acid residues in the CH3 domain of the first or both Fc monomers are deleted.

[0158] In a classical full-length antibody or immunoglobulin, each light (L) chain is linked to a heavy (H) chain by one covalent disulfide bond, while the two H chains are linked to each other by one or more disulfide bonds depending on the H chain isotype. The heavy chain constant (CH) domain most proximal to VH is usually designated as CH1. The constant ("C") domains are not directly involved in antigen binding, but exhibit various effector functions, such as antibody-dependent cell-mediated cytotoxicity (ADCC) and complement activation (complement dependent cytotoxicity, CDC). The Fc region of an antibody is the "tail" region of a classical antibody that interacts with cell surface receptors called Fc receptors and some proteins of the complement system. In IgG, IgA and IgD antibody isotypes, the Fc region is composed of two identical protein fragments, derived from the second and third constant domains (CH2 and CH3) of the antibody's two heavy chains. IgM and IgE Fc regions contain three heavy chain constant domains (CH2, CH3 and CH4) in each polypeptide chain. The Fc regions also contains part of the so-called "hinge" region held together by one or more disulfides and noncovalent interactions. The Fc region of a naturally occurring IgG bears a highly conserved N-glycosylation site. Glycosylation of the Fc fragment is essential for Fc receptor-mediated activity.

[0159] In some embodiments of the invention the CH2 domain of one or preferably each (both) polypeptide monomers of the third domain comprises an intra domain cysteine disulfide bridge. As known in the art the term "cysteine disulfide bridge" refers to a functional group with the general structure R—S—S—R. The linkage is also called an SS-bond or a disulfide bridge or a cysteine clamp and is derived by the coupling of two thiol groups of cysteine residues. In certain embodiments, the cysteines forming the cysteine disulfide bridge in the mature binding construct are introduced into the amino acid sequence of the CH2 domain corresponding to 309 and 321 (Kabat numbering). In other embodiments, the cysteine clamps are introduced in other domains of the binding constructs. See also, e.g. US 2016/0193295.

[0160] In one embodiment of the invention a glycosylation site in Kabat position 314 of the CH2 domain is removed. It

is preferred that this removal of the glycosylation site is achieved by a N314X substitution, wherein X is any amino acid excluding Q. Said substitution is preferably a N314G. In a more preferred embodiment, said CH2 domain additionally comprises the following substitutions (position according to Kabat) V321C and R309C (these substitutions introduce the intra domain cysteine disulfide bridge at Kabat positions 309 and 321).

[0161] Covalent modifications of the binding constructs are also included within the scope of this invention, and are generally, but not always, done post-translationally. For example, several types of covalent modifications of the binding construct are introduced into the molecule by reacting specific amino acid residues of the binding construct with an organic derivatizing agent that is capable of reacting with selected side chains or the N- or C-terminal residues.

[0162] Cysteiny residues most commonly are reacted with α -haloacetates (and corresponding amines), such as chloroacetic acid or chloroacetamide, to give carboxymethyl or carboxyamidomethyl derivatives. Cysteiny residues also are derivatized by reaction with bromotrifluoroacetone, α -bromo- β -(5-imidazolyl)propionic acid, chloroacetyl phosphate, N-alkylmaleimides, 3-nitro-2-pyridyl disulfide, methyl 2-pyridyl disulfide, p-chloromercuribenzoate, 2-chloromercuri-4-nitrophenol, or chloro-7-nitrobenzo-2-oxa-1,3-diazole.

[0163] Histidyl residues are derivatized by reaction with diethylpyrocarbonate at pH 5.5-7.0 because this agent is relatively specific for the histidyl side chain. Para-bromophenacyl bromide also is useful; the reaction is preferably performed in 0.1 M sodium cacodylate at pH 6.0. Lysiny and amino terminal residues are reacted with succinic or other carboxylic acid anhydrides. Derivatization with these agents has the effect of reversing the charge of the lysiny residues. Other suitable reagents for derivatizing alpha-amino-containing residues include imidoesters such as methyl picolinimidate; pyridoxal phosphate; pyridoxal; chloroborohydride; trinitrobenzenesulfonic acid; O-methylisourea; 2,4-pentanedione; and transaminase-catalyzed reaction with glyoxylate.

[0164] Arginy residues are modified by reaction with one or several conventional reagents, among them phenylglyoxal, 2,3-butanedione, 1,2-cyclohexanedione, and ninhydrin. Derivatization of arginine residues requires that the reaction be performed in alkaline conditions because of the high pKa of the guanidine functional group. Furthermore, these reagents may react with the groups of lysine as well as the arginine epsilon-amino group.

[0165] The specific modification of tyrosyl residues may be made, with particular interest in introducing spectral labels into tyrosyl residues by reaction with aromatic diazonium compounds or tetranitromethane. Most commonly, N-acetylimidazole and tetranitromethane are used to form O-acetyl tyrosyl species and 3-nitro derivatives, respectively. Tyrosyl residues are iodinated using ¹²⁵I or ¹³¹I to prepare labeled proteins for use in radioimmunoassay, the chloramine T method described above being suitable.

[0166] Carboxyl side groups (aspartyl or glutamyl) are selectively modified by reaction with carbodiimides (R'—N=C=N—R'), where R and R' are optionally different alkyl groups, such as 1-cyclohexyl-3-(2-morpholinyl-4-ethyl) carbodiimide or 1-ethyl-3-(4-azonia-4,4-dimethylpentyl) carbodiimide.

[0167] Furthermore, aspartyl and glutamyl residues are converted to asparaginyl and glutaminyl residues by reaction with ammonium ions.

[0168] Derivatization with bifunctional agents is useful for crosslinking the binding constructs of the present invention to a water-insoluble support matrix or surface for use in a variety of methods. Commonly used crosslinking agents include, e.g., 1,1-bis(diazoacetyl)-2-phenylethane, glutaraldehyde, N-hydroxysuccinimide esters, for example, esters with 4-azidosalicylic acid, homobifunctional imidoesters, including disuccinimidyl esters such as 3,3'-dithiobis(succinimidylpropionate), and bifunctional maleimides such as bis-N-maleimido-1,8-octane. Derivatizing agents such as methyl-3-[(p-azidophenyl)dithio]propioimide yield photo-activatable intermediates that are capable of forming crosslinks in the presence of light. Alternatively, reactive water-insoluble matrices such as cyanogen bromide-activated carbohydrates and the reactive substrates as described in U.S. Pat. Nos. 3,969,287; 3,691,016; 4,195,128; 4,247,642; 4,229,537; and 4,330,440 are employed for protein immobilization. Glutamyl and asparaginyl residues are frequently deamidated to the corresponding glutamyl and aspartyl residues, respectively. Alternatively, these residues are deamidated under mildly acidic conditions. Either form of these residues falls within the scope of this invention.

[0169] Other modifications include hydroxylation of proline and lysine, phosphorylation of hydroxyl groups of seryl or threonyl residues, methylation of the α -amino groups of lysine, arginine, and histidine side chains (T. E. Creighton, *Proteins: Structure and Molecular Properties*, W. H. Freeman & Co., San Francisco, 1983, pp. 79-86), acetylation of the N-terminal amine, and amidation of any C-terminal carboxyl group.

[0170] Another type of covalent modification of the binding constructs included within the scope of this invention comprises altering the glycosylation pattern of the protein. As is known in the art, glycosylation patterns can depend on both the sequence of the protein (e.g., the presence or absence of particular glycosylation amino acid residues, discussed below), or the host cell or organism in which the protein is produced. Particular expression systems are discussed below.

[0171] Glycosylation of polypeptides is typically either N-linked or O-linked. N-linked refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. The tri-peptide sequences asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline, are the recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain. Thus, the presence of either of these tri-peptide sequences in a polypeptide creates a potential glycosylation site. O-linked glycosylation refers to the attachment of one of the sugars N-acetylgalactosamine, galactose, or xylose, to a hydroxyamino acid, most commonly serine or threonine, although 5-hydroxyproline or 5-hydroxylysine may also be used.

[0172] Addition of glycosylation sites to the binding construct is conveniently accomplished by altering the amino acid sequence such that it contains one or more of the above-described tri-peptide sequences (for N-linked glycosylation sites). The alteration may also be made by the addition of, or substitution by, one or more serine or threonine residues to the starting sequence (for O-linked glycosylation sites). For ease, the amino acid sequence of a

binding construct is preferably altered through changes at the DNA level, particularly by mutating the DNA encoding the polypeptide at preselected bases such that codons are generated that will translate into the desired amino acids.

[0173] Another means of increasing the number of carbohydrate moieties on the binding construct is by chemical or enzymatic coupling of glycosides to the protein. These procedures are advantageous in that they do not require production of the protein in a host cell that has glycosylation capabilities for N- and O-linked glycosylation. Depending on the coupling mode used, the sugar(s) may be attached to (a) arginine and histidine, (b) free carboxyl groups, (c) free sulfhydryl groups such as those of cysteine, (d) free hydroxyl groups such as those of serine, threonine, or hydroxyproline, (e) aromatic residues such as those of phenylalanine, tyrosine, or tryptophan, or (f) the amide group of glutamine. These methods are described in WO 87/05330, and in Aplin and Wriston, 1981, *CRC Crit. Rev. Biochem.*, pp. 259-306.

[0174] Removal of carbohydrate moieties present on the starting binding construct may be accomplished chemically or enzymatically. Chemical deglycosylation requires exposure of the protein to the compound trifluoromethanesulfonic acid, or an equivalent compound. This treatment results in the cleavage of most or all sugars except the linking sugar (N-acetylglucosamine or N-acetylgalactosamine), while leaving the polypeptide intact. Chemical deglycosylation is described by Hakimuddin et al., 1987, *Arch. Biochem. Biophys.* 259:52 and by Edge et al., 1981, *Anal. Biochem.* 118:131. Enzymatic cleavage of carbohydrate moieties on polypeptides can be achieved by the use of a variety of endo- and exo-glycosidases as described by Thotakura et al., 1987, *Meth. Enzymol.* 138:350. Glycosylation at potential glycosylation sites may be prevented by the use of the compound tunicamycin as described by Duskin et al., 1982, *J. Biol. Chem.* 257:3105. Tunicamycin blocks the formation of protein-N-glycoside linkages.

[0175] Other modifications of the binding construct are also contemplated herein. For example, another type of covalent modification of the binding construct comprises linking the binding construct to various non-proteinaceous polymers, including, but not limited to, various polyols such as polyethylene glycol, polypropylene glycol, polyoxalkylenes, or copolymers of polyethylene glycol and polypropylene glycol, in the manner set forth in U.S. Pat. Nos. 4,640,835; 4,496,689; 4,301,144; 4,670,417; 4,791,192 or 4,179,337. In addition, as is known in the art, amino acid substitutions may be made in various positions within the binding construct, e.g. in order to facilitate the addition of polymers such as PEG.

[0176] In some embodiments, the covalent modification of the binding constructs of the invention comprises the addition of one or more labels. The labelling group may be coupled to the binding construct via spacer arms of various lengths to reduce potential steric hindrance. Various methods for labelling proteins are known in the art and can be used in performing the present invention. The term "label" or "labelling group" refers to any detectable label. In general, labels fall into a variety of classes, depending on the assay in which they are to be detected—the following examples include, but are not limited to:

[0177] isotopic labels, which may be radioactive or heavy isotopes, such as radioisotopes or radionuclides (e.g., ^3H , ^{14}C , ^{15}N , ^{35}S , ^{89}Zr , ^{90}Y , ^{99}Tc , ^{111}In , ^{125}I , ^{131}I)

[0178] magnetic labels (e.g., magnetic particles)

[0179] redox active moieties

[0180] optical dyes (including, but not limited to, chromophores, phosphors and fluorophores) such as fluorescent groups (e.g., FITC, rhodamine, lanthanide phosphors), chemiluminescent groups, and fluorophores which can be either "small molecule" fluors or proteinaceous fluors

[0181] enzymatic groups (e.g. horseradish peroxidase, β -galactosidase, luciferase, alkaline phosphatase)

[0182] biotinylated groups

[0183] predetermined polypeptide epitopes recognized by a secondary reporter (e.g., leucine zipper pair sequences, binding sides for secondary antibodies, metal binding domains, epitope tags, etc.)

[0184] By "fluorescent label" is meant any molecule that may be detected via its inherent fluorescent properties. Suitable fluorescent labels include, but are not limited to, fluorescein, rhodamine, tetramethylrhodamine, eosin, erythrosin, coumarin, methyl-coumarins, pyrene, Malachite green, stilbene, Lucifer Yellow, Cascade BlueJ, Texas Red, IAE-DANS, EDANS, BODIPY FL, LC Red 640, Cy 5, Cy 5.5, LC Red 705, Oregon green, the Alexa-Fluor dyes (Alexa Fluor 350, Alexa Fluor 430, Alexa Fluor 488, Alexa Fluor 546, Alexa Fluor 568, Alexa Fluor 594, Alexa Fluor 633, Alexa Fluor 660, Alexa Fluor 680), Cascade Blue, Cascade Yellow and R-phycoerythrin (PE) (Molecular Probes, Eugene, OR), FITC, Rhodamine, and Texas Red (Pierce, Rockford, IL), Cy5, Cy5.5, Cy7 (Amersham Life Science, Pittsburgh, PA). Suitable optical dyes, including fluorophores, are described in Molecular Probes Handbook by Richard P. Haugland.

[0185] Suitable proteinaceous fluorescent labels also include, but are not limited to, green fluorescent protein, including a *Renilla*, *Ptilosarcus*, or *Aequorea* species of GFP (Chalfie et al., 1994, Science 263:802-805), EGFP (Clontech Laboratories, Inc., Genbank Accession Number U55762), blue fluorescent protein (BFP, Quantum Biotechnologies, Inc. 1801 de Maisonneuve Blvd. West, 8th Floor, Montreal, Quebec, Canada H3H 1J9; Stauber, 1998, Biotechniques 24:462-471; Heim et al., 1996, Curr. Biol. 6:178-182), enhanced yellow fluorescent protein (EYFP, Clontech Laboratories, Inc.), luciferase (Ichiki et al., 1993, J. Immunol. 150:5408-5417), β galactosidase (Nolan et al., 1988, Proc. Natl. Acad. Sci. U.S.A. 85:2603-2607) and *Renilla* (WO92/15673, WO95/07463, WO98/14605, WO98/26277, WO99/49019, U.S. Pat. Nos. 5,292,658; 5,418,155; 5,683,888; 5,741,668; 5,777,079; 5,804,387; 5,874,304; 5,876,995; 5,925,558).

[0186] Leucine zipper domains are peptides that promote oligomerization of the proteins in which they are found. Leucine zippers were originally identified in several DNA-binding proteins (Landschulz et al., 1988, Science 240:1759), and have since been found in a variety of different proteins. Among the known leucine zippers are naturally occurring peptides and derivatives thereof that dimerize or trimerize. Examples of leucine zipper domains suitable for producing soluble oligomeric proteins are described in PCT application WO 94/10308, and the leucine zipper derived from lung surfactant protein D (SPD) described in Hoppe et al., 1994, FEBS Letters 344:191. The use of a modified leucine zipper that allows for stable trimerization of a heterologous protein fused thereto is described in Fanslow et al., 1994, Semin. Immunol. 6:267-78.

[0187] The binding construct of the invention may also comprise additional domains, which are e.g. helpful in the isolation of the molecule or relate to an adapted pharmacokinetic profile of the molecule. Domains helpful for the isolation of a binding construct may be selected from peptide motives or secondarily introduced moieties, which can be captured in an isolation method, e.g. an isolation column. Non-limiting embodiments of such additional domains comprise peptide motives known as Myc-tag, HAT-tag, HA-tag, TAP-tag, GST-tag, chitin binding domain (CBD-tag), maltose binding protein (MBP-tag), Flag-tag, Strep-tag and variants thereof (e.g. Strepli-tag) and His-tag. All herein disclosed binding constructs may comprise a His-tag domain, which is generally known as a repeat of consecutive His residues in the amino acid sequence of a molecule, preferably of five, and more preferably of six His residues (hexa-histidine) (SEQ ID NO: 569). The His-tag may be located e.g. at the N- or C-terminus of the binding construct, preferably it is located at the C-terminus. Most preferably, a hexa-histidine tag (HHHHHH) (SEQ ID NO: 569) is linked via peptide bond to the C-terminus of the binding construct according to the invention. Additionally, a conjugate system of PLGA-PEG-PLGA may be combined with a poly-histidine tag for sustained release application and improved pharmacokinetic profile.

Diagnostic Methods

[0188] The binding constructs that are provided herein are useful for detecting MAGEB2 in biological samples and can be used for diagnostic purposes to detect, diagnose, or monitor diseases and/or conditions associated with MAGEB2.

[0189] The disclosed binding constructs provide a means for the detection of the presence of MAGEB2 in a sample using, for example, classical immunohistological methods known to those of skill in the art (e.g., Tijssen, 1993, Practice and Theory of Enzyme Immunoassays, Vol 15 (Eds R. H. Burdon and P. H. van Knippenberg, Elsevier, Amsterdam); Zola, 1987, Monoclonal Antibodies: A Manual of Techniques, pp. 147-158 (CRC Press, Inc.); Jalkanen et al., 1985, J. Cell. Biol. 101:976-985; Jalkanen et al., 1987, J. Cell Biol. 105:3087-3096). The detection of MAGEB2 can be performed in vivo or in vitro, but is preferably done in vitro on a sample that comprises cells obtained from a patient. The methods used to detect MAGEB2 expression in human tissue, e.g., tumor tissue, can be qualitative, semi-quantitative, or quantitative. For example, art-recognized antibody-based methods for detecting protein levels in biological samples include, but are not limited to, enzyme-linked immunosorbent assay (ELISA), radioimmunoassays, electrochemiluminescence (ECL) assays, surface plasmon resonance, western blot, immunoprecipitation, fluorescence-activated cell sorting (FACS), immunofluorescence, immunohistochemistry, and the like. Methods for detecting MAGEB2 expression in a biological sample may include control samples (negative and positive controls). For example, a negative control sample may be a sample containing no MAGEB2 protein, and a positive controls sample is a sample containing MAGEB2 protein. Comparing the results with the negative and positive controls can confirm the presence or absence of MAGEB2 in the biological sample.

[0190] In a specific embodiment, the binding construct can be used in an immunohistochemistry (IHC) assay. These

IHC assays can be performed on modern, higher throughput platforms that in some instances are automated. For example, the Roche-Ventana BenchMark ULTRA system.

[0191] IHC is a commonly used assay that involves the use of binding constructs such as antibodies to specifically bind to a target antigen in a tissue sample. This allows the assay user to ascertain whether the target antigen is or is not present in the tissue sample.

[0192] Immunohistochemical methods are well-known in the art and are described in, e.g., *Immunohistochemical Staining Methods* 6th Edition 2013, Taylor C R, Rudbeck L, eds., Dako North America, available as an Education Guide at www.dako.com; “Immunohistochemical Staining Method guide”). Immunohistochemical staining may be conducted on fixed tissue sections or on frozen tissue sections. When using fixed tissue sections, e.g., formalin fixed paraffin embedded (FFPE) tissue sections, the procedure can generally use the following exemplary steps: obtaining a tumor tissue sample (e.g., by biopsy), fixation of the tumor sample; embedding (e.g., in paraffin); sectioning and mounting; antigen retrieval; incubation with a primary antibody (e.g., a MAGEB2 antibody described herein), detection (e.g., after amplification of the antigen/antibody complex signal), and interpretation by a skilled practitioner (e.g., using an art-recognized scoring system).

[0193] Suitable, non-limiting fixatives include, for example, paraformaldehyde, glutaraldehyde, formaldehyde, acetic acid, acetone, osmium tetroxide, chromic acid, mercuric chloride, picric acid, alcohols (e.g., methanol, ethanol), Gendre’s fluid, Rossman’s fluid, B5 fixative, Bouin’s fluid, Carnoy’s fixative, and methacarn. In one embodiment, the tumor sample is fixed in formaldehyde (e.g., 10% neutral buffered formalin, which corresponds to 4% formaldehyde in a buffered solution). Once fixed, the tumor sample can be serially dehydrated using alcohol or xylene, embedded in paraffin, and cut with a microtome to generate tumor tissue sections (e.g., with a thickness of about 4-5 μ m), which can then be mounted onto microscope slides (e.g., adhesive-coated glass slides). Exemplary embedding materials include paraffin, celloidin, OCT™ compound, agar, plastics, or acrylics. Another exemplary material, and as described in the Examples herein, is Histogel containing agarose/glycerine (ThermoFisher Scientific).

[0194] Provided herein are methods for detecting MAGEB2 expression in a biological sample, e.g., a human tumor tissue sample, by contacting the sample with an anti-MAGEB2 antigen binding construct, such as an anti-MAGEB2 antibody comprising the heavy and light chain variable region CDR sequences set forth herein, or the heavy and light chain variable region sequences set forth herein, wherein the antigen binding construct specifically binds to human MAGEB2, and detecting the binding of the antibody to MAGEB2 in the sample, e.g., by immunohistochemistry.

[0195] Accordingly, the invention provides a method of diagnosing a subject with a tumor, comprising: a) determining the level of MAGEB2 in a sample obtained from the subject; and b) diagnosing the subject as having a MAGEB2 positive tumor when the level of MAGEB2 is increased relative to a control. Provided herein is a method of identifying a subject having a MAGEB2 positive tumor comprising: a) determining the level of MAGEB2 in a sample obtained from the subject; and b) identifying the subject as having a MAGEB2 positive tumor when the level of MAGEB2 is increased relative to a control. Provided herein

is a method of identifying a subject as needing an anti-MAGEB2 therapeutic comprising: a) determining the level of MAGEB2 in a sample obtained from the subject; and b) identifying the subject as needing the anti-MAGEB2 therapeutic when the level of MAGEB2 is increased relative to a control.

[0196] Provided herein is a method of determining treatment for a subject with a MAGEB2 positive tumor comprising: determining the level of MAGEB2 in a sample obtained from the subject; and determining the treatment as comprising an anti-MAGEB2 therapeutic when the level of MAGEB2 is increased, relative to a control.

[0197] In certain embodiments, exemplary biological samples include blood samples, serum samples, cells, surgically-resected tissue, and biopsied tissue (e.g. cancer tissue) obtained from the cancer patient. Biological samples for use in the methods described herein can be fresh, frozen, or fixed. Biological samples, e.g., tumor samples, can be obtained from the patient using routine methods, such as, but not limited to, biopsy, surgical resection, or aspiration.

[0198] MAGEB2 protein expression in tissue, e.g., tumor, samples can be detected using direct or indirect methods. For example, in assays where the binding construct is an antibody, the primary antibody may comprise a detectable moiety, such as the enzyme horseradish peroxidase (HRP) or a fluorescent label (e.g., FITC, TRITC), as described herein. In certain embodiments, the primary antibody does not itself comprise a detectable moiety, but is instead detected by binding of a secondary antibody to it, for indirect immunohistochemistry. Accordingly, in certain embodiments, the secondary antibody comprises a detectable label, e.g., an enzymatic, chromogenic, or fluorescent label. In certain embodiments, the primary antibody is a chimeric antibody comprising human variable region sequences and a non-human Fc region. A secondary antibody can be used to recognize the non-human Fc region of the primary antibody in order to reduce background staining.

[0199] In other embodiments, MAGEB2 protein expression can be detected using the avidin-biotin complex method (ABC method). In these embodiments, for example when the binding construct is an antibody, the secondary antibody is biotinylated and can serve as a bridge between the primary antibody and biotin-avidin-peroxidase complex. Other suitable non-limiting methods for immunohistochemistry include those described in Chapter 6 of the *Immunohistochemical Staining Method guide*, e.g., methods described in Chilosi et al., *Biotech Histochem* 1994; 69:235; Sabatini et al., *J Clin Pathol* 1998; 51:506-11; and Gross et al., *JBC* 1959; 234:1622.

[0200] General methods for preparing and staining frozen tissue sections are well known in the art, and are described in, e.g., the *Immunohistochemical Staining Method guide*, Chapter 3 of *Immunohistochemistry and Methods* (Buchwalow and Bocker, Springer-Verlag Berlin Heidelberg 2010), and Chapter 21 of *Theory and Practice of Histological Techniques* (Bancroft and Gamble, 6.sup.th Edition, Elsevier Ltd., 2008). In some embodiments, slides stained with, e.g., the anti-MAGEB2 antibodies described herein can be further counterstained with, e.g., hematoxylin and/or eosin, using methods well known in the art.

[0201] MAGEB2 protein expression detected by immunohistochemistry can be digitally scanned and analyzed (e.g., as described herein in the Examples), or otherwise

evaluated and scored using art-recognized scoring methods. Non-limiting examples of scoring methods are described below.

[0202] For example, in certain embodiments, MAGEB2 expression can be evaluated using the H-score (histochemical score) system, which is widely used in the art and is useful given its dynamic range and use of weighted percentiles. The H-score is a semiquantitative scoring system based on the formula: $3 \times \text{percentage of strongly staining cells (3+ staining)} + 2 \times \text{percentage of moderately staining (2+ staining) cells} + 1 \times \text{percentage of weakly staining (1+ staining) cells} + 0 \times \text{percentage of non-stained (0 staining) cells}$, giving a score range of 0 to 300. See, e.g., McCarty et al., *Cancer Res* 1986; 46:4244-8; Bosman et al., *J Clin Pathol* 1992; 45:120-4; Dieset et al., *Analyt Quant Cytol Histol* 1996; 18:351-4. Control tissue can include, e.g., matching non-tumor tissue from the same subject. Accordingly, in certain embodiments, the number of cells staining positive for MAGEB2 using the binding constructs herein (e.g., antibodies), and the intensity of staining, can be used to determine an H-score.

[0203] In other embodiments, MAGEB2 expression can be evaluated using the Allred scoring system (Allred et al., *Mod Pathol* 1998; 11:155-68; Harvey et al., *J Clin Oncol* 1999; 17:1474-91). This scoring system involves adding proportion and intensity scores to obtain a total score. Accordingly, in some embodiments, the proportion score is obtained based on the estimated proportion of tumor cells that are positive for MAGEB2 (0: none, 1: $<1/100$, 2: $1/100$ to $1/10$, 3: $1/10$ to $1/5$, 4: $1/5$ to $2/3$, and 5: $>2/3$), and the intensity score is obtained based on the average intensity of MAGEB2 expression in positive tumor cells (0: none, 1: weak, 2: intermediate, 3: strong). Allred scores range from 0 to 8, with a score ranging from 3 to 8 considered positive (i.e., positive detection). Accordingly, in certain embodiments, tumors with an Allred score of 3 to 8, for example, an Allred score of 3, 4, 5, 6, 7, or 8, for MAGEB2 staining are considered MAGEB2-positive tumors.

[0204] In other embodiments, and as discussed herein above, the scoring system can be automated, e.g., computerized, and quantified by image analysis. Automated methods are well known in the art. For example, an average threshold measure (ATM) score, which obtains an average of 255 staining intensity levels, can be calculated as described in, e.g., Choudhury et al., *J Histochem Cytochem* 2010; 58:95-107; Rizzardi et al., *Diagnostic Pathology* 2012; 7:42-52. Another automated scoring system is AQUA® (automated quantitative analysis), which is performed using, e.g., tissue microarrays (TMAs), on a continuous scale. AQUA® is a hybrid of standard immunohistochemistry and flow cytometry that provides an objective numeric score ranging from 1-255, and involves antigen retrieval, use of primary and secondary antibodies, and multiplexed fluorescent detection. Optimal cutoff points can be determined as described in Camp et al. *Clin Cancer Res* 2004; 10:7252-9. The AQUA® scoring system is described in detail in Camp et al., *Nat Med* 2002; 8:1323-7; Camp et al., *Cancer Res* 2003; 63:1445-8; Ghosh et al., *Hum Pathol* 2008; 39:1835-43; Bose et al., *BMC Cancer* 2012; 12:332; Mascaux et al., *Clin Cancer Res* 2011; 17:7796-807. Other suitable automated immunohistochemistry platforms include commercially available platforms, such as the Leica BOND RX staining platform. The platform detects target protein expression in tumor tissue on the basis of staining intensity on the following scale: minimal, <1 cells per 20× objective

field; mild, 1-10 cells per 20× objective field; moderate, 10-50 cells per 20× objective field; marked, 50-200 cells per 20× objective; and intense, >200 cells per 20× objective field.

[0205] In some embodiments, a tumor is considered MAGEB2-positive and likely to benefit from a therapeutic that targets MAGEB2 if a sample of the tumor has more MAGEB2-positive cells than a matching normal tissue from the same cancer patient.

[0206] In certain embodiments, the cancer patient is likely to or predicted to benefit from a therapeutic that targets MAGEB2 if the number of MAGEB2-positive cells in the tumor sample exceeds a threshold level. A threshold level may be the lowest level of MAGEB2 as determined with a given detection system, e.g., immunohistochemistry, as further described herein, such as in the Examples. In certain embodiments, the cancer patient is likely to or predicted to benefit from a therapeutic that targets MAGEB2 if the number of MAGEB2-positive cells in the tumor sample exceeds a threshold number or proportion using an art-recognized scoring system described supra. Accordingly, in some embodiments, if the number or proportion of MAGEB2-positive cells exceeds a certain threshold number or proportion using a scoring system described herein, then the sample can be defined as being “MAGEB2-positive.”

[0207] In certain embodiments, a tumor sample having at least 1%, for example, at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 90%, or at least 95% of cells in the tumor expressing MAGEB2 is designated as being “MAGEB2-positive” and indicates that the cancer patient is likely to or predicted to respond to a therapeutic that targets MAGEB2.

[0208] In some embodiments, an H score can be calculated for the tumor sample. Accordingly, in certain embodiments, a tumor sample with an H score of at least 5, such as at least 10, at least 15, at least 20, at least 25, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 125, at least 150, at least 175, at least 200, at least 225, at least 250, at least 275, or at least 290 is designated as being “MAGEB2-positive” and indicates that the cancer patient is likely to or predicted to respond to a therapeutic that targets MAGEB2.

[0209] In some embodiments, an Allred score can be calculated for the tumor sample. Accordingly, in certain embodiments, a tumor sample with an Allred score of at least 3, such as at least 4, at least 5, at least 6, at least 7, or 8 is designated as being “MAGEB2-positive” and indicates that the cancer patient is likely to or predicted to respond to a therapeutic that targets MAGEB2.

[0210] In some embodiments, an AQUA score can be calculated. Accordingly, in certain embodiments, a tumor sample with an AQUA score of at least 5, such as at least 10, at least 15, at least 20, at least 25, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 125, at least 150, at least 175, at least 200, at least 225, or at least 250 is designated as being “MAGEB2-positive” and indicates that the cancer patient is likely to or predicted to respond to a therapeutic that targets MAGEB2.

[0211] In certain embodiments, a combination of one of the scoring systems above with another detection modality,

such as FISH, is used to increase the accuracy of determining whether a tumor is MAGEB2-positive or MAGEB2-negative.

[0212] In certain embodiments, the patient has a cancer selected from the group consisting of squamous cell carcinoma, small-cell lung cancer, non-small cell lung cancer, squamous non-small cell lung cancer (NSCLC), non squamous NSCLC, glioma, gastrointestinal cancer, renal cancer (e.g. clear cell carcinoma), ovarian cancer, liver cancer, colorectal cancer, endometrial cancer, kidney cancer (e.g., renal cell carcinoma (RCC)), prostate cancer (e.g. hormone refractory prostate adenocarcinoma), thyroid cancer, neuroblastoma, pancreatic cancer, glioblastoma (glioblastoma multiforme), cervical cancer, stomach cancer, bladder cancer, hepatoma, breast cancer, colon carcinoma, and head and neck cancer (or carcinoma), gastric cancer, germ cell tumor, pediatric sarcoma, sinonasal natural killer, melanoma (e.g., metastatic malignant melanoma, such as cutaneous or intraocular malignant melanoma), bone cancer, skin cancer, uterine cancer, cancer of the anal region, testicular cancer, carcinoma of the fallopian tubes, carcinoma of the endometrium, carcinoma of the cervix, carcinoma of the vagina, carcinoma of the vulva, cancer of the esophagus, cancer of the small intestine, cancer of the endocrine system, cancer of the parathyroid gland, cancer of the adrenal gland, sarcoma of soft tissue, cancer of the urethra, cancer of the penis, solid tumors of childhood, cancer of the ureter, carcinoma of the renal pelvis, neoplasm of the central nervous system (CNS), primary CNS lymphoma, tumor angiogenesis, spinal axis tumor, brain cancer, brain stem glioma, pituitary adenoma, Kaposi's sarcoma, epidermoid cancer, squamous cell cancer, T-cell lymphoma, environmentally-induced cancers including those induced by asbestos, virus-related cancers or cancers of viral origin (e.g., human papilloma virus (HPV)-related or -originating tumors)), and hematologic malignancies derived from either of the two major blood cell lineages, i.e., the myeloid cell line (which produces granulocytes, erythrocytes, thrombocytes, macrophages and mast cells) or lymphoid cell line (which produces B, T, NK and plasma cells), such as all types of leukemias, lymphomas, and myelomas, e.g., acute, chronic, lymphocytic and/or myelogenous leukemias, such as acute leukemia (ALL), acute myelogenous leukemia (AML), chronic lymphocytic leukemia (CLL), and chronic myelogenous leukemia (CML), undifferentiated AML (M0), myeloblastic leukemia (M1), myeloblastic leukemia (M2; with cell maturation), promyelocytic leukemia (M3 or M3 variant [M3V]), myelomonocytic leukemia (M4 or M4 variant with eosinophilia [M4E]), monocytic leukemia (M5), erythroleukemia (M6), megakaryoblastic leukemia (M7), isolated granulocytic sarcoma, and chloroma; lymphomas, such as Hodgkin's lymphoma (HL), non-Hodgkin's lymphoma (NHL), B cell hematologic malignancy, e.g., B-cell lymphomas, T-cell lymphomas, lymphoplasmacytoid lymphoma, monocytoid B-cell lymphoma, mucosa-associated lymphoid tissue (MALT) lymphoma, anaplastic (e.g., Ki 1+) large-cell lymphoma, adult T-cell lymphoma/leukemia, mantle cell lymphoma, angioimmunoblastic T-cell lymphoma, angiocentric lymphoma, intestinal T-cell lymphoma, primary mediastinal B-cell lymphoma, precursor T-lymphoblastic lymphoma, T-lymphoblastic; and lymphoma/leukaemia (T-Lbly/T-ALL), peripheral T-cell lymphoma, lymphoblastic lymphoma, post-transplantation lymphoproliferative disorder, true histiocytic lymphoma, primary central nervous system lymphoma, pri-

mary effusion lymphoma, B cell lymphoma, lymphoblastic lymphoma (LBL), hematopoietic tumors of lymphoid lineage, acute lymphoblastic leukemia, diffuse large B-cell lymphoma, Burkitt's lymphoma, follicular lymphoma, diffuse histiocytic lymphoma (DHL), immunoblastic large cell lymphoma, precursor B-lymphoblastic lymphoma, cutaneous T-cell lymphoma (CTLC) (also called mycosis fungoides or Sezary syndrome), and lymphoplasmacytoid lymphoma (LPL) with Waldenstrom's macroglobulinemia; myelomas, such as IgG myeloma, light chain myeloma, nonsecretory myeloma, smoldering myeloma (also called indolent myeloma), solitary plasmocytoma, and multiple myelomas, chronic lymphocytic leukemia (CLL), hairy cell lymphoma; hematopoietic tumors of myeloid lineage, tumors of mesenchymal origin, including fibrosarcoma and rhabdomyosarcoma; seminoma, teratocarcinoma, tumors of the central and peripheral nervous, including astrocytoma, schwannomas; tumors of mesenchymal origin, including fibrosarcoma, rhabdomyosarcoma, and osteosarcoma; and other tumors, including melanoma, xeroderma pigmentosum, keratoacanthoma, seminoma, thyroid follicular cancer and teratocarcinoma, hematopoietic tumors of lymphoid lineage, for example T-cell and B-cell tumors, including but not limited to T-cell disorders such as T-prolymphocytic leukemia (T-PLL), including of the small cell and cerebriform cell type; large granular lymphocyte leukemia (LGL) preferably of the T-cell type; a/d T-NHL hepatosplenic lymphoma; peripheral/post-thymic T cell lymphoma (pleomorphic and immunoblastic subtypes); angiocentric (nasal) T-cell lymphoma; cancer of the head or neck, renal cancer, rectal cancer, cancer of the thyroid gland; acute myeloid lymphoma, as well as any combinations of said cancers. In certain embodiments, the cancer is a metastatic cancer, refractory cancer, or recurrent cancer.

Treatment of Patients

[0213] Once a cancer patient has been identified as being likely to benefit from treatment with a therapeutic that targets MAGEB2 expressing tumors (i.e., has a MAGEB2-positive tumor), the patient can be treated with an anti-MAGEB2 therapeutic (e.g., an anti-MAGEB2 immunotherapeutic such as those found in U.S. Patent Application Nos. 63/027,148 filed on May 19, 2020 and entitled "MAGEB2 Binding Constructs" and 63/128,773 filed on Dec. 21, 2020 and entitled "MAGEB2 Binding Constructs." Accordingly, provided herein is a method for treating a tumor in a subject, said method comprising: determining the subject as responsive to treatment with an anti-MAGEB2 therapeutic by obtaining a sample from the subject, wherein the sample comprises a cell from the tumor, measuring the level of MAGEB2 in the sample, and determining the subject as responsive to treatment with an anti-MAGEB2 therapeutic, and administering to the subject an effective amount of the anti-MAGEB2 therapeutic. Provided herein are methods of treating a cancer patient comprising: (a) determining whether a tumor of a cancer patient is MAGEB2-positive using the methods described herein, e.g., contacting a tumor sample from the patient with binding construct described herein, wherein the antibody or antigen-binding fragment thereof specifically binds to human MAGEB2, detecting the binding of the antibody to MAGEB2 in the sample, and determining the level of MAGEB2 protein expression in the sample, wherein a level of MAGEB2 protein above a threshold level in tumor cells

indicates that the tumor is MAGEB2-positive, and (b) if the tumor is determined to be MAGEB2-positive, then administering a therapeutically effective amount of an agent that targets cells that express MAGEB2 and kills them. Provided herein is a method of treating a tumor in a subject, wherein the subject's tumor has been tested for MAGEB2 levels and the subject's tumor is MAGEB2 positive, comprising administering an anti-MAGEB2 therapeutic to the subject.

[0214] Provided herein is a method of determining efficacy of treatment with an anti-MAGEB2 therapeutic in a subject comprising: determining the level of MAGEB2 in a sample obtained from the subject before treatment with an anti-MAGEB2 therapeutic and after treatment with an anti-MAGEB2 therapeutic; and determining the treatment as effective when the level of MAGEB2 positive tumor cells is decreased after treatment with the anti-MAGEB2 therapeutic.

[0215] Provided herein is a method of monitoring treatment with an anti-MAGEB2 therapeutic in a subject comprising: determining the level of MAGEB2 in a sample obtained from the subject at a first timepoint; determining the level of MAGEB2 in a sample obtained from the subject at a second timepoint; and optionally continuing treatment with an anti-MAGEB2 therapeutic when the level of MAGEB2 is decreased at the second timepoint relative to the level at the first timepoint.

[0216] Provided herein is a method of identifying a subject as responsive to treatment with an anti-MAGEB2 therapeutic comprising: determining the level of MAGEB2 in a sample obtained from the subject; identifying the subject as responsive to treatment with an anti-MAGEB2 therapeutic when the level of MAGEB2 decreases upon treatment with an anti-MAGEB2 therapeutic.

[0217] In a specific embodiment of the above embodiments, the subject or patient is treated with a therapeutically-effective amount of any of the molecules disclosed in U.S. Patent Application Nos. 63/027,148 filed on May 19, 2020 and entitled "MAGEB2 Binding Constructs" and 63/128,773 filed on Dec. 21, 2020 and entitled "MAGEB2 Binding Constructs," the contents of which are herein incorporated by reference.

[0218] In a further specific embodiment of the above embodiments, various reference levels of MAGEB2 in a patient sample can be compared to each other. For example, a first reference level taken at a first time point can be compared to a second level taken at a second time point. Further, e.g., MAGEB2 levels can be compared prior to drug treatment and after drug treatment. In this example, it can be envisaged that successful treatment with a drug will yield a lower measured MAGEB2 level after treatment as compared to the reference MAGEB2 level measured before treatment. The first and the second tissue sample can be obtained, e.g., within 3-7 days, 1 week to 3 weeks, or 1 month to 3 months from each other, or even a longer duration.

[0219] In certain embodiments, a method for monitoring a MAGEB2-expressing tumor in a cancer patient comprises: (a) detecting MAGEB2 protein expression in a tissue (e.g., tumor) sample at a first time point by using an anti-MAGEB2 antibody, or antigen-binding portion thereof, (b) determining the level of MAGEB2 protein expression at the first time point, (c) detecting MAGEB2 protein expression in a tissue (e.g., tumor, which may be the same tumor as the one that was biopsied at the first time point) sample at a second time point using the same antibody used in step (a),

and (d) determining the level of MAGEB2 protein expression in the tumor from the second time point.

[0220] Provided herein are methods for monitoring the efficacy of anti-MAGEB2 immunotherapy (e.g., one of the molecules provided in U.S. Patent Application Nos. 63/027,148 filed on May 19, 2020 and entitled "MAGEB2 Binding Constructs" and 63/128,773 filed on Dec. 21, 2020 and entitled "MAGEB2 Binding Constructs") in a patient having a MAGEB2-positive tumor comprising: (a) detecting MAGEB2 protein expression in the MAGEB2-positive tumor at a first time point before or after initiating anti-MAGEB2 immunotherapy by using an anti-MAGEB2 antibody, (b) determining the level of MAGEB2 protein expression in the tumor from the first time point, (c) detecting MAGEB2 protein expression in the MAGEB2-positive tumor at a second time point after initiating anti-MAGEB2 immunotherapy using the same antibody from step (a), (d) determining the level of MAGEB2 protein expression in the tumor from the second time point; (e) comparing the levels of MAGEB2 protein expression determined at the first and second time points, wherein a higher level at the first time point relative to the second time point may be indicative of effective anti-MAGEB2 immunotherapy, a lower score at the first time point relative to the second time point is indicative of ineffective anti-MAGEB2 immunotherapy, and an unchanged score at the first time point relative to the second time point may be indicative of anti-MAGEB2 immunotherapy being stabilizing. MAGEB2 expression may be determined in the tumor, tumor tissue, or tumor cells.

[0221] In certain embodiments, the status of a MAGEB2-expressing tumor can be monitored repeatedly after initial diagnosis (i.e., after making a determination that the tumor is MAGEB2-positive), such as one month after initial diagnosis, two months after initial diagnosis, three months after initial diagnosis, four months after initial diagnosis, five months after initial diagnosis, six months after initial diagnosis, one year after initial diagnosis, etc. In other embodiments, the efficacy of anti-MAGEB2 immunotherapy against a MAGEB2-expressing tumor can be monitored repeatedly after initiating anti-MAGEB2 immunotherapy, such as one month after initiation of therapy, two months after initiation of therapy, three months after initiation of therapy, four months after initiation of therapy, five months after initiation of therapy, six months after initiation of therapy, one year after initiation of therapy, etc.

Sequence Modifications of MAGEB2 Binding Constructs

[0222] The term "amino acid" or "amino acid residue" typically refers to an amino acid having its art recognized definition such as an amino acid selected from the group consisting of: alanine (Ala or A); arginine (Arg or R); asparagine (Asn or N); aspartic acid (Asp or D); cysteine (Cys or C); glutamine (Gln or Q); glutamic acid (Glu or E); glycine (Gly or G); histidine (His or H); isoleucine (Ile or I); leucine (Leu or L); lysine (Lys or K); methionine (Met or M); phenylalanine (Phe or F); proline (Pro or P); serine (Ser or S); threonine (Thr or T); tryptophan (Trp or W); tyrosine (Tyr or Y); and valine (Val or V), although modified, synthetic, or rare amino acids may be used as desired. There are basically four different classes of amino acids determined by different side chains:

[0223] (1) non-polar and neutral (uncharged): Ala, Gly, Ile, Leu, Met, Phe, Pro, Val

[0224] (2) polar and neutral (uncharged): Asn, Cys (being only slightly polar), Gln, Ser, Thr, Trp (being only slightly polar), Tyr

[0225] (3) acidic and polar (negatively charged): Asp and Glu

[0226] (4) basic and polar (positively charged): Arg, His, Lys

[0227] Hydrophobic amino acids can be divided according to whether they have aliphatic or aromatic side chains. Phe and Trp (being very hydrophobic), Tyr and His (being less hydrophobic) are classified as aromatic amino acids. Strictly speaking, aliphatic means that the side chain contains only hydrogen and carbon atoms. By this strict definition, the amino acids with aliphatic side chains are alanine, isoleucine, leucine (also norleucine), proline and valine. Alanine's side chain, being very short, means that it is not particularly hydrophobic, and proline has an unusual geometry that gives it special roles in proteins. It is often convenient to consider methionine in the same category as isoleucine, leucine and valine, although it also contains a sulphur atom. The unifying theme is that these amino acids contain largely non-reactive and flexible side chains. The amino acids alanine, cysteine, glycine, proline, serine and threonine are often grouped together for the reason that they are all small in size. Gly and Pro may influence chain orientation.

[0228] Amino acid modifications include, for example, deletions of residues from, insertions of residues into, and/or substitutions of residues within the amino acid sequences of the binding constructs. Any combination of deletion, insertion, and/or substitution is made to arrive at a final binding construct, provided that the final construct possesses the desired characteristics, e.g. the biological activity of the unmodified parental molecule (such as binding to MAGEB2). The amino acid changes may also alter post-translational processes of the binding constructs, such as changing the number or position of glycosylation sites.

[0229] For example, 1, 2, 3, 4, 5, or 6 amino acids may be inserted, deleted and/or substituted in each of the CDRs (of course, dependent on their respective length), while 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or 25 amino acids may be inserted, deleted and/or substituted in each of the FRs. Amino acid sequence insertions also include N-terminal and/or C-terminal additions of amino acids ranging in length from e.g. 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 residues to polypeptides containing more than 10, e.g. one hundred or more residues, as well as intra-sequence insertions of single or multiple amino acid residues. An insertional variant of the binding construct of the invention includes the fusion of a polypeptide which increases or extends the serum half-life of the binding construct to the N-terminus or to the C-terminus of the binding construct. It is also conceivable that such insertion occurs within the binding construct, e.g. between the first and the second domain.

[0230] The sites of greatest interest for amino acid modifications, in particular for amino acid substitutions, include the hypervariable regions, in particular the individual CDRs of the heavy and/or light chain, but FR alterations in the heavy and/or light chain are also contemplated. The substitutions can be conservative substitutions as described herein. Preferably, 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids may be substituted in a CDR, while 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or 25 amino acids may be substituted in the framework regions (FRs), depend-

ing on the length of the CDR or FR, respectively. For example, if a CDR sequence encompasses 6 amino acids, it is envisaged that one, two or three of these amino acids are substituted. Similarly, if a CDR sequence encompasses 15 amino acids it is envisaged that one, two, three, four, five or six of these amino acids are substituted.

[0231] A useful method for the identification of certain residues or regions within the binding constructs that are preferred locations for mutagenesis is called "alanine scanning mutagenesis" and is described e.g. in Cunningham B. C. and Wells J. A. (Science. 1989 Jun. 2; 244(4908):1081-5). Here, a residue or group of residues within the binding construct is/are identified (e.g. charged residues such as Arg, His, Lys, Asp, and Glu) and replaced by a neutral or non-polar amino acid (most preferably alanine or polyalanine) to affect the interaction of the respective amino acid(s) with the epitope of the target protein. Alanine scanning is a technique used to determine the contribution of a specific residue to the stability or function of given protein. Alanine is used because of its non-bulky, chemically inert, methyl functional group that nevertheless mimics the secondary structure preferences that many of the other amino acids possess. Sometimes bulky amino acids such as valine or leucine can be used in cases where conservation of the size of mutated residues is needed. This technique can also be useful to determine whether the side chain of a specific residue plays a significant role in bioactivity. Alanine scanning is usually accomplished by site-directed mutagenesis or randomly by creating a PCR library. Furthermore, computational methods to estimate thermodynamic parameters based on a theoretical alanine substitutions have been developed. The data can be tested by IR, NMR Spectroscopy, mathematical methods, bioassays, etc.

[0232] Those amino acid locations demonstrating functional sensitivity to the substitutions (as determined e.g. by alanine scanning) can then be refined by introducing further or other variants at, or for, the sites of substitution. Thus, while the site or region for introducing an amino acid sequence variation is predetermined, the nature of the mutation per se needs not to be predetermined. For example, to analyze or optimize the performance of a mutation at a given site, alanine scanning or random mutagenesis may be conducted at a target codon or region, and the expressed binding construct variants are screened for the optimal combination of desired activity. Techniques for making substitution mutations at predetermined sites in the DNA having a known sequence are well known, for example, M13 primer mutagenesis and PCR mutagenesis. Screening of the mutants is done e.g. using assays of antigen (e.g. MAGEB2) binding activity and/or of cytotoxic activity.

[0233] Generally, if amino acids are substituted in one or more or all of the CDRs of the heavy and/or light chain, it is envisaged that the then-obtained "substituted" sequence is at least 60% or 65%, more preferably 70% or 75%, even more preferably 80% or 85%, and particularly preferably 90% or 95% identical/homologous to the "original" or "parental" CDR sequence. This means that the degree of identity/homology between the original and the substituted sequence depends on the length of the CDR. For example, a CDR having 5 amino acids in total and comprising one amino acid substitution is 80% identical to the "original" or "parental" CDR sequence, while a CDR having 10 amino acids in total and comprising one amino acid substitution is 90% identical to the "original" or "parental" CDR sequence.

Accordingly, the substituted CDRs of the binding construct of the invention may have different degrees of identity to their original sequences, e.g., CDRL1 may have 80%, while CDRL3 may have 90% of homology. The same considerations apply to the framework regions and to the entire VH and VL regions.

[0234] A “variant CDR” is a CDR with a specific sequence homology, similarity, or identity to the parent CDR of the invention, and shares biological function with the parent CDR, including, but not limited to, at least 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% of the specificity and/or activity of the parent CDR. Generally, the amino acid homology, similarity, or identity between individual variant CDRs is at least 60% to the parent sequences depicted herein, and more typically with increasing homologies, similarities or identities of at least 65% or 70%, preferably at least 75% or 80%, more preferably at least 85%, 90%, 91%, 92%, 93%, 94%, and most preferably 95%, 96%, 97%, 98%, 99%, and almost 100%. The same applies to “variant VH” and “variant VL”. According to one embodiment, the sequence variations within a “variant VH” and/or a “variant VL” do not extend to the CDRs. The present invention is hence directed to a binding construct as defined herein, comprising VH and VL sequences having a certain sequence homology (see above) to the specific sequences as defined herein (the “parental” VH and VL), wherein the CDR sequences are 100% identical to the specific CDR sequences as defined herein (the “parental” CDRs).

[0235] Preferred substitutions (or replacements) are conservative substitutions. However, any substitution (including non-conservative substitutions or one or more from the “exemplary substitutions” listed in Table 6, below) is envisaged, as long as the binding construct retains its capacity to bind to MAGEB2, and/or provided its CDRs, FRs, VH and/or VL sequences have a degree of identity to the original or parental sequence of at least 60% or 65%, more preferably at least 70% or 75%, even more preferably at least 80% or 85%, and particularly preferably at least 90% or 95%.

[0236] A conservative replacement (also called a conservative mutation or a conservative substitution) is an amino acid replacement that changes a given amino acid to a different amino acid with similar biochemical properties (e.g. charge, hydrophobicity, size). Conservative replacements in proteins often have a smaller effect on protein function than non-conservative replacements. Conservative substitutions are shown in Table 6. Exemplary conservative substitutions are shown as “exemplary substitutions”. If such substitutions result in a change in biological activity, then more substantial changes, as further described herein in reference to amino acid classes, may be introduced and the products screened for a desired characteristic.

TABLE 6

Amino acid substitutions (aa = amino acid)		
Original aa	Conservative substitutions	Exemplary Substitutions
Ala (A)	Small aa	Gly, Ser, Thr
Arg (R)	Polar aa, in particular Lys	Lys, Gln, Asn
Asn (N)	Polar aa, in particular Asp	Asp, Gln, His, Lys, Arg

TABLE 6-continued

Amino acid substitutions (aa = amino acid)		
Original aa	Conservative substitutions	Exemplary Substitutions
Asp (D)	Glu or other polar aa, in particular Asn	Glu, Asn
Cys (C)	Small aa	Ser, Ala
Gln (Q)	Polar aa, in particular Glu	Glu, Asn
Glu (E)	Asp or other polar aa, in particular Gln	Asp, Gln
Gly (G)	Small aa, such as Ala	Ala
His (H)		Asn, Gln, Arg, Lys, Tyr
Ile (I)	Hydrophobic, in particular aliphatic aa	Ala, Val, Met, Leu, Phe
Leu (L)	Hydrophobic, in particular aliphatic aa	Norleucine, Ile, Ala, Val, Met
Lys (K)	Polar aa, in particular Arg	Arg, Gln, Asn
Met (M)	Hydrophobic, in particular aliphatic aa	Leu, Ala, Ile, Val, Phe
Phe (F)	Aromatic or hydrophobic aa, in particular Tyr	Tyr, Trp, Leu, Val, Ile, Ala
Pro (P)	Small aa	Ala
Ser (S)	Polar or small aa, in particular Thr	Thr
Thr (T)	Polar aa, in particular Ser	Ser
Trp (W)	Aromatic aa	Tyr, Phe
Tyr (Y)	Aromatic aa, in particular Phe	Phe, Trp, Thr, Ser
Val (V)	Hydrophobic, in particular aliphatic aa	Leu, Ile, Ala, Met, Phe

[0237] Substantial modifications in the biological properties of the binding construct of the present invention are accomplished by selecting substitutions that differ significantly in their effect on maintaining (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a sheet or helical conformation, (b) the charge or hydrophobicity of the molecule at the target site, or (c) the bulk of the side chain. Non-conservative substitutions will usually entail exchanging a member of one of the above defined amino acid classes (such as polar, neutral, acidic, basic, aliphatic, aromatic, small . . .) for another class. Any cysteine residue not involved in maintaining the proper conformation of the binding construct may be substituted, generally with serine, to improve the oxidative stability of the binding construct.

[0238] In addition to the above described substitutions, other substitutions within the CDRs that contribute to binding can be made. For example, the consensus sequences set forth in Table 25 herein provide such substitutions. In certain embodiments, deletions as guided by the provided consensus sequences may also be made. With the guidance provided herein by the consensus sequences set forth in Table 25, various amino acid substitutions, and functional equivalents thereof, or deletions, can be readily made by one of ordinary skill in the art.

[0239] Sequence identity, homology and/or similarity of amino acid sequences is determined by using standard techniques known in the art, including, but not limited to, the local sequence identity algorithm of Smith and Waterman, 1981, Adv. Appl. Math. 2:482, the sequence identity alignment algorithm of Needleman and Wunsch (J Mol Biol. 1970 March; 48(3):443-53), the search for similarity method of Pearson and Lipman (Proc Natl Acad Sci USA. 1988 April; 85(8):2444-8), computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Drive, Madison, Wis.), the Best Fit sequence program described by Devereux et al. (Nucleic

Acids Res. 1984 Jan. 11; 12(1 Pt 1):387-95), preferably using the default settings, or by inspection. It is envisaged that percent identity is calculated by FastDB based upon the following parameters: mismatch penalty of 1; gap penalty of 1; gap size penalty of 0.33; and joining penalty of 30. See also "Current Methods in Sequence Comparison and Analysis," Macromolecule Sequencing and Synthesis, Selected Methods and Applications, pp 127-149 (1988), Alan R. Liss, Inc.

[0240] An example of a useful algorithm is PILEUP. PILEUP creates a multiple sequence alignment from a group of related sequences using progressive, pairwise alignments. It can also plot a tree showing the clustering relationships used to create the alignment. PILEUP uses a simplification of the progressive alignment method of Feng and Doolittle (J Mol Evol. 1987; 25(4):351-60); the method is similar to that described by Higgins and Sharp (Comput Appl Biosci. 1989 April; 5(2):151-3). Useful PILEUP parameters include a default gap weight of 3.00, a default gap length weight of 0.10, and weighted end gaps. Another example of a useful algorithm is the BLAST algorithm, described in: Altschul et al. (J Mol Biol. 1990 Oct. 5; 215(3):403-10.); Altschul et al., (Nucleic Acids Res. 1997 Sep. 1; 25(17):3389-402); and Karlin and Altschul (Proc Natl Acad Sci USA. 1993 Jun. 15; 90(12):5873-7). A particularly useful BLAST program is the WU-Blast-2 program which was obtained from Altschul et al., (Methods Enzymol. 1996; 266:460-80). WU-Blast-2 uses several search parameters, most of which are set to the default values. The adjustable parameters are set with the following values: overlap span=1, overlap fraction=0.125, word threshold (T)=II. The HSP S and HSP S2 parameters are dynamic values and are established by the program itself depending upon the composition of the particular sequence and composition of the particular database against which the sequence of interest is being searched; however, the values may be adjusted to increase sensitivity.

[0241] An additional useful algorithm is gapped BLAST as reported by Altschul et al. (Nucleic Acids Res. 1997 Sep. 1; 25(17):3389-402). Gapped BLAST uses BLOSUM-62 substitution scores; threshold T parameter set to 9; the two-hit method to trigger ungapped extensions, charges gap lengths of k a cost of 10+k; Xu set to 16, and Xg set to 40 for database search stage and to 67 for the output stage of the algorithms. Gapped alignments are triggered by a score corresponding to about 22 bits.

Nucleotides Encoding the Binding Constructs

[0242] The invention provides a polynucleotide/nucleic acid molecule encoding a binding construct of the invention. Nucleic acid molecules are biopolymers composed of nucleotides. A polynucleotide is a biopolymer composed of 13 or more nucleotide monomers covalently bonded in a chain. DNA (such as cDNA) and RNA (such as mRNA) are examples of polynucleotides/nucleic acid molecules with distinct biological function. Nucleotides are organic molecules that serve as the monomers or subunits of nucleic acid molecules like DNA or RNA. The nucleic acid molecule or polynucleotide of the present invention can be double stranded or single stranded, linear or circular. It is envisaged that the nucleic acid molecule or polynucleotide is comprised in a vector. It is furthermore envisaged that such vector is comprised in a host cell. Said host cell is, e.g. after transformation or transfection with the vector or the polynucleotide/nucleic acid molecule of the invention, capable

of expressing the binding construct. For this purpose, the polynucleotide or nucleic acid molecule is operatively linked with control sequences.

[0243] The genetic code is the set of rules by which information encoded within genetic material (nucleic acids) is translated into proteins. Biological decoding in living cells is accomplished by the ribosome which links amino acids in an order specified by mRNA, using tRNA molecules to carry amino acids and to read the mRNA three nucleotides at a time. The code defines how sequences of these nucleotide triplets, called codons, specify which amino acid will be added next during protein synthesis. With some exceptions, a three-nucleotide codon in a nucleic acid sequence specifies a single amino acid. Because the vast majority of genes are encoded with exactly the same code, this particular code is often referred to as the canonical or standard genetic code.

[0244] Degeneracy of codons is the redundancy of the genetic code, exhibited as the multiplicity of three-base pair codon combinations that specify an amino acid. Degeneracy results because there are more codons than encodable amino acids. The codons encoding one amino acid may differ in any of their three positions; however, more often than not, this difference is in the second or third position. For instance, codons GAA and GAG both specify glutamic acid and exhibit redundancy; but, neither specifies any other amino acid and thus demonstrate no ambiguity. The genetic codes of different organisms can be biased towards using one of the several codons that encode the same amino acid over the others—that is, a greater frequency of one will be found than expected by chance. For example, leucine is specified by six distinct codons, some of which are rarely used. Codon usage tables detailing genomic codon usage frequencies for most organisms are available. Recombinant gene technologies commonly take advantage of this effect by implementing a technique termed codon optimization, in which those codons are used to design a polynucleotide which are preferred by the respective host cell (such as a cell of human hamster origin, an *Escherichia coli* cell, or a *Saccharomyces cerevisiae* cell), e.g. in order to increase protein expression. It is hence envisaged that the polynucleotides/nucleic acid molecules of the present disclosure are codon optimized. Nevertheless, the polynucleotide/nucleic acid molecule encoding a binding construct of the invention may be designed using any codon that encodes the desired amino acid.

[0245] In line herewith, the term "percent (%) nucleic acid sequence identity/homology/similarity" with respect to the nucleic acid sequence encoding the binding constructs identified herein is defined as the percentage of nucleotide residues in a candidate sequence that are identical with the nucleotide residues in the coding sequence of the binding construct. One method to align two sequences and thereby determine their homology uses the BLASTN module of WU-Blast2 set to the default parameters, with overlap span and overlap fraction set to 1 and 0.125, respectively. Generally, the nucleic acid sequence homology, similarity, or identity between the nucleotide sequences encoding individual variant CDRs and the nucleotide sequences depicted herein are at least 60%, and more typically with increasing homologies, similarities or identities of at least 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, and almost 100%. Again, the same applies to nucleic acid sequence encoding the "variant VH" and/or "variant VL".

[0246] In one embodiment, the percentage of identity to human germline of the binding constructs according to the invention, or of the first and second domain (binding domains) of these binding constructs, is 70% or 75%, more preferably 80% or 85%, even more preferably 90%, and most preferably 91%, 92%, 93%, 94%, 95% or even 96%. Identity to human antibody germline gene products is thought to be an important feature to reduce the risk of therapeutic proteins to elicit an immune response against the drug in the patient during treatment. Hwang W. Y. and Foote J. (Methods. 2005 May; 36(1):3-10) demonstrate that the reduction of non-human portions of drug binding constructs leads to a decrease of risk of inducing anti-drug antibodies in the patients during treatment. By comparing an exhaustive number of clinically evaluated antibody drugs and the respective immunogenicity data, the trend is shown that humanization of the variable regions of antibodies/binding constructs makes the protein less immunogenic (average 5.1% of patients) than antibodies/binding constructs carrying unaltered non-human variable regions (average 23.59% of patients). A higher degree of identity to human sequences is hence desirable for protein therapeutics based on variable regions and in the form of binding constructs. For the purpose of determining the germline identity, the V-regions of VL can be aligned with the amino acid sequences of human germline V segments and J segments (<http://www2.mrc-imb.cam.ac.uk/vbase/>) using Vector NTI software and the amino acid sequence calculated by dividing the identical amino acid residues by the total number of amino acid residues of the VL in percent. The same can be done for the VH segments (<http://www2.mrc-imb.cam.ac.uk/vbase/>) with the exception that the VH CDR3 may be excluded due to its high diversity and a lack of existing human germline VH CDR3 alignment partners. Recombinant techniques can then be used to increase sequence identity to human antibody germline genes. According to one embodiment, the polynucleotide/nucleic acid molecule of the present invention encoding the binding construct of the invention is in the form of one single molecule or in the form of two or more separate molecules. If the binding construct of the present invention is a single chain binding construct, the polynucleotide/nucleic acid molecule encoding such construct will most likely also be in the form of one single molecule.

[0247] The same applies for the vector comprising a polynucleotide/nucleic acid molecule of the present invention. If the binding construct of the present invention is a single chain binding construct, one vector may comprise the polynucleotide which encodes the binding construct in one single location (as one single open reading frame, ORF). One vector may also comprise two or more polynucleotides/nucleic acid molecules at separate locations (with individual ORFs), each one of them encoding a different component of the binding construct of the invention. It is envisaged that the vector comprising the polynucleotide/nucleic acid molecule of the present invention is in the form of one single vector or two or more separate vectors. In one embodiment, and for the purpose of expressing the binding construct in a host cell, the host cell of the invention should comprise the polynucleotide/nucleic acid molecule encoding the binding construct or the vector comprising such polynucleotide/nucleic acid molecule in their entirety, meaning that all components of the binding construct—whether encoded as one single molecule or in separate molecules/locations—

will assemble after translation and form together the biologically active binding construct of the invention.

[0248] The invention also provides a vector comprising a polynucleotide/nucleic acid molecule of the invention. A vector is a nucleic acid molecule used as a vehicle to transfer (foreign) genetic material into a cell, usually for the purpose of replication and/or expression. The term “vector” encompasses—but is not restricted to—plasmids, viruses, cosmids, and artificial chromosomes. Some vectors are designed specifically for cloning (cloning vectors), others for protein expression (expression vectors). So-called transcription vectors are mainly used to amplify their insert. The manipulation of DNA is normally conducted on *E. coli* vectors, which contain elements necessary for their maintenance in *E. coli*. However, vectors may also have elements that allow them to be maintained in another organism such as yeast, plant or mammalian cells, and these vectors are called shuttle vectors. Insertion of a vector into the target or host cell is usually called transformation for bacterial cells and transfection for eukaryotic cells, while insertion of a viral vector is often called transduction.

[0249] In general, engineered vectors comprise an origin of replication, a multicloning site and a selectable marker. The vector itself is generally a nucleotide sequence, commonly a DNA sequence, that comprises an insert (transgene) and a larger sequence that serves as the “backbone” of the vector. While the genetic code determines the polypeptide sequence for a given coding region, other genomic regions can influence when and where these polypeptides are produced. Modern vectors may therefore encompass additional features besides the transgene insert and a backbone: promoter, genetic marker, antibiotic resistance, reporter gene, targeting sequence, protein purification tag. Vectors called expression vectors (expression constructs) specifically are for the expression of the transgene in the target cell, and generally have control sequences.

[0250] The term “control sequences” refers to DNA sequences necessary for the expression of an operably linked coding sequence in a particular host organism. The control sequences that are suitable for prokaryotes, for example, include a promoter, optionally an operator sequence, and a ribosome binding site. Eukaryotic cells are known to utilize promoters, polyadenylation signals, a Kozak sequence and enhancers.

[0251] A nucleic acid is “operably linked” when it is placed into a functional relationship with another nucleic acid sequence. For example, DNA for a presequence or secretory leader is operably linked to DNA for a polypeptide if it is expressed as a preprotein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence; or a ribosome binding site is operably linked to a coding sequence if it is positioned so as to facilitate translation. Generally, “operably linked” means that the nucleotide sequences being linked are contiguous, and, in the case of a secretory leader, contiguous and in reading phase. However, enhancers do not have to be contiguous. Linking is accomplished by ligation at convenient restriction sites. If such sites do not exist, the synthetic oligonucleotide adaptors or linkers are used in accordance with conventional practice.

[0252] “Transfection” is the process of deliberately introducing nucleic acid molecules or polynucleotides (including vectors) into target cells. The term is mostly used for

non-viral methods in eukaryotic cells. Transduction is often used to describe virus-mediated transfer of nucleic acid molecules or polynucleotides. Transfection of animal cells typically involves opening transient pores or “holes” in the cell membrane, to allow the uptake of material. Transfection can be carried out using biological particles (such as viral transfection, also called viral transduction), chemical-based methods (such as using calcium phosphate, lipofection, Fugene, cationic polymers, nanoparticles) or physical treatment (such as electroporation, microinjection, gene gun, cell squeezing, magnetofection, hydrostatic pressure, impalefection, sonication, optical transfection, heat shock).

[0253] The term “transformation” is used to describe non-viral transfer of nucleic acid molecules or polynucleotides (including vectors) into bacteria, and also into non-animal eukaryotic cells, including plant cells. Transformation is hence the genetic alteration of a bacterial or non-animal eukaryotic cell resulting from the direct uptake through the cell membrane(s) from its surroundings and subsequent incorporation of exogenous genetic material (nucleic acid molecules). Transformation can be effected by artificial means. For transformation to happen, cells or bacteria must be in a state of competence, which might occur as a time-limited response to environmental conditions such as starvation and cell density, and can also be artificially induced.

[0254] Moreover, the invention provides a host cell transformed or transfected with the polynucleotide/nucleic acid molecule of the invention or with the vector of the invention.

[0255] As used herein, the terms “host cell” or “recipient cell” are intended to include any individual cell or cell culture that can be or has been recipient of vectors, exogenous nucleic acid molecules and/or polynucleotides encoding the binding construct of the present invention; and/or recipients of the binding construct itself. The introduction of the respective material into the cell is carried out by way of transformation, transfection and the like (vide supra). The term “host cell” is also intended to include progeny or potential progeny of a single cell. Because certain modifications may occur in succeeding generations due to either natural, accidental, or deliberate mutation or due to environmental influences, such progeny may not, in fact, be completely identical (in morphology or in genomic or total DNA complement) to the parent cell, but is still included within the scope of the term as used herein. Suitable host cells include prokaryotic or eukaryotic cells, and also include—but are not limited to—bacteria (such as *E. coli*), yeast cells, fungi cells, plant cells, and animal cells such as insect cells and mammalian cells, e.g., hamster, murine, rat, macaque or human.

[0256] In addition to prokaryotes, eukaryotic microbes such as filamentous fungi or yeast are suitable cloning or expression hosts for the binding construct of the invention. *Saccharomyces cerevisiae*, or common baker’s yeast, is the most commonly used among lower eukaryotic host microorganisms. However, a number of other genera, species, and strains are commonly available and useful herein, such as *Schizosaccharomyces pombe*, *Kluyveromyces* hosts such as *K. lactis*, *K. fragilis* (ATCC 12424), *K. bulgaricus* (ATCC 16045), *K. wickerhamii* (ATCC 24178), *K. waltii* (ATCC 56500), *K. drosophilae* (ATCC 36906), *K. thermotolerans*, and *K. marxianus*; *Yarrowia* (EP 402 226); *Pichia pastoris* (EP 183 070); *Candida*; *Trichoderma reesia* (EP 244 234); *Neurospora crassa*; *Schwanniomyces* such as

Schwanniomyces occidentalis; and filamentous fungi such as *Neurospora*, *Penicillium*, *Tolypocladium*, and *Aspergillus* hosts such as *A. nidulans* and *A. niger*.

[0257] Suitable host cells for the expression of a glycosylated binding construct are derived from multicellular organisms. Examples of invertebrate cells include plant and insect cells. Numerous baculoviral strains and variants and corresponding permissive insect host cells from hosts such as *Spodoptera frugiperda* (caterpillar), *Aedes aegypti* (mosquito), *Aedes albopictus* (mosquito), *Drosophila melanogaster* (fruit fly), and *Bombyx mori* have been identified. A variety of viral strains for transfection are publicly available, e.g., the L-1 variant of *Autographa californica* NPV and the Bm-5 strain of *Bombyx mori* NPV, and such viruses may be used as the virus herein according to the present invention, particularly for transfection of *Spodoptera frugiperda* cells.

[0258] Plant cell cultures of cotton, corn, potato, soybean, *petunia*, tomato, *Arabidopsis* and tobacco can also be used as hosts. Cloning and expression vectors useful in the production of proteins in plant cell culture are known to those of skill in the art. See e.g. Hiatt et al., *Nature* (1989) 342: 76-78, Owen et al. (1992) *Bio/Technology* 10: 790-794, Artsaenko et al. (1995) *The Plant J* 8: 745-750, and Fecker et al. (1996) *Plant Mol Biol* 32: 979-986.

[0259] However, interest has been greatest in vertebrate cells, and propagation of vertebrate cells in culture (cell culture) has become a routine procedure. Examples of useful mammalian host cell lines are monkey kidney CV1 line transformed by SV40 (such as COS-7, ATCC CRL 1651); human embryonic kidney line (such as 293 or 293 cells subcloned for growth in suspension culture, Graham et al., *J. Gen. Virol.* 36: 59 (1977)); baby hamster kidney cells (such as BHK, ATCC CCL 10); Chinese hamster ovary cells/-DHFR (such as CHO, Urlaub et al., *Proc. Natl. Acad. Sci. USA* 77: 4216 (1980)); mouse sertoli cells (such as TM4, Mather, *Biol. Reprod.* 23: 243-251 (1980)); monkey kidney cells (such as CV1 ATCC CCL 70); African green monkey kidney cells (such as VERO-76, ATCC CRL1587); human cervical carcinoma cells (such as HELA, ATCC CCL 2); canine kidney cells (such as MDCK, ATCC CCL 34); buffalo rat liver cells (such as BRL 3A, ATCC CRL 1442); human lung cells (such as W138, ATCC CCL 75); human liver cells (such as Hep G2, 1413 8065); mouse mammary tumor (such as MMT 060562, ATCC CCL-51); TRI cells (Mather et al., *Annals N. Y. Acad. Sci.* (1982) 383: 44-68); MRC 5 cells; FS4 cells; and a human hepatoma line (such as Hep G2).

Production of Binding Constructs

[0260] In a further embodiment, the invention provides a process for producing a binding construct of the invention, said process comprising culturing a host cell of the invention under conditions allowing the expression of the binding construct of the invention and recovering the produced binding construct from the culture.

[0261] As used herein, the term “culturing” refers to the in vitro maintenance, differentiation, growth, proliferation and/or propagation of cells under suitable conditions in a medium. Cells are grown and maintained in a cell growth medium at an appropriate temperature and gas mixture. Culture conditions vary widely for each cell type. Typical growth conditions are a temperature of about 37° C., a CO₂ concentration of about 5% and a humidity of about 95%. Recipes for growth media can vary e.g. in pH, concentration

of the carbon source (such as glucose), nature and concentration of growth factors, and the presence of other nutrients (such as amino acids or vitamins). The growth factors used to supplement media are often derived from the serum of animal blood, such as fetal bovine serum (FBS), bovine calf serum (FCS), equine serum, and porcine serum. Cells can be grown either in suspension or as adherent cultures. There are also cell lines that have been modified to be able to survive in suspension cultures so they can be grown to a higher density than adherent conditions would allow.

[0262] The term “expression” includes any step involved in the production of a binding construct of the invention including, but not limited to, transcription, post-transcriptional modification, translation, folding, post-translational modification, targeting to specific subcellular or extracellular locations, and secretion. The term “recovering” refers to a series of processes intended to isolate the binding construct from the cell culture. The “recovering” or “purification” process may separate the protein and non-protein parts of the cell culture, and finally separate the desired binding construct from all other polypeptides and proteins. Separation steps usually exploit differences in protein size, physicochemical properties, binding affinity and biological activity. Preparative purifications aim to produce a relatively large quantity of purified proteins for subsequent use, while analytical purification produces a relatively small amount of a protein for a variety of research or analytical purposes.

[0263] When using recombinant techniques, the binding construct can be produced intracellularly, in the periplasmic space, or directly secreted into the medium. If the binding construct is produced intracellularly, as a first step, the particulate debris, either host cells or lysed fragments, are removed, for example, by centrifugation or ultrafiltration. The binding construct of the invention may e.g. be produced in bacteria such as *E. coli*. After expression, the construct is isolated from the bacterial cell paste in a soluble fraction and can be purified e.g. via affinity chromatography and/or size exclusion. Final purification can be carried out in a manner similar to the process for purifying a binding construct expressed in mammalian cells and secreted into the medium. Carter et al. (Biotechnology (NY) 1992 February; 10(2): 163-7) describe a procedure for isolating antibodies which are secreted to the periplasmic space of *E. coli*.

[0264] Where the antibody is secreted into the medium, supernatants from such expression systems are generally first concentrated using a commercially available protein concentration filter, for example, an ultrafiltration unit.

[0265] The binding construct of the invention prepared from the host cells can be recovered or purified using, for example, hydroxylapatite chromatography, gel electrophoresis, dialysis, and affinity chromatography. Other techniques for protein purification such as fractionation on an ion-exchange column, mixed mode ion exchange, HIC, ethanol precipitation, size exclusion chromatography, reverse phase HPLC, chromatography on silica, chromatography on heparin sepharose, chromatography on an anion or cation exchange resin (such as a polyaspartic acid column), immunoaffinity (such as Protein A/G/L) chromatography, chromatofocusing, SDS-PAGE, ultracentrifugation, and ammonium sulfate precipitation are also available depending on the binding construct to be recovered.

[0266] A protease inhibitor may be included in any of the foregoing steps to inhibit proteolysis, and antibiotics may be included to prevent the growth of contaminants.

Formulations

[0267] Moreover, the invention provides a composition or formulation comprising a binding construct of the invention or a binding construct produced according to the process of the invention.

[0268] Compositions of the invention include, but are not limited to, liquid, frozen, and lyophilized compositions.

[0269] The compositions may comprise a pharmaceutically acceptable carrier. In general, as used herein, “pharmaceutically acceptable carrier” means any and all aqueous and non-aqueous solutions, sterile solutions, solvents, buffers, e.g. phosphate buffered saline (PBS) solutions, water, suspensions, emulsions, such as oil/water emulsions, various types of wetting agents, liposomes, dispersion media and coatings, which are compatible with pharmaceutical administration, in particular with parenteral administration. The use of such media and agents in pharmaceutical compositions is well known in the art, and the compositions comprising such carriers can be formulated by well-known conventional methods.

[0270] Certain embodiments provide compositions comprising the binding construct of the invention and further one or more excipients such as those illustratively described in this section and elsewhere herein. Excipients can be used in the invention for a wide variety of purposes, such as adjusting physical, chemical, or biological properties of formulations, such as adjustment of viscosity, and or processes of the invention to improve effectiveness and/or to stabilize such formulations and processes against degradation and spoilage e.g. due to stresses that occur during manufacturing, shipping, storage, pre-use preparation, administration, and thereafter. Excipients should in general be used in their lowest effective concentrations.

[0271] In certain embodiments, the composition may contain formulation materials for the purpose of modifying, maintaining or preserving certain characteristics of the composition such as the pH, osmolarity, viscosity, clarity, color, isotonicity, odor, sterility, stability, rate of dissolution or release, adsorption or penetration (see, Remington's Pharmaceutical Sciences, 18th Edition, 1990, Mack Publishing Company). In such embodiments, suitable formulation materials may include, but are not limited to, e.g., amino acids, antimicrobials such as antibacterial and antifungal agents, antioxidants, buffers, buffer systems and buffering agents which are used to maintain the composition at physiological pH or at a slightly lower pH, typically within a pH range of from about 5 to about 8 or 9, non-aqueous solvents, vegetable oils, and injectable organic esters, aqueous carriers including water, alcoholic/aqueous solutions, emulsions or suspensions, including saline and buffered media, biodegradable polymers such as polyesters, bulking agents, chelating agents, isotonic and absorption delaying agents, complexing agents, fillers, carbohydrates, (low molecular weight) proteins, polypeptides or proteinaceous carriers, preferably of human origin, coloring and flavouring agents, sulfur containing reducing agents, diluting agents, emulsifying agents, hydrophilic polymers, salt-forming counter-ions, preservatives, metal complexes, solvents and co-solvents, sugars and sugar alcohols, suspending agents, surfactants or wetting agents, stability enhancing agents, tonicity enhancing agents, parenteral delivery vehicles, or intravenous delivery vehicles.

[0272] Different constituents of the composition can have different effects, for example, and amino acid can act as a

buffer, a stabilizer and/or an antioxidant; mannitol can act as a bulking agent and/or a tonicity enhancing agent; sodium chloride can act as delivery vehicle and/or tonicity enhancing agent; etc. Salts may be used in accordance with certain embodiments of the invention, e.g. in order to adjust the ionic strength and/or the isotonicity of a composition or formulation and/or to improve the solubility and/or physical stability of a binding construct or other ingredient of a composition in accordance with the invention. Ions can stabilize the native state of proteins by binding to charged residues on the protein's surface and by shielding charged and polar groups in the protein and reducing the strength of their electrostatic interactions, attractive, and repulsive interactions. Ions also can stabilize the denatured state of a protein by binding to, in particular, the denatured peptide linkages ($-\text{CONH}$) of the protein. Furthermore, ionic interaction with charged and polar groups in a protein also can reduce intermolecular electrostatic interactions and, thereby, prevent or reduce protein aggregation and insolubility.

[0273] Ionic species differ significantly in their effects on proteins. A number of categorical rankings of ions and their effects on proteins have been developed that can be used in formulating pharmaceutical compositions in accordance with the invention. One example is the Hofmeister series, which ranks ionic and polar non-ionic solutes by their effect on the conformational stability of proteins in solution. Stabilizing solutes are referred to as "kosmotropic". Destabilizing solutes are referred to as "chaotropic". Kosmotropes are commonly used at high concentrations to precipitate proteins from solution ("salting-out"). Chaotropes are commonly used to denature and/or to solubilize proteins ("salting-in"). The relative effectiveness of ions to "salt-in" and "salt-out" defines their position in the Hofmeister series.

[0274] Free amino acids can be used in formulations or compositions comprising the binding construct of the invention in accordance with various embodiments of the invention as bulking agents, stabilizers, and antioxidants, as well as for other standard uses. Certain amino acids can be used for stabilizing proteins in a formulation, others are useful during lyophilization to ensure correct cake structure and properties of the active ingredient. Some amino acids may be useful to inhibit protein aggregation in both liquid and lyophilized formulations, and others are useful as antioxidants.

[0275] Polyols are kosmotropic and are useful as stabilizing agents in both liquid and lyophilized formulations to protect proteins from physical and chemical degradation processes. Polyols are also useful for adjusting the tonicity of formulations and for protecting against freeze-thaw stresses during transport or the preparation of bulks during the manufacturing process. Polyols can also serve as cryoprotectants in the context of the present invention.

[0276] Certain embodiments of the formulation or composition comprising the binding construct of the invention can comprise surfactants. Proteins may be susceptible to adsorption on surfaces and to denaturation and resulting aggregation at air-liquid, solid-liquid, and liquid-liquid interfaces. These deleterious interactions generally scale inversely with protein concentration and are typically exacerbated by physical agitation, such as that generated during the shipping and handling of a product. Surfactants are routinely used to prevent, minimize, or reduce surface adsorption.

[0277] Surfactants also are commonly used to control protein conformational stability. The use of surfactants in this regard is protein specific, since one specific surfactant will typically stabilize some proteins and destabilize others.

[0278] Certain embodiments of the formulation or composition comprising the binding construct of the invention can comprise one or more antioxidants. To some extent deleterious oxidation of proteins can be prevented in pharmaceutical formulations by maintaining proper levels of ambient oxygen and temperature and by avoiding exposure to light. Antioxidant excipients can also be used to prevent oxidative degradation of proteins. It is envisaged that antioxidants for use in therapeutic protein formulations in accordance with the present invention can be water-soluble and maintain their activity throughout the shelf life of the product (the composition comprising the binding construct). Antioxidants can also damage proteins and should hence—among other things—be selected in a way to eliminate or sufficiently reduce the possibility of antioxidants damaging the binding construct or other proteins in the formulation.

[0279] Formulations in accordance with the invention may include metal ions that are protein co-factors and that are necessary to form protein coordination complexes, such as zinc necessary to form certain insulin suspensions. Metal ions also can inhibit some processes that degrade proteins. However, metal ions also catalyze physical and chemical processes that degrade proteins. Magnesium ions (10-120 mM) can be used to inhibit isomerization of aspartic acid to isoaspartic acid. Ca^{+2} ions (up to 100 mM) can increase the stability of human deoxyribonuclease. Mg^{+2} , Mn^{+2} , and Zn^{+2} , however, can destabilize rhDNase. Similarly, Ca^{+2} and Sr^{+2} can stabilize Factor VIII, it can be destabilized by Mg^{+2} , Mn^{+2} and Zn^{+2} , Cu^{+2} and Fe^{+2} , and its aggregation can be increased by Al^{+3} ions.

[0280] Certain embodiments of the formulation or composition comprising the binding construct of the invention can comprise one or more preservatives. Preservatives are necessary for example when developing multi-dose parenteral formulations that involve more than one extraction from the same container. Their primary function is to inhibit microbial growth and ensure product sterility throughout the shelf-life or term of use of the drug product. Although preservatives have a long history of use with small-molecule parenterals, the development of protein formulations that include preservatives can be challenging. Preservatives very often have a destabilizing effect (aggregation) on proteins, and this has become a major factor in limiting their use in multi-dose protein formulations. Several aspects need to be considered during the formulation and development of preserved dosage forms. The effective preservative concentration in the product must be optimized. This requires testing a given preservative in the dosage form with concentration ranges that confer anti-microbial effectiveness without compromising protein stability.

[0281] Development of liquid formulations containing preservatives are more challenging than lyophilized formulations. Freeze-dried products can be lyophilized without the preservative and reconstituted with a preservative containing diluent at the time of use. This shortens the time during which a preservative is in contact with the binding construct, significantly minimizing the associated stability risks. With liquid formulations, preservative effectiveness and stability should be maintained over the entire product shelf-life. An important point to note is that preservative effectiveness

should be demonstrated in the final formulation containing the active drug and all excipient components. Once the pharmaceutical composition has been formulated, it may be stored in sterile vials as a solution, suspension, gel, emulsion, solid, crystal, or as a dehydrated or lyophilized powder. Such formulations may be stored either in a ready-to-use form or in a form (e.g., lyophilized) that is reconstituted prior to administration.

Kits

[0282] In further embodiments, the invention provides a kit comprising a binding construct that binds to MAGEB2 and is useful for detection of MAGEB2 in a tumor sample, wherein the binding construct comprises any of the binding constructs disclosed herein. In one embodiment, the binding construct that is useful for detection of MAGEB2 is a monoclonal antibody that binds to MAGEB2. In further embodiments, the kit further comprises a therapeutic that is effective for treating cancer that expresses MAGEB2. In certain embodiments, the kit can optionally provide a package insert comprising instructions for detecting MAGEB2 and administration of a cancer therapeutic.

[0283] In further embodiments, the kit further comprises binding constructs (e.g., antibodies) that are useful to detect different biomarkers other than MAGEB2, e.g., further comprising a MAGEA4 or a MAGEA8 binding construct.

[0284] As used herein, the singular forms “a”, “an”, and “the” include plural references unless the context clearly indicates otherwise. Thus, for example, reference to “a reagent” includes one or more of such different reagents and reference to “the method” includes reference to equivalent steps and methods known to those of ordinary skill in the art that could be modified or substituted for the methods described herein.

[0285] Unless otherwise indicated, the term “at least” preceding a series of elements is to be understood to refer to every element in the series. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the present invention.

[0286] The term “and/or” wherever used herein includes the meaning of “and”, “or” and “all or any other combination of the elements connected by said term”.

[0287] The term “about” or “approximately” as used herein means within $\pm 20\%$, preferably within $\pm 15\%$, more preferably within $\pm 10\%$, and most preferably within $\pm 5\%$ of a given value or range. It also includes the concrete value, e.g., “about 50” includes the value “50”.

[0288] Throughout this specification and the claims, unless the context requires otherwise, the word “comprise”, and variations such as “comprises” and “comprising”, will be understood to imply the inclusion of a stated integer or step or group of integers or steps but not the exclusion of any other integer or step or group of integer or step. When used herein the term “comprising” can be substituted with the term “containing” or “including” or sometimes when used herein with the term “having”.

[0289] When used herein “consisting of” excludes any element, step, or ingredient not specified in the claim element. When used herein, “consisting essentially of” does not exclude materials or steps that do not materially affect the basic and novel characteristics of the claim.

[0290] In each instance herein, any of the terms “comprising”, “consisting essentially of” and “consisting of” may be replaced with either of the other two terms.

[0291] It should be understood that the above description and the below examples provide exemplary arrangements, but the present invention is not limited to the particular methodologies, techniques, protocols, material, reagents, substances, etc., described herein and as such can vary. The terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention, which is defined solely by the claims. Aspects of the invention are provided in the independent claims. Some optional features of the invention are provided in the dependent claims.

[0292] All publications and patents cited throughout the text of this specification (including all patents, patent applications, scientific publications, manufacturer’s specifications, instructions, etc.), whether supra or infra, are hereby incorporated by reference in their entirety. Nothing herein is to be construed as an admission that the invention is not entitled to antedate such disclosure by virtue of prior invention. To the extent the material incorporated by reference contradicts or is inconsistent with this specification, the specification will supersede any such material.

[0293] A better understanding of the present invention and of its advantages will be obtained from the following examples, offered for illustrative purposes only. The examples are not intended and should not be construed as to limit the scope of the present invention in any way.

EXAMPLES

Example 1: MAGEB2 Immunogen Design

[0294] Protein sequences of 35 *H. sapiens* MAGE family members were downloaded from UniProtKB (www.uniprot.org): MAGEA1 (P43355), MAGEA2 (P43356), MAGEA3 (P43357), MAGEA4 (P43358), MAGEA5 (P43359), MAGEA6 (P43360), MAGEA8 (P43361), MAGEA9 (P43362), MAGEA10 (P43363), MAGEA11 (P43364), MAGEA12 (P43365), MAGEB1 (P43366), MAGEB2 (015479), MAGEB3 (015479), MAGEB4 (015481), MAGEB5 (Q9BZ81), MAGEB6 (Q8N7X4), MAGEB10 (Q96LZ2), MAGEB16 (A2A368), MAGEB17 (A8MXT2), MAGEB18 (Q96M61), MAGEC1 (060732), MAGEC2 (Q9UBF10), MAGEC3 (Q8TD91), MAGED1 (Q9Y5V3), MAGED2 (Q9UNF1), MAGED4 (Q96JG8), MAGEE1 (Q9HC15), MAGEE2 (Q8TD90), MAGEH1 (Q9H213), MAGEF1 (Q9HAY2), MAGEL2 (Q9UI55), NSE3 (Q96MG7), TROP (Q12816), and NECD (Q99608). Clades were determined using the Neighbor-Joining Tree Builder algorithm in Geneious v10.2.

[0295] Shannon entropy, a quantitative metric describing the diversity of amino acids observed at a given position in an alignment, was used to select MAGEB2 peptide immunogens having low amino acid conservation across the entire MAGE family and MAGE-B clade (B1, B2, B3, B4, B5, B17). MAGE family or MAGE-B clade members were aligned using MUSCLE (R. C. Edgar. Nucleic Acids Res. 2004; 32(5): 1792-1797). The Shannon entropy was calculated at each position in the alignment using $H_s(j) = -\sum_i f_{ij} \log_2 f_{ij}$, where $HS(j)$ is the Shannon entropy at position j and f_{ij} is the frequency of AA i at position j . Shannon entropy was mapped onto the structure of the MAGE homology domain (MHD) of MAGEA4 (PDB ID 2wa0) or a Rosetta

homology model of the MHD of MAGEB2 using a custom script that replaces the PDB b-factor of each atom with the normalized Shannon entropy value for its corresponding residue and then visualizing in PyMOL v1.8 (The PyMOL Molecular Graphics System, Version 1.8 Schrödinger, LLC) using color by b-factor.

[0296] A custom script was used to deterministically rank regions of length n (n=5-40 amino acids) in the alignment from highest to lowest average Shannon entropy (lowest to highest conservation) using a sliding window approach. For the entire MAGE family, regions of length 10-30 with lowest conservation across the family cluster to three distinct areas: two areas N-terminal to the MHD and one area within the helical N-terminus of the MHD. Analysis of the MAGE-B clade revealed a clade-specific area of low conservation located on a loop at the interface of two globular lobes of the MHD. MAGEB2 peptide immunogens were selected from the following three identified regions of low conservation: the region N-terminal to the MHD, the helical N-terminus of the MHD, and the MDH interface loop region. The peptide ranges within these regions were refined based on average Shannon entropy score, visual inspection of sequence alignments, consideration of structural features such as solvent accessibility and conservation of secondary structure as an isolated peptide as predicted by PSIPRED (Jones D T. (1999) Protein secondary structure prediction based on position-specific scoring matrices. J. Mol. Biol. 292: 195-202), and whether a sequence blast against the non-reductant *H. sapiens* proteome revealed hits with high homology in non-MAGE proteins. Regions of MAGEB2 selected as peptides immunogens were MAGEB2 a.a.43-76: SSVSGGAASSSPAAGIPQEPQRAPTTAAAAAAGV (N-terminal region peptide) (SEQ ID NO: 2); MAGEB2 a.a.95-125: SSSQASTSTKSPSEDPLTRKSGSLVQFLLYK (MHD N-terminal helix peptide) (SEQ ID NO: 3); and MAGEB2 a.a.185-200: DLTDEESLLSSWDFPR (MHD middle loop peptide) (SEQ ID NO: 562).

[0297] Peptides from MAGE homologs with >50% pairwise identity to the MAGEB2 immunogens were selected for experimental counter-screening. Additional MAGEB2 homolog peptides were selected for counter-screening. For MAGEB2 a.a.43-76 (N-terminal region peptide), selected

counter screening peptides were derived from MAGE-B1, B4, B17, A4, A1, A5, A12, A6, A3, A2, A11, and A8. For MAGE B2 a.a.95-125 (MHD N-terminal helix peptide), selected counter screening peptides were derived from MAGE-B4, B5, B17, C3, A11, A8, A9, A3, A10, F1, A4, A5, A12, C2, A2, E1, and C1. For MAGEB2 a.a.185-200 (MHD middle loop peptide), a single counter screening peptide derived from MAGEB1 was selected.

Example 2: Generation and Selection of Antibodies

Immunizations

[0298] Rabbits were immunized according to standard protocols, and splenocytes were isolated and frozen from immune animals.

Preparation of Peptides for Antigen-Specific Sorting

[0299] 27 biotin-labelled MAGE counter-screening peptides (Table 7) were pooled together in equal molar concentration to create a working cocktail. The cocktail was complexed to neutravidin (Fisher: PI31000) at a molar ratio of 4:1 (neutravidin:peptide) in 2% FBS FACS buffer (calcium and magnesium free PBS with 2% v/v FBS). The mixture was left to incubate at 4° C. for 15 minutes. Each of the 3 biotin-labelled MAGEB2 target peptides (N-Terminal Region Peptide, Middle Loop Region Peptide, N-Terminal Helix Peptide, see Table 8) was complexed to a different fluorescently-tagged streptavidin using the same 4:1 (streptavidin:peptide) molar ratio in 2% FBS FACS buffer and then left to incubate at 4° C. for 15 minutes. MAGEB2 N-Terminal Peptide was complexed to BV421 streptavidin (BD: 563259), MAGEB2 Middle Loop Region Peptide was complexed to BV605 streptavidin (BD: 563855), and MAGEB2 N-Terminal Helix Peptide was complexed to Alexa Fluor 647 streptavidin (Jackson: 016-600-084).

[0300] Unbound streptavidin or neutravidin was quenched by adding excess amount of D-biotin (Fisher: BP232-1) at molar ratio of 600:1 (D-biotin:streptavidin or neutravidin), the peptides were then left to incubate for another 15 minutes at 4°C. The 3 MAGEB2 target peptides were pooled together just before use.

TABLE 7

27 MAGE counter-screening peptides. MAGE Counter-Screening Peptides		
Item	Peptide Name	Peptide Sequence
		SEQ ID NO:
1	A1_CTermHelix	{Biotin-NH} SYVKVLEYVIKVSARVRFFFPSSLREAA {COOH}
2	A1_NTerminalPeptide	{Biotin-NH} SSPLVLGTL EEVPTAGSTDPQPQSGASAFPTTINFT {COOH}
3	A2_NTermHelix	{Biotin-NH} PDLESEFQA AISRKMVELVHFLLLK {COOH}
4	A2_NTerminalPeptide	{Biotin-NH} SSTLVEVT LG EVPAA DSPSPHSPQGASSFSTTINYT {COOH}
5	A3_NTermHelix	{Biotin-NH} STFPDLESEFQA ALSRKVAELVHFLLLK {COOH}

TABLE 7-continued

27 MAGE counter-screening peptides. MAGE Counter-Screening Peptides			
Item	Peptide Name	Peptide Sequence	SEQ ID NO:
6	A3_NTerminalPeptide	{Biotin-NH} SSTLVEVTLGEVPAAESPDPPQSPQGASSLPTTMNY {COOH}	575
7	A5_NTermHelix	{Biotin-NH} STSPDPESVFRAALSKKVADLIHFLLLK {COOH}	576
8	A5_NTerminalPeptide	{Biotin-NH} SSPLVP GTLGEVPAAGSPGPLKSPQGASAIPTAIDFT {COOH}	577
9	A9_CTermHelix (C19S)	{Biotin-NH} SYEKVINYLVMNLNAREPISYPSLYEEV {COOH}	578
10	A9_NTermHelix	{Biotin-NH} SSSVDPAQLEFMFQEALKLKVAELVHFLLLK {COOH}	579
11	A10_NTermHelix	{Biotin-NH} STLQVLPDSESLPRSEIDEKVTDLVQFLLFK {COOH}	580
12	A11_NTermHelix	{Biotin-NH} STSPDLIDPESFSQDILHDKIIDLVHLLLRK {COOH}	581
13	A11_NTerminalPeptide	{Biotin-NH} SSTLNVGTLEELPAAESPPSPQSPQEEFSPTAMDAI {COOH}	582
14	A12_NTermHelix	{Biotin-NH} STFPDLETSPQVALSRKMAELVHFLLLK {COOH}	583
15	A12_NTerminalPeptide	{Biotin-NH} SSTLVEVTLREVPAAESPPSPHSPQGASTLPTTINYT {COOH}	584
16	B1_MiddleLoopRegion	{Biotin-NH} NLTNDGNLSNDWDFPR {COOH}	585
17	B1_NTerminalPeptide	{Biotin-NH} SSPVLGDTPTSSPAAGIPQKPQGAPPTTTAAAV {COOH}	586
18	B4_NTermHelix	{Biotin-NH} SSSQASTSTERSLKDSLTRKTKMLVQFLLYK {COOH}	587
19	B4_NTerminalPeptide	{Biotin-NH} SSVL RDTASSSLAFGIPQEPQREPPTTSAAAA {COOH}	588
20	B5_NtermHelix (C12S)	{Biotin-NH} SSEVSPSTESSSNFINIKVGLLEQFLLYK {COOH}	589
21	B17_NTermHelix	{Biotin-NH} NSFHGPSSESTGRDLLNTKTGELVQFLLNK {COOH}	590
22	B17_NTerminalPeptide (C5S)	{Biotin-NH} SSPASQSPQSPFNAGIPQESQRASYPSSPASAV {COOH}	591
23	C1_NTermHelix	{Biotin-NH} TDSESLIESEPLFTYTLDEKVDELARFLLLK {COOH}	592
24	C2_NTermHelix (C3S)	{Biotin-NH} GTSQGLPDSESSFTYTLDEKVAELVEFLLLK {COOH}	593

TABLE 7-continued

27 MAGE counter-screening peptides. MAGE Counter-Screening Peptides		
Item Peptide Name	Peptide Sequence	SEQ ID NO:
25 C3_NTermHelix	{Biotin-NH} HALPESESLPRYALDEKVAELVQFLLLK {COOH}	594
26 E1_NTermHelix	{Biotin-NH} NTSRVAITLKPQDPMEQNVAELLQFLLVK {COOH}	595
27 F1_NTermHelix	{Biotin-NH} ALAAKALARRRAYRRLNRTVAELVQFLLVK {COOH}	596

TABLE 8

3 MAGEB2 Target Peptides MAGEB2 Target Peptides		
Item Peptide Name	Peptide Sequence	SEQ ID NO
1 B2_MiddleLoopRegion	{Biotin-NH} DLTDEESLLSSWDFPR {COOH}	597
2 B2_NTermHelix	{Biotin-NH} SSSQASTSTKSPSEDPLTRKSGSLVQFLLYK {COOH}	598
3 B2_NTerminalPeptide	{Biotin-NH} SSVSGGAASSSPAAGIPQEPQRAPTAAAAAGV {COOH}	599

Preparation of Rabbit Splenocytes for Antigen-Specific Sorting

[0301] Rabbit splenocytes were thawed into 10% ICM (Immune Cell Culture Medium) and dead cells were removed using the Miltenyi Biotec MACS Dead Cell Removal Kit (Miltenyi: 130-090-101). The cells were then washed in 10 mL of cold 2% FBS FACS buffer. After a 4-minute centrifugation at 400G, the supernatant was removed using a serological pipette and the cocktail of neutravidin-complexed MAGE counter-screening peptides was added. The cells were left to incubate at 4° C. for 30 minutes, at which point the 3 streptavidin-complexed MAGEB2 target peptides along with 5 µg of FITC-conjugated goat anti-rabbit IgG Fc antibody (Serotec: STAR121F) were added to stain IgG-expressing, antigen-specific cells. The final concentrations of the peptides in the 1 mL reaction volume were: 120 nM for the cocktail of counter-screening peptides, and 16.5 nM for each target peptide. The cells were incubated for another 30 minutes at 4° C., then washed in 10 mL of cold 2% FBS FACS buffer and centrifuged again. After removing the supernatant, the cells were resuspended in 2 mL of 10% ICM and then put through a 40 µm cell strainer to remove any clumps. 10 µL of 7-AAD viability staining solution (Fisher: 00-6993-50) was added to the cells 10 minutes prior to sorting.

Sorting and Culturing of Antigen-Specific Cells

[0302] Cells were sorted on a BD FACSAria III equipped with 405 nm laser (detection: 450/50 nm, 610/20 nm), 488 nm laser (detection: 530/30 nm, 695/40 nm), and 640 nm

laser (detection: 730/45 nm) using a 100 micron nozzle at 20 Psi sheath pressure. Dead cells were excluded by gating on population not fluorescent in 7-AAD, and MAGEB2 target peptide binders were sorted by drawing 3 gates using the following selection criteria: FITC+/BV421+, FITC+/BV605+, FITC+/Alexa Fluor 647+(see FIG. 1). An initial sort was performed using Yield precision mode to enrich target populations from ~0.1% to ~50%, this was followed by a second round of sorting using Purity precision mode to single cell sort MAGEB2 target peptide binders into 384-well tissue culture plates pre-filled with 10% ICM supplemented with 4% rabbit T-cell supernatant, 2 µL/mL of anti-rabbit IgG microbeads (Miltenyi: 130-048-602), and gamma-ray-irradiated EL4 cells at a density of 437,500 cells/mL. The plates were stored in tissue culture incubator (37° C., 5% CO₂), and outgrowth was assessed 7 days later by visual scoring using an inverted microscope. After 8 days, supernatants were collected from the tissue culture plates and the cells were lysed in Buffer RLT (Qiagen: 79216) and frozen at -80° C.

Initial Detection of MAGEB2 Antibodies

[0303] B-cell culture supernatants were screened for the presence of positive secreting antibodies using a rabbit IgG capture ELISA. Goat anti rabbit IgG (Fc) (Jackson, Cat. No. 111-005-046) was coated in 1× phosphate buffered saline (PBS) overnight at 4° C. onto Corning 3702 384 well polystyrene plates. Plates were washed 4 cycles using a Titertek plate washer and blocked with 1×PBS/1% milk. Then the plates were aspirated and B-cell culture supernatants were diluted at 1:5 final in 1×PBS/1% milk and

incubated for 1 hour at room temperature. Plates were washed 4 cycles and then horseradish peroxidase (HRP) conjugated goat anti human IgG (Fc) (Jackson, Cat. No. 111-035-046) made up in 1×PBS/1% milk was added and incubated for 1 hour at room temperature. Plates were washed 4 cycles, then 1 step TMB (Neogen cat. No. 319177) was added for 30 minutes at room temperature and subsequently quenched with 1N hydrochloric acid. Plates were read on the Multiscan Ascent plate reader at OD 450 nm and a total of 129 rabbit antibodies were positive with the breakdown of hits per region represented in Table 9 below.

TABLE 9

Binding region peptide	Total # Positive
MAGEB2 middle loop region peptide	50
MAGEB2 N-term helix peptide	35
MAGEB2 N-term peptide	45

Identification of MAGEB2 Specific Antibodies

[0304] MAGEB2 antibodies were tested for binding using biotinylated MAGEB2 peptides coated on xMAP LumAvidin Microspheres or Luminex beads (Luminex Cat No. L100-LXXX-01). These avidin coupled microparticles are color coded into different spectrally distinct regions, with unique fluorescent dye combinations allowing for non-covalent binding of biotinylated proteins. To validate the rabbit IgG capture ELISA results in the first bead-based FACS assay the panel of 129 rabbit antibodies was tested against the same 3 peptides that the rabbits were immunized with: MAGEB2 middle loop region peptide, MAGEB2 N-terminal helix peptide and MAGEB2 N-terminal region peptide. The three biotinylated peptides were coated using three different LumAvidin beads in FACS buffer [1×PBS+2% fetal bovine serum (Hyclone Cat No. SH30396.03)] for 30 minutes at room temperature in the dark. The beads were then washed twice in FACS buffer by centrifugation for 2 min at 3500 RPM. The coated beads were resuspended and pooled together (3-plex) and added to the B-cell culture supernatants at a 1:2 final dilution onto Corning 3897 96 well V-bottom polystyrene plates and incubated for 1 hour at room temperature in the dark. Another 2 washes in FACS buffer were done before the addition of Alexa 488 conjugated goat anti rabbit IgG (Fc) (Jackson, Cat No. 111-545-046) made up in FACS buffer was added and incubated for 15 minutes at room temperature in the dark. A final wash was done before plates were read on the iQue FACS machine with Intellicyte autosampler to obtain flow cytometry measurements according to the manufacturer's recommendations; data analysis was done on Intellicyt ForeCyt® Enterprise Client Edition 6.2 (R3). A total of 82 wells were identified as MAGEB2 positive with the breakdown of hits per region as presented in Table 10 below.

TABLE 10

Binding region	Total # Positive
MAGEB2 middle loop region peptide	21
MAGEB2 N-term helix peptide	24
MAGEB2 N-term region peptide	37

[0305] In the second bead-based FACS assay the panel of 129 rabbit antibodies was tested against 9 other peptides of

similar sequence to determine MAGEB2 specificity: MAGEB1 middle loop region peptide, MAGEB4 N-term helix peptide, MAGEB5 N-term helix peptide, MAGE17 N-term helix peptide, MAGEF1 N-term helix peptide, MAGEE1 N-term helix peptide; MAGEB1 N-term peptide, MAGEB4 N-term region peptide and MAGEB17 N-term region peptide. The nine biotinylated peptides were coated using nine different LumAvidin beads in FACS buffer [1×PBS+2% fetal bovine serum (FBS)] for 30 minutes at room temperature in the dark. The beads were then washed twice in FACS buffer by centrifugation for 2 min at 3500 RPM. The coated beads were resuspended and pooled together (9-plex) and added to the rabbit B-cell supernatants at a 1:2 final dilution onto Corning 3897 96 well V-bottom polystyrene plates and incubated for 1 hour at room temperature in the dark. Another 2 washes in FACS buffer were done before the addition of Alexa 488 conjugated goat anti rabbit IgG (Fc) (Jackson, Cat No. 111-545-046) made up in FACS buffer was added and incubated for 15 minutes at room temperature in the dark. A final wash was done before plates were read on the iQue FACS machine with Intellicyte autosampler to obtain flow cytometry measurements according to the manufacturer's recommendations; data analysis was done on Intellicyt ForeCyt® Enterprise Client Edition 6.2 (R3). A total of 43 wells were identified as MAGEB2 specific with the breakdown of hits per region as represented in Table 11 below. All 43 wells underwent heavy chain and light chain sequencing.

TABLE 11

Binding region	Total # Positive
MAGEB2 middle loop region peptide	12
MAGEB2 N-term helix peptide	19
MAGEB2 N-term region peptide	12

Identification of MAGEB2 Highest Affinity Antibodies

[0306] To determine which of the 43 MAGEB2 specific antibodies to take forward to cloning, a limited antigen (LA) kinetics FACS assay done on LumAvidin beads using full length MAGEB2 protein, to rank the panel and select the highest affinity antibodies.

[0307] huMAGEB2(1-319) in pET23 was transformed into BL21(DE)-star pLysS strain (Invitrogen). Single colonies were used to inoculate 2 mL Luria Broth supplemented with carbenicillin (50 µg/mL) and chloramphenicol (34 µg/mL) and grown shaking at 37° C. for several hours until turbid. Two hundred microliters of culture were used to inoculate 400 mL of Luria Broth (Teknova) supplemented with selection antibiotics and antifoam 204 (Sigma). These cultures were grown shaking at 24° C. overnight. Turbid cultures were used to seed 1-liter Terrific Broth (Teknova) supplemented with 1% (w/v) glucose, 2 mM MgSO₄, 17 µg/L carbenicillin, 12 µg/L chloramphenicol and antifoam 204 (in 4 L baffled shake flasks) to an OD₆₀₀=0.2. These cultures were grown shaking at 37° C. until an OD₆₀₀=1.0, then were shifted to 16° C. and continued to grow to OD₆₀₀=2.0. At this point, protein expression was induced, with 100 µM IPTG, for 16 hrs. The cells were pelleted by centrifugation and stored at -80° C. until processing.

[0308] The MAGEB2 pellets were resuspended in Lysis Buffer (100 mM TRIS pH 8.0, 500 mM NaCl, 15 mM bME, 5 mM Benzamidine, 0.01 mg/mL lysozyme, 0.05 mg/mL

Dnase1, 5% (v/v) glycerol, and Halt Protease Inhibitor Tablets (Pierce)) using a Polytron PT2100 homogenizer. The homogenized cells were lysed using Microfluidizer M-110Y (Microfluidics) in three passes. The lysate was clarified by centrifugation at 13,000 rpm in a Beckman Centrifuge in a JLA-16.250 rotor. The clarified lysate was filtered through 0.22 μ m filter before being applied to HisTrap Excel column (Cytiva). The resin was washed with 25 mM TRIS pH 8.5, 0.5M NaCl, 10 mM BME, 5% glycerol, 10 mM imidazole and MAGEB2 was eluted with 25 mM TRIS pH 8.5, 0.5M NaCl, 10 mM BME, 5% glycerol, 500 mM imidazole.

[0309] The affinity captured MAGEB2 was dialyzed against 20 mM TRIS pH 8.5, 0.5M NaCl, 5% (v/v) glycerol, 1 mM TCEP and diluted 25-fold into 20 mM TRIS pH 8.5, 5% (v/v) glycerol, 1 mM TCEP before ion-exchange chromatography using Q-sepharose HP (Cytiva). Bound MAGEB2 was eluted from the column using a 20 mM to 500 mM NaCl gradient over 20 CV. Peak fractions were analyzed by SDS-PAGE and pooled based on the absence of low-molecular weight bands. The n-terminal 6xHis tag SEQ ID NO: 569) was removed from MAGEB2 using caspase 3 protease at a 1:100 (w/w) ratio, after the addition of 1 mM EDTA and 0.1% CHAPS. The digest proceeded overnight at 4° C. The digest was dialyzed against 20 mM TRIS pH 8.5, 0.25M NaCl, 1 mM TCEP to remove CHAPS and EDTA before ion-exchange chromatography.

[0310] Dialyzed MAGEB2 was diluted 10-fold into 20 mM TRIS pH 8.5, 1 mM TCEP and applied to Q-sepharose HP resin (Cytiva). Bound MAGEB2 was eluted from the column using a 20 mM to 500 mM NaCl gradient over 20 CV. Peak fractions were analyzed by SDS-PAGE and pooled based on the absence of high-molecular weight bands. This pool was biotinylated using recombinant BirA with 50 mM Bicine pH 8.3, 10 mM Mg-acetate, 10 mM ATP and 50 μ M d-biotin. The reaction proceeded overnight at 4° C. and 5 μ g of protein was analyzed by Mass spec (ask Dylan for the about instrument and methods) to monitor biotin addition to MAGEB2.

[0311] The biotinylation reaction was dialyzed against 20 mM TRIS pH 8.5, 150 mM NaCl, 1 mM TCEP to remove excess ATP and biotin. After dialysis the sample was diluted 10-fold, in 20 mM TRIS pH 8.5, 1 mM TCEP, and applied to Q-sepharose HP (Cytiva). Bound MAGEB2 was eluted off the column using a 20 mM to 500 mM NaCl gradient over 20 CV. Peak fractions were analyzed by SDS-PAGE and pooled based on the presence of a 37 Kd band. The pooled fractions were concentrated using 10 Kd MWCO spin filters (Sartorius). The concentrated MAGEB2 was dialyzed against 30 mM HEPES pH 8.2, 0.15M NaCl, 1 mM TCEP, filtered through 0.22 μ m Posidyne syringe filter (Pall) and stored at -80° C.

[0312] During purification, mass spec analysis was used to confirm the identity of intact MAGEB2, caspase-3 cleaved MAGEB2, and biotinylated MAGEB2. The homogeneity of purification intermediates and final protein lot was measured by sizing HPLC analysis on Agilent 1200 LC system using a YARA 3 μ m SEC 2000 column (Phenomenex). Protein concentration was determined by A280 absorbance using Spectramax spectrophotometer (Molecular Devices).

[0313] Biotinylated MAGEB2 full length protein was titrated 1:2 across 6 different limited antigen concentrations and then coated onto six different LumAvidin beads in FACS buffer [1xPBS+2% fetal bovine serum (Hyclone Cat No. SH30396.03)] for 30 minutes at room temperature in the

dark. The beads were then washed twice in FACS buffer by centrifugation for 2 min at 3500 RPM. The coated beads were resuspended and pooled together (6-plex) and added to the rabbit B-cell supernatants at a 1:2 final dilution onto Corning 3897 96 well V-bottom polystyrene plates and incubated overnight at room temperature in the dark. The next day 2 washes were done in FACS buffer before the addition of Alexa 488 conjugated goat anti rabbit IgG (Fc) (Jackson, Cat No. 111-545-046) made up in FACS buffer was added and incubated for 15 minutes at room temperature in the dark. A final wash was done before plates were read on the iQue FACS machine with Intellicyte autosampler to obtain flow cytometry measurements according to the manufacturer's recommendations; data analysis was done on Intellicyt ForeCyt® Enterprise Client Edition 6.2 (R3). The LA data was aligned with the sequence data and a total of 24 highest affinity, sequence diverse antibodies were identified, with the breakdown of hits per region as shown in Table 12 below and in FIG. 2 herein.

TABLE 12

Binding region	Total # Positive
MAGEB2 middle loop region peptide	6
MAGEB2 N-term helix peptide	16
MAGEB2 N-term region peptide	2

Heavy and Light Chain Sequencing of Antibodies

[0314] Messenger RNA (mRNA) was purified from wells containing the antibody-producing cells using the mRNA catcher plus kit (Invitrogen). Purified RNA was used to amplify the antibody heavy and light chain variable region (V) genes using cDNA synthesis via reverse transcription, followed by a polymerase chain reaction (RT-PCR). The rabbit antibody gamma heavy chain was obtained using the Qiagen One Step Reverse Transcriptase PCR kit (Qiagen). This method was used to generate the first strand cDNA from the RNA template and then to amplify the variable region of the gamma heavy chain using multiplex PCR. The 5' gamma chain-specific primer annealed to the signal sequence of the antibody heavy chain, while the 3' primer annealed to a region of the gamma constant domain. The rabbit kappa light chain was obtained using the One Step Reverse Transcriptase PCR kit (Qiagen). This method was used to generate the first strand cDNA from the RNA template and then to amplify the variable region of the kappa light chain using multiplex PCR. The 5' kappa light chain-specific primer annealed to the signal sequence of the antibody light chain while the 3' primer annealed to a region of the kappa constant domain.

[0315] The PCR product was sequenced and amino acid sequences were deduced from the corresponding nucleic acid sequences bioinformatically. Two additional, independent RT-PCR amplification and sequencing cycles were completed for each sample in order to confirm that any mutations observed were not a consequence of the PCR. The derived amino acid sequences were then analyzed to determine the germline sequence origin of the antibodies and to identify deviations from the germline sequence. A comparison of each of the heavy and light chain sequences to their original germline sequences are indicated in sequence Table 4 herein. The amino acid sequences were used to group the clones by similarity.

[0316] The variable region kappa light chain DNA sequences for the selected antibodies were made as a synthetic genes. These synthetic genes were cloned with the rabbit kappa constant region in to a pTTS based expression vector. The variable region heavy chain DNA sequences for the selected antibodies were made as synthetic genes. These synthetic genes were cloned with a rabbit IgG constant region in to a pTTS based expression vector. After cloning, these antibodies were expressed in HEK2936E cells transfected with PEI.

Example 3: Immunohistochemical Assays

Objectives

[0317] To assess the specificity of twenty-four rabbit anti-human MAGEB2 antibodies to human MAGEB2, and potential cross reactivity for other MAGE-B and MAGE-A family members by immunohistochemical (IHC)staining.

Overview Summary

[0318] Twenty-four rabbit-anti human MAGEB2 supernatants were assessed for binding to human MAGEB2 on transfected human MAGEB2+ control cells by IHC. After initial screening assays, small batch purified antibody aliquots from four MAGEB2 clones (4G17, 1J15, 1C3, 1I14) with desired attributes were assessed on transfected human MAGE-A+ and MAGE-B+ family member control cells, normal human testes tissues, normal human tissue TMAs, and thirteen human cancer cell lines with known endogenous MAGEB2 RNA expression level by IHC. MAGEB2 antibody clones 1I14 and 1C3 displayed properties that were desired for IHC.

Materials and Methods

Control Cells and Tissues

[0319] Negative control cells included parental Chinese hamster ovary (CHO) cells, CHO cells stably transfected with mCherry Vector Control (minus Myc/DDK tags), CHO cells stably transfected with mCherry-human DCAF4L2 (plus Myc/DDK tags). The cross-reactivity control cells included CHO cells transfected with mCherry Vector Control (plus Myc/DDK tags) with one of the human MAGE-B or MAGE-A family members (MAGEA1, MAGEA2, MAGEA3, MAGEA4, MAGEA5, MAGEA6, MAGEA8, MAGEA9, MAGEA11, MAGEA12, MAGEB1, MAGEB3, MAGEB4, MAGEB5, MAGEB6, MAGEB10, MAGEB16, MAGEB17, MAGEB18). Positive control cell lines included CHO cells transfected with mCherry Vector (plus Myc/DDK tags) with human MAGE-B2 or CHO cells transfected with mCherry Vector (open reading frame (ORF); minus Myc/DDK tags) with human MAGEB2, and thirteen human cancer cell lines with known endogenous RNA expression level of MAGEB2 (U266B1, T98G, KMM-1, U20S, CFPAC-1, SCaBER, UACC-257, CFPAC-1, VMRC-LCD, NCI-H1703, UM-UC-3, NCI-H1395, NCIH82).

[0320] Cell pellets were fixed in 10% neutral buffered formalin, suspended in Histogel™ containing agarose/glycerine (cat #HG-4000-12, Thermo Fisher Scientific, Waltham, MA), and processed to paraffin blocks. Two FFPE normal human testes tissues, and one FFPE normal human liver tissue were used as positive and negative controls, respectively, obtained from Human Tissue Science Center (HTSC).

TABLE 13

Cells and Tissues		
Species	HTSC Inventory Code (Block ID)	Descriptions
Hamster	694001	Amgen in house TMA contain CHO, CHO- mCherry, CHO- mCherry- human DCAF4L2+, CHO- MAGE-A1+, MAGE-A2+, MAGE-A3+, MAGE-A4+ (variant 2), MAGE-A5+, MAGE-A6+, and MAGE-A8+ cell line
Hamster	697782	Amgen in house TMA contain CHO, CHO- mCherry, CHO- mCherry- human DCAF4L2+, CHO- MAGE-A4+ (variant 4), MAGE-A9+, MAGE-A10+, MAGE-A11+ (variant 1), MAGE-A11+ (variant 2), MAGE-A12+ cell lines
Hamster	697548	Amgen in house TMA contain CHO, CHO- mCherry, CHO- mCherry- human DCAF4L2+, CHO- MAGE-B1+ (variant 1), MAGE-B1+ (variant 3), MAGE-B2+ (ORF = minus Myc/DDK tags), MAGE-B2 MAGE-B3+, MAGE-B5+ (variant 1) cell lines
Hamster	697639	Amgen in house TMA contain CHO, CHO- mCherry, CHO- mCherry- human DCAF4L2+, CHO- MAGE-B5+ (variant 2), MAGE-B6+, MAGE-B10+, MAGE-B16+, MAGE-17+, MAGE-18+ cell lines
Hamster	694686	Amgen in house TMA contain CHO, CHO- mCherry, CHO- mCherry- human DCAF4L2+, CHO- MAGE-B5+ (variant 2), MAGE-B6+, MAGE-B10+, MAGE-B16+, MAGE-17+, MAGE-18+ cell lines
Human	627446	Amgen in house TMA contain U266B1, KMM-1, NCI-H1703, SCaBER, T98G, UACC-257 and CFPAC-1 cell
Human	685358	Amgen in house TMA contain NCI-H1703, NCI-H1395, VMRC-LCD, SCaBER cell

TABLE 13-continued

Cells and Tissues		
Species	HTSC Inventory Code (Block ID)	Descriptions
Human	685359	Amgen in house TMA contain U2OS, UM-UC-3, NCF-H82, CFPAC-1 cell
Human	14749	Normal Testis
Human	8497	Normal Testis
Human	356137	Normal Liver

Tissues Microarray

[0321] FFPE normal human TMA slides (normal human multiple organ tissue microarray) were obtained from US BioMax (FDA999x, Crabbs Branch Way, Derwood, MD; <https://www.biomax.us/FDA999x>).

Antibodies

[0322] Twenty-four human MAGEB2 clone supernatants (monoclonal rabbit IgG1 antibody) were assessed for binding to human MAGEB2 and potential cross reactivity for other MAGE-A and MAGE-B family members on control cells by IHC staining. Four of the twenty-four human MAGEB2 clones (4G17, 1J15, 1C3, 1I14) were the most promising candidates for further assessment of MAGEB2 specificity and sensitivity by IHC staining. Small batch purified antibody aliquots from the four MAGEB2 clones (4G17, 1J15, 1C3, 1I14) were assessed on MAGE-A and MAGE-B family cross-reactivity control cells, control testes tissues, and normal human tissue TMAs by IHC. Rabbit Monoclonal IgG1 isotype control (EPR25A) was purchased from Abcam (cat #ab172730, Lot #GR3235749-11, Cambridge, United Kingdom; <https://www.abcam.com/rabbit-igg-monoclonal-epr25a-isotype-control-ab172730.html>).

Immunohistochemistry Assay

[0323] FFPE samples were sectioned at 4 μ m and mounted on positive charged glass slides. After mounting, paraffin sections were air-dried at room temperature. FFPE TMA samples were received as freshly sectioned and mounted on glass slides. The fully automated IHC assay was performed on a Discovery Ultra Stainer (Ventana Medical Systems, Tucson, AZ) using Ventana Anti-Rabbit HQ (cat #760-4815), Ventana Anti-HQ HRP (cat #760-4820), and Ventana Antibody Diluent with Casein (cat #760-219) (Ventana Medical Systems, Tucson, AZ). The slides were baked on the Discovery Ultra Stainer at 60° C. for 8 minutes, deparaffinized with Ventana Discovery wash (cat #950-510) for 3 cycles (8 minutes each) at 69° C., and underwent cell conditioning (target retrieval) at 95° C. for 16 minutes (for cell arrays) or 32 minutes (tissues or TMAs) in CC1 buffer (cat #950-500). Following this, slides were incubated with Background Sniper (cat #BS966M, Biocare Medical, Pacheco, CA) in a Ventana Option Prep kit (cat #771-751) for 28 minutes, incubated with CM inhibitor (a part of the Ventana DAB kit cat #760-159) for 12 minutes, and incubated for 1 h at RT with MAGE-B2 antibody diluted with Ventana diluent with Casein in a Ventana Prep kit (cat #770-001) (working concentration at 0.1 μ g/ml for cell pellet, or 1-5 μ g/ml for tissues) or with a matched concentration of isotype control Rabbit IgG1 antibody. Following

this, slides were incubated with the Ventana Anti-Rabbit HQ (cat #760-4815) for 28 minutes at RT, Ventana Anti-HQ HRP (cat #760-4820) for 28 minutes at RT, and signals detected with Ventana DAB chromogen kit (cat #760-159) for 5 minutes at RT. Slides were counterstained with Ventana Hematoxylin II (cat #5266726001) for 8 minutes and Ventana Bluing reagent (cat #526676900) for 4 minutes. Slides were coverslipped and digitized using an Aperio AT2 scanner (Leica Biosystems Inc, Buffalo Grove, IL).

TABLE 14

Assay steps.		
Steps on Ventana Ultra	Vendor/Catalog	Time
Baking	Ventana/NA	60° C., 8 minutes
Deparaffinization	Ventana/950-510	69° C., 3 cycles (8 minutes each)
Antigen Retrieval	Ventana/950-500	CC1, 95° C., 16 minutes (for FFPE cell pellets) or 32 minutes (for FFPE tissues)
Background Sniper	Biocare Medical/BS966M	RT, 28 minutes
Peroxidase inhibitor (CM)	Ventana/760-159	RT, 12 minutes
In house MAGEB2 antibody at 0.1 (cells) or 1-5 μ g/ml (tissues) in Ventana diluent with Casein	Ventana/760-219	RT, 60 minutes
Anti-Rabbit HQ	Ventana/760-4815	RT, 28 minutes
Anti-HQ HRP	Ventana/760-4820	RT, 28 minutes
DAB+ Chromogen	Ventana/760-159	RT, 5 minutes
Hematoxylin II	Ventana/5266726001	RT, 8 minutes
Bluing reagent	Ventana/526676900	RT, 4 minutes

Results

[0324] Twenty-four rabbit-anti human MAGEB2 antibody supernatants were assessed for binding to MAGEB2 and potential cross reactivity for other MAGE-A and MAGE-B family members by IHC. Five of twenty-four human MAGEB2 antibody supernatants (5H20, 1J13, 1J14, 714, 7J6) did not detect MAGEB2 by IHC on human MAGEB2+ transfected cells. Nineteen of the twenty-four antibody supernatants detected MAGEB2 by IHC on human MAGEB2+ transfected cells and were assessed on other human MAGE-A+ and MAGE-B+ family member cross-reactivity control cells. Four of nineteen supernatants (4G17, 1J15, 1C3, 1I14) showed the strongest IHC signal on human MAGEB2+ transfected control cells, and the lowest/or no cross reactivity on other MAGE-A and MAGE-B family members and selected as candidates for further assessment.

[0325] Note that the twenty-four MAGEB2 antibody supernatants were not purified antibodies. A false positive cross reactivity result with one of the discarded clones may be due to non-specific staining by the supernatants. Small batch purified antibody aliquots from four MAGEB2 clones (4G17, 1J15, 1C3, 1I14) were re-tested on transfected human MAGEA+ and MAGE-B+ family member control cells, and tested on control testes tissues, normal tissue TMAs, and thirteen human cancer cell lines with known endogenous RNA expression level of MAGEB2 by IHC.

[0326] There was intense membranous/cytoplasmic staining in the transfected human MAGEB2+ cells with the four anti-MAGEB2 antibodies (4G17, 1J15, 1C3, 1I14) (FIG. 3), no cross reactivity on other human MAGE-A+ and MAGE-B+ family member cross reactivity control cells (FIG. 4), (Table 15). There was intense MAGEB2 IHC staining in testis, characterized by cytoplasmic to membrane staining of spermatogonia cells. Nuclear staining in a subset of testis germ cells was observed with 4G17, 1J15, and 1C3 clones, and immunostaining in Leydig cells was observed with 1J15, and 4G17 clones. Leydig staining is likely non-specific (FIG. 5). Nuclear staining was considered likely specific because it was observed only in testis tissues, and nuclear localization is thought to play a role in enhancement of E2F transcriptional activity, cell proliferation, and resistance to ribotoxin stress (Peché et al 2015).

[0327] The four anti-MAGEB2 antibodies (4G17, 1J15, 1C3, 1I14) were assessed on normal tissue TMA. MAGEB2 specific immunostaining was observed only in testis cores. Immunostaining observed with 4G17, 1J15, and 1C3 clones in other tissues such as liver, adrenal gland, and intestines were presumed nonspecific. The MAGEB2 antibody (1I14) showed the lowest non-specific/background immunostaining when compared to Immunoglobulin G (IgG) isotype control (Tables 16, 17, 18, 19). Antibody clone 1I14 displays

properties that indicate it will be an excellent IHC reagent for detecting MAGEB2 (FIG. 6). Furthermore, down titration of MAGEB2 (1C3) antibody completely mitigated the nonspecific background immunostaining that was observed in liver tissues, while still retaining MAGEB2 specific immunostaining in testis (FIG. 7).

[0328] Anti-MAGEB2 antibodies 4G17, 1J15, 1C3, 1I14 detected varying levels of MAGEB2 expression in the thirteen human cancer cell lines. MAGEB2 IHC expression frequency was high (~90%) in U266B1 cells, moderate (~70%) in T98G, KMM-1, U20S cells, and undetectable in CFPAC-1, UACC-257, CFPAC-1, VMRC-LCD, NCI-H1703, UM-UC-3, NCI-H1395, NCI-H82 cell lines. MAGEB2 IHC expression frequency in *SCaBER* cells were inconsistent with the four MAGEB2 clones (4G17, 1J15, 1C3, 1I14) and maybe due to different binding intensity of each antibody or unoptimized IHC assay conditions (Table 20).

CONCLUSIONS

[0329] Twenty-four rabbit anti-human MAGEB2 antibodies were assessed for specificity to MAGEB2, and potential cross reactivity for other MAGE family members by IHC staining on FFPE tissues. Nineteen of the twenty-four detected MAGEB2 by IHC on control MAGEB2+ cells. Four of the nineteen antibodies (4G17, 1J15, 1C3, 1I14) were tested thoroughly and showed no cross reactivity to other MAGE-A or MAGE-B family members, strong IHC signal on control testis tissues, and detected the MAGEB2 IHC signal in thirteen human cancer cell lines with known endogenous RNA expression level of MAGEB2. Three of the four MAGEB2 antibodies (4G17, 1J15, 1C3) showed some non-specific background IHC staining on liver and adrenal gland in normal human tissue TMA. Data is summarized in Tables 15-20 herein.

SEQUENCES
Human MAGEB2 Amino Acid Sequence (SEQ ID NO: 1): >sp 015479 MAGEB2_HUMAN Melanoma-associated antigen B2 OS = Homo sapiens OX = 9606 GN = MAGEB2 PE = 1 SV = 3 MPRGQKSKLRAREKRRKARDETRGLNVPQVTEAEHEEAPCCSSSVSGGAASSSPAAGIPQ EPQRAPTTAAAGVSSSTKSKGAKSHQGEKNASSQASTSTKSPSEDLTRKSGSLVQ FLLYKYKIKKSVTKGEMLKIVGKRFRHFPEILKKASEGLSVVFGLELNKVNPNNGHTYTF IDKVDLTDEESLLSSWDFPRRKLMLPLLGVI FLNGNSATEEEIWEFLNMLGVYDGEHVS FGEFWKLITKDLVQEKYLEYKQVPSSDPPRFQFLWGPRAAETSKMKVLEFLAKVNGTTP CAFPTHYEALKDEEKAGV
Full amino acid sequences for three exemplary monoclonal antibodies light chains and heavy chains follow: >1C3_LC (SEQ ID NO: 555) MDMRVPAQLLGLLLWLRGARCQSLVQPEGSLTLTCTAFGVTLNYYICWVRQAPGKGLEWVGCIDNAN SGVPSRFKSGSGGTQFTLTISDLECAADATYYCQSYDDSRSSNFFYAFGGGTEVVVKGDPVAPTIVLLFPSSDEVATGT VTIVCVANKYFPDVTVTWEVDGTTQTGTGIENSKTPQNSADCTYNLSSLTLTSTQYNSHKEYTCKVTQGTTSVVQSF KNC
>1C3_HC (SEQ ID NO: 556) MDMRVPAQLLGLLLWLRGARCQSLVQPEGSLTLTCTAFGVTLNYYICWVRQAPGKGLEWVGCIDNAN GRYYASWAKGRFTISKTSSTTGTLMQTSLTADTATYFCARSLATPLWGPGLTVTVSSGQPKAPSVFPLAPCCGDT SSTVTGLGCLVKGYLPEPVTVTWNSGTLTNGVTRFPPSVRQSSGLYSLSVSVVTSSSQPVTCNVAPATNTKVDKTVAPS TCSKPTCPPELGGPSVFIFPPKPKDTLMISRTPEVTCVVVDVSDDDPEVQFTWYINNEQVRTARPPLEQQFNSTIR VVSTLPIAHQDWRGKEFKKVKVHNAKALPAPIEKTISKARGQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISVE WEKNGKAEDNYKTPAVLDSGYSFLYSKLSVPTSEWQRGDVFTCSVMHEALHNHYTQKSISRSPGK
>1I14_LC (SEQ ID NO: 557) MDMRVPAQLLGLLLWLRGARCQSLVQPEGSLTLTCTAFGVTLNYYICWVRQAPGKGLEWVGCIDNAN ASTLASGVPSRFASGSGTEFTLTISDLECAADATYYCQCTYYVSSYQNDFFGGGTEVVVKGDPVAPTIVLLFPSSDEVA TGTVTIVCVANKYFPDVTVTWEVDGTTQTGTGIENSKTPQNSADCTYNLSSLTLTSTQYNSHKEYTCKVTQGTTSVVQ SFSRKNK

-continued

SEQUENCES	
>1I14_HC (SEQ ID NO: 558) MDMRVPAQLLGLLLLWLRGARCQSVESGGRLVTPGTPLTLTCTISGFSLSSYAMSWVRQAPGKGLEWIGSIGGGGS AVYASWAKGRFTISKSTSTVDLRITSPTTEDTAMYFCGRGFYSIDLWGPGLTVTVSSGQPKAPSVFPLAPCCGDTPSST VTLGCLVKGYLPEPVTVTWNSGTLTNGVRTFPQSSGLYSLSSVSVTSSSQPVTCNVHPATNTKVDKTVAPSTC SKPTCPPPELLGGPSVFIFFPKPKDTLMIISRTPEVTCVVVDVSQDDPEVQFTWYINNEQVRTARPPLREQQFNSTIRVV STLPIAHQDWLRGKEFKCKVHNKALPAPIEKTISKARGQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISVEWE KNGKAEDNYKTTPAVLDSGSIYFLYSKLSVPTSEWQRGDVFTCSVMHEALHNHYTQKSISRSPGK	
>1H17_LC (SEQ ID NO: 559) MDMRVPAQLLGLLLLWLRGARCQEQLVESGGGLVKPGASLTLTCKASGFSFSSGQLMCWVRQAPGKGLEWIAICIGS STLASGVPSRFKSGSGTQFTLTISDVQCDDAATYYCAGGYSSSDTFAFGGGTEVVVKGDPVAPTLLFPSSSDEVAT GTVTIVCVANKYFPDVTVTWEVDGTTQTGTIENSKTPQNSADCTYNLSSLTTLTSTQYNSHKEYTCVKTQGTSSVVQS FSRKNC	
>1H17_HC (SEQ ID NO: 560) MDMRVPAQLLGLLLLWLRGARCQEQLVESGGGLVKPGASLTLTCKASGFSFSSGQLMCWVRQAPGKGLEWIAICIGS GSNAISTFYASWAQGRFTISKSSSTVTLQLTSLTAADTATYFCARVGSDDYGDSDVDFDPWGPGLTVTVSSGQPKAPS VFPLAPCCGDTPSSTVTLGCLVKGYLPEPVTVTWNSGTLTNGVRTFPQSSGLYSLSSVSVTSSSQPVTCNVHPA TNTKVDKTVAPSTCSKPTCPPPELLGGPSVFIFFPKPKDTLMIISRTPEVTCVVVDVSQDDPEVQFTWYINNEQVRTAR PPLREQQFNSTIRVVSTLPIAHQDWLRGKEFKCKVHNKALPAPIEKTISKARGQPLEPKVYTMGPPREELSSRSVSLTC MINGFYPSDISVEWEKNGKAEDNYKTTPAVLDSGSIYFLYSKLSVPTSEWQRGDVFTCSVMHEALHNHYTQKSISR PGK	

[0330] Additional exemplary sequences of the present invention are provided in Tables 21-25 herein.

TABLE 15						
MAGEB2 Immunohistochemistry Staining in MAGE-A/B control cells						
Antibody ID	HTSC Inventory Code (Block ID)	CHO cells	MAGE- B2 cells	Other	Other	MAGE-A family members
				MAGE-B family members	MAGE-A family members	
4G17	694001, 697782, 697548, 697639, 694686	negative	positive	negative	negative	
1J15	694001, 697782, 697548, 697639, 694686	negative	positive	negative	negative	

TABLE 15-continued						
MAGEB2 Immunohistochemistry Staining in MAGE-A/B control cells						
Antibody ID	HTSC Inventory Code (Block ID)	CHO cells	MAGE- B2 cells	Other MAGE-B family members	Other MAGE-A family members	
1C3	694001, 697782, 697548, 697639, 694686	negative	positive	negative	negative	
1i14	694001, 697782, 697548, 697639, 694686	negative	positive	negative	negative	
IgG	694001, 697782, 697548, 697639, 694686	negative	negative	negative	negative	

TABLE 16						
MAGE-B2 (1i14) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)						
Tissue identification				IHC Isotype Control		
Position	No.	Organ	Tissue			
			ID	Negative	Positive	Comment
A1	1	Cerebrum	Nct05N012	X		
A2	2	Cerebrum	Nct05N016	X		
A3	3	Cerebrum	Nct08N046	X		
A4	4	Cerebellum	Ncb05N008	X		
A5	5	Cerebellum	Ncb15N146	X		
A6	6	Cerebellum	Ncb05N011	X		
A7	7	Adrenal gland	Eag06N019	X		Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)

TABLE 16-continued

MAGE-B2 (1i14) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)					
A8	8	Adrenal gland	Eag06N010	X	Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)
A9	9	Adrenal gland	Eag15N145	X	Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)
A10	10	Ovary	Fov07N034	X	
A11	11	Ovary	Fov13N015	X	
A12	12	Ovary	Fov13N023	X	
B1	13	Pancreas	Dpa15N145	TI	Tissue autolyzed
B2	14	Pancreas	Dpa06N023	X	
B3	15	Pancreas	Dpa11N012	X	
B4	16	Lymph node	Ily06N015	X	
B5	17	Lymph node	Ily08N047	X	Non-specific staining in exfoliated cells
B6	18	Lymph node	Ily08N014	X	
B7	19	Hypophysis	Ept14N001	X	
B8	20	Hypophysis	Ept13N019	X	
B9	21	Hypophysis	Ept13N020	X	
B10	22	Testis	Mtt09N048	X	
B11	23	Testis	Mtt08N001	X	
B12	24	Testis	Mtt08N009	X	
C1	25	Thyroid gland	Etg07N002	X	Non-specific staining in colloid
C2	26	Thyroid gland	Etg08N047	X	Non-specific staining in colloid
C3	27	Thyroid gland	Etg06N003	X	Non-specific staining in colloid
C4	28	Breast	Fmg08N034	X	
C5	29	Breast	Fmg50N034	X	
C6	30	Breast	Fmg07N040	X	
C7	31	Spleen	Isp15N145	X	
C8	32	Spleen	Isp15N146	X	
C9	33	Spleen	Isp11N001	X	
C10	34	Tonsil	Rts13N011	TI	Only adipose tissue present
C11	35	Tonsil	Rts50N001	X	
C12	36	Tonsil	Rts08N042	X	Minimal lymphoid tissue present
D1	37	Thymus gland	Ith06N024	X	
D2	38	Thymus gland	Ith06N002	X	
D3	39	Thymus gland	Ith15N145	X	
D4	40	Bone marrow	Ibm06N024	X	Non-specific staining in exfoliated cells
D5	41	Bone marrow	Ibm11N015	X	Non-specific staining in exfoliated cells
D6	42	Bone marrow	Ibm12N001	X	Non-specific staining in exfoliated cells

TABLE 16-continued

MAGE-B2 (1i14) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)					
D7	43	Lung	Rln06N004	X	
D8	44	Lung	Rln06N027	X	
D9	45	Lung	Rln06N023	X	
D10	46	Heart	Cht11N009	X	
D11	47	Heart	Cht05N005	X	
D12	48	Heart	Cht11N001	X	
E1	49	Esophagus	Des06N027	X	
E2	50	Esophagus	Des17N004	X	Minimal epithelial tissue
E3	51	Esophagus	Des15N117	X	
E4	52	Stomach	Dst07N016	X	Hazy non- specific staining in gastric gland cells
E5	53	Stomach	Dst15N003	X	Hazy non- specific staining in gastric gland cells
E6	54	Stomach	Dst06N017	X	Hazy non- specific staining in gastric gland cells
E7	55	Small intestine	Din07N016	X	Weak non- specific mucosal staining
E8	56	Small intestine	Din08N014	X	
E9	57	Small intestine	Din08N007	X	
E10	58	Colon	Dco15N083	X	
E11	59	Colon	Dco06N022	X	
E12	60	Colon	Dco15N082	TI	No mucosal tissue
F1	61	Liver	Dlv05N015		X
F2	62	Liver	Dlv15N146	X	Non-specific particulate staining in hepatocytes
F3	63	Liver	Dlv15N145	X	
F4	64	Salivary gland	Doc11N007	X	
F5	65	Salivary gland	Doc06N023	X	
F6	66	Salivary gland	Doc06N022	X	
F7	67	Kidney	Ukn08N047	X	
F8	68	Kidney	Ukn08N035	X	
F9	69	Kidney	Ukn06N004	X	
F10	70	Prostate	Mpr08N004	X	
F11	71	Prostate	Mpr08N005	X	
F12	72	Prostate	Mpr11N013	X	
G1	73	Uterus	Fur17N003	X	
G2	74	Uterus	Fur50N029	X	
G3	75	Uterus	Fur07N040	X	
G4	76	Cervix	Fdu50N020	X	
G5	77	Cervix	Fdu17N003	X	
G6	78	Cervix	Fdu50N019	X	
G7	79	Bladder	Ubd08N007	X	
G8	80	Bladder	Ubd08N001	X	
G9	81	Bladder	Ubd08N009	X	
G10	82	Skeletal muscle	Smn15N008	X	
G11	83	Skeletal muscle	Smn15N009	X	
G12	84	Skeletal muscle	Smn05N014	X	
H1	85	Skin	Kin15N007	X	
H2	86	Skin	Kin07N025	X	
H3	87	Skin	Kin15N009	X	

TABLE 16-continued

MAGE-B2 (1i14) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)					
H4	88	Nerve	Scf10N003	X	
H5	89	Nerve	Scf08N010	X	
H6	90	Nerve	Scf11N005	X	
H7	91	Artery	Sbv07N029	X	
H8	92	Pericardium	Apr15N145	X	
H9	93	Pericardium	Apr15N146	X	
H10	94	Eye	Vey130002	TI	Only pigmented retina present
H11	95	Eye	Vey140001	TI	Only pigmented retina present
H12	96	Eye	Vey070007	X	Intense pigmentation
11	97	Larynx	Rla13N011	X	
12	98	Larynx	Rla13N020	X	
13	99	Larynx	Rla07N010	X	
Target IHC					
Negative	Cell Positive type		Intensity (0-3+)	Localization	Comment
X					
X					
X					Very weak non-specific staining in the vessels
X					
X					Very weak non-specific staining in the vessels
X					Very weak non-specific staining in the vessels
X					Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis) as observed in isotype
X					
X					
X					
X					
TI					Tissue autolyzed
X					
X					
X					
X					
X					Non-specific staining in exfoliated cells as observed in isotype
X					
X					
X					
X					
X	X	Spermatogonia cells	3	Cytoplasmic/ membrane	
X	X	Spermatogonia cells	3	Cytoplasmic/ membrane	
X	X	Spermatogonia cells	3	Cytoplasmic/ membrane	
					Non-specific staining as observed in

[illegible]

TABLE 16-continued

MAGE-B2 (1i14) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
X		
Tl		Only pigmented retina present
Tl		Only pigmented retina present
X		Intense pigmentation
X		
X		
X		

MAGE-B2 specific IHC staining with MAGE-B2 (1114) was observed only in testis tissues characterized by cytoplasmic to membrane staining of spermatogonia cells. Staining in other tissues were either observed in isotype or deemed non-specific.

TABLE 17

MAGE-B2 (IJ15) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)									
Tissue identification				Target IHC					
Position	No.	Organ	Tissue ID	Negative	Cell type		Intensity (0-3+)	Localization	Comment
A1	1	Cerebrum	Nct03N002	X					
A2	2	Cerebrum	Nct03N005	X					Non-specific staining in blood vessels
A3	3	Cerebrum	Nct04N001	X					
A4	4	Cerebrum	Nct11N001	X					Non-specific staining in blood vessels
A5	5	Cerebrum	Nct07N015	X					
A6	6	Cerebrum	Nct05N006	X					Non-specific staining in blood vessels
A7	7	Cerebellum	Ncb04N001	X					
A8	8	Cerebellum	Ncb05N006	X					Non-specific staining in blood vessels
A9	9	Cerebellum	Ncb05N011	X					
B1	10	Adrenal gland	Eag07N001		X	Epithelial cells	3+	Cytoplasmic/membrane	Intense staining in cortex and likely non-specific
B2	11	Adrenal gland	Eag06N010		X	Epithelial cells	3+	Cytoplasmic/membrane	
B3	12	Adrenal gland	Eag06N009		X	Epithelial cells	3+	Cytoplasmic/membrane	Intense staining in cortex and likely non-specific

TABLE 17-continued

MAGE-B2 (U15) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)								
Tissue identification				Target IHC				
Position	No.	Organ	Tissue ID	Negative	Cell Positive type	Intensity (0-3+)	Localization	Comment
B4	13	Ovary	Fov110136	X				
B5	14	Ovary	Fov160119	X				
B6	15	Ovary	Fdu100022	X				
B7	16	Pancreas	Dpa05N010		X Epithelial cells	2+	Cytoplasmic/membrane	Weak to mild staining in acinar epithelium and likely non-specific
B8	17	Pancreas	Dpa03N009	X				Non-specific staining in blood vessels
B9	18	Pancreas	Dpa08N040		X Epithelial cells	1+	Cytoplasmic/membrane	Very weak staining in acinar epithelium and likely non-specific
C1	19	Lymph node	Ily08N047	X				Non-specific staining in exfoliated cells and some blood vessels
C2	20	Lymph node	Ily11N004	X				Non-specific staining in exfoliated cells and some blood vessels
C3	21	Lymph node	Ily07N015	X				
C4	22	Hypophysis	Ept17N003	X				Non-specific staining in blood vessels
C5	23	Hypophysis	Ept17N003	X				Non-specific staining in blood vessels
C6	24	Hypophysis	Ept17N003	X				Non-specific staining in blood vessels
C7	25	Testis	Mtt07N016		X Spermatogonia cells; Leydig cells	3+	Cytoplasmic/membrane; nuclear staining in some	Leydig cells staining is likely non-specific
C8	26	Testis	Mtt11N004		X Spermatogonia cells; Leydig cells	2+	Cytoplasmic/membrane; nuclear staining in some	Leydig cells staining is likely non-specific
C9	27	Testis	Mtt07N026		X Spermatogonia cells; Leydig cells	3+	Cytoplasmic/membrane; nuclear staining in some	Leydig cells staining is likely non-specific
D1	28	Thyroid gland	Etg06N003	X				Non-specific staining in colloid and serum
D2	29	Thyroid gland	Etg06N010	X				Non-specific staining in colloid and serum
D3	30	Thyroid gland	Etg05N007	X				Non-specific staining in colloid and serum
D4	31	Breast	Fmg140091	X				
D5	32	Breast	Fmg07N034	X				
D6	33	Breast	Fmg12N003	X				
D7	34	Spleen	Isp06N023	X				Non-specific staining in exfoliated cells and some blood vessels
D8	35	Spleen	Isp05N009	X				Non-specific staining in exfoliated cells and some blood vessels
D9	36	Spleen	Isp07N028	X				Non-specific staining in exfoliated cells and some blood vessels
E1	37	Tonsil	Doc041188	X				
E2	38	Tonsil	Doc041188	X				
E3	39	Tonsil	Doc041188	X				
E4	40	Thymus gland	Ith06N002	X				Non-specific staining in endothelial cells
E5	41	Thymus gland	Ith06N024	X				Non-specific staining in endothelial cells
E6	42	Thymus gland	Ith06N025	X				Non-specific staining in endothelial cells
E7	43	Bone marrow	Ibm07N029	X				Non-specific staining in exfoliated cells

TABLE 17-continued

MAGE-B2 (IU15) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)								
Tissue identification				Target IHC				
Position	No.	Organ	Tissue ID	Negative	Cell Positive type	Intensity (0-3+)	Localization	Comment
E8	44	Bone marrow	Ibm07N023	X				Non-specific staining in exfoliated cells
E9	45	Bone marrow	Ibm07N026	X				Non-specific staining in exfoliated cells
F1	46	Lung	Rln05N023	X				Non-specific staining in alveolar septum, exfoliated cells
F2	47	Lung	Rln05N010	X				Non-specific staining in alveolar septum, exfoliated cells
F3	48	Lung	Rln08N014	X				Non-specific staining in alveolar septum, exfoliated cells
F4	49	Heart	Cht05N005	X				Non-specific staining in serum and blood vessels
F5	50	Heart	Cht03N009	X				Non-specific staining in serum and blood vessels
F6	51	Heart	Cht11N005	X				Non-specific staining in serum and blood vessels
F7	52	Esophagus	Des06N003		X	3+	Cytoplasmic/membrane	Staining in superficial epithelium, blood vessels and connective tissues are likely non-specific
F8	53	Esophagus	Des15N145		X	2+	Cytoplasmic/membrane	Staining in superficial epithelium, blood vessels and connective tissues are likely non-specific
F9	54	Esophagus	Des06N009	X				Non-specific staining in blood vessels and connective tissue
G1	55	Stomach	Dst08N033		X	2+	Cytoplasmic/membrane	Non-specific staining in glandular epithelial cells
G2	56	Stomach	Dst08N011		X Epithelium	3+	Cytoplasmic/membrane	Non-specific staining in glandular epithelial cells
G3	57	Stomach	Dst17N005	X				Non-specific staining in blood vessels; Weak hazy cytoplasmic staining in mucosal epithelium
G4	58	Small intestine	Din07N016	X				
G5	59	Small intestine	Din09N051	X				Non-specific staining in blood vessels; Weak hazy cytoplasmic staining in mucosal epithelium
G6	60	Small intestine	Din06N022	X				Non-specific staining in blood vessels; Weak hazy cytoplasmic staining in mucosal epithelium
G7	61	Colon	Dco15N142	X				Mild to intense staining and likely non-specific
G8	62	Colon	Dco15N103	X				
G9	63	Colon	Dco15N115	X				
H1	64	Liver	Dlv05N015	X	X Hepatocytes	2+	Cytoplasmic/membrane	Mild staining and likely non-specific
H2	65	Liver	Dlv15N145	X	X Hepatocytes	2+	Cytoplasmic/membrane	Very weak staining and likely non-specific
H3	66	Liver	Dlv03N005	X	X Hepatocytes	1+	Cytoplasmic/membrane	Non-specific staining in blood vessels and serum
H4	67	Salivary gland	Doc120045	X				Non-specific staining in blood vessels and serum
H5	68	Salivary gland	Doc05N009	X				Non-specific staining in blood vessels and serum

TABLE 17-continued

MAGE-B2 (IU15) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)								
Tissue identification				Target IHC				
Position	No.	Organ	Tissue ID	Negative	Cell Positive type	Intensity (0-3+)	Localization	Comment
H6	69	Salivary gland	Doc06N019	X				Non-specific staining in blood vessels and serum
H7	70	Kidney	Ukn06N006	X				Non-specific staining in vasculature serum proteins
H8	71	Kidney	Ukn05N003	X				Non-specific staining in vasculature serum proteins
H9	72	Kidney	Ukn06N023		X Tubular epithelium	2+	C/M	Staining in vasculature serum proteins and tubular epithelium likely non-specific
I1	73	Prostate	Mpr07N029	TI				Tissue inadequate
I2	74	Prostate	Mpr11N008	X				Weak hazy non-specific staining in stroma and glandular epithelium
I3	75	Prostate	Mpr09N052	X				Weak hazy non-specific staining in stroma and glandular epithelium
I4	76	Uterus	Fur07N010	X				Non-specific staining in blood vessels Non-specific nuclear staining in some stromal cells; hazy cytoplasmic staining in glandular epithelium that is likely non-specific
I5	77	Uterus	Fur07N001	X				
I6	78	Uterus	Fur150196	X				
I7	79	Cervix	Fur170174	X				Very weak non-specific hazy staining
I8	80	Cervix	Fdu060889	X				
I9	81	Cervix	Fdu170148	X				
J1	82	Skeletal muscle	Smm06N015	X				Very weak non-specific hazy staining
J2	83	Skeletal muscle	Smm05N005	X				
J3	84	Skeletal muscle	Smm05N007	X				
J4	85	Skin	Kin07N017	X				Intense pigmentation Tissue inadequate
J5	86	Skin	Kin06N025	X				
J6	87	Skin	Kin08N034	X				
J7	88	Nerve	Scf15N098	X				Non-specific staining in superficial epithelium
J8	89	Nerve	Scf15N009	X				
J9	90	Nerve	Scf07N017	X				
K1	91	Pericardium	Apr07N027	TI				Tissue inadequate
K2	92	Diaphragm	Smm17N006	X				
K3	93	Pericardium	Apr07N026	X				
K4	94	Eye	Vey090007	X				Non-specific staining in superficial epithelium
K5	95	Eye	Vey140007	TI				
K6	96	Eye	Vey090002	X				
K7	97	Larynx	Rla17N006	X				Tissue inadequate
K8	98	Larynx	Rla17N005	X				
K9	99	Larynx	Rla06N023	TI				

MAGE-B2 IHC staining with MAGE-B2 (IU15) is observed in testis tissues characterized by cytoplasmic to membrane staining of spermatogonia cells, nuclear staining in some and staining in Leydig cells. Leydig staining is likely non-specific. Staining in other tissues were either observed in isotype or deemed non-specific.

TABLE 18

MAGE-B2 (IC3) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)						
		Tissue	IHC Isotype Control			Target IHC
No.	Organ	ID	Negative	Positive	Comment	Negative Positive
1	Cerebrum	Nct05N012	X			X
2	Cerebrum	Nct05N016	X			X
3	Cerebrum	Nct08N046	X			X

TABLE 18-continued

MAGE-B2 (IC3) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)					
4	Cerebellum	Ncb05N008	X		X
5	Cerebellum	Ncb15N146	X		X
6	Cerebellum	Ncb05N011	X		
7	Adrenal gland	Eag06N019	X	Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)	X
8	Adrenal gland	Eag06N010	X	Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)	X
9	Adrenal gland	Eag15N145	X	Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)	X
10	Ovary	Fov07N034	X		X
11	Ovary	Fov13N015	X		X
12	Ovary	Fov13N023	X		X
13	Pancreas	Dpa15N145	TI	Tissue autolyzed	TI
14	Pancreas	Dpa06N023	X		X
15	Pancreas	Dpa11N012	X		X
16	Lymph node	Ily06N015	X		
17	Lymph node	Ily08N047	X	Non-specific staining in exfoliated cells	X
18	Lymph node	Ily08N014	X		X
19	Hypophysis	Ept14N001	X		X
20	Hypophysis	Ept13N019	X		X
21	Hypophysis	Ept13N020	X		X
22	Testis	Mtt09N048	X		X
23	Testis	Mtt08N001	X		X
24	Testis	Mtt08N009	X		X
25	Thyroid gland	Etg07N002	X	Hazy non-specific staining in colloid	
26	Thyroid gland	Etg08N047	X	Hazy non-specific staining in colloid	
27	Thyroid gland	Etg06N003	X	Hazy non-specific staining in colloid	
28	Breast	Fmg08N034	X		X
29	Breast	Fmg50N034	X		X
30	Breast	Fmg07N040	X		X
31	Spleen	Isp15N145	X		X
32	Spleen	Isp15N146	X		X
33	Spleen	Isp11N001	X		X
34	Tonsil	Rts13N011	TI	Only adipose tissue present	TI
35	Tonsil	Rts50N001	X		X
36	Tonsil	Rts08N042	X	Minimal lymphoid tissue present	X
37	Thymus gland	Ith06N024	X		X
38	Thymus gland	Ith06N002	X		X
39	Thymus gland	Ith15N145	X		X
40	Bone marrow	Ibm06N024	X	Non-specific staining in exfoliated cells	X
41	Bone marrow	Ibm11N015	X	Non-specific staining in exfoliated cells	X
42	Bone marrow	Ibm12N001	X	Non-specific staining in exfoliated cells	X
43	Lung	Rln06N004	X		X
44	Lung	Rln06N027	X		X
45	Lung	Rln06N023	X		X
46	Heart	Cht11N009	X		X

TABLE 18-continued

MAGE-B2 (IC3) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)					
47	Heart	Cht05N005	X		X
48	Heart	Cht11N001	X		X
49	Esophagus	Des06N027	X		X
50	Esophagus	Des17N004	X	Minimal epithelial tissue	X
51	Esophagus	Des15N117	X		X
52	Stomach	Dst07N016	X	Hazy non-specific staining in gastric gland cells	X
53	Stomach	Dst15N003	X	Hazy non-specific staining in gastric gland cells	X
54	Stomach	Dst06N017	X	Hazy non-specific staining in gastric gland cells	X
55	Small intestine	Din07N016	X	Weak non-specific mucosal staining	X
56	Small intestine	Din08N014	X		X
57	Small intestine	Din08N007	X		X
58	Colon	Dco15N083	X		X
59	Colon	Dco06N022	X		X
60	Colon	Dco15N082	TI	No mucosal tissue	TI
61	Liver	Dlv05N015	X		X
62	Liver	Dlv15N146	X	Non-specific particulate staining in hepatocytes	X
63	Liver	Dlv15N145	X		X
64	Salivary gland	Doc11N007	X		
65	Salivary gland	Doc06N023	X		X
66	Salivary gland	Doc06N022	X		X
67	Kidney	Ukn08N047	X		X
68	Kidney	Ukn08N035	X		X
69	Kidney	Ukn06N004	X		X
70	Prostate	Mpr08N004	X		X
71	Prostate	Mpr08N005	X		X
72	Prostate	Mpr11N013	X		X
73	Uterus	Fur17N003	X		X
74	Uterus	Fur50N029	X		X
75	Uterus	Fur07N040	X		X
76	Cervix	Fdu50N020	X		X
77	Cervix	Fdu17N003	X		X
78	Cervix	Fdu50N019	X		X
79	Bladder	Ubd08N007	X		X
80	Bladder	Ubd08N001	X		X
81	Bladder	Ubd08N009	X		X
82	Skeletal muscle	Srm15N008	X		X
83	Skeletal muscle	Srm15N009	X		X
84	Skeletal muscle	Srm05N014	X		X
85	Skin	Kin15N007	X		X
86	Skin	Kin07N025	X		X
87	Skin	Kin15N009	X		X
88	Nerve	Scf10N003	X		X
89	Nerve	Scf08N010	X		X
90	Nerve	Scf11N005	X		X
91	Artery	Sbv07N029	X		X
92	Pericardium	Apr15N145	X		X
93	Pericardium	Apr15N146	X		X
94	Eye	Vey130002	TI	Only pigmented retina present	TI
95	Eye	Vey140001	TI	Only pigmented retina present	TI
96	Eye	Vey070007	X	Intense pigmentation	X
97	Larynx	Rla13N011	X		X
98	Larynx	Rla13N020	X		X
99	Larynx	Rla07N010	X		X

TABLE 18-continued

MAGE-B2 (IC3) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)				
Target IHC				
No.	Cell type	Intensity (0-3+)	Localization	Comment
1				
2				Scattered glial cell staining that is likely non-specific due to high antibody concentration
3				Scattered glial cell staining that is likely non-specific due to high antibody concentration
4				
5				
6				
7	Epithelial cells	3+	Cytoplasmic/ membrane	Intense staining in cortex and likely non-specific
8	Epithelial cells	3+	Cytoplasmic/ membrane	Intense staining in cortex and likely non-specific
9	Epithelial cells	3+	Cytoplasmic/ membrane	Intense staining in cortex and likely non-specific
10				
11				
12				
13				Severe autolysis and non-specific staining
14	Epithelial cells	3+	C/M	Intense staining and likely non-specific
15	Epithelial cells	1+	C/M	Very weak staining and likely non-specific
16				
17				Non-specific staining in exfoliated cells
18				
19				
20				Non-specific nuclear staining in glandular cells
21				Non-specific nuclear staining in glandular cells
22	Spermatogonia cells; spermatids; Leydig cells	3+	C/M in Leydig; N in others	Intense nuclear staining suggests non-specific staining with this antibody clone at this high concentration
23	Spermatogonia cells; spermatids; Leydig cells	3+	C/M in Leydig; N in others	Intense nuclear staining suggests non-specific staining with this antibody clone at this high concentration
24	Spermatogonia cells; spermatids; Leydig cells	3+	C/M in Leydig; N in others	Intense nuclear staining suggests non-specific staining with this antibody clone at this high concentration
25				Non-specific staining in colloid and serum
26				Non-specific staining in colloid and serum
27				Non-specific staining in colloid and serum
28				
29				
30				
31				Hazy non-specific staining in red pulp
32				Hazy non-specific staining in red pulp
33				
34				Only adipose tissue present
35				Few nuclear staining in superficial epithelium
36				Minimal lymphoid tissue present
37				
38				
39				
40				Non-specific staining in exfoliated cells
41				Non-specific staining in exfoliated cells
42				Non-specific staining in exfoliated cells
43				Non-specific staining in exfoliated cells and alveolar macrophages
44				

TABLE 18-continued

MAGE-B2 (IC3) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)				
45				Non-specific staining in exfoliated cells and alveolar macrophages
46				Non-specific hazy staining in cardiomyocytes
47				Non-specific hazy staining in cardiomyocytes
48				
49				Non-specific cytoplasmic staining in squamous epithelial
50				Minimal epithelial tissue
51				
52				Intense non-specific staining in gastric gland cells
53				Weak non-specific staining in gastric gland cells
54				Intense non-specific staining in gastric gland cells
55				Non-specific cytoplasmic staining in epithelial cells as observed in isotype
56				
57				
58				
59				
60				No mucosal tissue
61	Hepatocytes	3+	C/M	Very intense staining and likely non-specific
62	Hepatocytes	3+	C/M	Very intense staining and likely non-specific
63	Hepatocytes	3+	C/M	Very intense staining and likely non-specific
64				
65				
66				
67				
68				
69				
70				
71				
72				
73				
74				on-specific nuclear staining in glandular epithelial cells and stromal cells
75				
76				Non-specific nuclear staining in epithelial and stromal cells
77				
78				
79				Non-specific nuclear staining in epithelial and stromal cells
80				Non-specific nuclear staining in epithelial and stromal cells
81				
82				
83				
84				Hazy non-specific staining in cytoplasm
85				
86				
87				
88				
89				
90				
91				
92				
93				
94				Only pigmented retina present
95				Only pigmented retina present
96				Intense pigmentation
97				
98				Non-specific nuclear staining in epithelial and stromal cells
99				Non-specific nuclear staining in epithelial and stromal cells

MAGE-B2 IHC staining with MAGE-B2 (IC3) is observed in testes tissues characterized by cytoplasmic to membrane staining of spermatogonia cells, and nuclear staining in some cells. There are MAGE-B2 IHC staining in other tissues such as liver, adrenal gland, intestines that are most likely non-specific.

TABLE 19

MAGE-B2 (4G17) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)						
Tissue identification				IHC Isotype Control		
Position	No.	Organ	Tissue ID	Negative	Positive	Comment
A1	1	Cerebrum	Nct05N012	X		
A2	2	Cerebrum	Nct05N016	X		
A3	3	Cerebrum	Nct08N046	X		
A4	4	Cerebellum	Ncb05N008	X		
A5	5	Cerebellum	Ncb15N146	X		
A6	6	Cerebellum	Ncb05N011	X		
A7	7	Adrenal gland	Eag06N019	X		Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)
A8	8	Adrenal gland	Eag06N010	X		Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)
A9	9	Adrenal gland	Eag15N145	X		Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis)
A10	10	Ovary	Fov07N034	X		
A11	11	Ovary	Fov13N015	X		
A12	12	Ovary	Fov13N023	X		
B1	13	Pancreas	Dpa15N145	TI		Tissue autolyzed
B2	14	Pancreas	Dpa06N023	X		
B3	15	Pancreas	Dpa11N012	X		
B4	16	Lymph node	Ily06N015	X		
B5	17	Lymph node	Ily08N047	X		Non-specific staining in exfoliated cells
B6	18	Lymph node	Ily08N014	X		
B7	19	Hypophysis	Ept14N001	X		
B8	20	Hypophysis	Ept13N019	X		
B9	21	Hypophysis	Ept13N020	X		
B10	22	Testis	Mtt09N048	X		
B11	23	Testis	Mtt08N001	X		
B12	24	Testis	Mtt08N009	X		
C1	25	Thyroid gland	Etg07N002	X		Hazy non-specific staining in colloid
C2	26	Thyroid gland	Etg08N047	X		Hazy non-specific staining in colloid
C3	27	Thyroid gland	Etg06N003	X		Hazy non-specific staining in colloid
C4	28	Breast	Fmg08N034	X		
C5	29	Breast	Fmg50N034	X		
C6	30	Breast	Fmg07N040	X		
C7	31	Spleen	Isp15N145	X		
C8	32	Spleen	Isp15N146	X		
C9	33	Spleen	Isp11N001	X		
C10	34	Tonsil	Rts13N011	TI		Only adipose tissue present
C11	35	Tonsil	Rts50N001	X		
C12	36	Tonsil	Rts08N042	X		Minimal lymphoid tissue present
D1	37	Thymus gland	Ith06N024	X		
D2	38	Thymus gland	Ith06N002	X		
D3	39	Thymus gland	Ith15N145	X		
D4	40	Bone marrow	Ibm06N024	X		Non-specific staining in exfoliated cells
D5	41	Bone marrow	Ibm11N015	X		Non-specific staining in exfoliated cells
D6	42	Bone marrow	Ibm12N001	X		Non-specific staining in exfoliated cells
D7	43	Lung	Rln06N004	X		
D8	44	Lung	Rln06N027	X		
D9	45	Lung	Rln06N023	X		

TABLE 19-continued

MAGE-B2 (4G17) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)					
D10	46	Heart	Cht11N009	X	
D11	47	Heart	Cht05N005	X	
D12	48	Heart	Cht11N001	X	
E1	49	Esophagus	Des06N027	X	
E2	50	Esophagus	Des17N004	X	Minimal epithelial tissue
E3	51	Esophagus	Des15N117	X	
E4	52	Stomach	Dst07N016	X	Hazy non-specific staining in gastric gland cells
E5	53	Stomach	Dst15N003	X	Hazy non-specific staining in gastric gland cells
E6	54	Stomach	Dst06N017	X	Hazy non-specific staining in gastric gland cells
E7	55	Small intestine	Din07N016	X	Weak non-specific mucosal staining
E8	56	Small intestine	Din08N014	X	
E9	57	Small intestine	Din08N007	X	
E10	58	Colon	Dco15N083	X	
E11	59	Colon	Dco06N022	X	
E12	60	Colon	Dco15N082	TI	No mucosal tissue
F1	61	Liver	Dlv05N015	X	
F2	62	Liver	Dlv15N146	X	Non-specific particulate staining in hepatocytes
F3	63	Liver	Dlv15N145	X	
F4	64	Salivary gland	Doc11N007	X	
F5	65	Salivary gland	Doc06N023	X	
F6	66	Salivary gland	Doc06N022	X	
F7	67	Kidney	Ukn08N047	X	
F8	68	Kidney	Ukn08N035		
F9	69	Kidney	Ukn06N004		
F10	70	Prostate	Mpr08N004		
F11	71	Prostate	Mpr08N005	X	
F12	72	Prostate	Mpr11N013	X	
G1	73	Uterus	Fur17N003	X	
G2	74	Uterus	Fur50N029	X	
G3	75	Uterus	Fur07N040	X	
G4	76	Cervix	Fdu50N020	X	
G5	77	Cervix	Fdu17N003	X	
G6	78	Cervix	Fdu50N019	X	
G7	79	Bladder	Ubd08N007	X	
G8	180	Bladder	Ubd08N001	X	
G9	81	Bladder	Ubd08N009	X	
G10	82	Skeletal muscle	Smn15N008	X	
G11	83	Skeletal muscle	Smn15N009	X	
G12	84	Skeletal muscle	Smn05N014	X	
H1	85	Skin	Kin15N007	X	
H2	86	Skin	Kin07N025	X	
H3	87	Skin	Kin15N009	X	
H4	188	Nerve	Scf10N003	X	
H5	89	Nerve	Scf08N010	X	
H6	90	Nerve	Scf11N005	X	
H7	91	Artery	Sbv07N029	X	
H8	92	Pericardium	Apr15N145	X	
H9	93	Pericardium	Apr15N146	X	
H10	94	Eye	Vey130002	TI	Only pigmented retina present
H11	95	Eye	Vey140001	TI	Only pigmented retina present
H12	196	Eye	Vey070007	X	Intense pigmentation
I1	197	Larynx	Rla13N011	X	
I2	98	Larynx	Rla13N020	X	
I3	199	Larynx	Rla07N010	X	
Target IHC					
Negative		Cell Positive type	Intensity (0-3+)	Local- ization	Comment
X					
X					Non-specific staining in few glial cells

TABLE 19-continued

MAGE-B2 (4G17) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)					
X					Non-specific staining in few capillaries
X					
X					Non-specific staining in few capillaries
X					Non-specific staining in few capillaries
X					Non-specific staining as observed in isotype but with increased intensity
X					Non-specific staining as observed in isotype but with increased intensity
X					Non-specific staining as observed in isotype but with increased intensity
X					Non-specific particulate staining in cortex (Zona fasciculata and zona reticularis) as observed in isotype
X					
X					
TI	X	Epithelial cells	1+	Cytoplasmic/ membrane	Tissue autolyzed Weak to mild staining and likely non-specific
X					
X					
X					Non-specific staining as in isotype, but with increased intensity
X					
X					
	X	Epithelial cells	2+	Cytoplasmic/ membrane	Predominantly in loosely adhered cells and likely non-specific
	X	Epithelial cells	2+	Cytoplasmic/ membrane	Predominantly in loosely adhered cells and likely non-specific
	X	Spermatogonia cells	3	Cytoplasmic/ membrane, Nuclear	There is nuclear staining in some spermatids and hazy cytoplasmic staining in some Leydig cells. Both are likely non-specific
	X	Spermatogonia cells	3	Cytoplasmic/ membrane, Nuclear	There is nuclear staining in some spermatids and hazy cytoplasmic staining in some Leydig cells. Both are likely non-specific
	X	Spermatogonia cells	3	C/M; N	There is nuclear staining in some spermatids and hazy cytoplasmic staining in some Leydig cells. Both are likely non-specific Non-specific staining as observed in isotype but with increased intensity Non-specific staining as observed in isotype but with increased intensity Non-specific staining as observed in isotype but with increased intensity
X					
X					
X					Weak non-specific staining in exfoliated cells in red pulp
X					Weak non-specific staining in exfoliated cells in red pulp
X					Weak non-specific staining in exfoliated cells in red pulp
TI					Only adipose tissue present
X					Non-specific nuclear staining in some cells

TABLE 19-continued

MAGE-B2 (4G17) Immunohistochemistry Staining in Normal Human Tissue TMA (FDA999x)	
X	
X	
X	
X	
X	
X	
X	
X	
X	
TI	Only pigmented retina present
TI	Only pigmented retina present
X	Intense pigmentation
X	
X	
X	

MAGE-B2 IHC staining with MAGE-B2 (4G17) was observed in testes, characterized by cytoplasmic to membrane staining of spermatogonia cells, nuclear staining in some and staining in Leydig cells. Leydig staining is likely non-specific. There was MAGE-B2 IHC staining in other tissues such as liver, adrenal gland, intestines that was most likely non-specific.

TABLE 20

Testing the 4 MAGE-B2 antibodies (4G17, 1J15, 1C3, 1i14) on thirteen human cancer cell lines with known endogenous RNA expression level by IHC.											
Cell line	Tron Database [RPKM]	RNASeq (exon specific FPKM)	CCL (FPKM)	IHC (1J15)		IHC (1i14)		IHC(1C3)		IHC (4G17)	
	MAGEB2 expression	MAGEB2 expression	MAGEB2 expression	Intensity	% Frequency	Intensity	% Frequency	Intensity	% Frequency	Intensity	% Frequency
T98G	0.05	0	0.01	1	70	2	70	2	70	2	70
UACC-257	0.04	0.01	0.01	0	0	0/1	0	0/1	0	0	0
KMM-1	56.91	31.58	42.69	0/1	70	1	70	1	70	2	70
U266B1	102.32	103.73	89.35	1	90	3	90	2	90	2	80
CFPAC-1	0.45	0.18	0.40	0/1	0	0/1	0	0/1	0	0/1	0
SCaBER	0.01	0.01	0.02	2	50	0	0	2	70	2	70
VMRC-LCD	0			0	0	0	0	0	0	0	0
NCI-H1703	0	0	0.00	0/1	0	0/1	0	0/1	0	0	0
UM-UC-3			0.02	0	0	0		0/1		0	0
NCI-H1395			3.52	0	0	0	0	0	0	0	0
U20S			45.42	1	50	2	70	2	60	2	60
NCI-H82		6.33	0.06	0	0	0	0	1	0	0	0

MAGEB2 IHC expression frequency with the four MAGE-B2 (4G17, 1J15, 1C3, 1i14) was high (~90%) in U266B1 cells, and moderate (~70%) in T98G, KMM-1, U20S cells; and undetectable in CFPAC-1, UACC-257, CFPAC-1, VMRC-LCD, NCI-H1703, UM-UC-3, NCI-H1395, NCI-H82 cell lines with the 4 MAGE-B2 clones (4G17, 1J15, 1C3, 1i14). MAGEB2 IHC expression frequencies in SCaBER cells were inconsistent with the four MAGE-B2 clones (4G17, 1J15, 1C3, 1i14).

TABLE 21

Standard IgG Antibody VL CDRs				
Ab	Type	CDR1	CDR2	CDR3
7M5	NA	CAGGCCAGTCAGAGCATTAGGAAT GAATTATTT (SEQ ID NO: 5)	GCTGCATCCAACTGGCCTCT (SEQ ID NO: 6)	CAATGCAGTTATGTTAGTAGTAGTGGTACT TATGGAAATGTT (SEQ ID NO: 7)
	AA	QASQSIRNELF (SEQ ID NO: 7)	AASKLAS (SEQ ID NO: 8)	QCSYVSSSGTYGNV (SEQ ID NO: 9)
1J14	NA	CAGGCCAGTGAAAGCATTAGCAAC TACTTATCC (SEQ ID NO: 10)	TGGGCATCCACTCTGGCATCT (SEQ ID NO: 11)	CAACAGGGTTATAGTAGTAGTAATGTTGAT AATCTT (SEQ ID NO: 12)
	AA	QASESISNYLS (SEQ ID NO: 13)	WASTLAS (SEQ ID NO: 14)	QQGYSSSNVDNL (SEQ ID NO: 15)

TABLE 21-continued

Standard IgG Antibody VL CDRs				
Ab	Type	CDR1	CDR2	CDR3
7F15	NA	CAGTCCAGTCAGAGTGTTTGGCATA ACGACTACTTATCC (SEQ ID NO: 16)	GGTGCATCCACTTTGGCATCT (SEQ ID NO: 17)	GCAGGCGGTTATGGACGTAGTAGTGAAAA TGGT (SEQ ID NO: 18)
	AA	QSSQSVWHNDYLS (SEQ ID NO: 19)	GASTLAS (SEQ ID NO: 20)	AGGYGRSSENG (SEQ ID NO: 21)
5I9	NA	CAGTCCAGTAAGAGTGTTTATAATA ACAACTGGTTAGCC (SEQ ID NO: 22)	GATGCATCGACTCTAGATTCT (SEQ ID NO: 23)	GTAGGCGGTTATAGTAGTCGTAGTGATAAT GGT (SEQ ID NO: 24)
	AA	QSSKSVYNNWLA (SEQ ID NO: 25)	DASTLDS (SEQ ID NO: 26)	VGGYSSRSNDG (SEQ ID NO: 27)
1J15	NA	CAGGCCAGTCAGAGTATTAGTAGTT ACTTATCC (SEQ ID NO: 28)	AGGGCATCCACTCTGGCATCT (SEQ ID NO: 29)	CAAAGCTATGATGATAGTAGTGATAATAAT TTTTTTTATGGT (SEQ ID NO: 30)
	AA	QASQSISSYLS (SEQ ID NO: 31)	RASTLAS (SEQ ID NO: 32)	QSYDDSSDNNFFYG (SEQ ID NO: 33)
4G17	NA	CAGGCCAGTCAGAACATTGATAGT TACTTAGCC (SEQ ID NO: 34)	AGGGCATCCACTCTGGCATCT (SEQ ID NO: 35)	CAAAGCTATGATGATAGTAGGAGTAGTAG TTTTTTTATGGT (SEQ ID NO: 36)
	AA	QASQNIDSYLA (SEQ ID NO: 37)	RASTLAS (SEQ ID NO: 38)	QSYDDSRSSSPFFYG (SEQ ID NO: 39)
4A15	NA	CAGGCCAGTCAGAACATTAATAGT TACTTAGCC (SEQ ID NO: 40)	AGGGCATCCACTCTGGCATCT (SEQ ID NO: 41)	CAAAGCTATGATGATAGTAGGAGTATTAGT TTTTTTTATGCT (SEQ ID NO: 42)
	AA	QASQNINSYLA SEQ ID NO: 43)	RASTLAS (SEQ ID NO: 44)	QSYDDSRSSISFFYA (SEQ ID NO: 45)
1J16	NA	CAGGCCAGTCAGAGCATTAGTAGC TACTTAGCC (SEQ ID NO: 46)	GCTGCATCCACTCTGGCATCT (SEQ ID NO: 47)	CAAAGCTATGATGATAGTAGGAGTAGTAG TTTTTTTATGCT (SEQ ID NO: 48)
	AA	QASQSISSYLA (SEQ ID NO: 49)	AASTLAS (SEQ ID NO: 50)	QSYDDSRSSSPFFYA (SEQ ID NO: 51)
1C18	NA	CAGGCCAGTCAGAGCATTAGCAGT TGGTTATCC (SEQ ID NO: 52)	AGGGCAACCACTCTGGCATCT (SEQ ID NO: 53)	CAAAGTTATGATGATAGTAGTAGTAGTAAT TTTTTTTATGCT (SEQ ID NO: 54)
	AA	QASQSISSWLS (SEQ ID NO: 55)	RATTLAS (SEQ ID NO: 56)	QSYDDSSSNFFYA SEQ ID NO: 57)
7H4	NA	CAGTCCAGTCAGAGTGTTTATAGTA ACAACCTCTTATCT (SEQ ID NO: 58)	AAGGCATCCACTCTGGCATCT (SEQ ID NO: 59)	CAAGGCTATTATAGTGGTGTATTATATG (SEQ ID NO: 60)
	AA	QSSQSVYSNNLLS (SEQ ID NO: 61)	KASTLAS (SEQ ID NO: 62)	QGYYSGLVIYM (SEQ ID NO: 63)
1H24	NA	CAGGCCAGTCAGAGCATTAGTAGT TACTTATCC (SEQ ID NO: 64)	AGGGCATCCACTCTGGCATCT (SEQ ID NO: 65)	CAAAGCTATGATGATAGTAGTAGTAATAAT TTTTTTTATGGT (SEQ ID NO: 66)
	AA	QASQSISSYLS (SEQ ID NO: 67)	RASTLAS (SEQ ID NO: 68)	QSYDDSSSNFFYG (SEQ ID NO: 69)
1C3	NA	CAGGCCAGTCAGAACATTAGTAGC TACTTAGCC (SEQ ID NO: 70)	AGGGCATCCACTCTGGCATCT (SEQ ID NO: 71)	CAAAGCTATGATGATAGTAGGAGTAGTAA TTTTTTTATGCT (SEQ ID NO: 72)
	AA	QASQNISSYLA (SEQ ID NO: 73)	RASTLAS (SEQ ID NO: 74)	QSYDDSRSSNFFYA (SEQ ID NO: 75)
5H20	NA	CAGTCCAGTGAGAGCGTTTATAATC ACAACTGGTTAGGC (SEQ ID NO: 76)	GATGCATCCACTCTGGCATCT (SEQ ID NO: 77)	CAAGGCTATTATCAAACTAGTGTTTGGGCT (SEQ ID NO: 78)
	AA	QSSSVYNNHWLG (SEQ ID NO: 79)	DASTLAS (SEQ ID NO: 80)	QGYQTTSVWA (SEQ ID NO: 81)
1I14	NA	CAGTCCAGTCAGAGTGTTTATGATA ACAATGCTTTAGCC (SEQ ID NO: 82)	GGTGCATCCACTCTGGCATCT (SEQ ID NO: 83)	CAATGTACTTATTATGTTAGTAGTTATCAA AATGAT (SEQ ID NO: 84)
	AA	QSSQSVYDNNALA (SEQ ID NO: 85)	GASTLAS (SEQ ID NO: 86)	QCTYYVSSYQND (SEQ ID NO: 87)

TABLE 21-continued

Standard IgG Antibody VL CDRs				
Ab	Type	CDR1	CDR2	CDR3
1H17	NA	CAGTCCAGTAAGAGTGTTTATAATA AGAACTGGTTATCC (SEQ ID NO: 88)	GGTGCATCCACTCTGGCATCT (SEQ ID NO: 89)	GCAGGCGGTTATAGTAGTAGTAGTGATACA TTTGCT (SEQ ID NO: 90)
	AA	QSSKSVYNKNWLS (SEQ ID NO: 91)	GASTLAS SEQ ID NO: 92)	AGGYSSSSDTFA (SEQ ID NO: 93)
7B20	NA	GCGACCTCTTATCC CAGTCCAGTCAGAGCGTTTATAGTA (SEQ ID NO: 94)	AAGGCATCCACTCTGGCATCT (SEQ ID NO: 95)	CAAGGCTACTATAGTGGTGTGGTTTATATT (SEQ ID NO: 96)
	AA	QSSQSVYSSDLLS (SEQ ID NO: 97)	KASTLAS SEQ ID NO: 98)	QGYSGVVYI (SEQ ID NO: 99)
7J6	NA	CAGTCCAGTCAGAGCGTTTATAGTA GCGACCTCTTATCC (SEQ ID NO: 100)	AAGGCATCCACTCTGGCATCT (SEQ ID NO: 101)	CAAGGCTACTATAGTGGTGTGGTTTATATT (SEQ ID NO: 102)
	AA	QSSQSVYSSDLLS (SEQ ID NO: 103)	KASTLAS (SEQ ID NO: 104)	QGYSGVVYI (SEQ ID NO: 105)
1D2	NA	CAGGCCAGTCAGAGCATTAGTAGT TACTTATCT (SEQ ID NO: 106)	AGGGCATCCCCTCTGGCATCT (SEQ ID NO: 107)	CAAAGCTACGATGATAGTAGTAGTAATAAT TTTTTTTATGGT (SEQ ID NO: 108)
	AA	QASQSISSYLS (SEQ ID NO: 109)	RASPLAS SEQ ID NO: 110)	QSYDDSSSNFFYG (SEQ ID NO: 111)
4G8	NA	AGAACTGGTTATCC CAGTCCAGTAAGAGTGTTTATAATA (SEQ ID NO: 112)	GGTGCATCCACTCTGGCATCT (SEQ ID NO: 113)	TTTGCT GCAGGCGGTTATAGTAGTAGTAGTGATACG (SEQ ID NO: 114)
	AA	QSSKSVYNKNWLS (SEQ ID NO: 115)	GASTLAS SEQ ID NO: 116)	AGGYSSSSDTFA (SEQ ID NO: 117)
7I4	NA	CAGTCCAGTCAGAGCCTTTATAATA GCGACCTCTTATCC (SEQ ID NO: 118)	AAGGCATCCACTCTGGCATCT (SEQ ID NO: 119)	CAAGGCTACTATAGTGGTGTGGTTTATATT (SEQ ID NO: 120)
	AA	QSSQSLYNSDLLS (SEQ ID NO: 121)	KASTLAS (SEQ ID NO: 122)	QGYSGVVYI (SEQ ID NO: 123)
1I23	NA	CAGTCCAGTAAGAGTGTTTATAATA AGAACTGGTTATCC (SEQ ID NO: 124)	GGTGCATCCACTCTGGCATCT (SEQ ID NO: 125)	GCAGGCGGTTATAGTAGTAGTAGTGATACA TTTGCT (SEQ ID NO: 126)
	AA	QSSKSVYNKNWLS (SEQ ID NO: 127)	GASTLAS (SEQ ID NO: 128)	AGGYSSSSDTFA (SEQ ID NO: 129)
1J13	NA	CAGGCCCTTCAGAGTGTTTATGATA ACAATGCTTTATCC (SEQ ID NO: 130)	GGTGCATCCACTCTGGCATCT (SEQ ID NO: 131)	CAATGTACTTATTATGTTAGTAGTTATCAA AATGAT (SEQ ID NO: 132)
	AA	QALQSVYDNNALS (SEQ ID NO: 133)	GASTLAS (SEQ ID NO: 134)	QCTYYVSSYQND (SEQ ID NO: 135)
1G24	NA	CAGGCCAGTCAGAGCATTGGCAAT GCATTAGCC (SEQ ID NO: 136)	AGGGCATCCACTCTGGCATCT (SEQ ID NO: 137)	CAAAGCTATGATGATAGTAGTAGTAGTAGT TTTTTTTATGCT (SEQ ID NO: 138)
	AA	QASESIGNALA (SEQ ID NO: 139)	RASTLAS (SEQ ID NO: 140)	QSYDDSSSSFFYA (SEQ ID NO: 141)
4A7	NA	CAGTCCAGTAAGAGTGTTTATAATA AGAACTGGTTATCC (SEQ ID NO: 142)	GGTGCATCCACTCTGGCATCT (SEQ ID NO: 143)	GCAGGCGGTTATAGTAGTAGTAGTGATACG TTTGCT (SEQ ID NO: 144)
	AA	QSSKSVYNKNWLS (SEQ ID NO: 145)	GASTLAS (SEQ ID NO: 146)	AGGYSSSSDTFA (SEQ ID NO: 147)

TABLE 22

Standard IgG Antibody VH CDRs				
Ab	Type	CDR1	CDR2	CDR3
7M5	NA	AGCCATGCAATGATC (SEQ ID NO: 148)	ACCATTGGGAGTCGTGATACTATATATTAT GCGAGCTGGGCGAAAGGC (SEQ ID NO: 149)	AACGCCTTG (SEQ ID NO: 150)
	AA	SHAMI (SEQ ID NO: 151)	TIGSRDTIYYASWAKG (SEQ ID NO: 152)	NAL (SEQ ID NO: 153)
1J14	NA	AGCTACGACATGAGC (SEQ ID NO: 154)	ATTATTTATGCTAGTGGTAGCACATACTAC GCGAGCTGGGCGAAAGGC (SEQ ID NO: 155)	GACCCTGCTGGTTATAGCATTAGCTTTGGC TTG (SEQ ID NO: 156)
	AA	SYDMS (SEQ ID NO: 157)	IIYASGSTYYASWAKG (SEQ ID NO: 158)	DPAGYSISFGL (SEQ ID NO: 159)
7F15	NA	AGCTACAACATGGGC (SEQ ID NO: 160)	ATCATTGGTGCTAGTGATAGCGCATTGTAC GCGAGCTGGGCGAAAGGC (SEQ ID NO: 161)	GGTGGTCTTGTTTGAGTACTGGTTTTCGCT TG (SEQ ID NO: 162)
	AA	SYNMG (SEQ ID NO: 163)	IIGASDSALYASWAKG (SEQ ID NO: 164)	GGLGLSTGFAL (SEQ ID NO: 165)
5I9	NA	AGCTACGACATGAGC (SEQ ID NO: 166)	TATATTGCTACTGATGGTAGGCCATATTAC GCGAGCTGGGCGAAAGGC (SEQ ID NO: 167)	GGGGGTATGCTGGTGGCTTG (SEQ ID NO: 168)
	AA	SYDMS (SEQ ID NO: 169)	YIATDGRPYASWAKG (SEQ ID NO: 170)	GGYAGGL (SEQ ID NO: 171)
1J15	NA	AATTATTATATTGTC (SEQ ID NO: 172)	TGCATTGACAATGCTAATGGTAGGACTTAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 173)	TCATTGTCTACTCCCTTG (SEQ ID NO: 174)
	AA	NYIIC (SEQ ID NO: 175)	CIDNANGRTYYASWAKG (SEQ ID NO: 176)	SLSTPL (SEQ ID NO: 177)
4G17	NA	AACTATTATATTGT (SEQ ID NO: 178)	TGCATTGACAATGTTAATGGTAGGACCTAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 179)	TCCTTGGCTACTCCCTTG (SEQ ID NO: 180)
	AA	NYIIC (SEQ ID NO: 181)	CIDNVNGRTYYASWAKG (SEQ ID NO: 182)	SLATPL (SEQ ID NO: 183)
4A15	NA	AACTACTACATCTGC (SEQ ID NO: 184)	TGCATTGACAATGTTAATGGTAGGACTTAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 185)	TCCTTGGCTACTCCCTTG (SEQ ID NO: 186)
	AA	NYIIC (SEQ ID NO: 187)	CIDNVNGRTYYASWAKG (SEQ ID NO: 188)	SLATPL (SEQ ID NO: 189)
1J16	NA	AACTACTACATCTGC (SEQ ID NO: 190)	TGCATTGACAATATTAATGGTAGGACTTAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 191)	TCCTTGGCTACTCCCTTG (SEQ ID NO: 192)
	AA	NYIIC (SEQ ID NO: 193)	CIDNINGRTYYASWAKG (SEQ ID NO: 194)	SLATPL (SEQ ID NO: 195)
1C18	NA	AATTATTATATATGT (SEQ ID NO: 196)	TGTATTGATAACGCTAATGGTAGGACTTAC TACGCGACCTGGGCGAAAGGC (SEQ ID NO: 197)	TCATTGTCTACTAAGTTG (SEQ ID NO: 198)
	AA	NYIIC (SEQ ID NO: 199)	CIDNANGRTYYATWAKG (SEQ ID NO: 200)	SLSTNL (SEQ ID NO: 201)
7H4	NA	AGCTACTACTACATATGC (SEQ ID NO: 202)	TGTATTGGTGGTGGTAATACCGATGCCACT GCCTACGCGAGGTGGGCGAAAGGC (SEQ ID NO: 203)	GGCGGTCTGATAATAATGTCCAATTTAAC TTG (SEQ ID NO: 204)
	AA	SYIIC (SEQ ID NO: 205)	CIGGGNTDATAYARWAKG (SEQ ID NO: 206)	GGPDNNVQFNL (SEQ ID NO: 207)
1H24	NA	AATTATTATATTGTC (SEQ ID NO: 208)	TGCATTGACAATAGTAATGGTAGGACTTAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 209)	TCATTGTCTACTCCCTTG (SEQ ID NO: 210)
	AA	NYIIC (SEQ ID NO: 211)	CIDNSNGRTYYASWAKG (SEQ ID NO: 212)	SLSTPL (SEQ ID NO: 213)
1C3	NA	AACTACTACATCTGC (SEQ ID NO: 214)	TGTATTGACAATGCTAATGGTAGGACTTAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 215)	TCCTTGGCTACTCCCTTG (SEQ ID NO: 216)
	AA	NYIIC (SEQ ID NO: 217)	CIDNANGRTYYASWAKG (SEQ ID NO: 218)	SLATPL (SEQ ID NO: 219)

TABLE 22-continued

Standard IgG Antibody VH CDRs				
Ab	Type	CDR1	CDR2	CDR3
5H20	NA	AGCAGTGCAGTGACC (SEQ ID NO: 220)	TTCTTCCAAGCTGGGGATGGTAGCGCATAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 221)	CATAAGGGTAATAGTTACGTGCCTAACTTG (SEQ ID NO: 222)
	AA	SSAVT (SEQ ID NO: 223)	FLQAGDGSAYYASWAKG (SEQ ID NO: 224)	HKGNSYVPNL (SEQ ID NO: 225)
1I14	NA	AGCTATGCAATGAGC (SEQ ID NO: 226)	AGCATTGGTGGTGGTGGTAGCGAGTCTAC GCGAGCTGGGCGAAAGGC (SEQ ID NO: 227)	GGATTTTATAGTATAGACTTG (SEQ ID NO: 228)
	AA	SYAMS (SEQ ID NO: 229)	SIGGGGSAVYASWAKG (SEQ ID NO: 230)	GFYSIDL (SEQ ID NO: 231)
1H17	NA	AGCGGCCAACTCATGTGC (SEQ ID NO: 232)	TGCATTGGTCTGTGGTAGTAATGCTATTAGC ACTTTCTACGCGAGCTGGGCGCAAGGC (SEQ ID NO: 233)	GTGGGCTCCGATGACTATGGTGACTCTGAT GTTTTTGATCCC (SEQ ID NO: 234)
	AA	SGQLMC (SEQ ID NO: 235)	CIGSGSNAISTFYASWAQG (SEQ ID NO: 236)	VGSDDYGDSDFDP (SEQ ID NO: 237)
7B20	NA	AGCGCCTACTACATATGC (SEQ ID NO: 238)	TGTATTGGTGGTGTAAATCGCGTTGCCACT GCCTACGCGACCTGGGCGAAAGGC (SEQ ID NO: 239)	GGCGGTCTGATAATAATGTCCAATTTAAC TTG (SEQ ID NO: 240)
	AA	SAYYIC (SEQ ID NO: 241)	CIGGVNVRATAYATWAKG (SEQ ID NO: 242)	GGPDNNVQFNL (SEQ ID NO: 243)
7J6	NA	AGGTACTACTACAGTTGC (SEQ ID NO: 244)	TGTGTTGGTGGTGTAAATCGCGATGCCACT GCCTACGCGACCTGGGCGAAAGGC (SEQ ID NO: 245)	GGCGGTCTGATAATAATGTCCAATTTAAC TTG (SEQ ID NO: 246)
	AA	RYYYSC (SEQ ID NO: 247)	CVGGVNRDATAYATWAKG (SEQ ID NO: 248)	GGPDNNVQFNL (SEQ ID NO: 249)
1D2	NA	AATTATTATATTTGT (SEQ ID NO: 250)	TGTATTGACAATGTTAATGGTAGGACTTAC TACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 251)	TCATTGTCTACTCCCTTG (SEQ ID NO: 252)
	AA	NYIIC (SEQ ID NO: 253)	CIDNVNGRTYYASWAKG (SEQ ID NO: 254)	SLSTPL (SEQ ID NO: 255)
4G8	NA	AGCAGCTACTTCATGTGC (SEQ ID NO: 256)	TGCATTGCTGTTGGTAGTAGTGGTAGCACT TACTACGCGAGCTGGGCGAAAGGC (SEQ ID NO: 257)	GTGGGCTACGATGACTATGGTGACTCTGAT GCTTTTGATCCC (SEQ ID NO: 258)
	AA	SSYFMC (SEQ ID NO: 259)	CIAGSSGSTYYASWAKG (SEQ ID NO: 260)	VGYYDDYGSDAFDP (SEQ ID NO: 261)
7I4	NA	AGGTACTACTACAGTTGC (SEQ ID NO: 262)	TGTGTTGGTGGTGTAAATCGCGATGCCACT GCCTACGCGACCTGGGCGAAAGGC (SEQ ID NO: 263)	GGCGGTCTGATAATAATGTCCAATTTAAC TTG (SEQ ID NO: 264)
	AA	RYYYSC (SEQ ID NO: 265)	CVGGVNRDATAYATWAKG (SEQ ID NO: 266)	GGPDNNVQFNL (SEQ ID NO: 267)
1I23	NA	AGCAGCTACTTCATGTGC (SEQ ID NO: 268)	TGCATTGGTCTGTGGTAGTAGTGTATTAGC ACTTTCTACGCGAGCTGGGCGCAAGGC (SEQ ID NO: 269)	GTGGGCTACGATGACTATGGTGACTCTGAT GCTTTTGATCCC (SEQ ID NO: 270)
	AA	SSYFMC (SEQ ID NO: 271)	CIGSGSSAISTFYASWAQG (SEQ ID NO: 272)	VGYYDDYGSDAFDP (SEQ ID NO: 273)
1J13	NA	AGCTATGGAATGACT (SEQ ID NO: 274)	TACATTGGAGTATGGGAGGACATACTAC GCAAACCTGGGCGAAAGGC (SEQ ID NO: 275)	CCCTTTGATGGTAATTATAGGGACATC (SEQ ID NO: 276)
	AA	SYGMT (SEQ ID NO: 277)	YIWTDGRTYYANWAKG (SEQ ID NO: 278)	PPDGNRYRDI (SEQ ID NO: 279)
1G24	NA	AACTACTACATCTGC (SEQ ID NO: 280)	TGTATTGATAATGCCAATGGTCGGACTTAC TACGCGAAGCTGGGCGAAAGGC (SEQ ID NO: 281)	TCCTTGGCTACTCCCTTG (SEQ ID NO: 282)
	AA	NYIIC (SEQ ID NO: 283)	CIDNANGRTYYANWAKG (SEQ ID NO: 284)	SLATPL (SEQ ID NO: 285)
4A7	NA	AGCAGCTACTTCATGTGC (SEQ ID NO: 286)	TGCATTGGTGTGGTGGTAGTGGTAGCACT TACTACGCGAAGCTGGGCGAAAGGC (SEQ ID NO: 287)	GTGGCCTACGATGACGATGGTGACTCTGAT GCTTTTGATCCC (SEQ ID NO: 288)
	AA	SSYFMC (SEQ ID NO: 289)	CIGVGGSGSTYYANWAKG (SEQ ID NO: 290)	VAYDDDGSDAFDP (SEQ ID NO: 291)

[illegible]

Ab	Type	Lc V-region	Hc V-region	Standard IgG Antibody Variable Region Sequences
4G17	NA	GCATCGACTAGATTCTGGGTTCTCATCGGGTTCAAA GGCAGTGAATCTGGACACAGATTCACTCTCACATCAG CCAGTCCAGTGTGGCATGGTCCCACTTACTACTGTG AGGCGTTATTAGTAGTCTAGTAGTAATAGTTTCGGCG GAGGACCGAGGTGGTGGTCAA (SEQ ID NO: 304) AAVLTQTPSPVSAAVGTVTSISCOSSKSVNNWLAWYQ QKQRPKLLIYDASTLDSGVSSRFKSGSGTQFTLISDVQ CDDGATYYCVGGYSRSDNRPFGGTEVVVK (SEQ ID NO: 306)	AGCCGATTACCATCTCCAAACCTCGTCGACACGGTGGATCTGAA AATCACAGTCCGACACAGGACACGGCCACCTATTTCTGTGTGTA AGGGGGTATGCTGGTGGTGTGGGCCCGACCCCTGGTCACC GTTTCTCA (SEQ ID NO: 305) QSLBESGRLVTPGGPLTLTCTVSGFSLSSYDMSVWRQAPKGLEWIGY IATDGRPVYASWAKGRFTISKPSSTTVDLKITPTTEDTATYFCVRGGYA GGLWGPGLTVTSS (SEQ ID NO: 307)	
	NA	GACATTGTGATGACCCAGACTCCAGCTCCGTGAGGC AGCTGTGGAGGCAGTCACTCAAGTCCGACGGCCA GTCAAGTATTAGTAGTACTTATCTGGTATCAGCAGA AACCAGGACGCCCTCCAAAGCTCTGATCTACAGGCCA TCCACTCTGGCAATCTGGGTTCCCATCGCGGTTACAGGC AGTGAATCTGGGACACAGTCTCACTCTCACCATCAGCCA CCTGAGTGTGCCATGTGCCACTTATTACTGTCAAAG CTATGATGATAGTAGTGAATAATTTTTTTTATGTTTCG TCTCTCAGCGAGGACCGAGGTGGTGGTCAGA (SEQ ID NO: 308) DIVMTQTPASVEAAVGTVTIKQASQSISSYLSWYQOKP GQPPKLLIYRASTLIASGVPFRFGSGSGTQFTLISLECAD AATYYCQSYDDSDNFFYFGFGGTEVVVR (SEQ ID NO: 310)	CAGTCGTTGGAGGAGTCCGGGGAGGCCTGGTCCAGCCTGAGGAT CCCTGACACTCACCTGCACAGCTTTTGGATTCCCTCAATAATATT ATATTGTGGGTCCCGCAGCTCCAGGGAAGGACGTGAGTGGAT CGCATGCAATCACATCTCTAATGCTAGGACTTACTACGCGAGCTGG CGAAAGCGGATTCACCATCTCCAAAACCTCTGTCGACACCGTGACTC TACAAATGACAGTCTGACAGCCGCGACACGCCACCTATTCTGTG CGAGTCATTCTCTACTCCCTTGTGGGGCCAGGCACCCCTGGTCAOCG TCTCTCA (SEQ ID NO: 309) QSLBESGGGLVQPSGLTLTCTARFGLTNVYICWVRQAPKGLEWIA IDNANGRIYASWAKGRFTISKTSSTTVTLQMTSLTAADTATYFCARSL STPLWGPGLTVTSS (SEQ ID NO: 311)	
4A15	NA	GACATTGTGATGACCCAGACTCCAGCTCCGTGAGGC AGCTGTGGAGGCAGTCACTCAAGTCCGACGGCCA GTCAAGTATTAGTAGTACTTATCTGGTATCAGCAGA AACCAGGACGCCCTCCAAAGCTCTGATCTACAGGCCA TCCACTCTGGCAATCTGGGTTCCCATCGCGGTTACAGGC AGTGAATCTGGGACACAGTCTCACTCTCACCATCAGCCA CCTGAGTGTGCCATGTGCCACTTATTACTGTCAAAG CTATGATGATAGTAGTGAATAATTTTTTTTATGTTTCG TCTCTCAGCGAGGACCGAGGTGGTGGTCAGA (SEQ ID NO: 312) DIVMTQTPASVEAAVGTVTIKQASQNISSYLAWYQOKP GQPPKLLIYRASTLIASGVPFRFGSGSGTQFTLISLECAD ATYYCQSYDDSDNFFYFGFGGTEVVVK (SEQ ID NO: 314)	CAGTCGTTGGAGGAGTCCGGGGAGGCCTGGTCCAGCCTGAGGAT CCCTGACACTCACCTGCACAGCTTCTGAGTCACCCTCAGTAACTAT TATATTGTTGGGTCCCGCAGGCTCCAGGGAAGGGCTGGAGTGGAT CGGGTGCAATGACAAATGATTAATGGTAGGACCTACTACGCGAGCTGG CGAAAGCCGATTCACCATCTCCAAAGCTCTGTCGACACAGGACTC TACAAATGACAGTCTGACAGCCGCGACACGCCACCTATTCTGTG CGAGTCCTGGCTACTCCCTTGTGGGGCCAGGCACCCCTAGTCAOC GTCTCTCA (SEQ ID NO: 313) QSLBESGGGLVQPSGLTLTCTASGVTLSNVYICWVRQAPKGLEWIG CIDVNGRTIYASWAKGRFTISKTSSTTGLTQMTSLTAADTATYFCAR SLATPLGPGPLTVTSS (SEQ ID NO: 315)	
	NA	GACATTGTGATGACCCAGACTCCAGCTCCGTGAGGC AGCTGTGGAGGCAGTCACTCAAGTCCGACGGCCA GTCAAGCAATTAAGTACTTACCTGGTATCAGCAGA AACCAGGACGCCCTCCAAAGCTCTGATCTACAGGCCA TCCACTCTGGCAATCTGGGTTCCCATCGCGGTTACAGGC AGTGAATCTGGGACACAGTCTCACTCTCACCATCAGCCA CCTGAGTGTGCCATGTGCCACTTATTACTGTCAAAG CTATGATGATAGTAGTGAATAATTTTTTTTATGTTTCG TCTCTCAGCGAGGACCGAGGTGGTGGTCAGA (SEQ ID NO: 316) DIVMTQTPASVEAAVGTVTIKQASQNISSYLAWYQOKP GQPPKLLIYRASTLIASGVPFRFGSGSGTQFTLISLECAD ATYYCQSYDDSDNFFYFGFGGTEVVVK (SEQ ID NO: 318)	CAGTCGTTGGAGGAGTCCGGGGAGGCCTGGTCCAGCCTGAGGAT CCCTGACACTCACCTGCACAGCTTCTGAGTCACCCTCAGTAACTAT TATATTGTTGGGTCCCGCAGGCTCCAGGGAAGGGCTGGAGTGGAT CGGGTGCAATGACAAATGATTAATGGTAGGACCTACTACGCGAGCTGG CGAAAGCCGATTCACCATCTCCAAAGCTCTGTCGACACAGGACTC TACAAATGACAGTCTGACAGCCGCGACACGCCACCTATTCTGTG CGAGTCCTGGCTACTCCCTTGTGGGGCCAGGCACCCCTAGTCAOC GTCTCTCA (SEQ ID NO: 317) QSLBESGGGLVQPSGLTLTCTASGVTLSNVYICWVRQAPKGLEWIG CIDVNGRTIYASWAKGRFTISKTSSTTGLTQMTSLTAADTATYFCAR SLATPLGPGPLTVTSS (SEQ ID NO: 319)	

TABLE 23-continued

Standard IgG Antibody Variable Region Sequences		
Ab	Type	LC V-region
1J16	NA	TCACCTCTGGCATCTGGGTCCCATCGCGTTCAAAGC AGTGGATCTGGGACAGAGTCACTCTCACCATCAGCGA CCTGGAGTGGCGATGCTGCCACTTACTACTGTCAAAG CATATGATGATGATGAGTATTATTTTTTTATGCTTTTCG GGGAGGACCCGAGGTGGTGGTCAAA (SEQ ID NO: 316) DIVMTQTPASVEAAVGGTVTIKQASQNSISYLAHWQK PGOPKLLIYRASTLASGVSPFKSGSGTEFTLTISDLECAD AATYYCQSYDSDRSISFFYAFGGTEVVVK (SEQ ID NO: 318)
1J16	NA	GACATTGTGATGACCCAGACTCCAGCTCCGTGGAGGC AGCTGTGGAGGCACAGTCACCATCAAGTCCGAGGCCA GT CAGAGCATTAGTACTACTTAGCTGTATCACCAGA AACCAGGCGAGCTCCCAAGCTCTGATCTATGCTGCATC CACTCTGGCATCTGGGTCCCATCGCGTTTCAAAGCA CTGGAGTGTCCGATGCTGCCACTTACTACTGTCAAAGC GTGGATCTGGGACACAGTTCACTCTCACCATCAGCGAC TATGATGATAGTAGGAGTAGTATTTTTTTATGCTTTC GGCGAGGAGCCGAGGTGGTGGTCAAA (SEQ ID NO: 320) DIVMTQTPASVEAAVGGTVTIKQASQNSISYLAHWQK GPPKLLIYAASTLAS GVSPREGSGGTQFTLTISDLECADAAATYYCQSYDSDRS SFFYAFGGTEVVVK (SEQ ID NO: 322)
1C18	NA	GACATTGTGATGACCCAGACTCCAGCTCCGTGGAGGC AGCTGTGGAGGCA CAGTCACCATCAAGTCCAGGCCAGTCAGAGCATTAGC AGTTGTTATCTGG TATCAGCAGAAAC CAGGCGAGCTCCCAAGCTCTGCTCT ACAGGGCAACCA CTCTGGCATCTGGGTCCCATCGCGTTCAAAGCAGT GGATCTGGGACACA GTTCACTCTCACCATCAGCGACCTGGAGTGTGCCGATG CTGCCACTTATTA GTCAAAGTTATGATGATAGTAGTGTAGTAAATTTTTTTA TGCTTTCCGCGGAG GGACCGAGGTGGTGGTCA (SEQ ID NO: 324) DIVMTQTPASVEAAVGGTVTIKQASQNSISYLAHWQK GPPKLLIYRASTLAS GVSPREGSGGTQFTLTISDLECADAAATYYCQSYDSDRS SNFFYAFGGTEVVVK (SEQ ID NO: 326)
7H4	NA	CAAGCGTGGTGACCCAGACTCCATCGTCCGTGTCTGC AGCTGTGGAGGCGAGTCACCATCAGTTCGCGAGTCA GT CAGAGTGTATTATAGTAACACCTCTTATCTTGGTATC

Standard IgG Antibody Variable Region Sequences

LC V-region

HC V-region

GAAAGCCGATTCACCATCTCCAAAGGCTCGTCGACACAGGAGCTC
TACAAATGACAGTCTCAGACCGCGACACGCCACCTATTCTGTG
CGAGTCCCTTGGGTACTCCCTTGTGGGGCCAGGACCCCTGGTCACC
GTCTCTCTCA
(SEQ ID NO: 317)

QSLSESGGLVQPEGSLLTCTASGVTLTNYYICWVRQAPKGLEWIG
CIDNVNGRTYYASWAKGRFTISKASSTTGLQMTSLTAADTATYFCAR
SLATPLWGPGLVTVSS
(SEQ ID NO: 319)

CAGTCGTTGAGGAGTCCGGGGAGGCTGGTCCAGCCTGAGGAT
CCCTGACAGTCACCTGACAGCTTTTGGAGTCACCTCACTAACTACT
ACATCTGCTGGTCCGCCAGGCTCCAGGGAAGGCTGGAGTGGATC
GGGTGATTCACATATATATGTTAGTACTACCGAGCTGGGC
GAAAGCCGATTCACCATCTCCAAAGCTCGTCGACACAGGAGCTC
TGAAATGACAGTCTCAGACCGCGACACGCCACCTATTCTGTG
CGAGTCCCTTGGGTACTCCCTTGTGGGGCCAGGACCCCTGGTCACC
GTCTCTCTCA
(SEQ ID NO: 321)

QSLSESGGLVQPEGSLLTCTASGVTLTNYYICWVRQAPKGLEWIG
IDNNGR
TYTASWAKGRFTISKSTTGLQMTSLTAADTATYFCARSLATPLWGP
GLVTVSS
(SEQ ID NO: 323)

CAGTCGTTGAGGAGTCCGGGGAGGCTGGTCCAGCCTGAGGAT
CCCTGAC
ACTCACCTGCACAGCTTCTGGATTCACTCACTAACTATTATATATG
TTGGGT
CCGCCAGGCTCCAGGGAAGGAGTGGAGTGGATCGCATGTATTGAT
AAGCTA
ATGTAGGACTTACTACGCGACCTGGGCGAAAGGCCGATTCACCATC
TCCAAA
ACCTCGTCGACACGCGTACTCTGCAAAATGCCAGTCTGACAGCCGG
GACAC
GGCCACCTATTCTGTGCGAGGTCAITGTCTACTAACTTGTGGGGCCCA
GGCAC CCTGTCACCGCTCTCTCA
(SEQ ID NO: 325)
QSLSESGGLVQPEGSLLTCTASGVTLTNYYICWVRQAPKGLEWIG
IDNNGR
TYTASWAKGRFTISKSTTGLQMTSLTAADTATYFCARSLATPLWGP
GLVTVSS
(SEQ ID NO: 327)

CAGTCGTTGAGGAGTCCGGGGAGGCTGGTCCAGCCTGAGGAT
CCCTGACACTCACCTGCAAGCTCTGGGTGAGACTTCAGAAGCTAC
TACTACATATGCTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGT

TABLE 23-continued

Standard IgG Antibody Variable Region Sequences		
Ab	Type	LC V-region
1H24	NA	AGCAGAAACAGGCGAGCCTCCCAAGCTCCTGATCTAC AAGGCATCAGCTCTGGCATCTGGGTCCTCATCGCGTTG AAGGACAGTGATCTGGGACACAGTTCACTCTCACAAT CAGCGAAGTACAGATGTGACGATCGTCCCACTATTACTG TCAAGGCTATTATAGTGGTGTATTATTATATGTCGCGG AGGACCGAGGTGGTGGTCAA (SEQ ID NO: 328)
		AA QAVVTCPTSSVAAGVGTITISQSSQSVSYNNLLSWYQK PGPPKLLIYKASTLASGVPSRLKSGSGTQFTLTISEVQCD DAATYYCQYISGVIIYMFGGGTEVVVK (SEQ ID NO: 330)
1C3	NA	GACATTGTGATGACCCAGACTCCATCTCCGTGGAGGC AGCTGTGGGAGCAGTCACTCACCATCAAGTGCCAGGCCA GTCAGAGCATTAGTAGTTACTTATCTCTGGTATCGGCAGA AACCAGGCGAGCTCCCAAGCTCCTGATCTACAGGGCA TCCACTCTGCATCTGGGTCCCATCGCGTTCAAAGGC AGTGATCTGGGACACAGTTCATCTCACCATCAGCGA CTTGGAGTGTGGCGATGCTGCCATTACTACTGTCAAAG CTATGATGATAGTAGTAGTAATAATTTTTTATGGTTTCG GGGAG GAGCAGGTGGTGGTACA (SEQ ID NO: 332)
		AA DIVMTQTPSVFAAVGGTVTIKQASQSISSYLSWYRQKP GQPPKLLIYRASTLASGVPSRPFKSGSGTQFTLTISDLECAD AATYYCQSYDDSSNNFFYFGGGTEVVVR (SEQ ID NO: 334)
5H20	NA	GACATTGTGATGACCCAGACTCCATCTCCCTGTGTCTGC AGCTGTGGGAGCAGTCACTCACCATCAAGTGCCAGGCCA GTCAGAACATTAGTAGCTACTTAGCTGTGATCAGCAG AAACAGGCGAGCTCCCAAGCTCCTGATCTACAGGGC ATCCACTCTGGCATCTGGGTCCCATCGCGTTCAAAG GCAGTGATCTGGGACACAGTTCATCTCACCATCAGCG ACCTGGAGTGTGGCGATGCTGCCATTACTACTGTCAAAG CTATGATGATAGTAGTAGTAATAATTTTTTATGGTTTCG GGGAGGACCGAGGTGGTGGTCAA (SEQ ID NO: 336)
		AA DIVMTQTPSVFAAVGGTVTIKQASQSISSYLSWYRQKP GQPPKLLIYRASTLASGVPSRPFKSGSGTQFTLTISDLECAD AATYYCQSYDDSSNNFFYFGGGTEVVVK (SEQ ID NO: 338)
5H20	NA	GGCAGAGCGCTGACCCAGACTCCATCTCCCTGTGTCTGC AGCTGTGGGAGCAGTCACTCACCATCAAGTGCCAGGCCA GTGAGAGCGCTTTATATACAACTAGTTAGCTGTGATC AGCAGAAACAGGCGAGCTCCCAACTCCTGATCTAT GATGATCAGCTCTGGCATCTGGGTCCCATCGCGGTTC AAGGACATGGATCTGGGACACAGTTCACCTCTCACAATCA (SEQ ID NO: 339)
		AA QSVLESGGLVQPEGLVLTCTAGVTLTNYIICWVRQAPGKLEWV CINANGRTYYASWAKGRFTISKTSITGLQMTSLTAADTATYFCARS LATPLWGPGLVTVSS (SEQ ID NO: 339)
1H24	NA	GGGTCCGATGATTGTGTGGTGAATAACGATGCCACTGCTACGGGA GGTGGCGAAAGCGGATTCACCATCTCCAAAACCTCGCGCACCGG TGGCTCTCAAATGACAGCTGTACAGCCCGGACGCGCACCTATT TCTGTGTGAGAGCGGTCCTGATAAATAGTCCAATTTAACTTGTGGG GCCAGGCACCTGGTCAACCGTCTCCTCA (SEQ ID NO: 329)
		AA QSVLESGGLVQPEGLVLTCTAGVTLTNYIICWVRQAPGKLEWV CINANGRTYYASWAKGRFTISKTSITGLQMTSLTAADTATYFC VRGPDNNVQFNLWGPGLVTVSS (SEQ ID NO: 331)
1C3	NA	CAGTCTGTGAGAGTCCGGGGAGGCTCGTCCAGCCTGAGGAT CCCTGACACTCACTGCACAGCTTTGGAGTCACTCACTAATAATTA TATTGTGGGTCCGCGAGGCTCCAGGAAGGAGCTGGAGTGGATCG CATGCAATTGACATAGTAATAGTACGACTTACGCGAGCTGGCG AAAGCGGTTTCACTCCAAAACCTCGTGCACCGACGCTGCTG CAATGACAGTCTGACAGCGCGGACACGGCCACCTATTCTGTGCG AGTCAATTGTCTACTCCCTTTGGGGCCAGGACCCCTGTGTCACCGTC TCCTCA (SEQ ID NO: 333)
		AA QSVLESGGLVQPEGLVLTCTAGVTLTNYIICWVRQAPGKLEWV IDNSNRTYYASWAKGRFTISKTSITGLQMTSLTAADTATYFCARS STPLWGPGLVTVSS (SEQ ID NO: 335)
1C3	NA	CAGTCTGTGAGAGTCCGGGGAGGCTCGTCCAGCCTGAGGAT CCCTGACACTCACTGCACAGCTTTGGAGTCACTCACTAATACTA CATCTGTGGGTCCGCGAGGCTCCAGGAAGGAGCTGGAGTGGTCTG GGTGTATTGACATAGTAAATGAGTACGACTTACGCGAGCTGGCG AAAGCGGATTCACCATCTCCAAAACCTCGTGCACCGACGAGGACTCT GAGTCTCTGCTACTCCCTTTGGGGCCAGGACCCCTGTGTCACCG GCAATGACAGTCTGACAGCGCGGACACGGCCACCTATTCTGTGCG TCTCTCA (SEQ ID NO: 337)
		AA QSVLESGGLVQPEGLVLTCTAGVTLTNYIICWVRQAPGKLEWV CINANGRTYYASWAKGRFTISKTSITGLQMTSLTAADTATYFCARS LATPLWGPGLVTVSS (SEQ ID NO: 339)
5H20	NA	CAGTCTGTGAGAGTCCGGGGAGGCTCGTCCAGCCTGAGGACCC CTGACACTCACTGCACAGTTCCTGGAGTCACTCACTAGTAGCAGTGA GTGACTGGGTCCGCGAGGCTCCAGGATGGGACTGGATACATCGGA TTCTCTCAAGCTGGGATGTTAGGACATACGCGAGCTGGCGGAAA GGCGGATTCACCATCTCCAAAACCTGTGACCACTGGATCTGAAA ATGACAGTCTCACAAACCGAGGACACGGCCACCTATTCTGTGCGCAGA (SEQ ID NO: 339)
		AA QSVLESGGLVQPEGLVLTCTAGVTLTNYIICWVRQAPGKLEWV CINANGRTYYASWAKGRFTISKTSITGLQMTSLTAADTATYFCARS LATPLWGPGLVTVSS (SEQ ID NO: 339)

Ab	Type	LC V-region	HC V-region
1H14	AA	GCGAAGTCAGTGTGACGATGCTGCCATTACTACTGTCA AGGCTATTATCAAACTAGTGTTGGGCTTTGCGGGAGG GACCG AGGTGGTGTCTCAA (SEQ ID NO: 340) AQAULTQTPSPVSAVGGTGTITSCOSSBSVNNHMLGWYQQ KPGPPKLLIYDASTLASGVPSPFKSGSGTQFTLTISQCD DAALYYCGYQTQSVWAFGGTEVVK (SEQ ID NO: 342)	CATAAGGGTATAGTTAGTGCTTAAGTGTGGGGCCAGGCACCTT GGTCAACGCTCTCCTCA (SEQ ID NO: 341) QSVESGGRLVTPGTPTLTLCVTSVGLSSAVTWVRAQPMGLEIYGF QAGDGSAYASWAKGRFTISKTSSTTVDLKMTSLTTEDTATYFCARHK NSYVPLWPGTLTVSS (SEQ ID NO: 343)
	NA	GATGTTGTGATGCCAGACTCCAGCTCCGTGGAGGC AACTGTGGAGGCACAGTCAACATCAAGTCCAGCTCA GTCAGTGTATATGATGAACAATCTTTAGCTGTGATC AGCAGATGCAGGACAGCGTCCAGACTCCTGTATCTAT GTGCTACCACTCTGGATCTGGGTTCCATCGGGTTCA GTGCAGTGGATCTGGACAGAGTCTCTCCACATCAG CGACCTGAGTGTCCGATGACAGCACTTACTACTGTCA ATGTACTTATTATGTATGATGTTATCAAAATGATTTCCG CGGAG GGACCGAGTGGTGTCTCAA (SEQ ID NO: 344)	CAGTCGGTGGAGGAGTCCGGGGTCCGCTGGTCAAGCCTGGACACCC CTGACACTCACTGCACAATCTTGGATCTTCCCTCAGTACTGATCAAA TGAAGTGGTCCCGCAGGCTCCAGGAAGGGCTGGAATGATCGGA AGCATTTGGTGGTGGTGGTAGCGCAGTCTACGCGAGCTGGCGAAAGC CGATTCACCATCTCCAAAACCTCCACACGGTGGATCTGAGAATCAAC AGTCCGACACCGAGGACAGCCATGATTTCTTGTGGCAGGGGATT TTATAGTATAGACTTGTGGGGCCAGGCACCTGGTCAACCGTCTCTCT CA (SEQ ID NO: 345)
1H17	AA	DVVMTQTPASVEATVGGTVTIKQSSQSVDDNNALAWYQ QNAGORPLLIYGASLTIASGVPSRFSAGSGTEPTLTISDLEC ADAATYYCQTYVSYSSQNDFGGGTEVVK (SEQ ID NO: 346)	GGGGSAYVASWAKGRFTISKTSSTTVDLRI TSPPTEDTAMYPGGRFYSI QSVESGGRLVTPGTPTLTCTISGFSLSVAMSWVRQAPGKLEWIGSI DLWPGTLTVSS (SEQ ID NO: 347)
	NA	GCCGCCGTGTCGCCAGACTCCATCTCCCGTGTCTGCAG CTGTGGGAGCACAGTACAGCCGCTTGTGCCAGTCCAGTA AGAGTGTATTAATAAGAACTGTGGTATCTCTGGTTTCAGC AGAAACAGGCGACGCTCCCAAGCTCTGATCTATGGTG CATCATCTCGCATCTGGGTCCCATCGGGTTTCAAAG CAGCTGATCTGGGACATGCTGACACTCTCAACATCAGCG ACGTGCAGTGTGACGATGCTGCCACTTACTACTGTGCAG GCGGTATTAGTAGTAGTAGTATACATTTGCTTTTCGGCG GCGG ACCGAGGTGGTGTCAAG (SEQ ID NO: 348)	CAGGAGCAGCTGGTGGAGTCCGGGGAGGCGCTGGTCAAGCCTGGGG CATCTCTACACTCACTGCAAAAGCTCTGGAATCTCTTCAGTAGCGG CCAATCATGTGCTGGGTCCGCGAGCTCCAGGAAGGGCTGGAGT GGATCGCATGCAATTTGGTCTGGTAGTAATGCTATTAGCACTTCTAC CGAGCTGGGCGCAAGCGCAATCACCATCTCCAAATCTCTGTCGA CCAAGTCACTGCTGCAATTCACCACTGACGCGCGGACACGCGCC ACCATTTCTGTGCGAGAGTGGGCTCCGATGACTATGGTGACTCTGAT GTTTTTGATCTCCTGGGCGCCAGCACCTGGTCAACCGTCTCCTCA (SEQ ID NO: 349)
7B20	AA	AAVLTQTPSPVSAVGGTGTVSASCOSSKSVNNKMLSWPQ QKPGPKLLIYGASLTIASGVPSRFSKSGSGTQFTLTISDV QCDDAATYYCAGGYSSSSDITPAFGGTEVVK (SEQ ID NO: 350)	QEQLVESGGGLVHPGASLTLTCKASGFSFSSGQLMCWVRQAPKGLE WIACTGGSNAISTFYASWAGRFTISKSSSTTVTLQLTLTAADTATYF CARVSDDYGDSVDFPDWPGPGLTVSS (SEQ ID NO: 351)
	NA	GCCCAAGTGTGACCCAGACTCCAGCTCCGTGTCTGC ACTGTGGGAGCACAGTCAACATCAAGTCTCCAGTCCAGTCAG TCAGAGCTTTATAGTACGACACCTCTTATCTCCGTGATCAG GTCGATGATTTGGTGTGTTATCGCTTGCCACTCCCTACCGCA	CTGACACTCACTCAAAAGCCTCTGGTCCAGACTCAGTACGCGCTA TCAATATGCTGGATCCCGCAGGCTCCAGGAAGGGCTGGAGTGG GTACGATGATTTGGTGTGTTATCGCTTGCCACTCCCTACCGCA

TABLE 23-continued

Standard IgG Antibody Variable Region Sequences		
Ab	Type	LC V-region HC V-region
7J6	AA	CAGAAACAGGGCAGCCTCCCAAGCTCTGATACAA GGCATCCACTCTGGCATCTGGGTCCCATCGCGTTCAAA GGCAGTGGATCTGGGACACAGTTCACTCTCAAAATCAGC GAACATGAGTGTGACATGCTGCCACTTATTACTGTCAA GGCTACTATAGTGTGTGTGTTATATTTTCGGCGGAGGG ACCGAGTGGTGGTCAAG (SEQ ID NO: 352) AÖVLTQTPASVAAVGGTVTISCQSSQSVYSSDLLSWYQQ KPGQPPKLLIYKASTLASGVPSRFKSGSGTQFTLTISELQC DDAATYCYGYSGVVYIFGGTEVVVK (SEQ ID NO: 354)
	NA	GCCCAAGTGTGACCCAGACTCCAGCTCCGTGTCTGC AGCTGTGGAGGCACAGTCACCATCACTTGCAGTCCAG TCAGACGGTTTATAGTAGCGACCTCTTATCTCGGTATCAG CAAAACCCAGGCGAGCCTCCCAAGCTCTGATACAA GGCATCCACTCTGGCATCTGGGTCCCATCGCGTTCAAA GGCAGTGGATCTGGGACACAGTTCACTCTCAAAATCAGC GAAGTACAGTGTGACATGCTGCCACTTATTAATCTGTCAA GGCTACTATAGTGTGTGTGTTATATTTTCGGCGGAGGG ACCGAGTGGTGGTCAAG (SEQ ID NO: 356)
1D2	AA	AÖVLTQTPASVAAVGGTVTISCQSSQSVYSSDLLSWYQQ KPGQPPKLLIYKASTLASGVPSRFKSGSGTQFTLTISEVQC DDAATYCYGYSGVVYIFGGTEVVVK (SEQ ID NO: 358)
	NA	GACATTGTGATGACCCAGACTCCATTCCTCCGTGGAGGC AGCTGTGGAGGCACAGTCACCATCAAGTCCAGGCGCA GTCAGAGCATTAGTAGTTACTTATCTTGTTATCAGCAGA AACCAGGCGAGCCTCCCAAGCTCCTGATCTACAGGCGAT CCCTCTGCGCATCTGGGTCCCATCGCGTTCAAGGCGAG TGATCTGGGACACAGTTCACTCTCACCATCAGCGACCTG GAGTGTCCGATGCTGCCACTTACTACTGTCAAAGCTAC GATGATAGTAGTAGTAATAATTTTATGTTTTCGGC GGAGGACCGAGTGGTGGTCAGA (SEQ ID NO: 360)
4G8	AA	DIVMTQPFVSFAAVGGTVTIKQASQSISSYLSWYQQKP GPPKLLIYRASPLASGVPSRFKSGSGTQFTLTISLECAD AATYCYQSDSSNNFFYFGGGTEVVVR (SEQ ID NO: 362)
	NA	GCCGCGTGTGACCCAGACTCCATCTCCGTGTCTGCAG CTGTGGAGGCACAGTCACCATCAATTCAGTCCAGTCAAGTA AGAGTGTATTATTAAGAATCTGGTTATCTCGTTTCAGC AGNAACAGGCGAGCCTCCCAAGCTCCTGATCTATGGTG CATCCACTCTGGCATCTGGGTGCCATCGCGTTCAAG

7J6	NA	CTGGCGAAAGGCGGATTCACCATCTCAAAACCTCGTCGACCAAGG CAGTCGTTGGAGAGTCCGGGGAGGCTGCTCAAGCCTGGGCATCC TGACTCTGCAATGACACAGTCTGACAGCCGCGACAGCCACTTATT TCTGTGTGAGAGCGGTCTCGATAATAATGTCCAATTTAACTTGTGG GGCCAGGCAACCTGGTCAACCGTCTCCTCA (SEQ ID NO: 353) QSLSEGGGLVKPASLTLTKASGDLSSAYYICWIRQAPGKLEWVA CIGGVNRVATAYATWAKGRFTISKTSSTTVTLQMTSLTAADTATYFCVR GGPDNNVQFNLMGPGTLVTVSS (SEQ ID NO: 355)
	NA	CAGTCGTTGGAGAGTCCGGGGAGGCTGCTCAAGCCTGGGCATCC CTGACACTCACCTGCAAAAGCCTCTGGGTCAAGCTCAGTAGTACTA CTACAGTTGCTGATCCGCCAGGCTCCAGGGAAGGGCTGGAGTGG GTGCAATGTTGGTGGTGTAAATCGCGATGCCACTGCTTACCGACCC TGGCGAAAGGCGGATTCACATCTCCAAAACCTCGTCGACCCAGGTG ACTCTGCAATGACAGTCTGACAGCCGCGGACACGGCCACTTATTTC TGTGTGAGAGCGGTCTGATAATAATGTCCTCAATTTAACTTGTGGGG CCAGGCAACCTGGTCAACCGTCTCCTCA (SEQ ID NO: 357) QSLSEGGGLVKPASLTLTKASGDLSSAYYICWIRQAPGKLEWV ACVGGVNRDATAYATWAKGRFTISKTSSTTVTLQMTSLTAADTATYFC VRGPDNNVQFNLMGPGTLVTVSS (SEQ ID NO: 359)
1D2	NA	CAGTCGTTGGAGAGTCCGGGGAGGCTGCTCAAGCCTGAGGAT CCCTGACACTCACTGCACAGCTTCTGGATTCACCTCAATAATATT ATATTGTTGGTCCGCCAGGCTCCAGGGAAGGACTGGAGTGGATC GCATGATGACAAATGTTAATGTAGGACTTACTACGAGCTGGGC GAAGGCGGATTCACATCTCCAGAACCTCGTCCGACCGGTGACTCT GCAAAACAGCAGTCTGACAGCCGCGGACACGCCACCTATTCTGTGC GAGTCAATGCTCTACTCCCTTTGTGGGCCACGACCCCTGGTCACCGT CTCCTCG (SEQ ID NO: 361) QSLSEGGGLVKPEGSLTLTKASGFTLNYYIICWVRQAPGKLEWVAC IDNVNGFTYYASWAKGRFTISRSTTVTLQMTSLTAADTATYFCARSL STPLWGPGLVTVSS (SEQ ID NO: 363)
	NA	CAGGACAGTGGTGGAGTCCGGGGAGGCTGCTCAAGCCTGAGG GATCCCTGACACTCACTGCAGCTTCTGGATTCTCTTCACTAGCA GCTACTTCACTGCTGGGTCCGCCAGGCTCCAGGGAAGGGCTGGAG TGGATCGCATGCATGCTGTTGGTAGTAGTACACTTACTACGC GAGTGGCGGAAAGGCGGATTCACCATCTCCAAAACCTCGTCCGACCA

TABLE 23-continued

Standard IgG Antibody Variable Region Sequences		
Ab	Type	LC V-region
714	NA	GCAGTGGATCTGGGACACAGATTCACTCTACCATCAGCG ACCTGCAGTGTGACGATGTTGCCACTTACTACTGTGCAGG CGGTTATAGTAGTAGTATGATACGTTTCTCTCGCGG AGGACCCGAGGTGGTGGTCAAA (SEQ ID NO: 364)
		AAVLQTSPVSAAVGGTVTINCSQSSKSVYNKNLWSFQ QKPGQPKLLIYGASTLASGVPSPKSGSGTQFTLTISDV QDDVATYYCAGYSSSDTFAPGGTEVVVK (SEQ ID NO: 366)
714	NA	GCCCAAGTGCTGACCCAGACTCCAGCCTCGGTGCTGCG AGCTGTGGAGGCACAGTCACCATCAGTTGCCAGTCCAG TCAGAGCCTTTTATAAGCAGCCTCTTATCTGGTATCAG CAAAACCCAGGCAGCCTCCCAAGCTCTGATCTACAA GGATCCACTCTGGCATCTGGGTCCTCCATCGCGGTTCAA GGCAGTGGATCTGGACACAGTTCACTCTCACAATCAGC GAAGTACAGTGTGACGATGCTGCCACTTATTACTGTCAA GGCTATAGTGGTGTGGTTTATATTTTCGCGGAGGG ACCG AGGTGGTGGTCAAG (SEQ ID NO: 368)
		AAVLQTSPASVAAG GVTISCSQSSQLYNSDLLSWYQKPGQPKLLIYKASTLA SGVPSRFKSGSGTQFTLTISEVQDDAATYYCQGYSGVW YIPGGTEVVVK (SEQ ID NO: 370)
1123	NA	GCGCGCGTGTGACCCAGACTCCATCTCCGCTGTGCGAG CTGTGGGAGGCACAGTCAGCATCAGTTGCCAGTCCAGTA AGAGTCTTTTATAAGAACTGGTTATCTGTTCTCAGC AGAAACAGGCAGCCTCCCAAGCTCTGATCTATGGTG CATCCACTCTGGCATCTGGGTCCTCCATCGCGGTTCAAAG GCAGTGGATCTGGACACAGTTCACTCTCAGCATCAGCG ACGTGCAGTGTGACGATGCTGCCACTTACTACTGTGCGAG GCGGTTATAGTAGTAGTAGTATGATACATTTGCTTTCCGGC GAGGG ACCGAGGTGGTGGTCAAA (SEQ ID NO: 372)
		AAVLQTSPVSAAVGGTVSISQSSKSVYNKNLWSFQ QKPGQPKLLIYGASTLASGVPSPKSGSGTQFTLTISDVQ CDDAATYYCAGYSSSDTFAPGGTEVVVK (SEQ ID NO: 374)
1J13	NA	GATGTTGTGATGACCCACACTCCAGCCTCCCTCTTTGAAA CTGTGGGAGGCACAGTCACATCAGTGCAGGCCCTTC AGAGTCTTTATGATAACAATGCTTTATCTGGTATCAACA AAATGCAGACAGCGTCCCATACTACTCTGATCTATGGTGCA

HC V-region
CGGTACTCTGCAAAATGACCAGTCTGACGCCGCGGACACGGCCACC TATTTCTGTGCGAGAGTGGCTACGATGACTATGGTGACTCTGATGC TTTTGATCCCTGGGCCAGGACCCCTGGTCACCGTCTCTCTCA (SEQ ID NO: 365)
QEQLVESGGGLVQPEGLTLITCAASGFSFSSSYFMCWVRQAPKGLEWI ACTAVSSGSITYYASNAKGRFTISKTSSTTVTLQMTSLTAADTATYFCA RVGYDDYDSDADFDPPWGPGLVTVSS (SEQ ID NO: 367)
CAGTCGTTTGAGGAGTCCGGGGAGGCGCTGGTCAAGCCTGGGGCATCC CTGACACTACCTGCAAGCCTCTGGGTCAGACTCAGTAGGTACTA CTACAGTTGCTGATCCCGCAGCTCCAGGAAAGGGCTGAGTGG GTCCGATGTTGGTGGTGTATTCGCGATGCCACTGCCCTACCGGACC TGGCGAAAGGCCGATTCACCATCTCCAAAACCTGTCGACCCAGGTG ACTCTCAAAATGACAGCTGTGACAGCCGCGACACGGCCACTTATTC TGTTGTGAGAGGCGTCTTGATATAATGTCCAATTAACTTGTGGGG CCGAGGCCCTGGTACCGTCTCTCTCA (SEQ ID NO: 369)
QSLSESGGLVKGASLTLTCKASGSDLRYYSYCWIRQAPKGLEWV ACVGGVNRDATATATWAKGRFTISKTSSTTVTLQMTSLTAADTATYFC VRGPDNNVQFNLWGPGLVTVSS (SEQ ID NO: 371)
CAGGAGCAGCTGGTGGAGTCCGGGGAGGCGCTGGTCCAGCCTAAGG GATCCCTCAGACTACCTGCGCAGCTTCTGGATTCTCTTCAGTAGCA GCTACTTTCATGTCTGGGTCGCCAGGCTCCAGGAAAGGGCTGGAG TGGATCCCATGCAATGGTCTGGTAGTAGTGTATTAGCACTTCTTAC GCGAGCTGGCGCAAGCGCATTCACCATCTCCAAAACCTCGTCGA CCACGGTGACTCTGCAAAATGACAGCTGACAGCCGCGGACACGGC CACTATTCTGTGCGAGAGTGGGTACGATGACTATGGTGAATCTG ATGCTTTTGATCTCCTGGGGCCAGGACCCCTGGTCAACCTCTCTCTCA (SEQ ID NO: 373)
QEQLVESGGGLVQPKSLTLITCAASGFSFSSSYFMCWVRQAPKGLEW IACIGSSSAISTFYASNAQGRFTISKTSSTTVTLQMTSLTAADTATYFCA RVGYDDYDSDADFDPPWGPGLVTVSS (SEQ ID NO: 375)
CAGTCGCTGGAGGAGTCCGGGGTCCGCTGGTAAACGCTGGGACACC CTGACACTACCTGCAAGCTCTGGAAATCGACTCAGTAGTATGGA ATGACTTTGGTCCCGCAGGCTCCAGGAAAGGGCTGGAGTGGATCGG ATACATTTGACTGATGGGAGGACATACTACGCAAACTGGGCGAAAG

TABLE 23-continued

Standard IgG Antibody Variable Region Sequences		
Ab	Type	LC V-region
		HC V-region
1G24	NA	TCACCTCTGGCATCTGGGGCCCAATCGCGGTTCAAGTGC AGTGGATCTGGGACAGATTTCACCTCAACATCATCGAC CTGGAGTGGCCGATGCTCCACTTACTACTGTCAATGT ACTTATATGTAGTAGTATCAAAATGATTGGCGGA GGG ACCGAGGTGGTGGTCAA (SEQ ID NO: 376)
		QSLVESGRLVTPPTPLTLTCTVSGIDLSYGMWVRQAPGKGLEWIGY IWDGRTYYANWAKGRFTISKSTTVDLKITSPTEADTATYFCARPPDG NYRDIWPGTLVTVSL (SEQ ID NO: 377)
1G24	NA	QSLVESGRLVTPPTPLTLTCTVSGIDLSYGMWVRQAPGKGLEWIGY IWDGRTYYANWAKGRFTISKSTTVDLKITSPTEADTATYFCARPPDG NYRDIWPGTLVTVSL (SEQ ID NO: 379)
		CAGTCGTTGGAGGAGTCCGGGGAGGCTGGTCCAGCCTGAGGAT CCCTGACACTCACTGCACAGCTTCTGGATTCACTCAATAACTACT ACATCTGCTGGTCCGCCAGGCTCCAGGGAAGGGCTGGAGTGGATC GCATGTATTGATATGCAATGGTTCGACTTACTACCGAAGTGGGC GAAAGGCCGATTCAACATCTCAAGACCTCGTCGACACCGTGATC TGCAATGACAGTCTGACAGCGCGGACACGCGCCACTATTCTGTG CGAGTCTTGGCTACTCCCTTGTGGGGCCAGGACCCCTGGTCACC GTCTCTCTCA (SEQ ID NO: 381)
AA	AA	QSLVESGRLVQPEGSLTLTCTASGFTLNNYIICWVRQAPGKGLEWIA IDNANGRTYYANWAKGRFTISKSTTVDLKITSPTEADTATYFCARSL ATPLWPGTLVTVSS (SEQ ID NO: 383)
		QSLVESGRLVQPEGSLTLTCTASGFTLNNYIICWVRQAPGKGLEWIA IDNANGRTYYANWAKGRFTISKSTTVDLKITSPTEADTATYFCARSL ATPLWPGTLVTVSS (SEQ ID NO: 385)
4A7	NA	CAGGAGCAGCTGTGGAGTCCGGGGAGGCTGGTCCAGCCTGAGG GATCCCTGACACTCACTGCAGCTTCTGGATTCTCTTCACTAGCA GCTACTTCACTGTGCTGGTCCGCCAGGCTCCAGGGAAGGGCTGGAG TGGATCGCATGCTATGCTGTGGTGTAGTGTAGGACTTACTACCG GAACTGGGCGAAGCGGATTCACATCTCAAAACCTCGTCGACCA CGGTGACTCTGCAATGACAGTCTGACAGCGCGGACACGCGCAC TATTCTGTGCGAGAGTGGCTACGATGACGATGGTGAATCTGATGC TTTTGATCCCTGGGGCCAGGACCCCTGGTCAACGCTCTCTCA (SEQ ID NO: 385)
		QSLVESGRLVQPEGSLTLTCAASGFSFSSYFMCWVRQAPGKLEWI ACIGVGGSGSTYYANWAKGRFTISKSTTVDLKITSPTEADTATYFC ARVAYDDGSDAFDPWPGTLVTVSS (SEQ ID NO: 387)
AA	AA	QSLVESGRLVQPEGSLTLTCAASGFSFSSYFMCWVRQAPGKLEWI ACIGVGGSGSTYYANWAKGRFTISKSTTVDLKITSPTEADTATYFC ARVAYDDGSDAFDPWPGTLVTVSS (SEQ ID NO: 387)
AA	AA	QSLVESGRLVQPEGSLTLTCAASGFSFSSYFMCWVRQAPGKLEWI ACIGVGGSGSTYYANWAKGRFTISKSTTVDLKITSPTEADTATYFC ARVAYDDGSDAFDPWPGTLVTVSS (SEQ ID NO: 387)

[illegible]

TABLE 24-continued

Standard IgG Antibody Variable Region Protein Alignment				
CDRs defined by Kabat				
Alignment numbering defined by ②				
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
HEAVY VARIABLE				
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②

TABLE 24-continued

[illegible]

TABLE 24-continued

Standard IgG Antibody Variable Region Protein Alignment				
CDRs defined by Kabat				
Alignment numbering defined by ②				
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②

TABLE 24-continued

Standard IgG Antibody Variable Region Protein Alignment CDRs defined by Kabat Alignment numbering defined by ⑦				
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦
H_FR2	H_CDR2	H_FR2	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦

⑦ indicates text missing or illegible when filed

TABLE 25				
below discloses SEQ ID NOS 600-644, 645, 644 and 646-664, respectively, in order of appearance.				
Standard IgG Antibody Variable Region Consensus Protein Alignment				
CDRs defined by Kabat				
Alignment numbering defined by ②				
KAPPA VARIABLE				
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②

TABLE 25-continued

below discloses SEQ ID NOS 600-644, 645, 644 and 646-664, respectively, in order of appearance.				
Standard IgG Antibody Variable Region Consensus Protein Alignment				
CDRs defined by Kabat				
Alignment numbering defined by ②				
②	②	②	②	②
②	②	②	②	②
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
K_FR2	K_CDR2	K_FR3	K_CDR3	K_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
Germline			K_FR1	K_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②

TABLE 25-continued

below discloses SEQ ID NOS 600-644, 645, 644 and 646-664, respectively, in order of appearance.

Standard IgG Antibody Variable Region Consensus Protein Alignment
CDRs defined by Kabat
Alignment numbering defined by ⑦

HEAVY_VARIABLE				
		Germline	H_FR1	H_CDR1
	?		?	?
?	?	?	?	?
		Germline	H_FR1	H_CDR1
	?		?	?
?	?	?	?	?
		Germline	H_FR1	H_CDR1
	?		?	?
?	?	?	?	?
		Germline	H_FR1	H_CDR1
	?		?	?
?	?	?	?	?
		Germline	H_FR1	H_CDR1
	?		?	?
?	?	?	?	?
		Germline	H_FR1	H_CDR1
	?		?	?
?	?	?	?	?
		Germline	H_FR1	H_CDR1
	?		?	?
?	?	?	?	?
	?		?	?
?	?	?	?	?
	?		?	?
?	?	?	?	?
	?		?	?
?	?	?	?	?
K_FR2	K_CDRR2	K_FR3	K_CDR3	K_FR4
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
H_FR2	H_CDRR2	H_FR3	H_CDR3	H_FR4
?	?	?	?	?
?	?	?	?	?
H_FR2	H_CDRR2	H_FR3	H_CDR3	H_FR4
?	?	?	?	?
?	?	?	?	?

TABLE 25-continued

below discloses SEQ ID NOS 600-644, 645, 644 and 646-664, respectively, in order of appearance.				
Standard IgG Antibody Variable Region Consensus Protein Alignment CDRs defined by Kabat Alignment numbering defined by ②				
H_FR2	H_CDRR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDRR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDRR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
		Germline	H_FR1	H_CDR1
	②		②	②
②	②	②	②	②

TABLE 25-continued

below discloses SEQ ID NOS 600-644, 645, 644 and 646-664, respectively, in order of appearance.				
Standard IgG Antibody Variable Region Consensus Protein Alignment				
CDRs defined by Kabat				
Alignment numbering defined by ②				
	Germline		H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
	Germline		H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
	Germline		H_FR1	H_CDR1
	②		②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②
H_FR2	H_CDR2	H_FR3	H_CDR3	H_FR4
②	②	②	②	②
②	②	②	②	②
②	②	②	②	②

TABLE 25-continued

below discloses SEQ ID NOS 600-644, 645, 644 and 646-664, respectively, in order of appearance.				
Standard IgG Antibody Variable Region Consensus Protein Alignment				
CDRs defined by Kabat				
Alignment numbering defined by ⑦				
H_FR2	H_CDR2	H_FR3	H_CDR3	H_FR4
⑦	⑦	⑦	⑦	⑦
⑦	⑦	⑦	⑦	⑦

⑦ indicates text missing or illegible when filed

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 664

<210> SEQ ID NO 1

<211> LENGTH: 319

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 1

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Met Pro Arg Gly Gln Lys Ser Lys Leu Arg Ala Arg Glu Lys Arg Arg
1          5          10          15

Lys Ala Arg Asp Glu Thr Arg Gly Leu Asn Val Pro Gln Val Thr Glu
20        25        30

Ala Glu Glu Glu Glu Ala Pro Cys Cys Ser Ser Ser Val Ser Gly Gly
35        40        45

Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg
50        55        60

Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly Val Ser Ser Thr Lys
65        70        75        80

Ser Lys Lys Gly Ala Lys Ser His Gln Gly Glu Lys Asn Ala Ser Ser
85        90        95

Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr
100       105       110

Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys Tyr Lys Ile
115       120       125

Lys Lys Ser Val Thr Lys Gly Glu Met Leu Lys Ile Val Gly Lys Arg
130       135       140

Phe Arg Glu His Phe Pro Glu Ile Leu Lys Lys Ala Ser Glu Gly Leu
145       150       155       160

Ser Val Val Phe Gly Leu Glu Leu Asn Lys Val Asn Pro Asn Gly His
165       170       175

Thr Tyr Thr Phe Ile Asp Lys Val Asp Leu Thr Asp Glu Glu Ser Leu
180       185       190

Leu Ser Ser Trp Asp Phe Pro Arg Arg Lys Leu Leu Met Pro Leu Leu
195       200       205

Gly Val Ile Phe Leu Asn Gly Asn Ser Ala Thr Glu Glu Glu Ile Trp
210       215       220

Glu Phe Leu Asn Met Leu Gly Val Tyr Asp Gly Glu Glu His Ser Val
225       230       235       240

Phe Gly Glu Pro Trp Lys Leu Ile Thr Lys Asp Leu Val Gln Glu Lys
245       250       255

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Tyr Leu Glu Tyr Lys Gln Val Pro Ser Ser Asp Pro Pro Arg Phe Gln
260 265 270
Phe Leu Trp Gly Pro Arg Ala Tyr Ala Glu Thr Ser Lys Met Lys Val
275 280 285
Leu Glu Phe Leu Ala Lys Val Asn Gly Thr Thr Pro Cys Ala Phe Pro
290 295 300
Thr His Tyr Glu Glu Ala Leu Lys Asp Glu Glu Lys Ala Gly Val
305 310 315

<210> SEQ ID NO 2
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 2

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1 5 10 15
Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Ala
20 25 30
Gly Val

<210> SEQ ID NO 3
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 3

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15
Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25 30

<210> SEQ ID NO 4
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 4

Asp Lys Thr His Thr Cys Pro Pro Cys Pro
1 5 10

<210> SEQ ID NO 5
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 5

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33

<210> SEQ ID NO 6
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 6

gctgcatcca aactggcctc t 21

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<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 7

caatgcagtt atgttagtag tagtggtact tatggaaatg tt 42

<210> SEQ ID NO 8

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 8

Ala Ala Ser Lys Leu Ala Ser
1 5

<210> SEQ ID NO 9

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 9

Gln Cys Ser Tyr Val Ser Ser Ser Gly Thr Tyr Gly Asn Val
1 5 10

<210> SEQ ID NO 10

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 10

caggccagtg aaagcattag caactactta tcc 33

<210> SEQ ID NO 11

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 11

tgggcatcca ctctggcatc t 21

<210> SEQ ID NO 12

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<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 12

caacaggggt atagtagtag taatgttgat aatctt 36

<210> SEQ ID NO 13
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 13

Gln Ala Ser Glu Ser Ile Ser Asn Tyr Leu Ser
1 5 10

<210> SEQ ID NO 14
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 14

Trp Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 15
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 15

Gln Gln Gly Tyr Ser Ser Ser Asn Val Asp Asn Leu
1 5 10

<210> SEQ ID NO 16
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 16

cagtccagtc agagtgtttg gcataacgac tacttatcc 39

<210> SEQ ID NO 17
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 17

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ggtgcatcca ctttggcatc t 21

<210> SEQ ID NO 18
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 18

gcaggcgggtt atggacgtag tagtgaaaat ggt 33

<210> SEQ ID NO 19
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 19

Gln Ser Ser Gln Ser Val Trp His Asn Asp Tyr Leu Ser
1 5 10

<210> SEQ ID NO 20
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 20

Gly Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 21
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 21

Ala Gly Gly Tyr Gly Arg Ser Ser Glu Asn Gly
1 5 10

<210> SEQ ID NO 22
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 22

cagtcacagta agagtgttta taataacaac tggttagcc 39

<210> SEQ ID NO 23
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 23

gatgcatcga ctctagattc t 21

<210> SEQ ID NO 24
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 24

gtaggcgggtt atagtagtcg tagtgataat ggt 33

<210> SEQ ID NO 25
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 25

Gln Ser Ser Lys Ser Val Tyr Asn Asn Asn Trp Leu Ala
1 5 10

<210> SEQ ID NO 26
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 26

Asp Ala Ser Thr Leu Asp Ser
1 5

<210> SEQ ID NO 27
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 27

Val Gly Gly Tyr Ser Ser Arg Ser Asp Asn Gly
1 5 10

<210> SEQ ID NO 28
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 28

caggccagtc agagtattag tagttactta tcc 33

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<210> SEQ ID NO 29
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 29

agggcatcca ctctggcatc t 21

<210> SEQ ID NO 30
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 30

caaagctatg atgatatag tagataataat tttttttatg gt 42

<210> SEQ ID NO 31
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 31

Gln Ala Ser Gln Ser Ile Ser Ser Tyr Leu Ser
1 5 10

<210> SEQ ID NO 32
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 32

Arg Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 33
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 33

Gln Ser Tyr Asp Asp Ser Ser Asp Asn Asn Phe Phe Tyr Gly
1 5 10

<210> SEQ ID NO 34
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

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<400> SEQUENCE: 34

caggccagtc agaacattga tagttactta gcc

33

<210> SEQ ID NO 35

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 35

agggcatcca ctctggcatc t

21

<210> SEQ ID NO 36

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 36

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42

<210> SEQ ID NO 37

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 37

Gln Ala Ser Gln Asn Ile Asp Ser Tyr Leu Ala

1

5

10

<210> SEQ ID NO 38

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 38

Arg Ala Ser Thr Leu Ala Ser

1

5

<210> SEQ ID NO 39

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 39

Gln Ser Tyr Asp Asp Ser Arg Ser Ser Ser Phe Phe Tyr Gly

1

5

10

<210> SEQ ID NO 40

<211> LENGTH: 33

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 40

caggccagtc agaacattaa tagttactta gcc 33

<210> SEQ ID NO 41
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 41

agggcatcca ctctggcatc t 21

<210> SEQ ID NO 42
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 42

caaagctatg atgatatgtag gagtattagt tttttttatg ct 42

<210> SEQ ID NO 43
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 43

Gln Ala Ser Gln Asn Ile Asn Ser Tyr Leu Ala
1 5 10

<210> SEQ ID NO 44
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 44

Arg Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 45
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 45

Gln Ser Tyr Asp Ser Arg Ser Ile Ser Phe Phe Tyr Ala

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<210> SEQ ID NO 46 <211> LENGTH: 33 <212> TYPE: DNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide			
<400> SEQUENCE: 46			
caggccagtc agagcattag tagctactta gcc			33
<210> SEQ ID NO 47 <211> LENGTH: 21 <212> TYPE: DNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide			
<400> SEQUENCE: 47			
gctgcatcca ctctggcatc t			21
<210> SEQ ID NO 48 <211> LENGTH: 42 <212> TYPE: DNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide			
<400> SEQUENCE: 48			
caaagctatg atgatatgtag gagtagtagt tttttttatg ct			42
<210> SEQ ID NO 49 <211> LENGTH: 11 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide			
<400> SEQUENCE: 49			
Gln Ala Ser Gln Ser Ile Ser Ser Tyr Leu Ala			
1	5	10	
<210> SEQ ID NO 50 <211> LENGTH: 7 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide			
<400> SEQUENCE: 50			
Ala Ala Ser Thr Leu Ala Ser			
1	5		
<210> SEQ ID NO 51 <211> LENGTH: 14 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic			

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peptide

<400> SEQUENCE: 51

Gln Ser Tyr Asp Asp Ser Arg Ser Ser Ser Phe Phe Tyr Ala
1 5 10

<210> SEQ ID NO 52
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 52

caggccagtc agagcattag cagttggta tcc 33

<210> SEQ ID NO 53
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 53

agggaacca ctctggcatc t 21

<210> SEQ ID NO 54
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 54

caaagttatg atgatagtag tagtagtaat tttttttatg ct 42

<210> SEQ ID NO 55
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 55

Gln Ala Ser Gln Ser Ile Ser Ser Trp Leu Ser
1 5 10

<210> SEQ ID NO 56
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 56

Arg Ala Thr Thr Leu Ala Ser
1 5

<210> SEQ ID NO 57

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<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 57

Gln Ser Tyr Asp Asp Ser Ser Ser Ser Asn Phe Phe Tyr Ala
1 5 10

<210> SEQ ID NO 58
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 58

cagtcagtc agagtgttta tagtaacaac ctcttatct 39

<210> SEQ ID NO 59
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 59

aaggcatcca ctctggcatc t 21

<210> SEQ ID NO 60
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 60

caaggctatt atagtgggtg ttttatatg 30

<210> SEQ ID NO 61
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 61

Gln Ser Ser Gln Ser Val Tyr Ser Asn Asn Leu Leu Ser
1 5 10

<210> SEQ ID NO 62
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 62

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Lys Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 63
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 63

Gln Gly Tyr Tyr Ser Gly Val Ile Tyr Met
1 5 10

<210> SEQ ID NO 64
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 64

caggccagtc agagcattag tagttactta tcc 33

<210> SEQ ID NO 65
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 65

agggcatcca ctctggcattc t 21

<210> SEQ ID NO 66
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 66

caaagctatg atgatatgtag tagtaataat tttttttatg gt 42

<210> SEQ ID NO 67
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 67

Gln Ala Ser Gln Ser Ile Ser Ser Tyr Leu Ser
1 5 10

<210> SEQ ID NO 68
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 68

Arg Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 69

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 69

Gln Ser Tyr Asp Asp Ser Ser Ser Asn Asn Phe Phe Tyr Gly
1 5 10

<210> SEQ ID NO 70

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 70

caggccagtc agaacattag tagctactta gcc 33

<210> SEQ ID NO 71

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 71

agggcatcca ctctggcatc t 21

<210> SEQ ID NO 72

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 72

caaagctatg atgatatgtag gagtagtaat tttttttatg ct 42

<210> SEQ ID NO 73

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 73

Gln Ala Ser Gln Asn Ile Ser Ser Tyr Leu Ala
1 5 10

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<210> SEQ ID NO 74
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 74

Arg Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 75
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 75

Gln Ser Tyr Asp Asp Ser Arg Ser Ser Asn Phe Phe Tyr Ala
1 5 10

<210> SEQ ID NO 76
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 76

cagtccagtg agagcggtta taatcacaac tggtaggc 39

<210> SEQ ID NO 77
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 77

gatgcatcca ctctggcatc t 21

<210> SEQ ID NO 78
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 78

caaggctatt atcaaactag tgtttgggct 30

<210> SEQ ID NO 79
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 79

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Gln Ser Ser Glu Ser Val Tyr Asn His Asn Trp Leu Gly
1 5 10

<210> SEQ ID NO 80
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 80

Asp Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 81
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 81

Gln Gly Tyr Tyr Gln Thr Ser Val Trp Ala
1 5 10

<210> SEQ ID NO 82
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 82

cagtccagtc agagtgttta tgataacaat gctttagcc 39

<210> SEQ ID NO 83
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 83

ggtgcatcca ctctggcatc t 21

<210> SEQ ID NO 84
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 84

caatgtactt attatgttag tagttatcaa aatgat 36

<210> SEQ ID NO 85
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 85

Gln Ser Ser Gln Ser Val Tyr Asp Asn Asn Ala Leu Ala
1 5 10

<210> SEQ ID NO 86
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 86

Gly Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 87
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 87

Gln Cys Thr Tyr Tyr Val Ser Ser Tyr Gln Asn Asp
1 5 10

<210> SEQ ID NO 88
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 88

cagtccagta agagtgttta taataagaac tggttatcc 39

<210> SEQ ID NO 89
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 89

ggtgcatcca ctctggcatc t 21

<210> SEQ ID NO 90
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 90

gcaggcggtt atagtagtag tagtgataca tttgct 36

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<210> SEQ ID NO 91
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 91

Gln Ser Ser Lys Ser Val Tyr Asn Lys Asn Trp Leu Ser
1 5 10

<210> SEQ ID NO 92
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 92

Gly Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 93
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 93

Ala Gly Gly Tyr Ser Ser Ser Ser Asp Thr Phe Ala
1 5 10

<210> SEQ ID NO 94
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 94

cagtccagtc agagcggttta tagtagcgac ctcttatcc 39

<210> SEQ ID NO 95
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 95

aaggcatcca ctctggcatc t 21

<210> SEQ ID NO 96
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

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<400> SEQUENCE: 96

caaggctact atagtgggtgt ggtttatatt

30

<210> SEQ ID NO 97

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 97

Gln Ser Ser Gln Ser Val Tyr Ser Ser Asp Leu Leu Ser
1 5 10

<210> SEQ ID NO 98

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 98

Lys Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 99

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 99

Gln Gly Tyr Tyr Ser Gly Val Val Tyr Ile
1 5 10

<210> SEQ ID NO 100

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 100

cagtcacagtc agagcgttta tagtagcgac ctcttatcc

39

<210> SEQ ID NO 101

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 101

aaggcatcca ctctggcacc t

21

<210> SEQ ID NO 102

<211> LENGTH: 30

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 102

caaggctact atagtgggtgt ggtttatatt 30

<210> SEQ ID NO 103
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 103

Gln Ser Ser Gln Ser Val Tyr Ser Ser Asp Leu Leu Ser
1 5 10

<210> SEQ ID NO 104
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 104

Lys Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 105
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 105

Gln Gly Tyr Tyr Ser Gly Val Val Tyr Ile
1 5 10

<210> SEQ ID NO 106
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 106

caggccagtc agagcattag tagttactta tct 33

<210> SEQ ID NO 107
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 107

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agggcacccc ctctggcatc t 21

<210> SEQ ID NO 108
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 108

caaagctacg atgatatagtag tagtaataat tttttttatg gt 42

<210> SEQ ID NO 109
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 109

Gln Ala Ser Gln Ser Ile Ser Ser Tyr Leu Ser
1 5 10

<210> SEQ ID NO 110
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 110

Arg Ala Ser Pro Leu Ala Ser
1 5

<210> SEQ ID NO 111
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 111

Gln Ser Tyr Asp Asp Ser Ser Ser Asn Asn Phe Phe Tyr Gly
1 5 10

<210> SEQ ID NO 112
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 112

cagtccagta agagtgttta taataagaac tgggttatcc 39

<210> SEQ ID NO 113
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 113

ggtgcatcca ctctggcatc t 21

<210> SEQ ID NO 114

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 114

gcaggcggtt atagtagtag tagtgatacg ttgct 36

<210> SEQ ID NO 115

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 115

Gln Ser Ser Lys Ser Val Tyr Asn Lys Asn Trp Leu Ser
1 5 10

<210> SEQ ID NO 116

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 116

Gly Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 117

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 117

Ala Gly Gly Tyr Ser Ser Ser Asp Thr Phe Ala
1 5 10

<210> SEQ ID NO 118

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 118

cagtccagtc agagccttta taatagcgac ctcttatcc 39

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<210> SEQ ID NO 119
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 119

aaggcatcca ctctggcatc t 21

<210> SEQ ID NO 120
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 120

caaggctact atagtgggtg ggtttatatt 30

<210> SEQ ID NO 121
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 121

Gln Ser Ser Gln Ser Leu Tyr Asn Ser Asp Leu Leu Ser
1 5 10

<210> SEQ ID NO 122
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 122

Lys Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 123
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 123

Gln Gly Tyr Tyr Ser Gly Val Val Tyr Ile
1 5 10

<210> SEQ ID NO 124
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

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<400> SEQUENCE: 124

cagtcacagta agagtgttta taataagaac tggttatcc 39

<210> SEQ ID NO 125

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 125

ggtgcatcca ctctggcatc t 21

<210> SEQ ID NO 126

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 126

gcaggcgggtt atagtagtag tagtgataca tttgct 36

<210> SEQ ID NO 127

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 127

Gln Ser Ser Lys Ser Val Tyr Asn Lys Asn Trp Leu Ser
1 5 10

<210> SEQ ID NO 128

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 128

Gly Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 129

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 129

Ala Gly Gly Tyr Ser Ser Ser Asp Thr Phe Ala
1 5 10

<210> SEQ ID NO 130

<211> LENGTH: 39

<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 130

caggcccttc agagtgttta tgataacaat gctttatcc 39

<210> SEQ ID NO 131
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 131

ggtgcatcca ctctggcatc t 21

<210> SEQ ID NO 132
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 132

caatgtactt attatgtag tagttatcaa aatgat 36

<210> SEQ ID NO 133
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 133

Gln Ala Leu Gln Ser Val Tyr Asp Asn Asn Ala Leu Ser
1 5 10

<210> SEQ ID NO 134
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 134

Gly Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 135
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 135

Gln Cys Thr Tyr Tyr Val Ser Ser Tyr Gln Asn Asp
1 5 10

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<210> SEQ ID NO 136
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 136

caggccagtg agagcattgg caatgcatta gcc 33

<210> SEQ ID NO 137
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 137

agggcatcca ctctggcatc t 21

<210> SEQ ID NO 138
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 138

caaagctatg atgatatagtag tagtagtagt tttttttatg ct 42

<210> SEQ ID NO 139
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 139

Gln Ala Ser Glu Ser Ile Gly Asn Ala Leu Ala
1 5 10

<210> SEQ ID NO 140
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 140

Arg Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 141
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

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<400> SEQUENCE: 141

Gln Ser Tyr Asp Asp Ser Ser Ser Ser Ser Phe Phe Tyr Ala
1 5 10

<210> SEQ ID NO 142

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 142

cagtccagta agagtgttta taataagaac tggttatcc 39

<210> SEQ ID NO 143

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 143

ggtgcatcca ctctggcatc t 21

<210> SEQ ID NO 144

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 144

gcaggcgggt atagtagtag tagtgatacg tttgct 36

<210> SEQ ID NO 145

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 145

Gln Ser Ser Lys Ser Val Tyr Asn Lys Asn Trp Leu Ser
1 5 10

<210> SEQ ID NO 146

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 146

Gly Ala Ser Thr Leu Ala Ser
1 5

<210> SEQ ID NO 147

<211> LENGTH: 12

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 147

Ala Gly Gly Tyr Ser Ser Ser Ser Asp Thr Phe Ala
1 5 10

<210> SEQ ID NO 148
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 148

agccatgcaa tgatc 15

<210> SEQ ID NO 149
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 149

accattggga gtcgtgatac tatatattat gcgagctggg cgaaaggc 48

<210> SEQ ID NO 150
<211> LENGTH: 9
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 150

aacgccttg 9

<210> SEQ ID NO 151
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 151

Ser His Ala Met Ile
1 5

<210> SEQ ID NO 152
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 152

Thr Ile Gly Ser Arg Asp Thr Ile Tyr Tyr Ala Ser Trp Ala Lys Gly

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1	5	10	15
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<210> SEQ ID NO 157 <211> LENGTH: 5 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide <400> SEQUENCE: 157 Ser Tyr Asp Met Ser 1 5			
<210> SEQ ID NO 158 <211> LENGTH: 16 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic			

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peptide

<400> SEQUENCE: 158

Ile Ile Tyr Ala Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly
1 5 10 15

<210> SEQ ID NO 159
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 159

Asp Pro Ala Gly Tyr Ser Ile Ser Phe Gly Leu
1 5 10

<210> SEQ ID NO 160
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 160

agctacaaca tgggc 15

<210> SEQ ID NO 161
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 161

atcattggtg ctagtgatag cgcattgtac gcgagctggg caaaaggc 48

<210> SEQ ID NO 162
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 162

ggtggtcttg gtttgagtac tggttttgcg ttg 33

<210> SEQ ID NO 163
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 163

Ser Tyr Asn Met Gly
1 5

<210> SEQ ID NO 164

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<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 164

Ile Ile Gly Ala Ser Asp Ser Ala Leu Tyr Ala Ser Trp Ala Lys Gly
1 5 10 15

<210> SEQ ID NO 165
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 165

Gly Gly Leu Gly Leu Ser Thr Gly Phe Ala Leu
1 5 10

<210> SEQ ID NO 166
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 166

agctacgaca tgagc 15

<210> SEQ ID NO 167
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 167

tatattgcta ctgatggtag gccatattac gcgagctggg cgaaaggc 48

<210> SEQ ID NO 168
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 168

gggggggtatg ctggtggcctt g 21

<210> SEQ ID NO 169
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 169

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Ser Tyr Asp Met Ser
1 5

<210> SEQ ID NO 170
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 170

Tyr Ile Ala Thr Asp Gly Arg Pro Tyr Tyr Ala Ser Trp Ala Lys Gly
1 5 10 15

<210> SEQ ID NO 171
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 171

Gly Gly Tyr Ala Gly Gly Leu
1 5

<210> SEQ ID NO 172
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 172

aattattata ttgtc 15

<210> SEQ ID NO 173
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 173

tgcattgaca atgctaattgg taggacttac tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 174
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 174

tcattgtcta ctcccttg 18

<210> SEQ ID NO 175
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 175

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 176

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 176

Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 177

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 177

Ser Leu Ser Thr Pro Leu
1 5

<210> SEQ ID NO 178

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 178

aactattata tttgt 15

<210> SEQ ID NO 179

<211> LENGTH: 51

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 179

tcgattgaca atgttaatgg taggacctac tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 180

<211> LENGTH: 18

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 180

tccttgcta ctcccttg 18

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<210> SEQ ID NO 181
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 181

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 182
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 182

Cys Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 183
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 183

Ser Leu Ala Thr Pro Leu
1 5

<210> SEQ ID NO 184
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 184

aactactaca tctgc 15

<210> SEQ ID NO 185
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 185

tgcattgaca atgttaatgg taggacttac tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 186
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 186

tccttggtta ctcccttg 18

<210> SEQ ID NO 187
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 187

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 188
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 188

Cys Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 189
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 189

Ser Leu Ala Thr Pro Leu
1 5

<210> SEQ ID NO 190
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 190

aactactaca tctgc 15

<210> SEQ ID NO 191
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 191

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tgcattgaca atattaatgg taggacttac tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 192
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 192

tccttggtta ctccttggtt 18

<210> SEQ ID NO 193
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 193

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 194
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 194

Cys Ile Asp Asn Ile Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 195
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 195

Ser Leu Ala Thr Pro Leu
1 5

<210> SEQ ID NO 196
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 196

aattattata tatgt 15

<210> SEQ ID NO 197
<211> LENGTH: 51
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 197

tgtattgata acgctaattgg taggacttac tacgcgacct gggcgaaagg c 51

<210> SEQ ID NO 198
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 198

tcattgtcta ctaacttg 18

<210> SEQ ID NO 199
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 199

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 200
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 200

Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Thr Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 201
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 201

Ser Leu Ser Thr Asn Leu
1 5

<210> SEQ ID NO 202
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 202

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agctactact acatatgc 18

<210> SEQ ID NO 203
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 203

tgtattggtg gtggtataac cgatgccact gcctacgcga ggtgggcgaa aggc 54

<210> SEQ ID NO 204
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 204

ggcggtcctg ataataatgt ccaatttaac ttg 33

<210> SEQ ID NO 205
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 205

Ser Tyr Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 206
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 206

Cys Ile Gly Gly Gly Asn Thr Asp Ala Thr Ala Tyr Ala Arg Trp Ala
1 5 10 15

Lys Gly

<210> SEQ ID NO 207
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 207

Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu
1 5 10

<210> SEQ ID NO 208
<211> LENGTH: 15

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 208

aattattata ttgtc 15

<210> SEQ ID NO 209
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 209

tgcattgaca atagtaatgg taggacttac tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 210
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 210

tcattgtcta ctcccttg 18

<210> SEQ ID NO 211
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 211

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 212
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 212

Cys Ile Asp Asn Ser Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 213
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 213

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Ser Leu Ser Thr Pro Leu
1 5

<210> SEQ ID NO 214
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 214

aactactaca tctgc 15

<210> SEQ ID NO 215
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 215

tgtattgaca atgctaattgg taggacttac tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 216
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 216

tccttggtgcta ctccttg 18

<210> SEQ ID NO 217
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 217

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 218
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 218

Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 219
<211> LENGTH: 6

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 219

Ser Leu Ala Thr Pro Leu
1 5

<210> SEQ ID NO 220
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 220

agcagtgacg tgacc 15

<210> SEQ ID NO 221
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 221

ttcctccaag ctgggggatgg tagcgcatatc tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 222
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 222

cataagggta atagttacgt gcctaacttg 30

<210> SEQ ID NO 223
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 223

Ser Ser Ala Val Thr
1 5

<210> SEQ ID NO 224
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 224

Phe Leu Gln Ala Gly Asp Gly Ser Ala Tyr Tyr Ala Ser Trp Ala Lys

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1	5	10	15
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Gly

<210> SEQ ID NO 225
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 225

His Lys Gly Asn Ser Tyr Val Pro Asn Leu
1 5 10

<210> SEQ ID NO 226
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 226

agctatgcaa tgagc 15

<210> SEQ ID NO 227
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 227

agcattggtg gtggtggtag cgcagctctac gcgagctggg cgaaaggc 48

<210> SEQ ID NO 228
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 228

ggattttata gtatagactt g 21

<210> SEQ ID NO 229
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 229

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 230
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 230

Ser Ile Gly Gly Gly Gly Ser Ala Val Tyr Ala Ser Trp Ala Lys Gly
1 5 10 15

<210> SEQ ID NO 231
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 231

Gly Phe Tyr Ser Ile Asp Leu
1 5

<210> SEQ ID NO 232
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 232

agcggccaac tcatgtgc 18

<210> SEQ ID NO 233
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 233

tgcattggtt ctggtagtaa tgctattagc actttctacg cgagctgggc gcaaggc 57

<210> SEQ ID NO 234
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 234

gtgggctccg atgactatgg tgactctgat gtttttgatc cc 42

<210> SEQ ID NO 235
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 235

Ser Gly Gln Leu Met Cys
1 5

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<210> SEQ ID NO 236
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 236

Cys Ile Gly Ser Gly Ser Asn Ala Ile Ser Thr Phe Tyr Ala Ser Trp
1 5 10 15
Ala Gln Gly

<210> SEQ ID NO 237
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 237

Val Gly Ser Asp Asp Tyr Gly Asp Ser Asp Val Phe Asp Pro
1 5 10

<210> SEQ ID NO 238
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 238

agcgccctact acatatgc 18

<210> SEQ ID NO 239
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 239

tgtattggtg gtgttaatcg cgttgccact gcctacgcga cctgggcgaa aggc 54

<210> SEQ ID NO 240
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 240

ggcggtcctg ataataatgt ccaatttaac ttg 33

<210> SEQ ID NO 241
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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peptide

<400> SEQUENCE: 241

Ser Ala Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 242
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 242

Cys Ile Gly Gly Val Asn Arg Val Ala Thr Ala Tyr Ala Thr Trp Ala
1 5 10 15

Lys Gly

<210> SEQ ID NO 243
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 243

Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu
1 5 10

<210> SEQ ID NO 244
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 244

aggtactact acagttgc 18

<210> SEQ ID NO 245
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 245

tgtgttggtg gtgttaatcg cgatgccact gcctacgcga cctgggcgaa aggc 54

<210> SEQ ID NO 246
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 246

ggcggtcctg ataataatgt ccaatttaac ttg 33

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<210> SEQ ID NO 247
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 247

Arg Tyr Tyr Tyr Ser Cys
1 5

<210> SEQ ID NO 248
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 248

Cys Val Gly Gly Val Asn Arg Asp Ala Thr Ala Tyr Ala Thr Trp Ala
1 5 10 15

Lys Gly

<210> SEQ ID NO 249
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 249

Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu
1 5 10

<210> SEQ ID NO 250
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 250

aattattata tttgt 15

<210> SEQ ID NO 251
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 251

tgtattgaca atgttaatgg taggacttac tacgcgagct gggcgaaagg c 51

<210> SEQ ID NO 252
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 252

tcattgtcta ctccttg 18

<210> SEQ ID NO 253

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 253

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 254

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 254

Cys Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 255

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 255

Ser Leu Ser Thr Pro Leu
1 5

<210> SEQ ID NO 256

<211> LENGTH: 18

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 256

agcagctact tcatgtgc 18

<210> SEQ ID NO 257

<211> LENGTH: 54

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 257

tgcattgctg ttgtagtag tggtagcact tactacgcga gctgggcgaa aggc 54

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<210> SEQ ID NO 258
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 258

gtgggctacg atgactatgg tgactctgat gcttttgatc cc 42

<210> SEQ ID NO 259
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 259

Ser Ser Tyr Phe Met Cys
1 5

<210> SEQ ID NO 260
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 260

Cys Ile Ala Val Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala
1 5 10 15

Lys Gly

<210> SEQ ID NO 261
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 261

Val Gly Tyr Asp Asp Tyr Gly Asp Ser Asp Ala Phe Asp Pro
1 5 10

<210> SEQ ID NO 262
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 262

agggtactact acagttgc 18

<210> SEQ ID NO 263
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 263

tgtgttggtg gtgttaatcg cgatgccact gcctacgcga cctgggcgaa aggc 54

<210> SEQ ID NO 264
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 264

ggcggtcctg ataataatgt ccaatttaac ttg 33

<210> SEQ ID NO 265
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 265

Arg Tyr Tyr Tyr Ser Cys
1 5

<210> SEQ ID NO 266
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 266

Cys Val Gly Gly Val Asn Arg Asp Ala Thr Ala Tyr Ala Thr Trp Ala
1 5 10 15

Lys Gly

<210> SEQ ID NO 267
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 267

Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu
1 5 10

<210> SEQ ID NO 268
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 268

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agcagctact tcatgtgc 18

<210> SEQ ID NO 269
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 269

tgcatgtggt ctggtagtag tgctattagc actttctacg cgagctgggc gcaaggc 57

<210> SEQ ID NO 270
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 270

gtgggctacg atgactatgg tgactctgat gcttttgatc cc 42

<210> SEQ ID NO 271
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 271

Ser Ser Tyr Phe Met Cys
1 5

<210> SEQ ID NO 272
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 272

Cys Ile Gly Ser Gly Ser Ser Ala Ile Ser Thr Phe Tyr Ala Ser Trp
1 5 10 15

Ala Gln Gly

<210> SEQ ID NO 273
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 273

Val Gly Tyr Asp Asp Tyr Gly Asp Ser Asp Ala Phe Asp Pro
1 5 10

<210> SEQ ID NO 274
<211> LENGTH: 15
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 274

agctatggaa tgact 15

<210> SEQ ID NO 275
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 275

tacatttggga ctgatgggag gacatactac gcaaactggg cgaaaggc 48

<210> SEQ ID NO 276
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 276

ccctttgatg gtaattatag ggacatc 27

<210> SEQ ID NO 277
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 277

Ser Tyr Gly Met Thr
1 5

<210> SEQ ID NO 278
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 278

Tyr Ile Trp Thr Asp Gly Arg Thr Tyr Tyr Ala Asn Trp Ala Lys Gly
1 5 10 15

<210> SEQ ID NO 279
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 279

Pro Phe Asp Gly Asn Tyr Arg Asp Ile
1 5

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<210> SEQ ID NO 280
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 280

aactactaca tctgc 15

<210> SEQ ID NO 281
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 281

tgtattgata atgccaatgg tcggacttac tacgcgaact gggcgaaagg c 51

<210> SEQ ID NO 282
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 282

tccttggcta ctccttg 18

<210> SEQ ID NO 283
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 283

Asn Tyr Tyr Ile Cys
1 5

<210> SEQ ID NO 284
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 284

Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Asn Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 285
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 285

Ser Leu Ala Thr Pro Leu
1 5

<210> SEQ ID NO 286

<211> LENGTH: 18

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 286

agcagctact tcatgtgc 18

<210> SEQ ID NO 287

<211> LENGTH: 54

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 287

tgcatgtgtg ttggtggtag tggtagcact tactacgcga actgggcgaa aggc 54

<210> SEQ ID NO 288

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide

<400> SEQUENCE: 288

gtggcctacg atgacgatgg tgactctgat gcttttgatc cc 42

<210> SEQ ID NO 289

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 289

Ser Ser Tyr Phe Met Cys
1 5

<210> SEQ ID NO 290

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 290

Cys Ile Gly Val Gly Gly Ser Gly Ser Thr Tyr Tyr Ala Asn Trp Ala
1 5 10 15

Lys Gly

-continued

<210> SEQ ID NO 291
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 291

Val Ala Tyr Asp Asp Asp Gly Asp Ser Asp Ala Phe Asp Pro
1 5 10

<210> SEQ ID NO 292
<211> LENGTH: 336
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 292

gatgttgtga tgacccagac tccagcctcc gtgtctgaac ctgtgggagg cacagtcacc 60
atcaagtgcc aggccagtca gaggcattagg aatgaattat tttggtggca gcagaaacca 120
gggcagcctc ccaagctcct gatctatgct gcatccaaac tggcctctgg ggtcccatcg 180
cgggttcacg gcagtggatc tgggacagag ttcactctca ccatcagcga cctggagtg 240
gcccgtgctg ccacttacta ctgtcaatgc agttatgtta gtagtagtgg tacttatgga 300
aatgttttcg gcggaggggac cgagggtggtg gtcaaaa 336

<210> SEQ ID NO 293
<211> LENGTH: 327
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 293

cagtcgctgg aggagtcagg gggtcgcctg gtaacgcctg gaggatccct gacactcacc 60
tgccacagtct ctggaatcga cctcagtagc catgcaatga tctgggtccg ccaggctcca 120
ggggaggggc tggaatggat cggaaccatt gggagtcgtg atactatata ttatgcgagc 180
tgggcgaaaag gccgattcac catctccaaa acctcgtcga ccacaatgga tctgaaaatg 240
accagtctga caatcgagga cagggccacc tatctctgtg tcagaaacgc cttgtggggc 300
ccaggcaccc tggtcaccgt ctctca 327

<210> SEQ ID NO 294
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 294

Asp Val Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly
1 5 10 15

Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Arg Asn Glu

-continued

20	25	30
Leu Phe Trp Trp Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile		
35	40	45
Tyr Ala Ala Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly		
50	55	60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys		
65	70	75
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Cys Ser Tyr Val Ser Ser Ser		
85	90	95
Gly Thr Tyr Gly Asn Val Phe Gly Gly Gly Thr Glu Val Val Val Lys		
100	105	110

<210> SEQ ID NO 295
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 295

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Gly Ser	
1	15
Leu Thr Leu Thr Cys Thr Val Ser Gly Ile Asp Leu Ser Ser His Ala	
20	30
Met Ile Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Ile Gly	
35	45
Thr Ile Gly Ser Arg Asp Thr Ile Tyr Tyr Ala Ser Trp Ala Lys Gly	
50	60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Met Asp Leu Lys Met	
65	80
Thr Ser Leu Thr Ile Glu Asp Thr Ala Thr Tyr Phe Cys Val Arg Asn	
85	95
Ala Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser	
100	105

<210> SEQ ID NO 296
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 296

gcctatgata tgaccagac tccagcctcc gtggaggcag ctgtgggagg cacagtcacc	60
atcaattgcc aggccagtga aagcattagc aactacttat cctgggtatca gcagaaacca	120
gggcagcctc ccaagctect gatctactgg gcatccactc tggcatctgg ggtctcatcg	180
cggttcaaa gcaaggatc tgggacacag ttcactctca ccatcagcgg cgtggagtgt	240
gccgatgctg ccacttacta ctgtcaacag gggttatagta gtagtaatgt tgataatctt	300
ttcggcggag ggaccgaggt ggtgggtcaaa	330

<210> SEQ ID NO 297
 <211> LENGTH: 348
 <212> TYPE: DNA

-continued

```

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 297

cagtcgctgg aggagtcgg gggtcgctg gtcacgcctg ggacaccct gacactcacc      60
tgcaaagtct ctggattctc cctcagcagc tacgacatga gctgggtccg ccagggtcca      120
gggaaggggc tggaatggat cggaattatt tatgctagt gtagcacata ctacgcgagc      180
tgggcgaaag gccgattcac catctccaaa acctcgacca cgggtggatct gaaaatcgcc      240
agtccgacaa ccgaggacac ggccacctat ttctgtgcca gagaccctgc tggttatagc      300
attagctttg gcttgtgggg cccaggcacc ctggtcaccg tctcctca      348


<210> SEQ ID NO 298
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 298

Ala Tyr Asp Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1           5           10          15
Gly Thr Val Thr Ile Asn Cys Gln Ala Ser Glu Ser Ile Ser Asn Tyr
          20          25          30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
          35          40          45
Tyr Trp Ala Ser Thr Leu Ala Ser Gly Val Ser Ser Arg Phe Lys Gly
          50          55          60
Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Gly Val Glu Cys
          65          70          75          80
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Gly Tyr Ser Ser Ser Asn
          85          90          95
Val Asp Asn Leu Phe Gly Gly Gly Thr Glu Val Val Val Lys
          100         105         110


<210> SEQ ID NO 299
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 299

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1           5           10          15
Leu Thr Leu Thr Cys Lys Val Ser Gly Phe Ser Leu Ser Ser Tyr Asp
          20          25          30
Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
          35          40          45
Ile Ile Tyr Ala Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly
          50          55          60
Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Ala
          65          70          75          80

```

-continued

Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Asp Pro
85 90 95

Ala Gly Tyr Ser Ile Ser Phe Gly Leu Trp Gly Pro Gly Thr Leu Val
100 105 110

Thr Val Ser Ser
115

<210> SEQ ID NO 300

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 300

```
gccgcgctgc tgacccagac tccatctccc gtgtctgcag ctgtgggagg cacagtcagc      60
atcagttgcc agtccagtc gagtggttg cataacgact acttatcctg gtatcaacag      120
aaaccagggc agcctcccaa gctcctgac tatggtgcat ccactttggc atctggggtc      180
ccatcgcggt tcaaaggcag tggatctggg acacagttca gtctcaccat cagcggcgtg      240
cagtgtagc atgctgccac ttactactgt gcaggcggtt atggacgtag tagtgaaat      300
ggtttcggcg gagggacga ggtggtggtc aaa                                  333
```

<210> SEQ ID NO 301

<211> LENGTH: 351

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 301

```
caggagcagc tgaaggagtc cggaggaggc ctggttgccg ctggaggaac cctgacactc      60
acctgcgcag tctctggatt ctccctcagt agctacaaca tgggctgggt ccgccaggct      120
ccaggggagg ggctggaata catcggaatc attggtgcta gtgatagcgc attgtacgcg      180
agctgggcaa aaggccgatt caccatctcc aaaacctga ccacggtgga tctgaaaatc      240
accagtccga caaccgagga caggccacc tatttctgtg ccagaggtgg tcttggttg      300
agtactggtt ttgcgttggt gggcccaggc accctggtea ccgtctctc a              351
```

<210> SEQ ID NO 302

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 302

Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1 5 10 15

Gly Thr Val Ser Ile Ser Cys Gln Ser Ser Gln Ser Val Trp His Asn
20 25 30

Asp Tyr Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45

-continued

Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
50 55 60

Lys Gly Ser Gly Ser Gly Thr Gln Phe Ser Leu Thr Ile Ser Gly Val
65 70 75 80

Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Ala Gly Gly Tyr Gly Arg
85 90 95

Ser Ser Glu Asn Gly Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

<210> SEQ ID NO 303
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 303

Gln Glu Gln Leu Lys Glu Ser Gly Gly Gly Leu Val Ala Pro Gly Gly
1 5 10 15

Thr Leu Thr Leu Thr Cys Ala Val Ser Gly Phe Ser Leu Ser Ser Tyr
20 25 30

Asn Met Gly Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Tyr Ile
35 40 45

Gly Ile Ile Gly Ala Ser Asp Ser Ala Leu Tyr Ala Ser Trp Ala Lys
50 55 60

Gly Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile
65 70 75 80

Thr Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Gly
85 90 95

Gly Leu Gly Leu Ser Thr Gly Phe Ala Leu Trp Gly Pro Gly Thr Leu
100 105 110

Val Thr Val Ser Ser
115

<210> SEQ ID NO 304
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 304

gccgccgtgc tgaccagac tccatctccc gtgtctgcag ctgtgggagg cacagtcagc 60

atcagttgcc agtccagtaa gagggtttat aataacaact ggtagcctg gtatcagcag 120

aaaccagggc agcgteccaa gctcctgac tacgatgcat cgactctaga ttctggggtc 180

tcatcgcggt tcaaaggcag tggatctggg acacagttca ctctaccat cagcgacgtg 240

cagtgtagcg atgggtgcc ttactactgt gtaggcgggt atagtagtcg tagtgataat 300

ggtttcggcg gagggaccga ggtgggtggtc aaa 333

<210> SEQ ID NO 305
<211> LENGTH: 339
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 305

```

cagtcgctgg aggagtcggt gggctgcctg gtaacgcctg gaggaccct gacactcacc    60
tgcacagtct ctggattctc cctcagcagc tacgacatga gctgggtccg ccaggctcca    120
gggaaggggc tggagtggat cggatatatt gctactgatg gtaggccata ttacgcgagc    180
tgggcgaaag gccgattcac catctccaaa ccctcgtcga ccacgggtgga tctgaaaatc    240
accagtccga caaccgagga cacggccacc tatttctgtg tcagaggggg gtatgctggt    300
ggcttggtgg gccaggcac cctggtcacc gtttcctca                               339

```

<210> SEQ ID NO 306

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 306

```

Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
 1             5             10            15
Gly Thr Val Ser Ile Ser Cys Gln Ser Ser Lys Ser Val Tyr Asn Asn
          20             25            30
Asn Trp Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Arg Pro Lys Leu
          35             40            45
Leu Ile Tyr Asp Ala Ser Thr Leu Asp Ser Gly Val Ser Ser Arg Phe
          50             55            60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Val
          65             70            75            80
Gln Cys Asp Asp Gly Ala Thr Tyr Tyr Cys Val Gly Gly Tyr Ser Ser
          85             90            95
Arg Ser Asp Asn Gly Phe Gly Gly Gly Thr Glu Val Val Val Lys
          100            105           110

```

<210> SEQ ID NO 307

<211> LENGTH: 113

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 307

```

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Gly Pro
 1             5             10            15
Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr Asp
          20             25            30
Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
          35             40            45
Tyr Ile Ala Thr Asp Gly Arg Pro Tyr Tyr Ala Ser Trp Ala Lys Gly
          50             55            60
Arg Phe Thr Ile Ser Lys Pro Ser Ser Thr Thr Val Asp Leu Lys Ile
          65             70            75            80
Thr Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Val Arg Gly

```

-continued

	85	90	95	
Gly Tyr Ala Gly Gly Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser				
	100	105	110	

Ser

<210> SEQ ID NO 308
 <211> LENGTH: 336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 308

gacattgtga tgacccagac tccagcctcc gtggaggcag ctgtgggagg cacagtcacc	60
atcaagtgcc aggccagtca gagtattagt agttacttat cctgggtatca gcagaaacca	120
gggcagcctc ccaagctect gatctacagg gcatccactc tggcatctgg ggtcccatcg	180
cggttcagag gcagtggatc tgggacacag ttcactctca ccatcagcga cctggagtgt	240
gccgatgctg ccacttatta ctgtcaaacg tatgatgata gtagtgataa taattttttt	300
tatgggtttcg gcggaggggac cgaggtggtg gtcaga	336

<210> SEQ ID NO 309
 <211> LENGTH: 339
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 309

cagtcgttgg aggagtccgg gggaggcctg gtccagcctg agggatccct gacactcacc	60
tgccacagctt ttggattcac cctcaataat tattatattt gctgggtccg ccagggtcca	120
gggaaggggac tggagtggat cgcatgcatt gacaatgcta atggtaggac ttactacgcg	180
agctggggcga aaggccgatt caccatctcc aaaacctcgt cgaccacggt gactctacaa	240
atgaccagtc tgacagccgc ggacacggcc acctatttct gtgcgaggtc attgtctact	300
cccttggtggg gccagggcac cctggtcacc gtctcctca	339

<210> SEQ ID NO 310
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 310

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly	
1 5 10 15	
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr	
20 25 30	
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile	
35 40 45	
Tyr Arg Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe Arg Gly	
50 55 60	

-continued

Ser	Gly	Ser	Gly	Thr	Gln	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys
65					70					75					80

Ala	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Ser	Tyr	Asp	Asp	Ser	Ser	Asp
				85					90					95	

Asn	Asn	Phe	Phe	Tyr	Gly	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Arg
		100					105						110		

<210> SEQ ID NO 311

<211> LENGTH: 113

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 311

Gln	Ser	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Glu	Gly	Ser
1			5						10					15	

Leu	Thr	Leu	Thr	Cys	Thr	Ala	Phe	Gly	Phe	Thr	Leu	Asn	Asn	Tyr	Tyr
		20						25					30		

Ile	Cys	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala
		35				40						45			

Cys	Ile	Asp	Asn	Ala	Asn	Gly	Arg	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	Lys
	50					55					60				

Gly	Arg	Phe	Thr	Ile	Ser	Lys	Thr	Ser	Ser	Thr	Val	Thr	Leu	Gln
65				70						75				80

Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg
			85						90					95	

Ser	Leu	Ser	Thr	Pro	Leu	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser
			100					105					110		

Ser

<210> SEQ ID NO 312

<211> LENGTH: 336

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 312

gacattgtga tgaccagac tccagcctcc gtggaggcag ctgtgggagg cacagtcacc	60
atcaagtgcc aggccagtca gaacattgat agttacttag cctgggtatca gcagaaacca	120
gggcagcctc ccaagctcct gatctacagg gcattccactc tggcatctgg ggtcccatcg	180
cgggttcaaag gcagtggatc tgggacagag ttcactctca ccatcagcga cctggagtgt	240
gccgatgctg ccacttacta ctgtcaaaag tatgatgata gtaggagtag tagttttttt	300
tatgggtttcg gcggaggagc cgagggtggtg gtcaaaa	336

<210> SEQ ID NO 313

<211> LENGTH: 339

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 313

-continued

```

cagtcgttgg aggagtcgagg gggaggcctg gtccagcctg agggatccct gacactcacc      60
tgcacagctt ctggagtcac cctcagtaac tattatattt gttgggtccg ccagggtcca      120
gggaagggggc tggagtggat cgggtgcatt gacaatgta atggtaggac ctactacgcg      180
agctggggcga aaggccgatt caccatctcc aagacctcgt cgaccacagg gactctacaa      240
atgaccagtc tgacagccgc ggacacggcc acctatttct gtgcgaggtc cttggctact      300
cccttgtggg gccccaggcac cctagtcacc gtctcctca                               339

```

```

<210> SEQ ID NO 314
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 314

```

```

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1             5             10             15

```

```

Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Asn Ile Asp Ser Tyr
      20             25             30

```

```

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
      35             40             45

```

```

Tyr Arg Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
      50             55             60

```

```

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
65             70             75             80

```

```

Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Arg Ser
      85             90             95

```

```

Ser Ser Phe Phe Tyr Gly Phe Gly Gly Thr Glu Val Val Val Lys
      100            105            110

```

```

<210> SEQ ID NO 315
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 315

```

```

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1             5             10             15

```

```

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Thr Leu Ser Asn Tyr Tyr
      20             25             30

```

```

Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
      35             40             45

```

```

Cys Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
      50             55             60

```

```

Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Gly Thr Leu Gln
65             70             75             80

```

```

Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
      85             90             95

```

```

Ser Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser
      100            105            110

```

-continued

Ser

<210> SEQ ID NO 316
 <211> LENGTH: 336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide
 <400> SEQUENCE: 316
 gacattgtga tgacccagac tccagcctcc gtggaggcag ctgtgggagg cacagtcacc 60
 atcaagtgcc aggccagtca gaacattaat agttacttag cctgggtatca gcagaaacca 120
 gggcagcctc ccaagctcct gatctacagg gcatccactc tggcatctgg ggtcccatcg 180
 cgggtcaaag gcagtggatc tgggacagag ttcactctca ccatcagcga cctggagtggt 240
 gccgatgctg ccacttacta ctgtcaaagc tatgatgata gtaggagtat tagttttttt 300
 tatgctttcg gcggaggggac cgaggtggtg gtcaaaa 336

<210> SEQ ID NO 317
 <211> LENGTH: 339
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide
 <400> SEQUENCE: 317
 cagtcggttg aggagtcagg gggaggcctg gtccagcctg agggatccct gacactcacc 60
 tgcacagctt ctggagtcac cctcactaac tactacatct gctgggtccg ccagggtcca 120
 gggaaggggc tggagtggat cgggtgcatt gacaatgta atggtaggac ttactacgcg 180
 agctgggcga aaggccgatt caccatctcc aaggcctcgt cgaccacagg gactctacaa 240
 atgaccagtc tgacagccgc ggacacggcc acctatttct gtgcgaggtc cttggctact 300
 cccttggtgg gcccaggcac cctggtcacc gtctcctca 339

<210> SEQ ID NO 318
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 318

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
 1 5 10 15
 Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Asn Ile Asn Ser Tyr
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
 35 40 45
 Tyr Arg Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
 65 70 75 80
 Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Arg Ser

-continued

	85	90	95
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Ile Ser Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

<210> SEQ ID NO 319
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 319

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1 5 10 15

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Thr Leu Thr Asn Tyr Tyr
20 25 30

Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
35 40 45

Cys Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
50 55 60

Gly Arg Phe Thr Ile Ser Lys Ala Ser Ser Thr Thr Gly Thr Leu Gln
65 70 75 80

Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85 90 95

Ser Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser
100 105 110

Ser

<210> SEQ ID NO 320
<211> LENGTH: 336
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 320

gacattgtga tgaccagac tccagcctcc gtggaggcag ctgtgggagg cacagtcacc	60
atcaagtgcc aggccagtca gacattagt agctacttag cctgggtatca ccagaaacca	120
gggcagcctc ccaagctcct gatctatgct gcatccactc tggcatctgg ggtcccatcg	180
cggttcgaag gcagtggatc tgggacacag ttcactctca ccatcagcga cctggagtgt	240
gccgatgctg ccacttacta ctgtcaaagc tatgatgata gtaggagtag tagttttttt	300
tatgctttcg gcggaggggac cgagggtggtg gtcaaaa	336

<210> SEQ ID NO 321
<211> LENGTH: 339
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 321

cagtcggttg aggagtccgg gggaggcctg gtccagcctg agggatccct gacagtcacc	60
tgcacagctt ttggagtcac cctcactaac tactacatct gctgggtccg ccagggtcca	120

-continued

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gggaaggggc tggagtggat cgggtgcatt gacaaatatta atggtaggac ttactacgcg    180
agctggggcga aaggccgatt caccatctcc aagacctcgt cgaccacagg gactctgcaa    240
atgaccagtc tgacagccgc ggacacggcc acctatttct gtgcgaggtc cttggctact    300
cccttggtggg gccacggcac cctggtcacc gtctcctca                               339

```

```

<210> SEQ ID NO 322
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

```

```

<400> SEQUENCE: 322

```

```

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1           5           10           15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr
20          25          30
Leu Ala Trp Tyr His Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35          40          45
Tyr Ala Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe Glu Gly
50          55          60
Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
65          70          75          80
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Arg Ser
85          90          95
Ser Ser Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100         105         110

```

```

<210> SEQ ID NO 323
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

```

```

<400> SEQUENCE: 323

```

```

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1           5           10           15
Leu Thr Val Thr Cys Thr Ala Phe Gly Val Thr Leu Thr Asn Tyr Tyr
20          25          30
Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
35          40          45
Cys Ile Asp Asn Ile Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
50          55          60
Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Gly Thr Leu Gln
65          70          75          80
Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85          90          95
Ser Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser
100         105         110

```

```

Ser

```

-continued

```

<210> SEQ ID NO 324
<211> LENGTH: 336
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 324

gacattgtga tgacccagac tccagcctcc gtggaggcag ctgtgggagg cacagtcacc      60
atcaagtgcc aggccagtca gagcattagc agttgggttat cctgggtatca gcagaaacca      120
gggcagcctc ccaagctcct gctctacagg gcaaccactc tggcatctgg ggtcccatcg      180
cggttcaaag gcagtggatc tgggacacag ttcactctca ccatcagcga cctggagtgt      240
gccgatgctg ccacttatta ctgtcaaagt tatgatgata gtagtagtag taattttttt      300
tatgctttcg gcggaggggac cgagggtggtg gtcaga                                336


<210> SEQ ID NO 325
<211> LENGTH: 339
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 325

cagtcgttgg aggagtccgg gggaggcctg gtccagcctg agggatccct gacactcacc      60
tgcacagctt ctggattcac cctcactaat tattatatat gttgggtccg ccaggctcca      120
gggaaggggac tggagtggat cgcattgtatt gataacgcta atggtaggac ttactacgcg      180
acctggggcga aaggccgatt caccatctcc aaaacctcgt cgaccacggg gactctgcaa      240
atgcccgatc tgacagccgc ggacacggcc acctatttct gtgcgaggtc attgtctact      300
aacttggtggg gccacggcac cctggtcacc gtctctctca                                339


<210> SEQ ID NO 326
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 326

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1         5         10        15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Trp
20        25        30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Leu
35        40        45
Tyr Arg Ala Thr Thr Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
50        55        60
Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
65        70        75        80
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Ser Ser
85        90        95
Ser Asn Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Arg
100       105       110

```

-continued

<210> SEQ ID NO 327
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 327

```
Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1          5          10          15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Thr Leu Thr Asn Tyr Tyr
20          25          30
Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala
35          40          45
Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Thr Trp Ala Lys
50          55          60
Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln
65          70          75          80
Met Pro Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85          90          95
Ser Leu Ser Thr Asn Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser
100         105         110
```

Ser

<210> SEQ ID NO 328
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 328

```
caagccgtgg tgaccagac tccatcgcc gtgtctgcag ctgtgggagg cacagtcacc      60
atcagttgcc agtccagtca gagtgtttat agtaacaacc tcttatcttg gtatcagcag      120
aaaccagggc agcctcccaa gctcctgata tacaaggcat ccactctggc atctggggtc      180
ccatcgcggt tgaaaggcag tggatctggg acacagttca ctctcacaat cagcgaagta      240
cagtgtagcg atgctgccac ttattactgt caaggtatt atagtgggtg tatttatatg      300
ttcgcgggag ggaccgaggt ggtgggtcaaa                                330
```

<210> SEQ ID NO 329
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 329

```
cagtcggttg aggagtcgg gggaggcctg gtcaagcctg gggcatccct gacactcacc      60
tgcaaagcct ctgggtcaga cttcagaagc tactactaca tatgctgggt cggcagggt      120
ccagggaagg ggctggagtg ggtcgcatgt attggtgggt gtaataccga tgccactgcc      180
tacgcgaggt gggcgaaagg ccgattcacc atctccaaaa cctcggcgac cacggtggct      240
```

-continued

ctccaaatga ccagttctgac agccgcggac acggccacct atttctgtgt gagaggcgggt 300

cctgataata atgtccaatt taacttgtgg ggcccaggca ccctggtcac cgtctcctca 360

<210> SEQ ID NO 330

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 330

Gln Ala Val Val Thr Gln Thr Pro Ser Ser Val Ser Ala Ala Val Gly
1 5 10 15

Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Val Tyr Ser Asn
20 25 30

Asn Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45

Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Leu
50 55 60

Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Val
65 70 75 80

Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Gln Gly Tyr Tyr Ser Gly
85 90 95

Val Ile Tyr Met Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

<210> SEQ ID NO 331

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 331

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala Ser
1 5 10 15

Leu Thr Leu Thr Cys Lys Ala Ser Gly Ser Asp Phe Arg Ser Tyr Tyr
20 25 30

Tyr Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Cys Ile Gly Gly Gly Asn Thr Asp Ala Thr Ala Tyr Ala Arg Trp
50 55 60

Ala Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ala Thr Thr Val Ala
65 70 75 80

Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
85 90 95

Val Arg Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu Trp Gly Pro
100 105 110

Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 332

<211> LENGTH: 336

<212> TYPE: DNA

-continued

<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 332

gacattgtga tgacccagac tccattctcc gtggaggcag ctgtgggagg cacagtcacc	60
atcaagtgcc aggccagtca gagcattagt agttacttat cctggtatcg gcagaaacca	120
gggcagcctc ccaagctcct gatctacagg gcattccactc tggcatctgg ggtcccatcg	180
cgggtcctaaag gcagtggatc tgggacacag ttcactctca ccatcagcga cctggagtgt	240
gccgatgctg ccacttacta ctgtcaaagc tatgatgata gtagtagtaa taattttttt	300
tatgggtttcg gcggaggggac cgagggtggg gtcaga	336

<210> SEQ ID NO 333
 <211> LENGTH: 339
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 333

cagtcggttg aggagtcagg gggaggcctg gtccagcctg agggatccct gacactcacc	60
tgacagagct ctggattcac cctcaataat tattatattt gctgggtccg ccaggctcca	120
gggaaggggac tggagtggat cgcattgcatt gacaatagta atggtaggac ttactacgcg	180
agctggggcga aaggccgttt caccatctcc aaaacctcgt cgaccacggg gactctgcaa	240
atgaccagtc tgacagccgc ggacacggcc acctatttct gtgcgaggtc attgtctact	300
cccttggtggg gccacggcac cctggtcacc gtctctca	339

<210> SEQ ID NO 334
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 334

Asp Ile Val Met Thr Gln Thr Pro Phe Ser Val Glu Ala Ala Val Gly	
1 5 10 15	
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr	
20 25 30	
Leu Ser Trp Tyr Arg Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile	
35 40 45	
Tyr Arg Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly	
50 55 60	
Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys	
65 70 75 80	
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Ser Ser	
85 90 95	
Asn Asn Phe Phe Tyr Gly Phe Gly Gly Gly Thr Glu Val Val Val Arg	
100 105 110	

<210> SEQ ID NO 335

<400> SEQUENCE: 335

Ser Leu Ser Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser
100 105 110

Ser

<400> SEQUENCE: 336

gacattgtga	tgaccagac	tccatctctc	gtggaggcag	ctgtgggagg	cacagtcacc	60
atcaagtgcc	aggccagtc	gaacattagt	agctacttag	cctgggatca	gcagaacca	120
gggcagcctc	ccaagctcct	gatctacagg	gcatccactc	tggcatctgg	ggccccatcg	180
cgggtcaaag	gcagtggtac	tgggacacag	ttcactctca	ccatcagcga	cctggagtg	240
gccgatgctg	ccacttacta	ctgtcaaagc	tatgatgata	gtaggagtag	taattttttt	300
tatgcttttc	gcggagggac	cgaagtggtg	gtcaaaa			336

<400> SEQUENCE: 337

cagtcgttgg	aggagtcagg	gggaggcctg	gtccagcctg	agggatccct	gacactcacc	60
tgcacagctt	ttggagtcac	cctcactaac	tactacatct	gctgggtccg	ccaggtccca	120
ggaaaggggc	tggagtgggt	cggtgttatt	gacaatgcta	atggtaggac	ttactacgcg	180
agctggggcg	aaggccgatt	caccatctcc	aagacctcgt	cgaccacagg	gactctgcaa	240
atgaccagtc	tqacagccgc	ggacacgccc	acctatttct	gtccgaagtc	cttgqctact	300

-continued

cccttggtggg gccacggcac cctggtcacc gtctctca

339

<210> SEQ ID NO 338

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 338

Asp Ile Val Met Thr Gln Thr Pro Ser Ser Val Glu Ala Ala Val Gly
1 5 10 15Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Asn Ile Ser Ser Tyr
20 25 30Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45Tyr Arg Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
50 55 60Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
65 70 75 80Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Arg Ser
85 90 95Ser Asn Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

<210> SEQ ID NO 339

<211> LENGTH: 113

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 339

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1 5 10 15Leu Thr Leu Thr Cys Thr Ala Phe Gly Val Thr Leu Thr Asn Tyr Tyr
20 25 30Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Gly
35 40 45Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
50 55 60Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Gly Thr Leu Gln
65 70 75 80Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85 90 95Ser Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser
100 105 110

Ser

<210> SEQ ID NO 340

<211> LENGTH: 330

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

-continued

<400> SEQUENCE: 340

```

gcgcaagcgc tgacccagac tccatcccct gtgtctgcag ctgtgggagg cacagtcacc    60
atcagttgcc agtccagtga gagcgtttat aatcacaact ggtaggctg gtatcagcag    120
aaaccagggc agcctcccaa actcctgacg tatgatgcat ccactctggc atctggggtc    180
ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctcacaat cagcgaaagt    240
cagtggtgacg atgctgccat ttactactgt caaggctatt atcaaaactag tgtttgggct    300
ttcggcgggag ggaccgaggt ggtggtcaaa    330

```

<210> SEQ ID NO 341

<211> LENGTH: 351

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 341

```

cagtcggtgg aggagtcggt gggtcgcctg gtcacgcctg ggacacccct gacactcacc    60
tgcacagttt ctggagtgcg cctcagtagc agtgcagtga cctgggtccg ccaggtccca    120
gggatgggac tggaatacat cggattcttc caagctgggg atggtagcgc atactacgcg    180
agctggggcga aaggccgatt caccatctcc aaaacctcgt cgaccacagt ggatctgaaa    240
atgaccagtc tcacaaccga ggacacggcc acctatttct gtgccagaca taagggtaat    300
agttacgtgc ctaactgtg gggcccaggc accctggtca ccgtctcttc a    351

```

<210> SEQ ID NO 342

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 342

```

Ala Gln Ala Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1           5           10          15
Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Glu Ser Val Tyr Asn His
20          25          30
Asn Trp Leu Gly Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35          40          45
Leu Ile Tyr Asp Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
50          55          60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Ser
65          70          75          80
Gln Cys Asp Asp Ala Ala Ile Tyr Tyr Cys Gln Gly Tyr Tyr Gln Thr
85          90          95
Ser Val Trp Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100         105         110

```

<210> SEQ ID NO 343

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 343

```

Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1          5          10          15
Leu Thr Leu Thr Cys Thr Val Ser Gly Val Asp Leu Ser Ser Ser Ala
20        25        30
Val Thr Trp Val Arg Gln Ala Pro Gly Met Gly Leu Glu Tyr Ile Gly
35        40        45
Phe Leu Gln Ala Gly Asp Gly Ser Ala Tyr Tyr Ala Ser Trp Ala Lys
50        55        60
Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Asp Leu Lys
65        70        75        80
Met Thr Ser Leu Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85        90        95
His Lys Gly Asn Ser Tyr Val Pro Asn Leu Trp Gly Pro Gly Thr Leu
100       105       110
Val Thr Val Ser Ser
115

```

<210> SEQ ID NO 344

<211> LENGTH: 336

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 344

```

gatgttgatga tgaccagac tccagcctcc gtggaggcaa ctgtgggagg cacagtcacc      60
atcaagtgcc agtccagtca gagggtttat gataacaatg ctttagcctg gtatcagcag      120
aatgcaggac agcgctccag actcctgatac tatggtgcat ccactctggc atctgggggc      180
ccatcgcggt tcaagtccag tggatctggg acagagttca ctctcaccat cagcgacctg      240
gagtggtccg atgcagccac ttactactgt caatgtactt attatgtag tagttatcaa      300
aatgatttcg gcggaggagc cgagggtggtg gtcaaaa                                336

```

<210> SEQ ID NO 345

<211> LENGTH: 336

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 345

```

cagtcggtgg aggagtcagg gggtcgctg gtcacgcctg ggacaccct gacactcacc      60
tgcaaatct ctggattctc cctcagtagc tatgcaatga gctgggtccg ccagggtcca      120
gggaaggggc tggaatggat cggaagcatt ggtggtggtg gtagcgcagt ctacgcgagc      180
tgggcgaaag gccgattcac catctccaaa acctcgacca cgggtggatct gagaatcacc      240
agtccgacaa ccgaggacac ggccatgtat ttctgtggca ggggatttta tagtatagac      300
ttgtggggcc caggcacccct ggtcacccgc tcctca                                336

```

-continued

<210> SEQ ID NO 346
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 346

```

Asp Val Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Thr Val Gly
1          5          10          15
Gly Thr Val Thr Ile Lys Cys Gln Ser Ser Gln Ser Val Tyr Asp Asn
20          25          30
Asn Ala Leu Ala Trp Tyr Gln Gln Asn Ala Gly Gln Arg Pro Arg Leu
35          40          45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
50          55          60
Ser Ala Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Asp Leu
65          70          75          80
Glu Cys Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Cys Thr Tyr Tyr Val
85          90          95
Ser Ser Tyr Gln Asn Asp Phe Gly Gly Gly Thr Glu Val Val Val Lys
100         105         110

```

<210> SEQ ID NO 347
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 347

```

Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1          5          10          15
Leu Thr Leu Thr Cys Thr Ile Ser Gly Phe Ser Leu Ser Ser Tyr Ala
20          25          30
Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
35          40          45
Ser Ile Gly Gly Gly Gly Ser Ala Val Tyr Ala Ser Trp Ala Lys Gly
50          55          60
Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Arg Ile Thr
65          70          75          80
Ser Pro Thr Thr Glu Asp Thr Ala Met Tyr Phe Cys Gly Arg Gly Phe
85          90          95
Tyr Ser Ile Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
100         105         110

```

<210> SEQ ID NO 348
 <211> LENGTH: 336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 348

gccgcgctgc tgaccacagac tccatctccc gtgtctgcag ctgtgggagg cacagtcagc 60

-continued

```

gccagttgcc agtccagtaa gagtgtttat aataagaact gggtatcctg gtttcagcag 120
aaaccagggc agcctcccaa gctcctgac tatggtgcat ccactctggc atctggggtc 180
ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctcaccat cagcgacgtg 240
cagtgtgacg atgctgccac ttactactgt gcaggcgggt atagtagtag tagtgataca 300
tttgctttcg gcggaggggac cgagggtggtg gtcaag 336

```

```

<210> SEQ ID NO 349
<211> LENGTH: 375
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
    polynucleotide

```

```

<400> SEQUENCE: 349
caggagcagc tgggtggagtc cgggggaggc ctggtcaagc ctggggcatc cctgacactc 60
acctgcaaag cctctggatt ctccctcagt agcggccaac tcatgtgctg ggtccgcag 120
gctccaggga aggggctgga gtggatcgca tgcattggtt ctggtagtaa tgctattagc 180
actttctacg cgagctgggc gcaaggccga ttcaccatct ccaaaccctc gtcgaccacg 240
gtgactctgc aattgaccag tctgacagcc gcggacacgg ccacctattt ctgtgcgaga 300
gtgggctccg atgactatgg tgactctgat gtttttgatc cctggggccc aggcaccctg 360
gtcacctctc cctca 375

```

```

<210> SEQ ID NO 350
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
    polypeptide

```

```

<400> SEQUENCE: 350
Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1          5          10         15
Gly Thr Val Ser Ala Ser Cys Gln Ser Ser Lys Ser Val Tyr Asn Lys
20         25         30
Asn Trp Leu Ser Trp Phe Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35         40         45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
50         55         60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Val
65         70         75         80
Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Ala Gly Gly Tyr Ser Ser
85         90         95
Ser Ser Asp Thr Phe Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100        105        110

```

```

<210> SEQ ID NO 351
<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
    polypeptide

```

-continued

<400> SEQUENCE: 351

Gln Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala
 1 5 10 15
 Ser Leu Thr Leu Thr Cys Lys Ala Ser Gly Phe Ser Phe Ser Ser Gly
 20 25 30
 Gln Leu Met Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
 35 40 45
 Ile Ala Cys Ile Gly Ser Gly Ser Asn Ala Ile Ser Thr Phe Tyr Ala
 50 55 60
 Ser Trp Ala Gln Gly Arg Phe Thr Ile Ser Lys Ser Ser Ser Thr Thr
 65 70 75 80
 Val Thr Leu Gln Leu Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr
 85 90 95
 Phe Cys Ala Arg Val Gly Ser Asp Asp Tyr Gly Asp Ser Asp Val Phe
 100 105 110
 Asp Pro Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 352

<211> LENGTH: 330

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 352

gcccgaagtgc tgaccagac tccagcctcc gtgtctgcag ctgtgggagg cacagtcacc 60
 atcagttgcc agtccagtca gacggtttat agtagcgacc tcttatcctg gtatcagcag 120
 aaaccagggc agcctcccaa gctcctgac tacaaggcat ccactctggc atctggggtc 180
 ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctcacaat cagcgaacta 240
 cagtgtgacg atgctgccac ttattactgt caaggetact atagtgggtg ggtttatatt 300
 tcggcgaggag ggaccgaggt ggtgggtcaag 330

<210> SEQ ID NO 353

<211> LENGTH: 360

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 353

cagtcggttg aggagtcagg gggaggcctg gtcaagcctg gggcatccct gacactcacc 60
 tgcaaagcct ctgggtcaga cctcagtagc gcctactaca tatgctggat ccgccaggct 120
 ccagggaagg ggctggagtg ggtcgcatgt attggtgggt ttaatcgctg tgccactgcc 180
 tacgcgacct gggcgaaagg ccgattcacc atctccaaaa cctcgctcgac caggtgact 240
 ctgcaaatga ccagtctgac agccgcggac acggccactt attctgtgt gagaggcgg 300
 cctgataata atgtccaatt taactgtgg ggcccaggca cctgggtcac cgtctcctca 360

<210> SEQ ID NO 354

<211> LENGTH: 110

<212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 354

```

Ala Gln Val Leu Thr Gln Thr Pro Ala Ser Val Ser Ala Ala Val Gly
 1             5             10             15
Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Val Tyr Ser Ser
 20             25             30
Asp Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
 35             40             45
Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
 50             55             60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Leu
 65             70             75             80
Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Gln Gly Tyr Tyr Ser Gly
 85             90             95
Val Val Tyr Ile Phe Gly Gly Gly Thr Glu Val Val Val Lys
 100            105            110

```

<210> SEQ ID NO 355
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 355

```

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala Ser
 1             5             10             15
Leu Thr Leu Thr Cys Lys Ala Ser Gly Ser Asp Leu Ser Ser Ala Tyr
 20             25             30
Tyr Ile Cys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35             40             45
Ala Cys Ile Gly Gly Val Asn Arg Val Ala Thr Ala Tyr Ala Thr Trp
 50             55             60
Ala Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr
 65             70             75             80
Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
 85             90             95
Val Arg Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu Trp Gly Pro
 100            105            110
Gly Thr Leu Val Thr Val Ser Ser
 115            120

```

<210> SEQ ID NO 356
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 356

gcccaagtgc tgaccacagac tccagcctcc gtgtctgcag ctgtgggagg cacagtcacc 60

-continued

```

atcagttgcc agtccagtcag gacggtttat agtagcgacc tcttatcctg gtatcagcaa 120
aaaccagggc agcctcccaa gctcctgatac tacaaggcat ccactctggc atctggggtc 180
ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctcacaat cagcgaagta 240
cagtgtgacg atgctgccac ttattactgt caaggctact atagtgggtg ggtttatatt 300
ttcggcggag ggaccgaggt ggtgggtcaag 330

```

```

<210> SEQ ID NO 357
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

```

```

<400> SEQUENCE: 357

cagtcggttg aggagtcagg gggaggcctg gtcaagcctg gggcatccct gacactcacc 60
tgcaaagcct ctgggtcaga cctcagtagg tactactaca gttgctggat ccgccaggct 120
ccagggaagg ggctggagtg ggtcgcatgt gttggtgggtg ttaatcgaga tgccactgcc 180
tacgcgacct gggcgaaagg ccgattcacc atctccaaaa ctcgctcgac cacggtgact 240
ctgcaaatga ccagtctgac agccgcggac acggccactt attctgtgtg gagaggcggt 300
cctgataata atgtccaatt taactgttg ggcccaggca ccctggtcac cgtctctca 360

```

```

<210> SEQ ID NO 358
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

```

```

<400> SEQUENCE: 358

Ala Gln Val Leu Thr Gln Thr Pro Ala Ser Val Ser Ala Ala Val Gly
1      5      10      15
Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Val Tyr Ser Ser
20     25     30
Asp Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35     40     45
Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
50     55     60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Val
65     70     75     80
Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Gln Gly Tyr Tyr Ser Gly
85     90     95
Val Val Tyr Ile Phe Gly Gly Gly Thr Glu Val Val Val Lys
100    105    110

```

```

<210> SEQ ID NO 359
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

```

```

<400> SEQUENCE: 359

```

-continued

Gln	Ser	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Gly	Ala	Ser
1			5						10					15	
Leu	Thr	Leu	Thr	Cys	Lys	Ala	Ser	Gly	Ser	Asp	Leu	Ser	Arg	Tyr	Tyr
		20						25					30		
Tyr	Ser	Cys	Trp	Ile	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
		35				40						45			
Ala	Cys	Val	Gly	Gly	Val	Asn	Arg	Asp	Ala	Thr	Ala	Tyr	Ala	Thr	Trp
	50					55					60				
Ala	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Lys	Thr	Ser	Ser	Thr	Thr	Val	Thr
65					70				75					80	
Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys
			85						90					95	
Val	Arg	Gly	Gly	Pro	Asp	Asn	Asn	Val	Gln	Phe	Asn	Leu	Trp	Gly	Pro
		100						105					110		
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser								
		115					120								

<210> SEQ ID NO 360
 <211> LENGTH: 336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 360

gacattgtga tgacccagac tccattctcc gtggaggcag ctgtgggagg cacagtcacc	60
atcaagtgcc aggccagtca gaggcattagt agttacttat cttggtatca gcagaaacca	120
gggcagcctc ccaagctcct gatctacagg gcatcccctc tggcatctgg ggtcccatcg	180
cggttcaaag gcagtggatc tgggacacag ttcactctca ccatcagcga cctggagtggt	240
gcccgatgctg ccacttacta ctgtcaaagc tacgatgata gtagtagtaa taattttttt	300
tatggtttcg gcggaggggac cgaggtggtg gtcaga	336

<210> SEQ ID NO 361
 <211> LENGTH: 339
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 361

cagtcgttgg aggagtccgg gggaggcctg gtccagcctg agggatccct gacactcacc	60
tgacacagctt ctggattcac cctcaataat tattatattt gttgggtccg ccagggtcca	120
gggaaggggac tggagtggat cgcattgtatt gacaatgtta atggtaggac ttactacgcg	180
agctggggcga aaggccgatt caccatctcc agaacctcgt cgaccacgggt gactctgcaa	240
atgaccagtc tgacagccgc ggacacggcc acctatttct gtgcgaggtc attgtctact	300
cccttggtggg gccagggcac cctggtcacc gtctcctcg	339

<210> SEQ ID NO 362
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 362

```

Asp Ile Val Met Thr Gln Thr Pro Phe Ser Val Glu Ala Ala Val Gly
1         5             10             15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr
20        25        30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35        40        45
Tyr Arg Ala Ser Pro Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
50        55        60
Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
65        70        75        80
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Ser Ser
85        90        95
Asn Asn Phe Phe Tyr Gly Phe Gly Gly Gly Thr Glu Val Val Val Arg
100       105       110

```

<210> SEQ ID NO 363

<211> LENGTH: 113

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 363

```

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1         5             10             15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Thr Leu Asn Asn Tyr Tyr
20        25        30
Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala
35        40        45
Cys Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys
50        55        60
Gly Arg Phe Thr Ile Ser Arg Thr Ser Ser Thr Thr Val Thr Leu Gln
65        70        75        80
Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85        90        95
Ser Leu Ser Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser
100       105       110

```

Ser

<210> SEQ ID NO 364

<211> LENGTH: 336

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 364

```

gccgccgtgc tgaccagac tccatctccc gtgtctgcag ctgtgggagg cacagtcacc      60
atcaattgcc agtccagtaa gagggtttat aataagaact gggtatcctg gtttcagcag      120
aaaccagggc agcctcccaa gctcctgata tatggtgcat ccactctggc atctggggtc      180

```

-continued

ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctcaccat cagcgacgtg 240
cagtgtagac atgttgccac ttactactgt gcaggcgggt atagtagtag tagtgatacg 300
tttgctttcg gcgaggggac cgagggtggtg gtcaaaa 336

<210> SEQ ID NO 365
<211> LENGTH: 372
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 365

caggagcagc tgggtggagtc cgggggaggc ctggtccagc ctgagggatc cctgacactc 60
acctgcgcag cttctggatt ctcttcagt agcagctact tcatgtgctg ggtccgcag 120
gctccaggga aggggctgga gtggatcgca tgcattgctg ttggtagtag tggtagcact 180
tactacgcga gctgggcgaa aggccgattc accatctcca aaacctcgtc gaccacggtg 240
actctgcaaa tgaccagtct gacagccgcy gacacggcca cctatttctg tgcgagagtg 300
ggctacgatg actatggtga ctctgatgct ttgatacct ggggcccagg caccctggtc 360
accgtctcct ca 372

<210> SEQ ID NO 366
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 366

Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Thr Ile Asn Cys Gln Ser Ser Lys Ser Val Tyr Asn Lys
20 25 30
Asn Trp Leu Ser Trp Phe Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
50 55 60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Val
65 70 75 80
Gln Cys Asp Asp Val Ala Thr Tyr Tyr Cys Ala Gly Gly Tyr Ser Ser
85 90 95
Ser Ser Asp Thr Phe Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

<210> SEQ ID NO 367
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 367

Gln Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly

-continued

1	5	10	15
Ser Leu Thr	Leu Thr Cys Ala Ala	Ser Gly Phe Ser Phe Ser Ser	
	20	25	30
Tyr Phe Met	Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp		
	35	40	45
Ile Ala Cys	Ile Ala Val Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser		
	50	55	60
Trp Ala Lys	Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val		
	65	70	75
Thr Leu Gln	Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe		
	85	90	95
Cys Ala Arg	Val Gly Tyr Asp Asp Tyr Gly Asp Ser Asp Ala Phe Asp		
	100	105	110
Pro Trp Gly	Pro Gly Thr Leu Val Thr Val Ser Ser		
	115	120	

<210> SEQ ID NO 368
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 368

gccccagtgc tgaccagac tccagcctcc gtgtctgcag ctgtgggagg cacagtcacc	60
atcagttgcc agtccagtca gagcctttat aatagegacc tcttatcctg gtatcagcaa	120
aaaccagggc agcctcccaa gctcctgac tacaaggcat ccactctggc atctggggtc	180
ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctcacaat cagcgaagta	240
cagtgtagc atgctgccac ttattactgt caaggctact atagtgggtg ggtttatatt	300
ttcggcggag ggaccgaggt ggtgggtcaag	330

<210> SEQ ID NO 369
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 369

cagtcggttg aggagtcgg gggaggcctg gtcaagcctg gggcatccct gacactcacc	60
tgcaaagcct ctgggtcaga cctcagtagg tactactaca gttgctggat ccgccaggct	120
ccagggaagg ggctggagtg ggtcgcatgt gttggtggty ttaatcgga tgccactgcc	180
tacgcgacct gggcgaaagg ccgattcacc atctccaaaa cctcgtcgac cacggtgact	240
ctgcaaatga ccagtctgac agccgcggac acggccactt attctgtgt gagaggcgg	300
cctgataata atgtccaatt taacttgtg ggcccaggca ccctggtcac cgtctcctca	360

<210> SEQ ID NO 370
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

polypeptide

<400> SEQUENCE: 370

Ala Gln Val Leu Thr Gln Thr Pro Ala Ser Val Ser Ala Ala Val Gly
 1 5 10 15
 Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Leu Tyr Asn Ser
 20 25 30
 Asp Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
 35 40 45
 Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
 50 55 60
 Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Val
 65 70 75 80
 Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Gln Gly Tyr Tyr Ser Gly
 85 90 95
 Val Val Tyr Ile Phe Gly Gly Gly Thr Glu Val Val Val Lys
 100 105 110

<210> SEQ ID NO 371

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 371

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala Ser
 1 5 10 15
 Leu Thr Leu Thr Cys Lys Ala Ser Gly Ser Asp Leu Ser Arg Tyr Tyr
 20 25 30
 Tyr Ser Cys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ala Cys Val Gly Gly Val Asn Arg Asp Ala Thr Ala Tyr Ala Thr Trp
 50 55 60
 Ala Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr
 65 70 75 80
 Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
 85 90 95
 Val Arg Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu Trp Gly Pro
 100 105 110
 Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 372

<211> LENGTH: 336

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 372

gccgccgtgc tgaccagac tccatctccc gtgtctgcag ctgtgggagg cacagtcagc 60
 atcagttgcc agtccagtaa gagtgtttat aataagaact gggtatcctg gtttcagcag 120
 aaaccagggc agcctcccaa gctcctgatac tatggtgcat ccactctggc atctggggtc 180

-continued

```
ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctcaccat cagcgacgtg    240
cagtggtgacg atgctgccac ttactactgt gcaggcgggt atagtagtag tagtgataca    300
tttgctttcg gcgaggggac cgagggtggtg gtcaaaa                            336
```

```
<210> SEQ ID NO 373
<211> LENGTH: 375
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide
```

```
<400> SEQUENCE: 373
```

```
caggagcagc tgggtggagtc cgggggagggc ctgggtccagc ctaagggtac cctgacactc    60
acctgcgcag cttctggatt ctcttcagc agcagctact tcatgtgctg ggtccgcag      120
gtccaggga aggggctgga gtggatcgca tgcattggtt ctggtagtag tgctattagc      180
actttctacg cgagctgggc gcaaggccga ttcaccatct ccaaacctc gtcgaccacg      240
gtgactctgc aaatgaccag tctgacagcc gcggacacgg ccacctattt ctgtgcgaga      300
gtgggctacg atgactatgg tgactctgat gcttttgatc cctggggccc aggcaccctg      360
gtcacgctct cctca                            375
```

```
<210> SEQ ID NO 374
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide
```

```
<400> SEQUENCE: 374
```

```
Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
 1             5             10            15
Gly Thr Val Ser Ile Ser Cys Gln Ser Ser Lys Ser Val Tyr Asn Lys
      20            25            30
Asn Trp Leu Ser Trp Phe Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
      35            40            45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
      50            55            60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Val
      65            70            75            80
Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Ala Gly Gly Tyr Ser Ser
      85            90            95
Ser Ser Asp Thr Phe Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
      100           105           110
```

```
<210> SEQ ID NO 375
<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide
```

```
<400> SEQUENCE: 375
```

```
Gln Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Lys Gly
```

-continued

1	5	10	15
Ser Leu Thr	Leu Thr Cys Ala Ala	Ser Gly Phe Ser Phe Ser Ser	
	20	25	30
Tyr Phe Met	Cys Trp Val Arg Gln Ala Pro Gly Lys	Gly Leu Glu Trp	
	35	40	45
Ile Ala Cys	Ile Gly Ser Gly Ser Ser Ala Ile Ser Thr Phe Tyr Ala		
	50	55	60
Ser Trp Ala	Gln Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr		
65	70	75	80
Val Thr Leu	Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr		
	85	90	95
Phe Cys Ala	Arg Val Gly Tyr Asp Asp Tyr Gly Asp Ser Asp Ala Phe		
	100	105	110
Asp Pro Trp	Gly Pro Gly Thr Leu Val Thr Val Ser Ser		
	115	120	125

<210> SEQ ID NO 376
 <211> LENGTH: 336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 376

```

gagtgtgtga tgaccacac tccagcctcc ctctttgaaa ctgtgggagg cacagtcacc      60
atcaagtgcc aggcccttca gagtgtttat gataacaatg ctttatcctg gtatcaacaa      120
aatgcaggac agcgteccat actcctgatac tatggtgcat ccactctggc atctggggcc      180
ccatcgcggt tcaagtccag tggatctggg acagatttca ctctcaccat catcgacctg      240
gagtgtgccg atgcttcac ttactactgt caatgtactt attatgtag tagttatcaa      300
aatgatttcg gcggaggagc cgagggtgtg gtcaaaa                               336

```

<210> SEQ ID NO 377
 <211> LENGTH: 342
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 377

```

cagtcgctgg aggagtccgg gggtcgcctg gtaacgcctg ggacacccct gacactcacc      60
tgacagttct ctggaatcga cctcagtagc tatggaatga cttgggtccg ccaggctcca      120
gggaaggggc tggagtggat cggatacatt tggactgatg ggaggacata ctacgcaaac      180
tgggcgaaaag gccgattcac catctccaaa acctcgacca cagtggatct caagatcacc      240
agtccgacag ccgaggacac ggccacctat ttctgtgcca gaccctttga tggaattat      300
agggacatct ggggccaggg caccctggtc acgggtctct ta                               342

```

<210> SEQ ID NO 378
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

polypeptide

<400> SEQUENCE: 378

Asp Val Val Met Thr His Thr Pro Ala Ser Leu Phe Glu Thr Val Gly
 1 5 10 15
 Gly Thr Val Thr Ile Lys Cys Gln Ala Leu Gln Ser Val Tyr Asp Asn
 20 25 30
 Asn Ala Leu Ser Trp Tyr Gln Gln Asn Ala Gly Gln Arg Pro Ile Leu
 35 40 45
 Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Gly Ala Pro Ser Arg Phe
 50 55 60
 Ser Ala Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ile Asp Leu
 65 70 75 80
 Glu Cys Ala Asp Ala Ser Thr Tyr Tyr Cys Gln Cys Thr Tyr Tyr Val
 85 90 95
 Ser Ser Tyr Gln Asn Asp Phe Gly Gly Gly Thr Glu Val Val Val Lys
 100 105 110

<210> SEQ ID NO 379

<211> LENGTH: 114

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 379

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
 1 5 10 15
 Leu Thr Leu Thr Cys Thr Val Ser Gly Ile Asp Leu Ser Ser Tyr Gly
 20 25 30
 Met Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
 35 40 45
 Tyr Ile Trp Thr Asp Gly Arg Thr Tyr Tyr Ala Asn Trp Ala Lys Gly
 50 55 60
 Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Thr
 65 70 75 80
 Ser Pro Thr Ala Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Pro Phe
 85 90 95
 Asp Gly Asn Tyr Arg Asp Ile Trp Gly Pro Gly Thr Leu Val Thr Val
 100 105 110
 Ser Leu

<210> SEQ ID NO 380

<211> LENGTH: 336

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 380

gacattgtga tgaccagac tccagcctcc gtggaggcag ctgtgggagg cacagtcacc 60
 atcaagtgcc aggccagtga gagcattggc aatgcattag cctgggtatca gcagaaacca 120
 gggcagcctc ccagtctcct gatctacagg gcateccactc tggcatctgg aatcccatcg 180

-continued

```

cgggtcaaa gcaagtggatc tgggacacag ttcactctca ccatcagcga cctggagtgt    240
gccgatgctg ccacttacta ttgtcaaaagc tatgatgata gtagtagtag tagttttttt    300
tatgcttttcg gcggaggggac cgagggtggtg gtcaaaa                            336

```

```

<210> SEQ ID NO 381
<211> LENGTH: 339
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polynucleotide

```

```

<400> SEQUENCE: 381
cagtcgttgg aggagtccgg gggaggcctg gtccagcctg agggatccct gacactcacc    60
tgcacagctt ctggattcac cctcaataac tactacatct gctgggtccg ccaggctcca    120
gggaaggggc tggagtggat cgcattgtatt gataatgcca atggtcggac ttactacgcg    180
aactggggcg aagcccgatt caccatctcc aagacctcgt cgaccacggt gactctgcaa    240
atgaccagtc tgacagccgc ggacacggcc acctatttct gtgcgaggtc cttggctact    300
cccttggtgg gccacggcac cctggtcacc gtctcctca                            339

```

```

<210> SEQ ID NO 382
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

```

```

<400> SEQUENCE: 382
Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1           5           10          15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Glu Ser Ile Gly Asn Ala
20          25          30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Ser Leu Leu Ile
35          40          45
Tyr Arg Ala Ser Thr Leu Ala Ser Gly Ile Pro Ser Arg Val Lys Gly
50          55          60
Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
65          70          75          80
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Asp Asp Ser Ser Ser
85          90          95
Ser Ser Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100         105         110

```

```

<210> SEQ ID NO 383
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

```

```

<400> SEQUENCE: 383
Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1           5           10          15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Thr Leu Asn Asn Tyr Tyr

```

-continued

	20		25		30	
Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala						
	35		40		45	
Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Asn Trp Ala Lys						
	50		55		60	
Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln						
	65		70		75	80
Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg						
	85		90		95	
Ser Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser						
	100		105		110	

Ser

<210> SEQ ID NO 384
 <211> LENGTH: 336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 384

gccgccgtgc tgaccagac tccatctccc gtgtctgcag ctgtgggagg cacagtcagc	60
atcagttgcc agtccagtaa gagtgtttat aataagaact ggttatcctg gtttcagcag	120
aaaccagggc agcctcccaa gctcctgac tatggtgcat ccactctggc atctggggtc	180
ccatcgcggt tcaaaggcag tggatctggg acacagttca ctctaccat cagcgcggtg	240
cagtgtagc atgctgccac ttactactgt gcaggcggtt atagtagtag tagtgatacg	300
tttgctttcg gcggaggga caggtggtg gtcaaaa	336

<210> SEQ ID NO 385
 <211> LENGTH: 372
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 385

caggagcagc tggtaggagc cgggggaggc ctggtccagc ctgagggatc cctgacactc	60
acctgcgcag cttctggatt ctcttcagt agcagctact tcatgtgctg ggteccgcag	120
gctccaggga aggggctgga gtggatcgca tgcattggtg ttggtggtag tggtagcact	180
tactacgcga actgggcgaa aggccgattc accatctcca aaacctcgtc gaccacgggtg	240
actctgcaaa tgaccagtct gacagccgcg gacacggcca cctatttctg tgcgagagtg	300
gcctacgatg acgatggtga ctctgatgct ttgatccct ggggccagg caccctggtc	360
accgtctcct ca	372

<210> SEQ ID NO 386
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 386

```

Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1           5           10           15
Gly Thr Val Ser Ile Ser Cys Gln Ser Ser Lys Ser Val Tyr Asn Lys
20           25           30
Asn Trp Leu Ser Trp Phe Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35           40           45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Gly Val Pro Ser Arg Phe
50           55           60
Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Gly Val
65           70           75           80
Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys Ala Gly Gly Tyr Ser Ser
85           90           95
Ser Ser Asp Thr Phe Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100          105          110

```

<210> SEQ ID NO 387

<211> LENGTH: 124

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 387

```

Gln Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly
1           5           10           15
Ser Leu Thr Leu Thr Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Ser
20           25           30
Tyr Phe Met Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
35           40           45
Ile Ala Cys Ile Gly Val Gly Gly Ser Gly Ser Thr Tyr Tyr Ala Asn
50           55           60
Trp Ala Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val
65           70           75           80
Thr Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe
85           90           95
Cys Ala Arg Val Ala Tyr Asp Asp Asp Gly Asp Ser Asp Ala Phe Asp
100          105          110
Pro Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115          120

```

<210> SEQ ID NO 388

<211> LENGTH: 33

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 388

```

Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro
1           5           10           15
Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly
20           25           30
Val

```

<210> SEQ ID NO 389

-continued

<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 389

Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln
1 5 10 15

Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25 30

<210> SEQ ID NO 390
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 390

Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu
1 5 10 15

Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25 30

<210> SEQ ID NO 391
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 391

Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu
1 5 10 15

Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25 30

<210> SEQ ID NO 392
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 392

Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro
1 5 10 15

Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25 30

<210> SEQ ID NO 393
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 393

Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln
1 5 10 15

Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25

<210> SEQ ID NO 394
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 394

-continued

Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg
1 5 10 15

Ala Pro Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25

<210> SEQ ID NO 395
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 395

Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala
1 5 10 15

Pro Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25

<210> SEQ ID NO 396
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 396

Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro
1 5 10 15

Thr Thr Ala Ala Ala Ala Ala Gly Val
20 25

<210> SEQ ID NO 397
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 397

Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr
1 5 10 15

Thr Ala Ala Ala Ala Ala Gly Val
20 25

<210> SEQ ID NO 398
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 398

Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr
1 5 10 15

Ala Ala Ala Ala Ala Gly Val
20

<210> SEQ ID NO 399
<211> LENGTH: 23
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 399

Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala
1 5 10 15

Ala Ala Ala Ala Ala Gly Val
20

-continued

<210> SEQ ID NO 400
<211> LENGTH: 22
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 400

Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala
1 5 10 15

Ala Ala Ala Ala Gly Val
20

<210> SEQ ID NO 401
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 401

Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala
1 5 10 15

Ala Ala Ala Gly Val
20

<210> SEQ ID NO 402
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 402

Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala
1 5 10 15

Ala Ala Gly Val
20

<210> SEQ ID NO 403
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 403

Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala
1 5 10 15

Ala Gly Val

<210> SEQ ID NO 404
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 404

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Ala
1 5 10 15

Gly Val

<210> SEQ ID NO 405
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 405

-continued

Gln	Glu	Pro	Gln	Arg	Ala	Pro	Thr	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Gly
1				5					10						15

Val

<210> SEQ ID NO 406

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 406

Glu	Pro	Gln	Arg	Ala	Pro	Thr	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Gly	Val
1				5					10						15

<210> SEQ ID NO 407

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 407

Pro	Gln	Arg	Ala	Pro	Thr	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Gly	Val
1				5					10					15

<210> SEQ ID NO 408

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 408

Gln	Arg	Ala	Pro	Thr	Thr	Ala	Ala	Ala	Ala	Ala	Gly	Val
1				5					10			

<210> SEQ ID NO 409

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 409

Arg	Ala	Pro	Thr	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Gly	Val
1				5					10			

<210> SEQ ID NO 410

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 410

Ala	Pro	Thr	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Gly	Val
1				5					10		

<210> SEQ ID NO 411

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 411

Pro	Thr	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Gly	Val
1				5					10	

<210> SEQ ID NO 412

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

-continued

<400> SEQUENCE: 412

Thr Thr Ala Ala Ala Ala Ala Ala Gly Val
1 5 10

<210> SEQ ID NO 413

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 413

Thr Ala Ala Ala Ala Ala Ala Gly Val
1 5

<210> SEQ ID NO 414

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 414

Ala Ala Ala Ala Ala Ala Gly Val
1 5

<210> SEQ ID NO 415

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 415

Ala Ala Ala Ala Ala Gly Val
1 5

<210> SEQ ID NO 416

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 416

Ala Ala Ala Ala Gly Val
1 5

<210> SEQ ID NO 417

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 417

Ala Ala Ala Gly Val
1 5

<210> SEQ ID NO 418

<211> LENGTH: 33

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 418

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala
20 25 30

Gly

-continued

<210> SEQ ID NO 419

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 419

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Ala
20 25 30

<210> SEQ ID NO 420

<211> LENGTH: 31

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 420

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala
20 25 30

<210> SEQ ID NO 421

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 421

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala
20 25 30

<210> SEQ ID NO 422

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 422

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala
20 25

<210> SEQ ID NO 423

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 423

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala
20 25

<210> SEQ ID NO 424

<211> LENGTH: 27

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

-continued

<400> SEQUENCE: 424

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala
20 25

<210> SEQ ID NO 425

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 425

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr
20 25

<210> SEQ ID NO 426

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 426

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr
20 25

<210> SEQ ID NO 427

<211> LENGTH: 24

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 427

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro
20

<210> SEQ ID NO 428

<211> LENGTH: 23

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 428

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala
20

<210> SEQ ID NO 429

<211> LENGTH: 22

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 429

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

-continued

Pro Gln Glu Pro Gln Arg
20

<210> SEQ ID NO 430
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 430

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln
20

<210> SEQ ID NO 431
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 431

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro
20

<210> SEQ ID NO 432
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 432

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu

<210> SEQ ID NO 433
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 433

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln

<210> SEQ ID NO 434
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 434

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro

<210> SEQ ID NO 435
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

-continued

<400> SEQUENCE: 435

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

<210> SEQ ID NO 436

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 436

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly
1 5 10 15

<210> SEQ ID NO 437

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 437

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala
1 5 10

<210> SEQ ID NO 438

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 438

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala
1 5 10

<210> SEQ ID NO 439

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 439

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro
1 5 10

<210> SEQ ID NO 440

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 440

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser
1 5 10

<210> SEQ ID NO 441

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 441

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser
1 5 10

<210> SEQ ID NO 442

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 442

Ser Ser Val Ser Gly Gly Ala Ala Ser
1 5

<210> SEQ ID NO 443

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 443

Ser Ser Val Ser Gly Gly Ala Ala
1 5

<210> SEQ ID NO 444

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 444

Ser Ser Val Ser Gly Gly Ala
1 5

<210> SEQ ID NO 445

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 445

Ser Ser Val Ser Gly Gly
1 5

<210> SEQ ID NO 446

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 446

Ser Ser Val Ser Gly
1 5

<210> SEQ ID NO 447

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 447

Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro
1 5 10 15

Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Ala Gly
20 25 30

<210> SEQ ID NO 448

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 448

Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln
1 5 10 15

Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala Ala

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20 25 30

<210> SEQ ID NO 449
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 449

Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu
1 5 10 15

Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala
20 25

<210> SEQ ID NO 450
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 450

Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro
1 5 10 15

Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala
20 25

<210> SEQ ID NO 451
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 451

Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln
1 5 10 15

Arg Ala Pro Thr Thr Ala Ala Ala
20

<210> SEQ ID NO 452
<211> LENGTH: 22
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 452

Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg
1 5 10 15

Ala Pro Thr Thr Ala Ala
20

<210> SEQ ID NO 453
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 453

Ala Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala
1 5 10 15

Pro Thr Thr Ala
20

<210> SEQ ID NO 454
<211> LENGTH: 18
<212> TYPE: PRT

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<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 454

Ser Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro
1 5 10 15

Thr Thr

<210> SEQ ID NO 455

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 455

Ser Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro Thr
1 5 10 15

<210> SEQ ID NO 456

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 456

Ser Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala Pro
1 5 10

<210> SEQ ID NO 457

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 457

Pro Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg Ala
1 5 10

<210> SEQ ID NO 458

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 458

Ala Ala Gly Ile Pro Gln Glu Pro Gln Arg
1 5 10

<210> SEQ ID NO 459

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 459

Ala Gly Ile Pro Gln Glu Pro Gln
1 5

<210> SEQ ID NO 460

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 460

Gly Ile Pro Gln Glu Pro
1 5

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<210> SEQ ID NO 461
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 461

Ile Pro Gln Glu
1

<210> SEQ ID NO 462
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 462

Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu
1 5 10 15

Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25 30

<210> SEQ ID NO 463
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 463

Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr
1 5 10 15

Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25

<210> SEQ ID NO 464
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 464

Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg
1 5 10 15

Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25

<210> SEQ ID NO 465
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 465

Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys
1 5 10 15

Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25

<210> SEQ ID NO 466
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 466

Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser
1 5 10 15

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Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25

<210> SEQ ID NO 467
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 467

Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly
1 5 10 15

Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25

<210> SEQ ID NO 468
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 468

Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser
1 5 10 15

Leu Val Gln Phe Leu Leu Tyr Lys
20

<210> SEQ ID NO 469
<211> LENGTH: 23
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 469

Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser Leu
1 5 10 15

Val Gln Phe Leu Leu Tyr Lys
20

<210> SEQ ID NO 470
<211> LENGTH: 22
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 470

Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser Leu Val
1 5 10 15

Gln Phe Leu Leu Tyr Lys
20

<210> SEQ ID NO 471
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 471

Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser Leu Val Gln
1 5 10 15

Phe Leu Leu Tyr Lys
20

<210> SEQ ID NO 472

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<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 472

Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe
1 5 10 15

Leu Leu Tyr Lys
20

<210> SEQ ID NO 473
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 473

Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu
1 5 10 15

Leu Tyr Lys

<210> SEQ ID NO 474
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 474

Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu
1 5 10 15

Tyr Lys

<210> SEQ ID NO 475
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 475

Asp Pro Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr
1 5 10 15

Lys

<210> SEQ ID NO 476
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 476

Pro Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5 10 15

<210> SEQ ID NO 477
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 477

Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5 10 15

<210> SEQ ID NO 478
<211> LENGTH: 14

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<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 478

Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5 10

<210> SEQ ID NO 479

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 479

Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5 10

<210> SEQ ID NO 480

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 480

Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5 10

<210> SEQ ID NO 481

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 481

Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5 10

<210> SEQ ID NO 482

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 482

Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5 10

<210> SEQ ID NO 483

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 483

Ser Leu Val Gln Phe Leu Leu Tyr Lys
1 5

<210> SEQ ID NO 484

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 484

Leu Val Gln Phe Leu Leu Tyr Lys
1 5

<210> SEQ ID NO 485

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<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 485

Val Gln Phe Leu Leu Tyr Lys
1 5

<210> SEQ ID NO 486
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 486

Gln Phe Leu Leu Tyr Lys
1 5

<210> SEQ ID NO 487
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 487

Phe Leu Leu Tyr Lys
1 5

<210> SEQ ID NO 488
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 488

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr
20 25 30

<210> SEQ ID NO 489
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 489

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu
20 25

<210> SEQ ID NO 490
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 490

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu
20 25

<210> SEQ ID NO 491
<211> LENGTH: 27

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<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 491

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe
 20 25

<210> SEQ ID NO 492

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 492

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu Val Gln
 20 25

<210> SEQ ID NO 493

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 493

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu Val
 20 25

<210> SEQ ID NO 494

<211> LENGTH: 24

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 494

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu
 20

<210> SEQ ID NO 495

<211> LENGTH: 23

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 495

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser
 20

<210> SEQ ID NO 496

<211> LENGTH: 22

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 496

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro

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1	5	10	15
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Leu Thr Arg Lys Ser Gly
20

<210> SEQ ID NO 497
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 497

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser
20

<210> SEQ ID NO 498
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 498

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys
20

<210> SEQ ID NO 499
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 499

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg

<210> SEQ ID NO 500
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 500

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr

<210> SEQ ID NO 501
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 501

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu

<210> SEQ ID NO 502
<211> LENGTH: 16
<212> TYPE: PRT

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<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 502

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

<210> SEQ ID NO 503

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 503

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp
1 5 10 15

<210> SEQ ID NO 504

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 504

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu
1 5 10

<210> SEQ ID NO 505

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 505

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser
1 5 10

<210> SEQ ID NO 506

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 506

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro
1 5 10

<210> SEQ ID NO 507

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 507

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser
1 5 10

<210> SEQ ID NO 508

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 508

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys
1 5 10

<210> SEQ ID NO 509

<211> LENGTH: 9

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<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 509

Ser Ser Ser Gln Ala Ser Thr Ser Thr
1 5

<210> SEQ ID NO 510
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 510

Ser Ser Ser Gln Ala Ser Thr Ser
1 5

<210> SEQ ID NO 511
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 511

Ser Ser Ser Gln Ala Ser Thr
1 5

<210> SEQ ID NO 512
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 512

Ser Ser Ser Gln Ala Ser
1 5

<210> SEQ ID NO 513
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 513

Ser Ser Ser Gln Ala
1 5

<210> SEQ ID NO 514
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 514

Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu
1 5 10 15

Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr
20 25

<210> SEQ ID NO 515
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 515

Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr
1 5 10 15

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Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu
20 25

<210> SEQ ID NO 516
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 516

Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg
1 5 10 15

Lys Ser Gly Ser Leu Val Gln Phe Leu
20 25

<210> SEQ ID NO 517
<211> LENGTH: 23
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 517

Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys
1 5 10 15

Ser Gly Ser Leu Val Gln Phe
20

<210> SEQ ID NO 518
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 518

Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser
1 5 10 15

Gly Ser Leu Val Gln
20

<210> SEQ ID NO 519
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 519

Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly
1 5 10 15

Ser Leu Val

<210> SEQ ID NO 520
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 520

Ser Thr Lys Ser Pro Ser Glu Asp Pro Leu Thr Arg Lys Ser Gly Ser
1 5 10 15

Leu

<210> SEQ ID NO 521
<211> LENGTH: 15
<212> TYPE: PRT

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<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 521

Thr	Lys	Ser	Pro	Ser	Glu	Asp	Pro	Leu	Thr	Arg	Lys	Ser	Gly	Ser
1				5					10					15

<210> SEQ ID NO 522

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 522

Lys	Ser	Pro	Ser	Glu	Asp	Pro	Leu	Thr	Arg	Lys	Ser	Gly
1				5					10			

<210> SEQ ID NO 523

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 523

Ser	Pro	Ser	Glu	Asp	Pro	Leu	Thr	Arg	Lys	Ser
1				5					10	

<210> SEQ ID NO 524

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 524

Pro	Ser	Glu	Asp	Pro	Leu	Thr	Arg	Lys
1				5				

<210> SEQ ID NO 525

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 525

Ser	Glu	Asp	Pro	Leu	Thr	Arg
1				5		

<210> SEQ ID NO 526

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 526

Glu	Asp	Pro	Leu	Thr
1				5

<210> SEQ ID NO 527

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 527

Leu	Thr	Asp	Glu	Glu	Ser	Leu	Leu	Ser	Ser	Trp	Asp	Phe	Pro	Arg
1				5						10				15

<210> SEQ ID NO 528

<211> LENGTH: 14

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<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 528

Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5 10

<210> SEQ ID NO 529

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 529

Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5 10

<210> SEQ ID NO 530

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 530

Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5 10

<210> SEQ ID NO 531

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 531

Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5 10

<210> SEQ ID NO 532

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 532

Ser Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5 10

<210> SEQ ID NO 533

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 533

Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5

<210> SEQ ID NO 534

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 534

Leu Ser Ser Trp Asp Phe Pro Arg
1 5

<210> SEQ ID NO 535

-continued

<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 535

Ser Ser Trp Asp Phe Pro Arg
1 5

<210> SEQ ID NO 536
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 536

Ser Trp Asp Phe Pro Arg
1 5

<210> SEQ ID NO 537
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 537

Trp Asp Phe Pro Arg
1 5

<210> SEQ ID NO 538
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 538

Asp Leu Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro
1 5 10 15

<210> SEQ ID NO 539
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 539

Asp Leu Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe
1 5 10

<210> SEQ ID NO 540
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 540

Asp Leu Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp
1 5 10

<210> SEQ ID NO 541
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 541

Asp Leu Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp
1 5 10

-continued

<210> SEQ ID NO 542
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 542

Asp Leu Thr Asp Glu Glu Ser Leu Leu Ser Ser
1 5 10

<210> SEQ ID NO 543
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 543

Asp Leu Thr Asp Glu Glu Ser Leu Leu Ser
1 5 10

<210> SEQ ID NO 544
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 544

Asp Leu Thr Asp Glu Glu Ser Leu Leu
1 5

<210> SEQ ID NO 545
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 545

Asp Leu Thr Asp Glu Glu Ser Leu
1 5

<210> SEQ ID NO 546
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 546

Asp Leu Thr Asp Glu Glu Ser
1 5

<210> SEQ ID NO 547
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 547

Asp Leu Thr Asp Glu Glu
1 5

<210> SEQ ID NO 548
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 548

Asp Leu Thr Asp Glu
1 5

-continued

<210> SEQ ID NO 549
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 549

Leu Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro
1 5 10

<210> SEQ ID NO 550
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 550

Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe
1 5 10

<210> SEQ ID NO 551
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 551

Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp
1 5 10

<210> SEQ ID NO 552
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 552

Glu Glu Ser Leu Leu Ser Ser Trp
1 5

<210> SEQ ID NO 553
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 553

Glu Ser Leu Leu Ser Ser
1 5

<210> SEQ ID NO 554
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 554

Ser Leu Leu Ser
1

<210> SEQ ID NO 555
<211> LENGTH: 238
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 555

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```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
 1          5          10          15
Leu Arg Gly Ala Arg Cys Asp Ile Val Met Thr Gln Thr Pro Ser Ser
          20          25          30
Val Glu Ala Ala Val Gly Gly Thr Val Thr Ile Lys Cys Gln Ala Ser
          35          40          45
Gln Asn Ile Ser Ser Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
          50          55          60
Pro Pro Lys Leu Leu Ile Tyr Arg Ala Ser Thr Leu Ala Ser Gly Val
65          70          75          80
Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr
          85          90          95
Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser
          100          105          110
Tyr Asp Asp Ser Arg Ser Ser Asn Phe Phe Tyr Ala Phe Gly Gly Gly
          115          120          125
Thr Glu Val Val Val Lys Gly Asp Pro Val Ala Pro Thr Val Leu Leu
          130          135          140
Phe Pro Pro Ser Ser Asp Glu Val Ala Thr Gly Thr Val Thr Ile Val
145          150          155          160
Cys Val Ala Asn Lys Tyr Phe Pro Asp Val Thr Val Thr Trp Glu Val
          165          170          175
Asp Gly Thr Thr Gln Thr Thr Gly Ile Glu Asn Ser Lys Thr Pro Gln
          180          185          190
Asn Ser Ala Asp Cys Thr Tyr Asn Leu Ser Ser Thr Leu Thr Leu Thr
          195          200          205
Ser Thr Gln Tyr Asn Ser His Lys Glu Tyr Thr Cys Lys Val Thr Gln
          210          215          220
Gly Thr Thr Ser Val Val Gln Ser Phe Ser Arg Lys Asn Cys
225          230          235

```

<210> SEQ ID NO 556

<211> LENGTH: 458

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 556

```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
 1          5          10          15
Leu Arg Gly Ala Arg Cys Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu
          20          25          30
Val Gln Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Phe Gly Val
          35          40          45
Thr Leu Thr Asn Tyr Tyr Ile Cys Trp Val Arg Gln Ala Pro Gly Lys
          50          55          60
Gly Leu Glu Trp Val Gly Cys Ile Asp Asn Ala Asn Gly Arg Thr Tyr
65          70          75          80
Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser
          85          90          95
Thr Thr Gly Thr Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala

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100					105					110					
Thr	Tyr	Phe	Cys	Ala	Arg	Ser	Leu	Ala	Thr	Pro	Leu	Trp	Gly	Pro	Gly
	115						120					125			
Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gln	Pro	Lys	Ala	Pro	Ser	Val	Phe
	130						135				140				
Pro	Leu	Ala	Pro	Cys	Cys	Gly	Asp	Thr	Pro	Ser	Ser	Thr	Val	Thr	Leu
145					150					155					160
Gly	Cys	Leu	Val	Lys	Gly	Tyr	Leu	Pro	Glu	Pro	Val	Thr	Val	Thr	Trp
				165					170					175	
Asn	Ser	Gly	Thr	Leu	Thr	Asn	Gly	Val	Arg	Thr	Phe	Pro	Ser	Val	Arg
			180					185						190	
Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Ser	Val	Thr	Ser
	195						200					205			
Ser	Ser	Gln	Pro	Val	Thr	Cys	Asn	Val	Ala	His	Pro	Ala	Thr	Asn	Thr
	210					215					220				
Lys	Val	Asp	Lys	Thr	Val	Ala	Pro	Ser	Thr	Cys	Ser	Lys	Pro	Thr	Cys
225					230					235					240
Pro	Pro	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro
				245					250					255	
Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys
			260					265					270		
Val	Val	Val	Asp	Val	Ser	Gln	Asp	Asp	Pro	Glu	Val	Gln	Phe	Thr	Trp
	275						280					285			
Tyr	Ile	Asn	Asn	Glu	Gln	Val	Arg	Thr	Ala	Arg	Pro	Pro	Leu	Arg	Glu
	290					295					300				
Gln	Gln	Phe	Asn	Ser	Thr	Ile	Arg	Val	Val	Ser	Thr	Leu	Pro	Ile	Ala
305					310					315					320
His	Gln	Asp	Trp	Leu	Arg	Gly	Lys	Glu	Phe	Lys	Cys	Lys	Val	His	Asn
			325						330					335	
Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Arg	Gly
		340						345					350		
Gln	Pro	Leu	Glu	Pro	Lys	Val	Tyr	Thr	Met	Gly	Pro	Pro	Arg	Glu	Glu
	355					360						365			
Leu	Ser	Ser	Arg	Ser	Val	Ser	Leu	Thr	Cys	Met	Ile	Asn	Gly	Phe	Tyr
	370					375					380				
Pro	Ser	Asp	Ile	Ser	Val	Glu	Trp	Glu	Lys	Asn	Gly	Lys	Ala	Glu	Asp
385					390					395					400
Asn	Tyr	Lys	Thr	Thr	Pro	Ala	Val	Leu	Asp	Ser	Asp	Gly	Ser	Tyr	Phe
			405						410					415	
Leu	Tyr	Ser	Lys	Leu	Ser	Val	Pro	Thr	Ser	Glu	Trp	Gln	Arg	Gly	Asp
		420						425					430		
Val	Phe	Thr	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr
	435						440					445			
Gln	Lys	Ser	Ile	Ser	Arg	Ser	Pro	Gly	Lys						
	450					455									

<210> SEQ ID NO 557

<211> LENGTH: 238

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 557

```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
1           5           10           15
Leu Arg Gly Ala Arg Cys Asp Val Val Met Thr Gln Thr Pro Ala Ser
20           25           30
Val Glu Ala Thr Val Gly Gly Thr Val Thr Ile Lys Cys Gln Ser Ser
35           40           45
Gln Ser Val Tyr Asp Asn Asn Ala Leu Ala Trp Tyr Gln Gln Asn Ala
50           55           60
Gly Gln Arg Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser
65           70           75           80
Gly Val Pro Ser Arg Phe Ser Ala Ser Gly Ser Gly Thr Glu Phe Thr
85           90           95
Leu Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Thr Tyr Tyr Cys
100          105          110
Gln Cys Thr Tyr Tyr Val Ser Ser Tyr Gln Asn Asp Phe Gly Gly Gly
115          120          125
Thr Glu Val Val Val Lys Gly Asp Pro Val Ala Pro Thr Val Leu Leu
130          135          140
Phe Pro Pro Ser Ser Asp Glu Val Ala Thr Gly Thr Val Thr Ile Val
145          150          155          160
Cys Val Ala Asn Lys Tyr Phe Pro Asp Val Thr Val Thr Trp Glu Val
165          170          175
Asp Gly Thr Thr Gln Thr Thr Gly Ile Glu Asn Ser Lys Thr Pro Gln
180          185          190
Asn Ser Ala Asp Cys Thr Tyr Asn Leu Ser Ser Thr Leu Thr Leu Thr
195          200          205
Ser Thr Gln Tyr Asn Ser His Lys Glu Tyr Thr Cys Lys Val Thr Gln
210          215          220
Gly Thr Thr Ser Val Val Gln Ser Phe Ser Arg Lys Asn Cys
225          230          235

```

<210> SEQ ID NO 558

<211> LENGTH: 457

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 558

```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
1           5           10           15
Leu Arg Gly Ala Arg Cys Gln Ser Val Glu Glu Ser Gly Gly Arg Leu
20           25           30
Val Thr Pro Gly Thr Pro Leu Thr Leu Thr Cys Thr Ile Ser Gly Phe
35           40           45
Ser Leu Ser Ser Tyr Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys
50           55           60
Gly Leu Glu Trp Ile Gly Ser Ile Gly Gly Gly Gly Ser Ala Val Tyr
65           70           75           80
Ala Ser Trp Ala Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr
85           90           95

```

```
<210> SEQ ID NO 559
<211> LENGTH: 238
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
```

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 559

```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
1      5      10      15
Leu Arg Gly Ala Arg Cys Ala Ala Val Leu Thr Gln Thr Pro Ser Pro
20     25     30
Val Ser Ala Ala Val Gly Gly Thr Val Ser Ala Ser Cys Gln Ser Ser
35     40     45
Lys Ser Val Tyr Asn Lys Asn Trp Leu Ser Trp Phe Gln Gln Lys Pro
50     55     60
Gly Gln Pro Pro Lys Leu Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser
65     70     75     80
Gly Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Gln Phe Thr
85     90     95
Leu Thr Ile Ser Asp Val Gln Cys Asp Asp Ala Ala Thr Tyr Tyr Cys
100    105    110
Ala Gly Gly Tyr Ser Ser Ser Ser Asp Thr Phe Ala Phe Gly Gly Gly
115    120    125
Thr Glu Val Val Val Lys Gly Asp Pro Val Ala Pro Thr Val Leu Leu
130    135    140
Phe Pro Pro Ser Ser Asp Glu Val Ala Thr Gly Thr Val Thr Ile Val
145    150    155    160
Cys Val Ala Asn Lys Tyr Phe Pro Asp Val Thr Val Thr Trp Glu Val
165    170    175
Asp Gly Thr Thr Gln Thr Thr Gly Ile Glu Asn Ser Lys Thr Pro Gln
180    185    190
Asn Ser Ala Asp Cys Thr Tyr Asn Leu Ser Ser Thr Leu Thr Leu Thr
195    200    205
Ser Thr Gln Tyr Asn Ser His Lys Glu Tyr Thr Cys Lys Val Thr Gln
210    215    220
Gly Thr Thr Ser Val Val Gln Ser Phe Ser Arg Lys Asn Cys
225    230    235

```

<210> SEQ ID NO 560

<211> LENGTH: 470

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 560

```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
1      5      10      15
Leu Arg Gly Ala Arg Cys Gln Glu Gln Leu Val Glu Ser Gly Gly Gly
20     25     30
Leu Val Lys Pro Gly Ala Ser Leu Thr Leu Thr Cys Lys Ala Ser Gly
35     40     45
Phe Ser Phe Ser Ser Gly Gln Leu Met Cys Trp Val Arg Gln Ala Pro
50     55     60
Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Gly Ser Gly Ser Asn Ala
65     70     75     80

```

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Ile	Ser	Thr	Phe	Tyr	Ala	Ser	Trp	Ala	Gln	Gly	Arg	Phe	Thr	Ile	Ser	85	90	95
Lys	Ser	Ser	Ser	Thr	Thr	Val	Thr	Leu	Gln	Leu	Thr	Ser	Leu	Thr	Ala	100	105	110
Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Val	Gly	Ser	Asp	Asp	Tyr	115	120	125
Gly	Asp	Ser	Asp	Val	Phe	Asp	Pro	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	130	135	140
Val	Ser	Ser	Gly	Gln	Pro	Lys	Ala	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	145	150	155
Cys	Cys	Gly	Asp	Thr	Pro	Ser	Ser	Thr	Val	Thr	Leu	Gly	Cys	Leu	Val	165	170	175
Lys	Gly	Tyr	Leu	Pro	Glu	Pro	Val	Thr	Val	Thr	Trp	Asn	Ser	Gly	Thr	180	185	190
Leu	Thr	Asn	Gly	Val	Arg	Thr	Phe	Pro	Ser	Val	Arg	Gln	Ser	Ser	Gly	195	200	205
Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Ser	Val	Thr	Ser	Ser	Ser	Gln	Pro	210	215	220
Val	Thr	Cys	Asn	Val	Ala	His	Pro	Ala	Thr	Asn	Thr	Lys	Val	Asp	Lys	225	230	235
Thr	Val	Ala	Pro	Ser	Thr	Cys	Ser	Lys	Pro	Thr	Cys	Pro	Pro	Pro	Glu	245	250	255
Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Pro	Lys	Asp	260	265	270
Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	275	280	285
Val	Ser	Gln	Asp	Asp	Pro	Glu	Val	Gln	Phe	Thr	Trp	Tyr	Ile	Asn	Asn	290	295	300
Glu	Gln	Val	Arg	Thr	Ala	Arg	Pro	Pro	Leu	Arg	Glu	Gln	Gln	Phe	Asn	305	310	315
Ser	Thr	Ile	Arg	Val	Val	Ser	Thr	Leu	Pro	Ile	Ala	His	Gln	Asp	Trp	325	330	335
Leu	Arg	Gly	Lys	Glu	Phe	Lys	Cys	Lys	Val	His	Asn	Lys	Ala	Leu	Pro	340	345	350
Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Arg	Gly	Gln	Pro	Leu	Glu	355	360	365
Pro	Lys	Val	Tyr	Thr	Met	Gly	Pro	Pro	Arg	Glu	Glu	Leu	Ser	Ser	Arg	370	375	380
Ser	Val	Ser	Leu	Thr	Cys	Met	Ile	Asn	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	385	390	395
Ser	Val	Glu	Trp	Glu	Lys	Asn	Gly	Lys	Ala	Glu	Asp	Asn	Tyr	Lys	Thr	405	410	415
Thr	Pro	Ala	Val	Leu	Asp	Ser	Asp	Gly	Ser	Tyr	Phe	Leu	Tyr	Ser	Lys	420	425	430
Leu	Ser	Val	Pro	Thr	Ser	Glu	Trp	Gln	Arg	Gly	Asp	Val	Phe	Thr	Cys	435	440	445
Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Ile	450	455	460
Ser	Arg	Ser	Pro	Gly	Lys											465	470	

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<210> SEQ ID NO 561
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 561

Gly Val Tyr Asp Gly Glu Glu His Ser Val
1 5 10

<210> SEQ ID NO 562
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 562

Asp Leu Thr Asp Glu Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5 10 15

<210> SEQ ID NO 563
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 563

Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro
1 5 10

<210> SEQ ID NO 564
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 564

Glu Leu Lys Thr Pro Leu Asp Thr Thr His Thr Cys Pro Arg Cys Pro
1 5 10 15

<210> SEQ ID NO 565
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 565

Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro Arg Cys
1 5 10 15

Pro

<210> SEQ ID NO 566
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 566

-continued

Gln Ala Ser Gln Ser Ile Arg Asn Glu Leu Phe
1 5 10

<210> SEQ ID NO 567
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 567

Glu Ser Lys Tyr Gly Pro Pro Cys Pro Ser Cys Pro
1 5 10

<210> SEQ ID NO 568
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 568

Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro
1 5 10 15

<210> SEQ ID NO 569
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
6xHis tag

<400> SEQUENCE: 569

His His His His His His
1 5

<210> SEQ ID NO 570
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 570

Ser Tyr Val Lys Val Leu Glu Tyr Val Ile Lys Val Ser Ala Arg Val
1 5 10 15

Arg Phe Phe Phe Pro Ser Leu Arg Glu Ala Ala
20 25

<210> SEQ ID NO 571
<211> LENGTH: 37
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 571

Ser Ser Pro Leu Val Leu Gly Thr Leu Glu Glu Val Pro Thr Ala Gly
1 5 10 15

-continued

Ser	Thr	Asp	Pro	Pro	Gln	Ser	Pro	Gln	Gly	Ala	Ser	Ala	Phe	Pro	Thr
			20					25					30		

Thr	Ile	Asn	Phe	Thr
				35

<210> SEQ ID NO 572
 <211> LENGTH: 25
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 572

Pro	Asp	Leu	Glu	Ser	Glu	Phe	Gln	Ala	Ala	Ile	Ser	Arg	Lys	Met	Val
1			5					10					15		

Glu	Leu	Val	His	Phe	Leu	Leu	Leu	Lys
			20				25	

<210> SEQ ID NO 573
 <211> LENGTH: 37
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 573

Ser	Ser	Thr	Leu	Val	Glu	Val	Thr	Leu	Gly	Glu	Val	Pro	Ala	Ala	Asp
1			5					10					15		

Ser	Pro	Ser	Pro	Pro	His	Ser	Pro	Gln	Gly	Ala	Ser	Ser	Phe	Ser	Thr
			20					25					30		

Thr	Ile	Asn	Tyr	Thr
				35

<210> SEQ ID NO 574
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 574

Ser	Thr	Phe	Pro	Asp	Leu	Glu	Ser	Glu	Phe	Gln	Ala	Ala	Leu	Ser	Arg
1			5					10					15		

Lys	Val	Ala	Glu	Leu	Val	His	Phe	Leu	Leu	Leu	Lys
			20					25			

<210> SEQ ID NO 575
 <211> LENGTH: 36
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 575

Ser	Ser	Thr	Leu	Val	Glu	Val	Thr	Leu	Gly	Glu	Val	Pro	Ala	Ala	Glu
1			5					10					15		

Ser	Pro	Asp	Pro	Pro	Gln	Ser	Pro	Gln	Gly	Ala	Ser	Ser	Leu	Pro	Thr
			20					25					30		

-continued

Thr Met Asn Tyr
35

<210> SEQ ID NO 576
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 576

Ser Thr Ser Pro Asp Pro Glu Ser Val Phe Arg Ala Ala Leu Ser Lys
1 5 10 15

Lys Val Ala Asp Leu Ile His Phe Leu Leu Leu Lys
20 25

<210> SEQ ID NO 577
<211> LENGTH: 37
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 577

Ser Ser Pro Leu Val Pro Gly Thr Leu Gly Glu Val Pro Ala Ala Gly
1 5 10 15

Ser Pro Gly Pro Leu Lys Ser Pro Gln Gly Ala Ser Ala Ile Pro Thr
20 25 30

Ala Ile Asp Phe Thr
35

<210> SEQ ID NO 578
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 578

Ser Tyr Glu Lys Val Ile Asn Tyr Leu Val Met Leu Asn Ala Arg Glu
1 5 10 15

Pro Ile Ser Tyr Pro Ser Leu Tyr Glu Glu Val
20 25

<210> SEQ ID NO 579
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 579

Ser Ser Ser Val Asp Pro Ala Gln Leu Glu Phe Met Phe Gln Glu Ala
1 5 10 15

Leu Lys Leu Lys Val Ala Glu Leu Val His Phe Leu Leu His Lys
20 25 30

-continued

<210> SEQ ID NO 580
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 580

Ser Thr Leu Gln Val Leu Pro Asp Ser Glu Ser Leu Pro Arg Ser Glu
1 5 10 15

Ile Asp Glu Lys Val Thr Asp Leu Val Gln Phe Leu Leu Phe Lys
20 25 30

<210> SEQ ID NO 581
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 581

Ser Thr Ser Pro Asp Leu Ile Asp Pro Glu Ser Phe Ser Gln Asp Ile
1 5 10 15

Leu His Asp Lys Ile Ile Asp Leu Val His Leu Leu Leu Arg Lys
20 25 30

<210> SEQ ID NO 582
<211> LENGTH: 37
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 582

Ser Ser Thr Leu Asn Val Gly Thr Leu Glu Glu Leu Pro Ala Ala Glu
1 5 10 15

Ser Pro Ser Pro Pro Gln Ser Pro Gln Glu Glu Ser Phe Ser Pro Thr
20 25 30

Ala Met Asp Ala Ile
35

<210> SEQ ID NO 583
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 583

Ser Thr Phe Pro Asp Leu Glu Thr Ser Phe Gln Val Ala Leu Ser Arg
1 5 10 15

Lys Met Ala Glu Leu Val His Phe Leu Leu Leu Lys
20 25

<210> SEQ ID NO 584
<211> LENGTH: 37
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 584

Ser Ser Thr Leu Val Glu Val Thr Leu Arg Glu Val Pro Ala Ala Glu
1 5 10 15

Ser Pro Ser Pro Pro His Ser Pro Gln Gly Ala Ser Thr Leu Pro Thr
20 25 30

Thr Ile Asn Tyr Thr
35

<210> SEQ ID NO 585

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 585

Asn Leu Thr Asn Asp Gly Asn Leu Ser Asn Asp Trp Asp Phe Pro Arg
1 5 10 15

<210> SEQ ID NO 586

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 586

Ser Ser Pro Val Leu Gly Asp Thr Pro Thr Ser Ser Pro Ala Ala Gly
1 5 10 15

Ile Pro Gln Lys Pro Gln Gly Ala Pro Pro Thr Thr Thr Ala Ala Ala
20 25 30

Ala Val

<210> SEQ ID NO 587

<211> LENGTH: 31

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 587

Ser Ser Ser Gln Ala Ser Thr Ser Thr Glu Arg Ser Leu Lys Asp Ser
1 5 10 15

Leu Thr Arg Lys Thr Lys Met Leu Val Gln Phe Leu Leu Tyr Lys
20 25 30

<210> SEQ ID NO 588

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 588

Ser Ser Val Leu Arg Asp Thr Ala Ser Ser Ser Leu Ala Phe Gly Ile

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1	5	10	15
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Pro Gln Glu Pro Gln Arg Glu Pro Pro Thr Thr Ser Ala Ala Ala Ala
20 25 30

<210> SEQ ID NO 589
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 589

Ser Ser Glu Val Ser Pro Ser Thr Glu Ser Ser Ser Ser Asn Phe Ile	
1 5 10 15	

Asn Ile Lys Val Gly Leu Leu Glu Gln Phe Leu Leu Tyr Lys
20 25 30

<210> SEQ ID NO 590
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 590

Asn Ser Phe His Gly Pro Ser Ser Ser Glu Ser Thr Gly Arg Asp Leu	
1 5 10 15	

Leu Asn Thr Lys Thr Gly Glu Leu Val Gln Phe Leu Leu Asn Lys
20 25 30

<210> SEQ ID NO 591
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 591

Ser Ser Pro Ala Ser Gln Ser Pro Pro Gln Ser Phe Pro Asn Ala Gly	
1 5 10 15	

Ile Pro Gln Glu Ser Gln Arg Ala Ser Tyr Pro Ser Ser Pro Ala Ser
20 25 30

Ala Val

<210> SEQ ID NO 592
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 592

Thr Asp Ser Glu Ser Leu Ile Glu Ser Glu Pro Leu Phe Thr Tyr Thr	
1 5 10 15	

Leu Asp Glu Lys Val Asp Glu Leu Ala Arg Phe Leu Leu Lys
20 25 30

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<210> SEQ ID NO 593
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 593

Gly Thr Ser Gln Gly Leu Pro Asp Ser Glu Ser Ser Phe Thr Tyr Thr
1 5 10 15
Leu Asp Glu Lys Val Ala Glu Leu Val Glu Phe Leu Leu Leu Lys
20 25 30

<210> SEQ ID NO 594
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 594

His Ala Leu Pro Glu Ser Glu Ser Leu Pro Arg Tyr Ala Leu Asp Glu
1 5 10 15
Lys Val Ala Glu Leu Val Gln Phe Leu Leu Leu Lys
20 25

<210> SEQ ID NO 595
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 595

Asn Thr Ser Arg Val Ala Ile Thr Leu Lys Pro Gln Asp Pro Met Glu
1 5 10 15
Gln Asn Val Ala Glu Leu Leu Gln Phe Leu Leu Val Lys
20 25

<210> SEQ ID NO 596
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 596

Ala Leu Ala Ala Lys Ala Leu Ala Arg Arg Arg Ala Tyr Arg Arg Leu
1 5 10 15
Asn Arg Thr Val Ala Glu Leu Val Gln Phe Leu Leu Val Lys
20 25 30

<210> SEQ ID NO 597
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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<400> SEQUENCE: 597

Asp Leu Thr Asp Glu Ser Leu Leu Ser Ser Trp Asp Phe Pro Arg
1 5 10 15

<210> SEQ ID NO 598

<211> LENGTH: 31

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 598

Ser Ser Ser Gln Ala Ser Thr Ser Thr Lys Ser Pro Ser Glu Asp Pro
1 5 10 15

Leu Thr Arg Lys Ser Gly Ser Leu Val Gln Phe Leu Leu Tyr Lys
20 25 30

<210> SEQ ID NO 599

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 599

Ser Ser Val Ser Gly Gly Ala Ala Ser Ser Ser Pro Ala Ala Gly Ile
1 5 10 15

Pro Gln Glu Pro Gln Arg Ala Pro Thr Thr Ala Ala Ala Ala Ala
20 25 30

Gly Val

<210> SEQ ID NO 600

<211> LENGTH: 103

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 600

Asp Gly Val Met Thr Gln Thr Pro Ala Pro Val Ser Ala Ala Val Gly
1 5 10 15

Gly Thr Val Thr Ile Asn Cys Gln Ala Ser Gln Ser Ile Gly Ser Asp
20 25 30

Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Ser Lys Leu Ala Thr Val Pro Ser Arg Phe Asn Gly Ser
50 55 60

Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Gly Val Gln Cys Asp
65 70 75 80

Asp Ala Ala Thr Tyr Gln Cys Gly Trp Tyr Tyr Asn Ala Phe Gly Gly
85 90 95

Gly Thr Glu Val Val Val Lys
100

<210> SEQ ID NO 601

<211> LENGTH: 110

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 601

Asp Val Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly
1 5 10 15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Arg Asn Glu
20 25 30
Leu Phe Trp Trp Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45
Tyr Ala Ala Ser Lys Leu Ala Ser Val Pro Ser Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
65 70 75 80
Asp Ala Ala Thr Tyr Tyr Gln Cys Ser Tyr Val Ser Ser Ser Gly Thr
85 90 95
Tyr Gly Asn Val Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

<210> SEQ ID NO 602
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 602

Gln Ala Val Val Thr Gln Thr Pro Ser Ser Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Val Tyr Ser Asn
20 25 30
Asn Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Leu Lys
50 55 60
Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Val Gln
65 70 75 80
Cys Asp Asp Ala Ala Thr Tyr Gln Gly Tyr Tyr Ser Gly Val Ile Tyr
85 90 95
Met Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 603
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 603

Asp Val Val Met Thr His Thr Pro Ala Ser Leu Phe Glu Thr Val Gly
1 5 10 15
Gly Thr Val Thr Ile Lys Cys Gln Ala Leu Gln Ser Val Tyr Asp Asn

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20	25	30
Asn Ala Leu Ser Trp Tyr Gln Gln Asn Ala Gly Gln Arg Pro Ile Leu		
35	40	45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Ala Pro Ser Arg Phe Ser		
50	55	60
Ala Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ile Asp Leu Glu		
65	70	75
Cys Ala Asp Ala Ser Thr Tyr Tyr Gln Cys Thr Tyr Tyr Val Ser Ser		
85	90	95
Tyr Gln Asn Asp Phe Gly Gly Gly Thr Glu Val Val Val Lys		
100	105	110

<210> SEQ ID NO 604
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 604

Ala Ile Asp Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1 5 10 15
Gly Thr Ile Thr Ile Asn Cys Gln Ala Ser Glu Ser Ile Ser Ser Trp
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45
Tyr Glu Ala Ser Lys Leu Ala Ser Val Pro Ser Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Gly Val Gln Cys Asp
65 70 75 80
Asp Ala Ala Thr Tyr Gln Gly Gly Tyr Tyr Gly Trp Tyr Tyr Asn Ala
85 90 95
Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 605
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 605

Ala Tyr Asp Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1 5 10 15
Gly Thr Val Thr Ile Asn Cys Gln Ala Ser Glu Ser Ile Ser Asn Tyr
20 25 30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45
Tyr Trp Ala Ser Thr Leu Ala Ser Val Ser Ser Arg Phe Lys Gly Ser
50 55 60
Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Gly Val Glu Cys Ala
65 70 75 80
Asp Ala Ala Thr Tyr Gln Gln Gly Tyr Ser Ser Ser Asn Val Asp Asn

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85	90	95
Leu Phe Gly Gly Thr Glu Val Val Lys		
100	105	
<210> SEQ ID NO 606		
<211> LENGTH: 110		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 606		
Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly		
1	5	10 15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr		
20	25	30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile		
35	40	45
Tyr Arg Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Arg Gly Ser		
50	55	60
Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala		
65	70	75 80
Asp Ala Ala Thr Tyr Tyr Gln Ser Tyr Asp Asp Ser Ser Asp Asn Asn		
85	90	95
Phe Phe Tyr Gly Phe Gly Gly Gly Thr Glu Val Val Val Arg		
100	105	110
<210> SEQ ID NO 607		
<211> LENGTH: 110		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 607		
Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly		
1	5	10 15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr		
20	25	30
Leu Ala Trp Tyr His Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile		
35	40	45
Tyr Ala Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Glu Gly Ser		
50	55	60
Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala		
65	70	75 80
Asp Ala Ala Thr Tyr Tyr Gln Ser Tyr Asp Asp Ser Arg Ser Ser Ser		
85	90	95
Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys		
100	105	110
<210> SEQ ID NO 608		
<211> LENGTH: 110		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic		

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polypeptide

<400> SEQUENCE: 608

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Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1           5           10           15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Trp
                20           25           30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Leu
            35           40           45
Tyr Arg Ala Thr Thr Leu Ala Ser Val Pro Ser Arg Phe Lys Gly Ser
            50           55           60
Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
65           70           75           80
Asp Ala Ala Thr Tyr Tyr Gln Ser Tyr Asp Asp Ser Ser Ser Ser Asn
            85           90           95
Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Arg
            100           105           110

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<210> SEQ ID NO 609

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 609

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Asp Ile Val Met Thr Gln Thr Pro Phe Ser Val Glu Ala Ala Val Gly
1           5           10           15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr
                20           25           30
Leu Ser Trp Tyr Arg Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
            35           40           45
Tyr Arg Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys Gly Ser
            50           55           60
Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
65           70           75           80
Asp Ala Ala Thr Tyr Tyr Gln Ser Tyr Asp Asp Ser Ser Ser Asn Asn
            85           90           95
Phe Phe Tyr Gly Phe Gly Gly Gly Thr Glu Val Val Val Arg
            100           105           110

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<210> SEQ ID NO 610

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 610

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Asp Ile Val Met Thr Gln Thr Pro Ser Ser Val Glu Ala Ala Val Gly
1           5           10           15
Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Asn Ile Ser Ser Tyr
                20           25           30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
            35           40           45

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Tyr Arg Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys Gly Ser
50 55 60

Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
65 70 75 80

Asp Ala Ala Thr Tyr Tyr Gln Ser Tyr Asp Asp Ser Arg Ser Ser Asn
85 90 95

Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

<210> SEQ ID NO 611

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 611

Asp Ile Val Met Thr Gln Thr Pro Phe Ser Val Glu Ala Ala Val Gly
1 5 10 15

Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr
20 25 30

Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45

Tyr Arg Ala Ser Pro Leu Ala Ser Val Pro Ser Arg Phe Lys Gly Ser
50 55 60

Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
65 70 75 80

Asp Ala Ala Thr Tyr Tyr Gln Ser Tyr Asp Asp Ser Ser Ser Asn Asn
85 90 95

Phe Phe Tyr Gly Phe Gly Gly Gly Thr Glu Val Val Val Arg
100 105 110

<210> SEQ ID NO 612

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 612

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Ala Ala Val Gly
1 5 10 15

Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Glu Ser Ile Gly Asn Ala
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Ser Leu Leu Ile
35 40 45

Tyr Arg Ala Ser Thr Leu Ala Ser Ile Pro Ser Arg Val Lys Gly Ser
50 55 60

Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
65 70 75 80

Asp Ala Ala Thr Tyr Tyr Gln Ser Tyr Asp Asp Ser Ser Ser Ser Ser
85 90 95

Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105 110

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<210> SEQ ID NO 613
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 613

Ala Gln Val Leu Thr Gln Thr Glu Ser Pro Val Ser Ala Pro Val Gly
1 5 10 15
Gly Thr Val Thr Ile Asn Cys Gln Ala Ser Gln Ser Val Tyr Asp Asn
20 25 30
Asn Tyr Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Asp Ala Ser Lys Leu Ala Ser Val Pro Ser Arg Phe Ser
50 55 60
Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Gly Val Gln
65 70 75 80
Cys Asp Asp Ala Ala Thr Tyr Gln Gly Ser Tyr Tyr Gly Trp Tyr Tyr
85 90 95
Asn Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 614
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 614

Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Ser Ile Ser Cys Gln Ser Ser Gln Ser Val Trp His Asn
20 25 30
Asp Tyr Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys
50 55 60
Gly Ser Gly Ser Gly Thr Gln Phe Ser Leu Thr Ile Ser Gly Val Gln
65 70 75 80
Cys Asp Asp Ala Ala Thr Tyr Ala Gly Gly Tyr Gly Arg Ser Ser Glu
85 90 95
Asn Gly Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 615
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 615

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Ala	Gln	Val	Leu	Thr	Gln	Thr	Glu	Ser	Pro	Val	Ser	Ala	Pro	Val	Gly
1				5					10					15	
Gly	Thr	Val	Thr	Ile	Asn	Cys	Gln	Ala	Ser	Gln	Ser	Val	Tyr	Asp	Asn
			20					25					30		
Asn	Trp	Leu	Ser	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu
		35					40					45			
Leu	Ile	Tyr	Asp	Ala	Ser	Lys	Leu	Ala	Ser	Val	Pro	Ser	Arg	Phe	Ser
		50				55					60				
Gly	Ser	Gly	Ser	Gly	Thr	Gln	Phe	Thr	Leu	Thr	Ile	Ser	Gly	Val	Gln
65					70					75					80
Cys	Asp	Asp	Ala	Ala	Thr	Tyr	Gln	Gly	Ser	Tyr	Tyr	Gly	Trp	Tyr	Tyr
				85					90					95	
Asn	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys				
			100					105							

<210> SEQ ID NO 616

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 616

Ala	Ala	Val	Leu	Thr	Gln	Thr	Pro	Ser	Pro	Val	Ser	Ala	Ala	Val	Gly
1				5					10					15	
Gly	Thr	Val	Ser	Ile	Ser	Cys	Gln	Ser	Ser	Lys	Ser	Val	Tyr	Asn	Asn
			20					25					30		
Asn	Trp	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Arg	Pro	Lys	Leu
		35					40					45			
Leu	Ile	Tyr	Asp	Ala	Ser	Thr	Leu	Asp	Ser	Val	Ser	Ser	Arg	Phe	Lys
		50				55					60				
Gly	Ser	Gly	Ser	Gly	Thr	Gln	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Val	Gln
65					70					75					80
Cys	Asp	Asp	Gly	Ala	Thr	Tyr	Val	Gly	Gly	Tyr	Ser	Ser	Arg	Ser	Asp
				85					90					95	
Asn	Gly	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys				
			100					105							

<210> SEQ ID NO 617

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 617

Ala	Gln	Ala	Leu	Thr	Gln	Thr	Pro	Ser	Pro	Val	Ser	Ala	Ala	Val	Gly
1				5					10					15	
Gly	Thr	Val	Thr	Ile	Ser	Cys	Gln	Ser	Ser	Glu	Ser	Val	Tyr	Asn	His
			20					25					30		
Asn	Trp	Leu	Gly	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu
		35					40					45			
Leu	Ile	Tyr	Asp	Ala	Ser	Thr	Leu	Ala	Ser	Val	Pro	Ser	Arg	Phe	Lys
		50				55					60				

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Gly	Ser	Gly	Ser	Gly	Thr	Gln	Phe	Thr	Leu	Thr	Ile	Ser	Glu	Ser	Gln
65					70					75					80

Cys	Asp	Asp	Ala	Ala	Ile	Tyr	Tyr	Gln	Gly	Tyr	Tyr	Gln	Thr	Ser	Val
			85						90					95	

Trp	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys
	100						105				

<210> SEQ ID NO 618

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 618

Ala	Ala	Val	Leu	Thr	Gln	Thr	Pro	Ser	Pro	Val	Ser	Ala	Ala	Val	Gly
1			5						10					15	

Gly	Thr	Val	Ser	Ala	Ser	Cys	Gln	Ser	Ser	Lys	Ser	Val	Tyr	Asn	Lys
		20					25						30		

Asn	Trp	Leu	Ser	Trp	Phe	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu
	35					40					45				

Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Leu	Ala	Ser	Val	Pro	Ser	Arg	Phe	Lys
	50				55					60					

Gly	Ser	Gly	Ser	Gly	Thr	Gln	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Val	Gln
65					70					75					80

Cys	Asp	Asp	Ala	Ala	Thr	Tyr	Ala	Gly	Gly	Tyr	Ser	Ser	Ser	Ser	Asp
			85						90					95	

Thr	Phe	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys
		100					105					

<210> SEQ ID NO 619

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 619

Ala	Ala	Val	Leu	Thr	Gln	Thr	Pro	Ser	Pro	Val	Ser	Ala	Ala	Val	Gly
1			5						10					15	

Gly	Thr	Val	Thr	Ile	Asn	Cys	Gln	Ser	Ser	Lys	Ser	Val	Tyr	Asn	Lys
		20						25					30		

Asn	Trp	Leu	Ser	Trp	Phe	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu
	35					40					45				

Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Leu	Ala	Ser	Val	Pro	Ser	Arg	Phe	Lys
	50				55					60					

Gly	Ser	Gly	Ser	Gly	Thr	Gln	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Val	Gln
65					70					75					80

Cys	Asp	Asp	Val	Ala	Thr	Tyr	Ala	Gly	Gly	Tyr	Ser	Ser	Ser	Ser	Asp
			85						90					95	

Thr	Phe	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys
		100					105					

<210> SEQ ID NO 620

<211> LENGTH: 109

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 620

Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Ser Ile Ser Cys Gln Ser Ser Lys Ser Val Tyr Asn Lys
20 25 30
Asn Trp Leu Ser Trp Phe Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys
50 55 60
Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Asp Val Gln
65 70 75 80
Cys Asp Asp Ala Ala Thr Tyr Ala Gly Gly Tyr Ser Ser Ser Ser Asp
85 90 95
Thr Phe Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 621
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 621

Ala Ala Val Leu Thr Gln Thr Pro Ser Pro Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Ser Ile Ser Cys Gln Ser Ser Lys Ser Val Tyr Asn Lys
20 25 30
Asn Trp Leu Ser Trp Phe Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Gly Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys
50 55 60
Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Gly Val Gln
65 70 75 80
Cys Asp Asp Ala Ala Thr Tyr Ala Gly Gly Tyr Ser Ser Ser Ser Asp
85 90 95
Thr Phe Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 622
<211> LENGTH: 105
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 622

Ala Gln Val Met Thr Gln Thr Pro Ala Ser Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Thr Ile Ile Cys Gln Ala Ser Glu Ser Ile Ser Ser Tyr

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20	25	30
Leu Asn Trp Tyr Gln Gln Lys	Leu Gly Gln Pro Pro Lys	Leu Leu Ile
35	40	45
Tyr Tyr Ala Ser Thr Leu Ala	Ser Val Pro Ser Arg Phe Lys Gly Ser	
50	55	60
Gly Ser Gly Thr Glu Tyr Thr	Leu Thr Ile Ser Gly Val Gln Cys Asp	
65	70	75
Asp Ala Ala Thr Tyr Gln His	Gly Tyr Gly Trp Tyr Tyr Asn Ala Phe	
85	90	95
Gly Gly Gly Thr Glu Val Val	Val Lys	
100	105	

<210> SEQ ID NO 623
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 623

Asp Ile Val Met Thr Gln Thr	Pro Ala Ser Val Glu Ala Ala Val Gly
1	15
Gly Thr Val Thr Ile Lys Cys	Gln Ala Ser Gln Asn Ile Asp Ser Tyr
20	30
Leu Ala Trp Tyr Gln Gln Lys	Pro Gly Gln Pro Pro Lys Leu Leu Ile
35	45
Tyr Arg Ala Ser Thr Leu Ala	Ser Val Pro Ser Arg Phe Lys Gly Ser
50	60
Gly Ser Gly Thr Glu Phe Thr	Leu Thr Ile Ser Asp Leu Glu Cys Ala
65	80
Asp Ala Ala Thr Tyr Tyr Gln	Ser Tyr Asp Asp Ser Arg Ser Ser Ser
85	95
Phe Phe Tyr Gly Phe Gly Gly	Gly Thr Glu Val Val Val Lys
100	110

<210> SEQ ID NO 624
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 624

Asp Ile Val Met Thr Gln Thr	Pro Ala Ser Val Glu Ala Ala Val Gly
1	15
Gly Thr Val Thr Ile Lys Cys	Gln Ala Ser Gln Asn Ile Asn Ser Tyr
20	30
Leu Ala Trp Tyr Gln Gln Lys	Pro Gly Gln Pro Pro Lys Leu Leu Ile
35	45
Tyr Arg Ala Ser Thr Leu Ala	Ser Val Pro Ser Arg Phe Lys Gly Ser
50	60
Gly Ser Gly Thr Glu Phe Thr	Leu Thr Ile Ser Asp Leu Glu Cys Ala
65	80
Asp Ala Ala Thr Tyr Tyr Gln	Ser Tyr Asp Asp Ser Arg Ser Ile Ser

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	85		90		95
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Phe Phe Tyr Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys
 100 105 110

<210> SEQ ID NO 625
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 625

Asp	Val	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Glu	Ala	Thr	Val	Gly
1				5					10					15	
Gly	Thr	Val	Thr	Ile	Lys	Cys	Gln	Ser	Ser	Gln	Ser	Val	Tyr	Asp	Asn
			20					25					30		
Asn	Ala	Leu	Ala	Trp	Tyr	Gln	Gln	Asn	Ala	Gly	Gln	Arg	Pro	Arg	Leu
	35					40						45			
Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Leu	Ala	Ser	Val	Pro	Ser	Arg	Phe	Ser
	50					55				60					
Ala	Ser	Gly	Ser	Gly	Thr	Glu	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu
65					70					75				80	
Cys	Ala	Asp	Ala	Ala	Thr	Tyr	Tyr	Gln	Cys	Thr	Tyr	Tyr	Val	Ser	Ser
			85						90					95	
Tyr	Gln	Asn	Asp	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys		
	100							105					110		

<210> SEQ ID NO 626
 <211> LENGTH: 105
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 626

Ala	Gln	Val	Leu	Thr	Gln	Thr	Pro	Ala	Ser	Val	Ser	Ala	Ala	Val	Gly
1				5					10					15	
Gly	Thr	Val	Thr	Ile	Asn	Cys	Gln	Ala	Ser	Glu	Ser	Ile	Ser	Ser	Tyr
			20					25					30		
Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Leu	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile
	35					40						45			
Tyr	Tyr	Ala	Ser	Thr	Leu	Ala	Ser	Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser
	50				55						60				
Gly	Ser	Gly	Thr	Glu	Tyr	Thr	Leu	Thr	Ile	Ser	Gly	Val	Gln	Cys	Asp
65				70					75					80	
Asp	Ala	Ala	Thr	Tyr	Gln	His	Gly	Tyr	Gly	Trp	Tyr	Tyr	Asn	Ala	Phe
			85					90					95		
Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys							
	100							105							

<210> SEQ ID NO 627
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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polypeptide

<400> SEQUENCE: 627

Ala Gln Val Leu Thr Gln Thr Pro Ala Ser Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Val Tyr Ser Ser
20 25 30
Asp Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys
50 55 60
Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Leu Gln
65 70 75 80
Cys Asp Asp Ala Ala Thr Tyr Tyr Gln Gly Tyr Tyr Ser Gly Val Val
85 90 95
Tyr Ile Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 628

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 628

Ala Gln Val Leu Thr Gln Thr Pro Ala Ser Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Val Tyr Ser Ser
20 25 30
Asp Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45
Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys
50 55 60
Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Val Gln
65 70 75 80
Cys Asp Asp Ala Ala Thr Tyr Gln Gly Tyr Tyr Ser Gly Val Val Tyr
85 90 95
Ile Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 629

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 629

Ala Gln Val Leu Thr Gln Thr Pro Ala Ser Val Ser Ala Ala Val Gly
1 5 10 15
Gly Thr Val Thr Ile Ser Cys Gln Ser Ser Gln Ser Leu Tyr Asn Ser
20 25 30
Asp Leu Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
35 40 45

-continued

Leu Ile Tyr Lys Ala Ser Thr Leu Ala Ser Val Pro Ser Arg Phe Lys
50 55 60

Gly Ser Gly Ser Gly Thr Gln Phe Thr Leu Thr Ile Ser Glu Val Gln
65 70 75 80

Cys Asp Asp Ala Ala Thr Tyr Gln Gly Tyr Tyr Ser Gly Val Val Tyr
85 90 95

Ile Phe Gly Gly Gly Thr Glu Val Val Val Lys
100 105

<210> SEQ ID NO 630

<211> LENGTH: 123

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 630

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Gly Ser
1 5 10 15

Leu Thr Leu Thr Cys Thr Val Ser Gly Ile Asp Leu Ser Tyr Ala Met
20 25 30

Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Tyr Ile Gly Ile
35 40 45

Ile Ser Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
50 55 60

Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Asp Leu Lys Met Thr
65 70 75 80

Ser Leu Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Gly Ala Tyr
85 90 95

Thr Ser Ser Ser Gly Tyr Tyr Ile Gly Tyr Tyr Ile Tyr Phe Asn Leu
100 105 110

Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 631

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 631

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Gly Ser
1 5 10 15

Leu Thr Leu Thr Cys Thr Val Ser Gly Ile Asp Leu Ser His Ala Met
20 25 30

Ile Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Ile Gly Thr
35 40 45

Ile Gly Ser Arg Asp Thr Ile Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
50 55 60

Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Met Asp Leu Lys Met Thr
65 70 75 80

Ser Leu Thr Ile Glu Asp Thr Ala Thr Tyr Phe Cys Val Arg Asn Ala
85 90 95

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Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
100 105

<210> SEQ ID NO 632

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 632

Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1 5 10 15

Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Tyr Ala Met
20 25 30

Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Ile
35 40 45

Ile Ser Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
50 55 60

Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Thr Ser
65 70 75 80

Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser Tyr Asp
85 90 95

Asp Tyr Gly Asp Tyr Gly Tyr Tyr Ile Tyr Phe Asn Leu Trp Gly Pro
100 105 110

Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 633

<211> LENGTH: 115

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 633

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1 5 10 15

Leu Thr Leu Thr Cys Lys Val Ser Gly Phe Ser Leu Ser Tyr Asp Met
20 25 30

Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Ile
35 40 45

Ile Tyr Ala Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
50 55 60

Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Ala Ser
65 70 75 80

Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Asp Pro Ala
85 90 95

Gly Tyr Ser Ile Ser Phe Gly Leu Trp Gly Pro Gly Thr Leu Val Thr
100 105 110

Val Ser Ser
115

<210> SEQ ID NO 634

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<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 634

Glu Gln Gln Lys Glu Ser Gly Gly Arg Leu Val Met Pro Gly Gly Ser
1 5 10 15
Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Ser Tyr Asn Met Gly
20 25 30
Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Tyr Ile Gly Trp Ile
35 40 45
Ser Thr Gly Gly Ser Ala Tyr Tyr Ala Ser Trp Val Asn Gly Arg Phe
50 55 60
Thr Ile Ser Lys Thr Ser Thr Thr Met Asp Leu Lys Met Thr Ser Leu
65 70 75 80
Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Phe Gly Glu Gly
85 90 95
Pro Thr His His Val Ala Met Val Val Gly Tyr Tyr Ile Tyr Phe Asn
100 105 110
Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 635
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 635

Glu Gln Leu Lys Glu Ser Gly Gly Gly Leu Val Ala Pro Gly Gly Thr
1 5 10 15
Leu Thr Leu Thr Cys Ala Val Ser Gly Phe Ser Ser Tyr Asn Met Gly
20 25 30
Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Tyr Ile Gly Ile Ile
35 40 45
Gly Ala Ser Asp Ser Ala Leu Tyr Ala Ser Trp Ala Lys Gly Arg Phe
50 55 60
Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Thr Ser Pro
65 70 75 80
Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Gly Gly Leu Gly
85 90 95
Leu Ser Thr Gly Phe Ala Leu Trp Gly Pro Gly Thr Leu Val Thr Val
100 105 110
Ser Ser

<210> SEQ ID NO 636
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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<400> SEQUENCE: 636

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Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1           5           10           15
Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Tyr Ala Met
20          25          30
Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Ile
35          40          45
Ile Ser Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
50          55          60
Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Thr Ser
65          70          75          80
Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Tyr Gly Val
85          90          95
Leu Val Pro Met Ala Met Gly Tyr Tyr Ile Tyr Tyr Gly Met Asp Leu
100         105         110
Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115         120

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<210> SEQ ID NO 637

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 637

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Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Gly Pro
1           5           10           15
Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Tyr Asp Met
20          25          30
Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Tyr
35          40          45
Ile Ala Thr Asp Gly Arg Pro Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
50          55          60
Phe Thr Ile Ser Lys Pro Ser Ser Thr Thr Val Asp Leu Lys Ile Thr
65          70          75          80
Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Val Arg Gly Gly
85          90          95
Tyr Ala Gly Gly Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
100         105         110

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<210> SEQ ID NO 638

<211> LENGTH: 124

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 638

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Gln Ser Leu Glu Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Ala Ser
1           5           10           15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Phe Ser Ser Tyr Tyr
20          25          30
Met Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala

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35	40	45
Cys Ile Tyr Ala Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala		
50	55	60
Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu		
65	70	75
Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala		
85	90	95
Arg Ser Tyr Asp Asp Tyr Gly Asp Tyr Gly Tyr Tyr Ile Tyr Phe Asn		
100	105	110
Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser		
115	120	

<210> SEQ ID NO 639
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 639

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser		
1	5	10
Leu Thr Leu Thr Cys Thr Ala Phe Gly Phe Thr Leu Asn Tyr Tyr Ile		
20	25	30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys		
35	40	45
Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys Gly		
50	55	60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln Met		
65	70	75
Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser		
85	90	95
Leu Ser Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser		
100	105	110

<210> SEQ ID NO 640
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 640

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser		
1	5	10
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Thr Leu Asn Tyr Tyr Ile		
20	25	30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys		
35	40	45
Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Thr Trp Ala Lys Gly		
50	55	60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln Met		
65	70	75
Pro Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser		

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85	90	95
Leu Ser Thr Asn Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser		
100	105	110

<210> SEQ ID NO 641
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 641

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala Ser		
1	5	10 15
Leu Thr Leu Thr Cys Lys Ala Ser Gly Ser Asp Phe Ser Tyr Tyr Tyr		
20	25	30
Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ala		
35	40	45
Cys Ile Gly Gly Gly Asn Thr Asp Ala Thr Ala Tyr Ala Arg Trp Ala		
50	55	60
Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ala Thr Thr Val Ala Leu		
65	70	75 80
Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Val		
85	90	95
Arg Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu Trp Gly Pro Gly		
100	105	110
Thr Leu Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 642
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 642

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser		
1	5	10 15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Thr Leu Asn Tyr Tyr Ile		
20	25	30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys		
35	40	45
Ile Asp Asn Ser Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys Gly		
50	55	60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln Met		
65	70	75 80
Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser		
85	90	95
Leu Ser Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser		
100	105	110

<210> SEQ ID NO 643
 <211> LENGTH: 119
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 643

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala Ser
1 5 10 15
Leu Thr Leu Thr Cys Lys Ala Ser Gly Ser Asp Leu Ser Ala Tyr Tyr
20 25 30
Ile Cys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ala
35 40 45
Cys Ile Gly Gly Val Asn Arg Val Ala Thr Ala Tyr Ala Thr Trp Ala
50 55 60
Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu
65 70 75 80
Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Val
85 90 95
Arg Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu Trp Gly Pro Gly
100 105 110
Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 644
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 644

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala Ser
1 5 10 15
Leu Thr Leu Thr Cys Lys Ala Ser Gly Ser Asp Leu Arg Tyr Tyr Tyr
20 25 30
Ser Cys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ala
35 40 45
Cys Val Gly Gly Val Asn Arg Asp Ala Thr Ala Tyr Ala Thr Trp Ala
50 55 60
Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu
65 70 75 80
Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Val
85 90 95
Arg Gly Gly Pro Asp Asn Asn Val Gln Phe Asn Leu Trp Gly Pro Gly
100 105 110
Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 645
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 645

-continued

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1 5 10 15

Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Thr Leu Asn Tyr Tyr Ile
20 25 30

Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
35 40 45

Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys Gly
50 55 60

Arg Phe Thr Ile Ser Arg Thr Ser Ser Thr Thr Val Thr Leu Gln Met
65 70 75 80

Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser
85 90 95

Leu Ser Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
100 105 110

<210> SEQ ID NO 646
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 646

Gln Ser Leu Glu Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Ala Ser
1 5 10 15

Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Phe Ser Ser Tyr Tyr
20 25 30

Met Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala
35 40 45

Cys Ile Tyr Ala Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala
50 55 60

Lys Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu
65 70 75 80

Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala
85 90 95

Arg Ser Tyr Asp Asp Tyr Gly Asp Tyr Tyr Tyr Ile Asp Ala Phe Asp
100 105 110

Pro Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 647
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 647

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1 5 10 15

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Thr Leu Asn Tyr Tyr Ile
20 25 30

Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Cys
35 40 45

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Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys Gly
  50          55          60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Gly Thr Leu Gln Met
  65          70          75          80
Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser
          85          90          95
Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
      100          105          110

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<210> SEQ ID NO 648
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 648

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Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
  1          5          10          15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Thr Leu Asn Tyr Tyr Ile
      20          25          30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Cys
      35          40          45
Ile Asp Asn Val Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys Gly
  50          55          60
Arg Phe Thr Ile Ser Lys Ala Ser Ser Thr Thr Gly Thr Leu Gln Met
  65          70          75          80
Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser
      85          90          95
Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
      100          105          110

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<210> SEQ ID NO 649
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 649

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Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
  1          5          10          15
Leu Thr Val Thr Cys Thr Ala Phe Gly Val Thr Leu Asn Tyr Tyr Ile
      20          25          30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Cys
      35          40          45
Ile Asp Asn Ile Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys Gly
  50          55          60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Gly Thr Leu Gln Met
  65          70          75          80
Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser
      85          90          95
Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
      100          105          110

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<210> SEQ ID NO 650
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 650

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1 5 10 15
Leu Thr Leu Thr Cys Thr Ala Phe Gly Val Thr Leu Asn Tyr Tyr Ile
20 25 30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Gly Cys
35 40 45
Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Ser Trp Ala Lys Gly
50 55 60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Gly Thr Leu Gln Met
65 70 75 80
Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser
85 90 95
Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
100 105 110

<210> SEQ ID NO 651
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 651

Gln Ser Leu Glu Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1 5 10 15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Thr Leu Asn Tyr Tyr Ile
20 25 30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
35 40 45
Ile Asp Asn Ala Asn Gly Arg Thr Tyr Tyr Ala Asn Trp Ala Lys Gly
50 55 60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln Met
65 70 75 80
Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser
85 90 95
Leu Ala Thr Pro Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
100 105 110

<210> SEQ ID NO 652
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 652

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Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Gly Ser
1          5          10          15
Leu Thr Leu Thr Cys Thr Val Ser Gly Ile Asp Leu Ser Tyr Ala Met
          20          25          30
Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Tyr Ile Gly Ile
          35          40          45
Ile Ser Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
          50          55          60
Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Asp Leu Lys Met Thr
          65          70          75          80
Ser Leu Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Gly Tyr Ala
          85          90          95
Gly Ser Ser Trp Asp Gly Tyr Tyr Ile Tyr Phe Asn Leu Trp Gly Pro
          100          105          110
Gly Thr Leu Val Thr Val Ser Ser
          115          120

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<210> SEQ ID NO 653
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

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<400> SEQUENCE: 653

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Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1          5          10          15
Leu Thr Leu Thr Cys Thr Val Ser Gly Val Asp Leu Ser Ser Ala Val
          20          25          30
Thr Trp Val Arg Gln Ala Pro Gly Met Gly Leu Glu Tyr Ile Gly Phe
          35          40          45
Leu Gln Ala Gly Asp Gly Ser Ala Tyr Tyr Ala Ser Trp Ala Lys Gly
          50          55          60
Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Asp Leu Lys Met
          65          70          75          80
Thr Ser Leu Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg His
          85          90          95
Lys Gly Asn Ser Tyr Val Pro Asn Leu Trp Gly Pro Gly Thr Leu Val
          100          105          110
Thr Val Ser Ser
          115

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<210> SEQ ID NO 654
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

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<400> SEQUENCE: 654

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Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1          5          10          15
Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Tyr Ala Met
          20          25          30

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Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Ile
   35           40           45

Ile Ser Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
   50           55           60

Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Thr Ser
   65           70           75           80

Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Tyr Thr
           85           90           95

Ser Ser Ser Gly Tyr Tyr Ile Tyr Tyr Ile Tyr Tyr Gly Met Asp Leu
   100           105           110

Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
   115           120

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<210> SEQ ID NO 655
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 655

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Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
 1           5           10           15

Leu Thr Leu Thr Cys Thr Ile Ser Gly Phe Ser Leu Ser Tyr Ala Met
 20           25           30

Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Ser
 35           40           45

Ile Gly Gly Gly Gly Ser Ala Val Tyr Ala Ser Trp Ala Lys Gly Arg
 50           55           60

Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Arg Ile Thr Ser
 65           70           75           80

Pro Thr Thr Glu Asp Thr Ala Met Tyr Phe Cys Gly Arg Gly Phe Tyr
           85           90           95

Ser Ile Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
 100           105           110

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<210> SEQ ID NO 656
<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 656

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Glu Gln Leu Glu Glu Ser Gly Gly Asp Leu Val Lys Pro Glu Gly Ser
 1           5           10           15

Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Ser Ser Tyr Trp Ile
 20           25           30

Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
 35           40           45

Ile Tyr Ala Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys
 50           55           60

Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln
 65           70           75           80

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-continued

Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85 90 95

Tyr Tyr Ser Tyr Gly Tyr Ala Tyr Ala Thr Tyr Tyr Ile Asp Ala Phe
100 105 110

Asp Pro Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 657
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 657

Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Ala Ser
1 5 10 15

Leu Thr Leu Thr Cys Lys Ala Ser Gly Phe Ser Ser Gly Gln Leu Met
20 25 30

Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
35 40 45

Ile Gly Ser Gly Ser Asn Ala Ile Ser Thr Phe Tyr Ala Ser Trp Ala
50 55 60

Gln Gly Arg Phe Thr Ile Ser Lys Ser Ser Ser Thr Thr Val Thr Leu
65 70 75 80

Gln Leu Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala
85 90 95

Arg Val Gly Ser Asp Asp Tyr Gly Asp Ser Asp Val Phe Asp Pro Trp
100 105 110

Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 658
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 658

Glu Gln Leu Glu Glu Ser Gly Gly Asp Leu Val Lys Pro Glu Gly Ser
1 5 10 15

Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Ser Ser Tyr Trp Ile
20 25 30

Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
35 40 45

Ile Tyr Ala Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys
50 55 60

Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln
65 70 75 80

Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85 90 95

Ser Tyr Asp Asp Tyr Gly Asp Tyr Tyr Tyr Ile Asp Ala Phe Asp Pro
100 105 110

-continued

Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 659
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 659

Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1 5 10 15
 Leu Thr Leu Thr Cys Ala Ala Ser Gly Phe Ser Ser Ser Tyr Phe Met
20 25 30
 Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
35 40 45
 Ile Ala Val Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys
50 55 60
 Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln
65 70 75 80
 Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
85 90 95
 Val Gly Tyr Asp Asp Tyr Gly Asp Ser Asp Ala Phe Asp Pro Trp Gly
100 105 110
 Pro Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 660
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 660

Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Lys Gly Ser
1 5 10 15
 Leu Thr Leu Thr Cys Ala Ala Ser Gly Phe Ser Ser Ser Tyr Phe Met
20 25 30
 Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
35 40 45
 Ile Gly Ser Gly Ser Ser Ala Ile Ser Thr Phe Tyr Ala Ser Trp Ala
50 55 60
 Gln Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu
65 70 75 80
 Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala
85 90 95
 Arg Val Gly Tyr Asp Asp Tyr Gly Asp Ser Asp Ala Phe Asp Pro Trp
100 105 110
 Gly Pro Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 661
 <211> LENGTH: 120

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 661

Gln Ser Val Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1 5 10 15
Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Tyr Ala Met
20 25 30
Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Ile
35 40 45
Ile Ser Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg
50 55 60
Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Thr Ser
65 70 75 80
Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ser Tyr Asp
85 90 95
Asp Tyr Gly Asp Tyr Gly Tyr Tyr Ile Tyr Phe Asn Ile Trp Gly Pro
100 105 110
Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 662
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 662

Gln Ser Leu Glu Glu Ser Gly Gly Arg Leu Val Thr Pro Gly Thr Pro
1 5 10 15
Leu Thr Leu Thr Cys Thr Val Ser Gly Ile Asp Leu Ser Tyr Gly Met
20 25 30
Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly Tyr
35 40 45
Ile Trp Thr Asp Gly Arg Thr Tyr Tyr Ala Asn Trp Ala Lys Gly Arg
50 55 60
Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Asp Leu Lys Ile Thr Ser
65 70 75 80
Pro Thr Ala Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Pro Phe Asp
85 90 95
Gly Asn Tyr Arg Asp Ile Trp Gly Pro Gly Thr Leu Val Thr Val Ser
100 105 110
Leu

<210> SEQ ID NO 663
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 663

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Glu Gln Leu Glu Glu Ser Gly Gly Asp Leu Val Lys Pro Glu Gly Ser
1           5           10           15
Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Ser Ser Tyr Trp Ile
          20           25           30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
          35           40           45
Ile Tyr Ala Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys
          50           55           60
Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln
65           70           75           80
Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
          85           90           95
Tyr Ala Gly Ser Ser Trp Asp Tyr Tyr Ile Asp Ala Phe Asp Pro Trp
          100          105          110
Gly Pro Gly Thr Leu Val Thr Val Ser Ser
          115          120

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<210> SEQ ID NO 664
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
<400> SEQUENCE: 664

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Glu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Glu Gly Ser
1           5           10           15
Leu Thr Leu Thr Cys Ala Ala Ser Gly Phe Ser Ser Ser Tyr Phe Met
          20           25           30
Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys
          35           40           45
Ile Gly Val Gly Gly Ser Gly Ser Thr Tyr Tyr Ala Asn Trp Ala Lys
          50           55           60
Gly Arg Phe Thr Ile Ser Lys Thr Ser Ser Thr Thr Val Thr Leu Gln
65           70           75           80
Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg
          85           90           95
Val Ala Tyr Asp Asp Asp Gly Asp Ser Asp Ala Phe Asp Pro Trp Gly
          100          105          110
Pro Gly Thr Leu Val Thr Val Ser Ser
          115          120

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What is claimed is:

1. An isolated antigen binding construct that binds MAGEB2, wherein the binding construct binds to an epitope comprising a sequence selected from SEQ ID NOs: 2, 3, 4, 562, or 388-554.

2. The isolated antigen binding construct of claim 1, wherein:

a) the antigen binding construct comprises a CDRL1, a CDRL2, a CDRL3, a CDRH1, a CDRH2, and a CDRH3, wherein the CDRL1 comprises a sequence set forth in SEQ ID NO: 85; the CDRL2 comprises a sequence set forth in SEQ ID NO: 86; the CDRL3 comprises a sequence set forth in SEQ ID NO: 87; the

CDRH1 comprises a sequence set forth in SEQ ID NO: 229; the CDRH2 comprises a sequence set forth in SEQ ID NO: 230; and the CDRH3 comprises a sequence set forth in SEQ ID NO: 231; or

b) the antigen binding construct comprises a CDRL1, a CDRL2, a CDRL3, a CDRH1, a CDRH2, and a CDRH3, wherein the CDRL1 comprises a sequence set forth in SEQ ID NO: 73; the CDRL2 comprises a sequence set forth in SEQ ID NO: 74; the CDRL3 comprises a sequence set forth in SEQ ID NO: 75; the CDRH1 comprises a sequence set forth in SEQ ID NO: 217; the CDRH2 comprises a sequence set forth in SEQ

- ID NO: 218; and the CDRH3 comprises a sequence set forth in SEQ ID NO: 219; or
- c) the antigen binding construct comprises a CDRL1, a CDRL2, a CDRL3, a CDRH1, a CDRH2, and a CDRH3, wherein the CDRL1 comprises a sequence set forth in SEQ ID NO: 91; the CDRL2 comprises a sequence set forth in SEQ ID NO: 92; the CDRL3 comprises a sequence set forth in SEQ ID NO: 93; the CDRH1 comprises a sequence set forth in SEQ ID NO: 235; the CDRH2 comprises a sequence set forth in SEQ ID NO: 236; and the CDRH3 comprises a sequence set forth in SEQ ID NO: 237.
3. (canceled)
4. (canceled)
5. The isolated antigen binding construct of claim 1, wherein:
- a) the antigen binding construct comprises a light chain variable region comprising a sequence set forth in SEQ ID NO: 346 and a heavy chain variable region comprising a sequence set forth in SEQ ID NO: 347; or
- b) the antigen binding construct comprises a light chain variable region comprising a sequence set forth in SEQ ID NO: 338 and a heavy chain variable region comprising a sequence set forth in SEQ ID NO: 339; or
- c) the antigen binding construct comprises a light chain variable region comprising a sequence set forth in SEQ ID NO: 350 and a heavy chain variable region comprising a sequence set forth in SEQ ID NO: 351.
6. (canceled)
7. (canceled)
8. The isolated antigen binding construct of claim 1, wherein the antigen binding construct is a monoclonal antibody or an antibody fragment thereof.
9. The isolated antibody of claim 8, wherein the antibody is of the IgG1 or the IgG2 isotype.
10. The isolated antigen binding construct of claim 1, wherein the antigen binding construct is coupled to a labeling group.
11. A nucleic acid molecule encoding the antigen binding construct, or fragment thereof, according to claim 1.
12. A vector comprising a nucleic acid molecule according to claim 11.
13. A host cell comprising the nucleic acid molecule according to claim 12.
14. An antigen binding construct, or fragment thereof, produced by the host cell of claim 13.
15. A method of making the antigen binding construct, or fragment thereof, according to claim 1, comprising the step of preparing the antibody or fragment thereof from a host cell that secretes the antibody.
16. An isolated antibody or fragment thereof that competes for binding to MAGEB2 with an antibody or fragment thereof of claim 1.
17. A method of making an antibody that binds to MAGEB2 comprising immunizing an animal with a peptide comprising a sequence selected from SEQ ID NO: 2, 3, 4, or 562, and isolating from said animal antibodies that bind to MAGEB2.
18. An antibody made by the method of claim 17.
19. A method for treating a tumor in a subject, said method comprising: determining the subject as responsive to

treatment with an anti-MAGEB2 therapeutic by obtaining a sample from the subject, wherein the sample comprises a cell from the tumor, measuring the level of MAGEB2 in the sample using an antigen binding construct of claim 1, and determining the subject as responsive to treatment with an anti-MAGEB2 therapeutic, and administering to the subject an effective amount of the anti-MAGEB2 therapeutic.

20. A method of identifying a subject as needing an anti-MAGEB2 therapeutic comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct of claim 1; and b) identifying the subject as needing the anti-MAGEB2 therapeutic when the level of MAGEB2 is increased relative to a control.

21. A method of determining treatment for a subject with a MAGEB2 positive tumor comprising: determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct of claim 1; and determining the treatment as comprising an anti-MAGEB2 therapeutic when the level of MAGEB2 is increased, relative to a control.

22. A method of determining efficacy of treatment with an anti-MAGEB2 therapeutic in a subject comprising: determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct of claim 1 before treatment with an anti-MAGEB2 therapeutic and after treatment with an anti-MAGEB2 therapeutic; and determining the treatment as effective when the level of MAGEB2 positive tumor cells is decreased after treatment with the anti-MAGEB2 therapeutic.

23. A method of diagnosing a subject with a tumor, comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct of claim 1; and b) diagnosing the subject as having a MAGEB2 positive tumor when the level of MAGEB2 is increased relative to a control.

24. A method of identifying a subject having a MAGEB2 positive tumor comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct of claim 1; and b) identifying the subject as having a MAGEB2 positive tumor when the level of MAGEB2 is increased relative to a control.

25. A method of identifying a subject as needing an anti-MAGEB2 therapeutic comprising: a) determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct of claim 1; and b) identifying the subject as needing the anti-MAGEB2 therapeutic when the level of MAGEB2 is increased relative to a control.

26. A method of determining treatment for a subject with a MAGEB2 positive tumor comprising: determining the level of MAGEB2 in a sample obtained from the subject using an antigen binding construct of claim 1; and determining the treatment as comprising an anti-MAGEB2 therapeutic when the level of MAGEB2 is increased, relative to a control.

27. The method of claim 19, wherein MAGEB2 is detected using an IHC assay.

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